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LearNova

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Chapter 1: Introduction

1.1 Background

The global education sector is undergoing a fundamental structural transformation, characterized by a pronounced shift from institution-centered, synchronous delivery models toward digitally mediated learning environments that prioritize flexibility, scalability, and continuous skill acquisition. This transition has been driven by rapid advances in information and communication technologies and by the intensifying labor market demand for adaptive, lifelong learning pathways capable of accommodating diverse professional trajectories across geographic and socioeconomic boundaries [31], [80]. Consequently, digital education platforms have assumed a central role within modern learning ecosystems, progressively displacing traditional institutional models as the primary mechanism through which technical competencies are acquired, updated, and verified.

Despite this substantial expansion, persistent concerns remain regarding the pedagogical efficacy of prevailing digital delivery models. The majority of contemporary e-learning platforms are architecturally designed as content distribution systems repositories providing learners with extensive collections of recorded lectures, static readings, and pre-packaged instructional materials. While such platforms have successfully addressed systemic barriers related to accessibility and geographical reach, they tend to overlook critical dimensions of learner cognition, individual difference, and dynamic engagement. Empirical analyses of learner behavior in video-based instruction indicate that engagement deteriorates significantly after only a few minutes of uninterrupted passive exposure, a pattern that reveals a fundamental incompatibility between broadcast-style delivery and the neurocognitive constraints governing effective knowledge acquisition a limitation central to Cognitive Load Theory, which posits that working memory capacity is inherently limited and highly sensitive to instructional design [1], [77], [81]. The resulting disengagement is not attributable to learner apathy; rather, it reflects a deep mismatch between the medium and the learner's cognitive architecture.

Beyond cognitive disengagement, asynchronous digital learning environments consistently lack the instructional scaffolding and social interaction inherent to traditional classroom settings. This fundamental absence contributes to learner isolation, diminished intrinsic motivation, and reduced persistence conditions that collectively undermine course completion at scale. Empirical evidence drawn from large-scale online learning initiatives, particularly Massive Open Online Courses (MOOCs), indicates completion rates consistently reported within the range of 10 to 15 percent, despite characteristically high initial enrollment figures [9], [24], [26]. These findings highlight a systemic deficiency in content-centric models that neither adapt to individual behavioral patterns nor provide timely, proactive intervention when disengagement emerges.

Concurrently, the accelerating pace of technological transformation has deepened the disconnect between formal educational outputs and evolving labor market requirements. University curricula, constrained by accreditation cycles and institutional inertia, frequently fail to keep pace with the rapidly shifting competency demands of high-growth technical domains particularly data analytics, artificial intelligence, and software engineering [31], [38]. This curricular lag has produced a paradoxical labor market condition in which elevated graduate unemployment coexists with acute shortages of job-

ready technical professionals. Within the Egyptian and broader MENA context, this paradox is particularly acute: tertiary degree holders exhibit a youth unemployment rate of 23.1 percent more than double the 11.3 percent recorded among secondary school graduates and Egypt's graduate skill-set ranks 133rd out of 141 countries globally [32], [33], [34]. This counterintuitive pattern signals a systemic misalignment between academic output and occupational demand, not a deficit of educational participation.

Addressing these compounding failures requires moving beyond static content delivery toward adaptive, learner-centered architectures that align individual cognitive profiles, vocational aptitudes, and behavioral tendencies with real-world occupational requirements. Within this context, there is an increasing demand for intelligent educational systems that integrate psychometric-informed guidance, personalized learning pathways, and continuous, proactive learner support systems capable of monitoring progress, detecting early indicators of disengagement, and dynamically calibrating the instructional experience in response to individual need. These capabilities constitute the foundational motivation for the development of LearNova, an adaptive educational ecosystem engineered to address the persistent gap between content-rich but learner-blind platforms and the evidence-based, personalized guidance that empirical evidence suggests contemporary learners require ultimately guiding them into high-demand specializations through continuous, data-driven alignment between individual profile and occupational demand.

1.2 Motivation

The motivation for this research originates in persistent socio-economic structural challenges observed within the labor market of the Middle East and North Africa (MENA) region specifically, the widening disconnect between formal academic qualifications and demonstrable professional employability. Despite a steady increase in the number of graduates from technical and computing-related disciplines, regional industries continue to report pronounced difficulties in recruiting candidates who possess the applied competencies required for immediate workforce integration [32], [33]. This situation has produced a well-documented paradox in which higher levels of educational attainment are associated not with reduced unemployment, but with its elevation: tertiary degree holders in Egypt exhibit a youth unemployment rate of 23.1 percent, compared to 11.3 percent among secondary school leavers, and the country's graduate skill-set ranks 133rd out of 141 nations globally [34].

A primary factor underlying this paradox is the systematic misalignment between university curricula and the rapidly evolving competency requirements of the technology sector. Academic programs are frequently structured around theoretical foundations and standardized learning outcomes, while industry roles in fast-developing fields particularly data analytics, artificial intelligence, and software engineering increasingly demand applied skills, familiarity with contemporary toolchains, and the capacity to adapt within continuously changing technological environments [31], [38]. Curricular updates, constrained by accreditation cycles and institutional inertia, frequently lag several years behind real-world practice, rendering graduates technically credentialed but occupationally underprepared.

Compounding this institutional limitation, learners navigating the expanding landscape of technical career pathways face a distinct and equally consequential challenge: the absence of structured, evidence-based guidance at the point of specialization selection. The proliferation of available tracks encompassing data analytics, data engineering, machine learning, and cybersecurity, among others confronts novice learners with a decision environment of considerable complexity. In the absence of diagnostic or advisory mechanisms grounded in individual aptitude assessment, decision-making tends to be guided by incomplete information, social influence, or prevailing market trends rather than evidence-based self-evaluation. This phenomenon, described within decision theory as the "paradox of choice," has been shown to increase decision anxiety, reduce long-term satisfaction with outcomes, and elevate the probability of costly specialization mismatches [4], [31].

In response to these converging failures, this project is motivated by the need to reconceptualize digital education systems as active, data-driven decision-support environments rather than passive content repositories. Specifically, the motivation centers on three imperatives: first, to operationalize psychometric and cognitive assessment as a functional routing mechanism that guides learners toward specializations aligned with their individual aptitude profiles; second, to deliver instructional content through an adaptive, modality-aware architecture that dynamically calibrates presentation format to the learner's detected cognitive style; and third, to sustain learner engagement and well-being through a proactive AI-assisted mentorship framework capable of detecting early

disengagement signals and triggering timely, human-supervised intervention. These imperatives underpin the design and implementation of LearNova an adaptive educational ecosystem developed specifically to address the education-to-employment gap through personalized, research-driven guidance rather than uniform instructional delivery.

1.3 Problem Statement

Despite the rapid growth and widespread adoption of educational technology platforms, significant architectural limitations persist that continue to undermine learner success in digital learning environments. While access to educational content has expanded at an unprecedented scale, the capacity of independent learners to effectively navigate, persist within, and derive measurable occupational value from online learning pathways has not advanced commensurately. This discrepancy reflects not a failure of content availability but a fundamental deficiency in the pedagogical assumptions and system architectures underlying contemporary e-learning platforms.

The first and most pervasive of these deficiencies is the homogeneity of instructional delivery models. The majority of platforms adopt a standardized pedagogical approach in which identical learning materials commonly pre-recorded video lectures and static textual resources are distributed uniformly to all learners, irrespective of individual differences in cognitive preference, information processing style, or prior knowledge [77], [80]. This "one-size-fits-all" model requires learners to adapt to rigid system structures rather than benefiting from systems that dynamically adapt to their learning needs. Such misalignment introduces measurable friction into the learning process, contributing directly to the engagement deterioration and attritional dropout patterns documented extensively across large-scale MOOC studies [9], [24], [26].

A second critical deficiency arises from the absence of structured diagnostic and career guidance mechanisms within existing platforms. The proliferation of specializations and learning tracks within technical disciplines has created an environment in which learners are confronted with an overwhelming volume of choices. Without psychometrically-grounded advisory systems, most students lack the analytical tools necessary to assess which pathways align with their individual aptitudes, behavioral tendencies, and long-term professional objectives. Decision-making consequently defaults to incomplete information or externally imposed preferences, producing inefficient learning trajectories, elevated regret rates, and suboptimal career outcomes a pattern empirically observed in both regional and global graduate cohorts [34], [39], [70].

A third, equally consequential deficiency is the near-universal reliance on reactive rather than proactive engagement models, creating a critical architectural void herein termed the "Missing Middle." Existing platforms offer limited intervention capacity unless learners explicitly initiate a support request a condition that systematically fails populations most at risk of attrition. Furthermore, these platforms frequently lack intrinsic motivation mechanics to combat the inherent social isolation of asynchronous digital learning. Empirical observations across large-scale online learning initiatives confirm that completion rates remain persistently low, commonly reported in the range of 10 to 15 percent, with social isolation, insufficient instructional support, and the absence of early-warning indicators cited as dominant contributing factors [9], [24], [26]. The absence of continuous behavioral monitoring and timely intervention frameworks significantly undermines learner persistence and renders the educational investment inefficient.

Collectively, these three deficiencies content homogeneity, guidance vacuum, and reactive support architecture delineate a systemic failure in the prevailing e-learning paradigm: one that prioritizes content availability over learner-centered adaptivity and

treats engagement as a passive outcome rather than an actively managed process. Addressing these deficiencies requires not merely incremental improvements to existing platform features, but a fundamental architectural rethinking one that integrates adaptive multimodal content delivery, psychometric-informed pathway recommendation, and proactive AI-assisted mentorship into a unified, evidence-based educational ecosystem.

1.4 Project Objectives

The primary objective of this project is to design, implement, and validate **LearNova** an intelligent, adaptive educational ecosystem engineered to address the structural limitations inherent in conventional, content-centric e-learning platforms. Rather than treating content availability as a terminal goal, the system is architected around learner-centered guidance, evidence-driven pathway recommendation, and continuous adaptive support. The following technical and functional objectives collectively operationalize this overarching aim, each grounded in established theoretical frameworks and realized through a production-grade, multi-tier technical implementation.

Objective 1: Design and Implementation of a Five-Point Psychometric Diagnostic and the Smart Compatibility Engine

The first objective is the design and full implementation of a **Smart Compatibility Engine** capable of generating informed, evidence-based academic pathway recommendations. This component operates through a multi-stage, five-point psychometric calibration process comprising five structurally distinct assessment instruments: a 120-item personality inventory derived from the IPIP implementation of the Five-Factor Model (OCEAN) [58]; a 20-item Situational Judgement Test (SJT) evaluating behavioral competencies including communication, teamwork, and ethical reasoning; a 16-item learning style assessment grounded in the VARK framework [18]; a 60-item career interest inventory structured according to the O*NET RIASEC taxonomy; and a 35-item cognitive aptitude battery spanning logical and abstract reasoning sections. Each instrument contributes distinct feature scores to a unified learner profile. Those features are subsequently normalized and processed by a weighted scoring algorithm to produce three final track scores corresponding to the available specialization pathways Data Analytics (DA), Data Engineering (DE), and Data Science (DS). The track assignment is determined by the argmax of these scores, wherein a base score derived from O*NET vocational anchors is combined with a modifier score computed from the full feature vector according to empirically calibrated, per-track weights. Importantly, this output is framed as a transparent, decision-support recommendation rather than an irreversible algorithmic assignment, thereby preserving learner agency in accordance with the autonomy provisions of Self-Determination Theory [12].

Objective 2: Development of an Adaptive Content Delivery System Grounded in Multimodal Learning Theory

The learner profile yielded by the calibration process serves as the primary input to the second objective: the development of an adaptive learning environment that dynamically configures content presentation modalities in response to each learner's established cognitive preferences. Drawing upon the modality alignment principle articulated in Mayer's Multimedia Learning theory [2] and the VARK framework [18], the system associates each curricular topic with three structurally parallel content representations a Visual slot (video lectures and annotated diagrams), an Auditory slot (podcast-style narrative explanations), and a Textual slot (structured transcripts and annotated articles) stored within a polymorphic content architecture in the production database. At runtime, the system resolves the learner's calibrated learning style from the student_profiles table

and serves the corresponding content variant, thereby minimizing extraneous cognitive load in accordance with Sweller's Cognitive Load Theory [1]. The learning style model is further refined over time through a Bayesian update mechanism, wherein qualifying content engagement sessions update posterior probability parameters, enabling the system to transition from initial calibration-based delivery toward an increasingly evidence-grounded adaptive state as behavioral data accumulates [6].

Objective 3: Implementation of an AI-Assisted Mentorship Framework with Affective Risk Detection

The project further aims to implement a production-grade, AI-assisted mentorship framework providing continuous learner support throughout the educational lifecycle. This framework is instantiated through **Mitchy**, a conversational pedagogical agent powered by a Large Language Model (LLM) backend accessed via a hardened, fault-tolerant client architecture incorporating model fallback chains, configurable timeout thresholds, and mandatory non-empty response guarantees to ensure service continuity irrespective of upstream API failures. The agent operates through a Retrieval-Augmented Generation (RAG) architecture, wherein curriculum content is chunked, vector-embedded, and indexed in a dedicated knowledge store, enabling contextually grounded, curriculum-aligned pedagogical responses grounded in the scaffolding principles theorized by Vygotsky [4]. Beyond its instructional function, Mitchy serves concurrently as an affective risk-detection mechanism, performing sentiment analysis on each inbound learner message and triggering escalating intervention protocols when computed sentiment scores breach configurable thresholds. These protocols are logged as structured records and escalated to human administrators via the administrative dashboard and push notification infrastructure, instantiating a human-in-the-loop safety architecture informed by research on the integration of sentiment analysis within learning analytics systems [25] and machine learning-based dropout prediction [26].

Objective 4: Delivery of a Cross-Platform Mobile Interface and Supporting Administrative Infrastructure

Given the documented reliance of the target demographic on mobile devices as primary learning environments [29], a further objective is the delivery of a production-ready, cross-platform mobile application ensuring equitable accessibility across heterogeneous device ecosystems. The learner-facing interface is implemented using the Flutter framework [82], targeting Android and iOS operating systems, and adheres to the Material Design 3 specification [86] to ensure interface familiarity and ergonomic efficiency. Local state management via Riverpod [83] and session persistence mechanisms are implemented to safeguard learner progress against network interruptions and application lifecycle events. In parallel, a web-based administrative dashboard exposes a dedicated control surface for curriculum management, real-time risk monitoring, and manual intervention dispatch to flagged learners. The full communication stack is secured through a layered architecture wherein the mobile client exclusively invokes authenticated Supabase Edge Functions [84], which in turn forward verified, server-to-server requests to the Railway-hosted scoring and AI backend ensuring that service credentials, scoring logic, and sensitive ML outputs remain entirely concealed from the client layer.

Objective 5: Integration of a Gamification and Learner Retention Ecosystem

The final objective is the design and implementation of a gamification framework engineered to sustain intrinsic motivation and mitigate the learner attrition patterns documented extensively in large-scale online learning research [9]. This ecosystem comprises several interoperating mechanisms: a tiered Experience Points (XP) system awarding performance-differentiated rewards across topic completion and module assessment events; a streak-tracking subsystem exploiting the behavioral-economic principle of loss aversion to sustain daily engagement continuity [51]; a snapshot-based leaderboard architecture enabling track-scoped peer comparison; a weekly challenges system providing time-bounded, module-linked assessment events with difficulty-tiered reward structures; and an achievement and badge framework that materializes progression milestones as persistent, auditable records. The design is grounded in Self-Determination Theory [12], with competitive mechanics balanced against autonomy-preserving and collaborative features informed by research on the comparative effects of competitive and collaborative gamification on student learning outcomes [10], and the motivational architecture is further guided by the Octalysis Framework [52], which governs the deliberate calibration of urgency-based and empowerment-based motivational mechanics. Collectively, these retention mechanisms are designed to promote sustained, intrinsically motivated learning persistence constituting a measurable contribution to the platform's overarching goal of transforming passive content consumption into an engaged, accountable educational journey.

1.5 Project Scope

This section delineates the scope of the LearNova project by establishing its functional, curricular, and technical boundaries as realized in the current implementation. Defining a precise scope is essential not only to ensure the feasibility of the system within the constraints of an academic graduation project, but also to situate its contributions accurately within the broader landscape of educational technology research. The boundaries presented here reflect deliberate design decisions that balance architectural ambition with the pedagogical and engineering realities of a production-deployed, multi-tier platform.

1.5.1 Functional Scope

The functional scope of LearNova is organized around four interlocking subsystems that collectively address the learner's full educational lifecycle: a diagnostic calibration and pathway assignment engine, an adaptive content delivery environment, an AI-assisted mentorship and affective monitoring framework, and a gamified engagement retention ecosystem. Each subsystem is architecturally distinct yet operationally interdependent, with outputs from earlier stages informing the behavior of subsequent ones.

The platform's decision-support architecture is centered on the **Smart Compatibility Engine**, which operationalizes a five-point psychometric calibration process to generate an evidence-based specialization assignment for each learner. Learner responses across personality, soft skills, learning style, career interest, and cognitive aptitude assessments are processed by a backend scoring service implemented as a FastAPI application deployed on Railway which computes normalized feature scores, applies empirically calibrated per-track modifier weights, and determines the learner's specialization assignment (DA, DE, or DS) via an argmax decision function. All enrolled learners commence on a shared **Foundation** path, which is architecturally distinct from the assigned specialization track: the Foundation phase establishes a common technical baseline, while the assigned track written to student_profiles.assigned_track following diagnostic completion represents the system's forward-projected specialization recommendation. These recommendations are framed as decision-support outputs rather than irreversible algorithmic assignments, in alignment with the autonomy provisions of Self-Determination Theory [12].

The **Adaptive Learning Environment** dynamically configures instructional modality presentation in accordance with each learner's calibrated learning style. Each topic within the curriculum is associated with up to three structurally parallel content representations Visual, Auditory, and Textual stored in the topic_resources table and served at runtime based on the learner's profile. Content engagement data is captured through the content_engagement_logs table and fed into a Bayesian update mechanism that incrementally refines the system's estimate of the learner's dominant modality as behavioral evidence accumulates across sessions, enabling the adaptive model to evolve beyond static initial calibration toward a progressively evidence-grounded state, consistent with the adaptive scaffolding principles articulated by Wang et al. [6].

The **AI Mentorship Module**, instantiated as the agent **Mitchy**, operates across two concurrent functional layers. At the learner interaction level, the agent provides real-time pedagogical guidance including curriculum navigation support, concept clarification, and Socratic inquiry grounded in retrieved curriculum content via a Retrieval-Augmented Generation (RAG) architecture backed by pgvector embeddings stored in the document_chunks table. At the monitoring level, every learner message is subjected to sentiment analysis, with computed scores persisted to the student_sentiment_history table and aggregated into structured risk records within the risk_scores table. When risk thresholds are breached, the system automatically triggers in-app notifications and, where device tokens are registered, Firebase Cloud Messaging (FCM) push alerts to the administrative interface, initiating a human-led intervention workflow logged in the intervention_logs table a design informed by Gupta's research on the integration of sentiment analysis within learning analytics frameworks [25].

Learner retention is further sustained through a gamification ecosystem encompassing XP accumulation, streak tracking, time-bounded weekly challenges, snapshot-based leaderboards, and a persistent achievement and badge framework. These mechanisms are implemented without requiring synchronous peer-to-peer communication, thereby preserving learner privacy while sustaining the social comparative motivation associated with relatedness fulfillment as theorized within Self-Determination Theory [12] and validated empirically in research on social gamification elements [11].

1.5.2 Content Scope

The functional capabilities described above operate within a deliberately constrained content domain. The content scope of the LearNova platform is explicitly limited to the field of Computer Science and data-oriented technical disciplines. In its current iteration, the system is structured around a **Data Analytics diploma** as the representative use case through which the adaptive and recommendation mechanisms are validated. This specialization was selected on the basis of its documented relevance within the contemporary regional technology labor market, its suitability for modular, competency-based instruction, and the availability of verified skill cluster data derived from Egyptian job market analyses conducted via Wuzzuf job posting aggregation [97], [98].

The curriculum is organized as a strict four-level hierarchy Courses → Levels → Modules → Topics with all active content records governed by is_active flags and role-based access policies enforced through Row Level Security at the database layer [84]. All learners are initially enrolled in the shared **Literacy Phase** (Foundation track), which establishes a common baseline of technical competency before branching into one of three structured specialization sub-tracks: Data Analysis, Data Engineering, or Data Science. Progression through the curriculum is governed by a mastery-based unlocking mechanism requiring a passing score threshold on module assessments before subsequent content is made accessible, in alignment with Bloom's Taxonomy-informed assessment design principles [2]. While the system architecture is designed with horizontal expansion in mind and skill cluster data for adjacent technical domains, including Cybersecurity, AI Engineering, and Software Development, has been

documented through validated labor market analysis such expansions remain outside the boundaries of the current project phase.

1.5.3 Technical Scope

From a technical perspective, the LearNova system is implemented as a multi-tier, service-oriented architecture comprising three primary layers: a cross-platform mobile client, a Backend-as-a-Service data and authentication layer, and a Railway-hosted serverless scoring and intelligence backend.

The learner-facing interface is a production-deployed mobile application developed using the Flutter framework [82], targeting Android (API Level 28 and above) and iOS (version 15 and above), with state management handled via Riverpod [83] and local persistence mechanisms implemented to ensure session continuity across network interruptions. All client-server communication is mediated exclusively through Supabase Edge Functions, authenticated via JSON Web Tokens issued by the Supabase GoTrue identity gateway [84], which forward trusted server-to-server requests to the Railway-hosted backend using private API keys ensuring that scoring logic, LLM credentials, and sensitive ML outputs remain entirely isolated from the client layer.

The administrative interface accessible as a privileged module within the same application binary provides dedicated capabilities for curriculum lifecycle management, real-time learner risk monitoring, engagement analytics, and content format effectiveness reporting, all surfaced through purpose-built database functions operating over the platform's relational schema [84]. Administrative decisions are enacted through a structured intervention workflow supported by manual escalation capabilities dispatched via the in-app notification system and FCM push infrastructure.

The backend intelligence layer hosts a daily Machine Learning pipeline executed as a Railway cron service, which computes per-student and per-topic metrics including concept decay scores, engagement velocity, topic struggle indices, exponential moving averages of engagement, and probabilistic retention and struggle estimates persisted to dedicated analytics tables. A secondary daily cron process synthesizes these metrics into administrative insight briefings, in accordance with the learning analytics principles articulated by Wilson [29]. Storage for media assets including calibration question images, topic content materials, and student-uploaded submissions is managed through three dedicated Supabase storage buckets served via Content Delivery Network for low-latency mobile access.

1.5.4 Limitations and Delimitations

Several boundaries and known constraints govern the current implementation, the explicit acknowledgment of which is essential for accurate contextualization of the project's contributions.

With respect to social interaction, the platform supports indirect peer engagement through leaderboard-based comparison and achievement visibility, but deliberately excludes

synchronous peer-to-peer messaging, discussion forums, and social networking functionalities. This delimitation reflects both scope constraints and an intentional design emphasis on individualized learning trajectories, consistent with the privacy-preserving autonomy provisions outlined within Self-Determination Theory [12].

Regarding instructional content, LearNova functions as a content curation and adaptive delivery platform rather than a primary instructional authoring system. Educational resources are structured within the platform's polymorphic content architecture by administrators but are sourced from existing, credible instructional providers, consistent with an Open Educational Resources curation model [80]. Advanced immersive learning technologies specifically Virtual Reality and Augmented Reality laboratory environments supported by platforms such as Unity [88] are identified as high-priority future enhancements grounded in the kinesthetic modality gap identified in the system's future research directions, but are explicitly excluded from the current implementation.

With respect to known technical constraints, the system's adaptive Bayesian model is subject to a cold-start latency period during which personalization quality is necessarily limited, as posterior learning style estimates require a minimum volume of qualifying engagement sessions before diverging meaningfully from initial calibration-derived priors a limitation consistent with the cold-start problem documented in recommender systems and adaptive learning architectures [29]. Additionally, the mobile-first architecture presents an inherent pedagogical boundary for hands-on technical skill development: the platform is optimized for knowledge consumption and formative assessment, but does not currently support in-platform code execution environments required for authentic programming practice a constraint that future cross-platform expansion incorporating cloud-hosted development environments is intended to address [91]. Finally, while Mitchy's hardened fallback architecture guarantees non-empty responses under LLM API failure conditions, the pedagogical quality of those fallback responses during periods of extended API unavailability remains bounded by the local affective analysis module rather than the full LLM reasoning pipeline [25].

1.6 Significance of the Project

The significance of the **LearNova** project extends beyond the delivery of a digital learning application, addressing structural failures that operate simultaneously at the macroeconomic, pedagogical, and individual levels. By integrating production-deployed adaptive learning mechanisms, psychometric decision-support, real-time affective monitoring, and market-validated curriculum design into a single cohesive ecosystem, the system contributes to ongoing efforts to reform educational effectiveness, workforce readiness, and individual career alignment. The impact of this contribution may be examined across three primary dimensions: economic relevance, pedagogical advancement, and individual and social significance.

1.6.1 Economic Significance

From a macroeconomic perspective, the inefficiencies arising from skills mismatch and workforce disengagement constitute one of the most consequential and preventable sources of global economic loss. The Gallup State of the Global Workplace Report estimates that low employee engagement costs the global economy approximately $8.8 trillion annually equivalent to roughly nine percent of global GDP [35]. At the sectoral level, research by Korn Ferry projects that a talent shortage of 85 million people could leave $8.5 trillion in annual revenues unrealized by 2030 [37], while BCG has estimated that skills mismatch currently imposes a six percent productivity loss on the global economy, a figure projected to approach ten percent of GDP without structural intervention [38]. The World Economic Forum further identifies the skills gap as the single most significant barrier to business transformation cited by 63 percent of employers, with 40 percent of core job skills expected to undergo substantive change in the coming years and 78 million net new roles projected globally by 2030, concentrated in AI, data, and technology domains [31].

These pressures are acutely intensified within the MENA region. The International Labour Organization reports a youth unemployment rate of 24.4 percent across the region nearly double the global average alongside a NEET rate of 31.5 percent among young people not in employment, education, or training [32]. In Egypt specifically, the paradox of higher education is particularly pronounced: tertiary degree holders face an unemployment rate of 23.1 percent, exceeding that of secondary school graduates at 11.3 percent, while the country ranks 133rd out of 141 nations on the WEF's graduate skill-set index [32], [33]. This pattern is not indicative of insufficient educational participation; it reflects a deep structural misalignment between academic outputs and labor market requirements precisely the failure that **LearNova's** curriculum architecture, grounded in validated skill cluster data derived from Egyptian job market analysis [97], [98], is designed to address.

From an organizational perspective, the system's competency-based, market-aligned graduate profile offers a direct contribution to recruitment efficiency. Industry analyses consistently report that replacing a technical professional costs between 80 and 150 percent of annual salary, while replacing senior leadership may reach 200 to 400 percent [73], [74]. By producing graduates whose psychometric profiles, soft skill competencies, and technical aptitudes are verified and documented through the platform's diagnostic architecture, **LearNova** reduces the information asymmetry that drives costly role

mismatches functioning as a preventative mechanism for both learners and the organizations that employ them.

1.6.2 Pedagogical Significance

Pedagogically, this project contributes to the evolving paradigm shift from passive, broadcast-style content delivery toward adaptive, learner-responsive instructional architectures. Large-scale MOOC research consistently reports course completion rates in the range of 10 to 15 percent [9], outcomes attributed not primarily to learner apathy but to the structural limitations of homogeneous content delivery: prolonged passive exposure, insufficient adaptive scaffolding, and the absence of timely support mechanisms. Analysis of 6.9 million video sessions on edX demonstrated that median learner engagement with educational video collapses after approximately six minutes regardless of content length [81], a finding that establishes the cognitive case for the micro-structured, modality-adaptive content architecture implemented in **LearNova**.

The platform's adaptive delivery system addresses this challenge through mechanisms validated by peer-reviewed research. Controlled studies comparing Adaptive Microlearning environments against conventional instruction have reported cognitive load reductions with a mean difference of −20.02 ($p < 0.05$), learning adaptability improvements of +40.72 ($p < 0.05$), and time-to-mastery reductions of up to 50 percent [6]. Multimodal instructional delivery, grounded in Mayer's Multimedia Learning theory [2] and Paivio's Dual Coding framework [3], has been shown to produce significantly superior delayed recall, with multimedia learners outperforming text-only counterparts at a two-week interval ($M = 26.8$ vs. $M = 16.12$, $F = 15.58$, $P < 0.001$) [7]. These findings directly validate the polymorphic content architecture deployed in the platform, wherein each learning objective is represented across three distinct modality variants served dynamically in accordance with each learner's calibrated profile [18].

The inclusion of psychometric-informed pathway guidance further constitutes a contribution to pedagogical inclusivity. Data from a global VARK study encompassing over one million participants indicates that 66 percent of learners exhibit multimodal preferences, and that the kinesthetic modality almost entirely absent from conventional e-learning interfaces is present in 87.6 percent of learner profiles [18], [19]. By designing an adaptive system that acknowledges and responds to this cognitive diversity rather than assuming the uniform read/write preference for which most academic content is presently structured, the platform seeks to deliver more equitable learning experiences across heterogeneous cognitive profiles, consistent with the modality alignment principles articulated by Mayer [2].

Finally, **LearNova's** affective computing layer constitutes a contribution to proactive learner support at a scale and granularity not available in current commercial e-learning contexts. Research on sentiment analysis integrated within learning analytics frameworks demonstrates that machine learning-based dropout prediction models achieve accuracy rates exceeding 84 percent, with the first two weeks of engagement identified as the strongest single predictive window [25], [26]. By operationalizing these findings through a structured sentiment monitoring, risk scoring, and human-led intervention architecture

and by ensuring that risk indicators surface to administrators rather than triggering automated enforcement the system implements an evidence-based, human-in-the-loop intervention model that addresses a functional gap unserved by any current commercial platform [25].

1.6.3 Individual and Social Significance

At the individual level, the project addresses a well-documented psychological burden that disproportionately affects learners at the earliest and most consequential stages of their educational trajectories. The proliferation of available specializations and technical pathways has been associated with heightened decision anxiety and choice paralysis: 63 percent of students globally report feeling overwhelmed by the process of selecting a major [34], and 51 percent indicate a lack of confidence in their chosen career path at the point of enrollment [33]. The long-term consequences of this guidance deficit are substantial. The Strada Institute reports that 52 percent of graduates are underemployed one year after graduation, and that 45 percent remain so a decade later with those who begin underemployed being 3.5 times more likely to remain in that state [40]. Federal Reserve data further indicates that graduates in the most regretted major categories earn only 25 percent more than secondary school graduates, compared to the 88 percent wage premium associated with correctly matched roles [39].

By providing structured, evidence-based pathway recommendations generated through a multi-dimensional psychometric calibration process, the **LearNova** system seeks to reduce the probability of initial misalignment and its compounding downstream consequences. Rather than relying on anecdotal guidance or externally imposed social expectations, learners are guided toward specializations that reflect the intersection of their demonstrated aptitudes, behavioral tendencies, and verified labor market demand. This tripartite alignment is a state that individual self-assessment and conventional academic advising rarely achieve simultaneously. In this respect, the platform's individual significance is inseparable from its social significance: each correctly guided learner represents a reduction in systemic human capital waste.

At a broader social level, the project addresses structural equity concerns within regional educational and labor market systems. The documented aspiration-market mismatch in the MENA region—wherein less than one percent of surveyed high school students plan to study Computer Science or Artificial Intelligence, despite these fields representing the fastest-growing and highest-compensated technical roles [34] represents a form of information asymmetry whose consequences fall disproportionately on learners without access to professional networks or informed mentorship. Brookings research further indicates that only 15 percent of young women in the MENA region participate in the formal labor force [34], a figure that reflects, in part, the absence of accessible, credible guidance infrastructure. By delivering a psychometrically grounded, market-anchored career navigation tool accessible on consumer-grade mobile devices without institutional affiliation requirements, the platform contributes to the democratization of professional guidance. It converts a resource historically concentrated among socially and economically privileged learners into a scalable, algorithmically delivered public good.

Chapter 2:
Literature Review and Related Work

2.1 Overview of the E-Learning Landscape

The global educational technology sector has undergone a significant structural transformation across the preceding two decades, characterized by two successive and conceptually distinct paradigmatic phases. The first phase, which dominated the early-to-mid portion of this period, was defined by the rapid proliferation of Massive Open Online Courses (MOOCs), exemplified by large-scale platforms such as Coursera, edX, and Udemy. These systems constituted a pivotal milestone in the democratization of instructional access, enabling unprecedented geographical reach by distributing content from elite academic institutions to learners irrespective of their physical or socioeconomic location. Architecturally, this phase operated under what may be described as a **"Library Model"** of learning delivery a paradigm in which educational value was derived almost exclusively from the aggregation and distribution of static instructional content [80].

Within this model, courses functioned as repositories of pre-recorded lectures, standardized assessments, and fixed reading materials, accessible on demand but structurally identical for every enrolled learner. While this approach successfully resolved barriers related to geographical access and institutional exclusivity, it rested upon a reductive assumption: that the provision of content is, in itself, sufficient for the facilitation of learning. As a consequence, the Library Model largely decoupled instructional delivery from the learner's individual cognitive state, prior competencies, motivational context, and preferred processing modality, giving rise to persistent and well-documented challenges in retention, sustained engagement, and the translation of content exposure into verifiable competency [6], [29].

2.1.1 Limitations of Content-Centric Models

Empirical evaluations of the MOOC ecosystem consistently reveal a widening and structurally significant gap between content availability and meaningful educational outcomes. A particularly persistent finding across large-scale observational and analytical studies is the prevalence of extremely low course completion rates; evidence from adaptive AI systems research and machine learning-based dropout analyses indicates that the substantial majority of learners enrolled in non-credentialed or open-enrollment MOOCs disengage long before course completion, with reported rates frequently falling within the ten-to-fifteen percent range [6], [9]. Critically, this phenomenon is not attributable to learner apathy or a generalized absence of motivation; rather, it reflects deeper architectural and pedagogical limitations inherent in content-centric instructional models.

Research by Sadki et al. (2025) directly attributes elevated attrition rates to the structural reliance on passive, broadcast-style pedagogical delivery, wherein diverse learner populations are systematically exposed to identical, linearly sequenced curricula irrespective of their prior competencies, individual learning velocity, or cognitive preferences [4]. This "one-size-fits-all" approach imposes a disproportionate cognitive burden on the self-directed learner, who is required to manage pacing, sustain intrinsic motivation, and self-assess comprehension without the benefit of adaptive scaffolding, timely formative feedback, or responsive instructional adjustment. Bandura's foundational work on self-efficacy further contextualizes this failure: when learners receive no

calibrated feedback confirming their growing competence, perceived self-efficacy declines, and with it, the persistence required to complete long-form curricula [5].

A compounding structural limitation concerns the organization and scale of available content. The expansive catalogues characteristic of dominant MOOC platforms introduce a demonstrable layer of cognitive strain that extends beyond instructional delivery itself. While extensive course choice ostensibly empowers the learner, it simultaneously induces what researchers and industry analysts have termed the "paradox of choice," wherein the abundance of available learning pathways generates decision paralysis, heightened selection anxiety, and diminished long-term commitment to any single learning trajectory [31]. This phenomenon is of particular relevance in the context of technical education, where learners lacking domain expertise are structurally least equipped to evaluate which of the available pathways aligns most effectively with their aptitudes, career objectives, or cognitive profile. Because these platforms lack adaptive, learner-aware mechanisms, they represent a foundational architectural deficiency of first-wave e-learning systems one that open access to content alone cannot resolve [5], [6], [80].

2.1.2 The Emergence of Adaptive Architectures

In response to the structural deficiencies outlined above, the e-learning domain has undergone a discernible conceptual transition toward a second paradigmatic phase, one centered on the personalization and adaptive calibration of learning experiences rather than the mere provision of instructional material. This shift is reflected, at the commercial level, in the rise of Learning Experience Platforms (LXPs), which diverge from traditional Learning Management Systems (LMS) by reorienting their core design philosophy around learner agency, contextual relevance, and dynamic personalization, rather than course administration and completion tracking [80]. Unlike LMS architectures, which are primarily designed to manage institutional content lifecycles and generate compliance records, LXPs reconceptualize the primary unit of design as the individual learner's evolving cognitive and behavioral state.

This learner-centric reorientation has been rendered increasingly feasible by concurrent advances in Artificial Intelligence and machine learning methodologies, which enable platforms to construct computational models of learner behavior, infer latent cognitive states from interaction data, and dynamically adjust content sequencing, instructional modality, and assessed difficulty with minimal latency relative to learner interaction events [29], [13]. Furthermore, progress in learning analytics and affective computing has expanded the capacity of adaptive systems to detect not merely performance-level indicators, but also affective and motivational signals such as declining engagement velocity, elevated session abandonment rates, or negatively trending sentiment in learner communications that can be leveraged to anticipate and proactively mitigate dropout risk before disengagement becomes irreversible [25], [26].

The accelerating integration of these capabilities reflects a maturing consensus within learning science: effective digital education demands closed-loop adaptive mechanisms capable of responding to learner engagement, demonstrated performance, and expressed preference in a continuous and contextualized manner. Next-generation platforms are, accordingly, expected to operate not as passive content repositories but as

responsive instructional environments that continuously calibrate the learning experience to the individual. It is at the underserved intersection of diagnostic psychometric profiling and adaptive content delivery the convergence point that existing systems have failed to operationalize that the architectural proposition of LearNova is situated.

2.2 Theoretical Foundations

To ensure pedagogical validity and a genuinely learner-centered system design, the proposed LearNova architecture is grounded in established frameworks from cognitive science, educational psychology, and assessment theory. This section examines the theoretical models that directly inform the platform's core mechanisms the polymorphic content delivery system, the scaffolded curriculum progression architecture, and the stratified assessment structure providing a formal justification for the transition from static, broadcast-style instruction toward an adaptive, multimodal learning environment governed by evidence-based pedagogical principles.

2.2.1 The VARK Model and Multimodal Efficacy

The content delivery architecture of LearNova draws its foundational conceptual guidance from the VARK framework, which categorizes individual learning preferences according to four sensory-cognitive modalities: Visual, Aural, Read/Write, and Kinesthetic [18]. Empirical investigations into the prevalence of these preferences across diverse learner populations indicate substantial heterogeneity in dominant modality profiles; a study by Kharb (2016) examining learning styles across student cohorts found that kinesthetic and visual preferences predominate at the population level, while read/write preferences which form the implicit default of most lecture-based instruction represent a comparatively small minority [19]. A broader study by Pradhan (2023) corroborates this distribution, reinforcing the inference that the prevailing design assumption of uniform text-and-audio delivery serves the majority of learners suboptimally [20].

It is necessary to acknowledge, however, that the empirically contested rigid learning styles hypothesis the claim that learners possess fixed, trait-like modality preferences and learn exclusively or best through their dominant modality — has been subject to sustained empirical critique. Contemporary research in multimedia learning, particularly Mayer's Cognitive Theory of Multimedia Learning, supports a more nuanced and defensible formulation: the principle of modality alignment [2]. This principle holds that instructional formats can differentially influence cognitive processing efficiency depending on the nature of the content, the learner's prior knowledge, and the representational demands of the task without requiring the assumption of fixed learner types. Mayer's work demonstrates that learners who receive information through congruent representational channels demonstrate superior immediate and delayed recall compared to those receiving content through a single, format-uniform channel [2]. This finding is supported by Al-Shehri (2024), whose investigation into multimedia instruction confirms that format-content congruence produces measurable retention advantages relative to text-only or video-only delivery [7]. Paivio's Dual Coding Theory provides the underlying cognitive mechanism: meaningful learning is most efficiently achieved when verbal and non-verbal representational systems are simultaneously and coherently engaged, enabling the construction of richer, more interconnected memory traces [3].

Rather than adopting a prescriptive typology model, LearNova operationalizes the defensible core of the VARK framework through a **polymorphic content strategy**. Under this architecture, each atomic learning unit represented in the database as a topic is associated with up to three parallel content representations: a Visual slot, an Auditory slot, and a Textual slot. The adaptive engine resolves the learner's dominant modality from

their calibration profile and serves the corresponding variant as the default presentation, while retaining access to alternative representations to preserve learner agency. This approach exploits modality alignment as a cognitive efficiency mechanism rather than a prescriptive classification system. Critically, the initial calibration is not treated as a permanent fixed assignment; Bayesian posterior scores stored in student_profiles are updated continuously from observed engagement data, enabling the system to refine its model of each learner's preferences through behavioral evidence rather than relying solely on self-reported questionnaire responses a distinction with meaningful implications for adaptive accuracy that is elaborated in the system design chapter [18].

2.2.2 Cognitive Load Theory (CLT) and Scaffolding

Curriculum sequencing and instructional pacing within LearNova are fundamentally informed by Sweller's Cognitive Load Theory (CLT), which represents one of the most extensively validated frameworks in instructional design research [1]. CLT posits that working memory capacity is inherently limited in both duration and throughput, and that the effectiveness of any instructional design is critically dependent upon how it manages three distinct categories of cognitive demand: intrinsic load, arising from the inherent complexity of the learning material; extraneous load, imposed by poorly designed instructional presentation; and germane load, reflecting cognitive effort directed toward schema construction and the integration of new knowledge with existing mental models [1], [14].

When learners encounter material whose intrinsic complexity substantially exceeds their current competency level, or when extraneous load is poorly controlled through cluttered or format-inconsistent presentation, available working memory capacity is rapidly exhausted and schema construction is impeded. Under these conditions, learners experience the subjective states frustration, disorientation, and diminished self-efficacy that empirical research consistently identifies as immediate precursors to disengagement and dropout in self-directed online learning environments [5], [15].

To address these constraints systematically, the LearNova architecture employs a scaffolded progression strategy grounded in Vygotsky's concept of the Zone of Proximal Development (ZPD) [4]. Within the ZPD framework, optimal learning occurs when instructional tasks are positioned beyond what the learner can accomplish independently but within what they can achieve with appropriate structural support. In LearNova, this zone is approximated through system-level constraints that control the rate at which content complexity is introduced. The platform enforces a mandatory shared Foundation phase, during which all learners acquire a common baseline of technical and analytical competencies before advancing to any of the three specialization tracks. Progression through each level is further gated by a mastery threshold the passing_score constraint within module_assessments ensuring that learners cannot advance until they have demonstrated adequate schema formation at the current complexity tier. This mastery-gated architecture directly operationalizes the CLT objective of maintaining instructional activity within the learner's estimated ZPD, preventing both the cognitive overload of premature advancement and the disengagement of excessive repetition [1], [15]. Additionally, the polymorphic content delivery mechanism described in Section 2.2.1 functions as an instrument of extraneous load reduction: serving content in the format

most congruent with the learner's calibrated processing profile minimizes the cognitive effort expended on translating between mismatched representational systems effort that contributes to extraneous load without advancing schema construction [2], [6].

2.2.3 Bloom's Taxonomy and Assessment Strategy

The assessment architecture of LearNova is structured in accordance with the hierarchical cognitive framework of Bloom's Taxonomy in its revised formulation, which organizes cognitive processes across six progressive levels of complexity: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating. This taxonomy provides a principled and widely validated basis for calibrating assessment demands to the learner's current developmental stage, ensuring that evaluation instruments measure cognitive operations that are genuinely attainable and pedagogically appropriate at each phase of the learning trajectory. Research on adaptive AI learning systems confirms that assessment instruments calibrated to match the learner's demonstrated cognitive tier produce superior learning outcomes compared to fixed-format evaluations that do not account for developmental progression [15].

The system implements this principle through a structurally differentiated two-tier assessment architecture. At the module level, formative assessments implemented as module_assessments with MCQ question banks are calibrated to the lower-order cognitive objectives of the taxonomy: Remembering and Understanding. These assessments function as comprehension checkpoints, employing recognition-based and concept-verification formats that confirm whether learners have successfully encoded foundational schemata without imposing the additional demands of application or synthesis. A minimum passing threshold enforced at the database level ensures that advancement is predicated on demonstrated foundational competence rather than mere content exposure.

At the level boundary, a second and more cognitively demanding tier is encountered in level_assessments, which escalate evaluation demands to encompass the higher-order processes of Applying, Analyzing, and Evaluating. Notably, the schema for level_assessment_questions includes ai_grading_rubric, mitchy_hint, and mitchy_explanation fields, indicating support for open-ended response formats evaluated through AI-assisted rubric grading an architectural capability that would be unnecessary for purely recognition-based items and that reflects the higher-order cognitive demands this tier is designed to assess. This graduated, two-tier structure enhances construct validity by aligning the cognitive complexity of evaluation instruments with the learner's developmental stage, thereby reducing both the false negative of failing competent learners through premature demands and the false positive of advancing learners who have demonstrated only surface-level pattern recognition without genuine schema formation [1], [15].

2.3 Psychometric and Affective Frameworks

The decision-support logic of the LearNova architecture is informed by two complementary theoretical domains: established models from differential psychology and vocational science for long-term trait and interest assessment, and principles from affective computing for continuous, real-time monitoring of learner emotional and cognitive states. These two domains converge on a single functional output a personalized learning path through parallel but interdependent analytical pipelines, as illustrated in Figure 2.1 a psychometric assessment layer that operates during onboarding to establish the learner's stable profile, and an affective state inference layer that operates continuously throughout the learning journey to detect and respond to transient risk signals. This section delineates the theoretical foundations of both pipelines, providing formal justification for the platform's approach to proactive, individualized learner support.

2.3.1 Trait Theory, Vocational Interest Mapping, and Specialization Alignment

The specialization recommendation architecture of LearNova is grounded in two complementary psychometric traditions: trait-based personality assessment through the Five-Factor Model (FFM), and vocational interest profiling through an assessment instrument organized according to Holland's RIASEC typology, operationalized via the O*NET Interest Profiler framework.

The Five-Factor Model, also known as the OCEAN model, represents one of the most extensively validated structural frameworks in differential psychology for characterizing stable, cross-situational dimensions of personality. The five dimensions Openness to Experience, Conscientiousness, Extraversion, Agreeableness, and Neuroticism have been demonstrated across decades of empirical research to exhibit meaningful correlations with occupational performance, work-style preferences, and long-term career satisfaction. Within technical domains specifically, high Openness to Experience, characterized by intellectual curiosity, preference for novelty, and comfort with abstract reasoning, has been associated with aptitude for exploratory, hypothesis-driven roles such as Data Science. Conversely, high Conscientiousness, reflecting organizational diligence, systematic planning, and process adherence, is empirically linked to success in structured, pipeline-oriented roles such as Data Engineering [29]. LearNova operationalizes these correlations through the **IPIP personality instrument** a 120-item assessment employing a five-point Likert response scale with reverse scoring applied to negatively keyed items which generates normalized scores across all five OCEAN dimensions as weighted features in the track assignment calculation.

Complementing the FFM assessment, the **O*NET Career Interest assessment** comprising 60 items organized across the six RIASEC categories of Realistic, Investigative, Artistic, Social, Enterprising, and Conventional provides a vocational interest profile that directly anchors the specialization assignment decision. Holland's foundational proposition, that individuals are drawn toward and perform best in occupational environments congruent with their dominant interest type, has been extensively validated in career development research. Within the LearNova scoring architecture, three RIASEC categories function as direct specialization anchors: the Artistic interest dimension anchors the Data Analytics track, reflecting the communicative and interpretive nature of analytical work; the Realistic dimension anchors the Data

Engineering track, reflecting its emphasis on concrete system construction and procedural problem-solving; and the Investigative dimension anchors the Data Science track, reflecting its foundation in hypothesis generation and abstract pattern recognition. These O*NET base scores contribute thirty percent of the final track assignment score, with the remaining seventy percent derived from weighted modifier scores computed from the full feature vector spanning personality, soft skills, learning style, and cognitive aptitude dimensions [29].

This dual-instrument approach addresses a fundamental limitation of single-instrument recommendation systems: personality alone predicts work-style fit without capturing intrinsic motivation, while interest profiling captures motivation without characterizing behavioral tendencies or cognitive preferences. By synthesizing both dimensions, the **Smart Compatibility Engine** generates a multidimensional learner profile that is considerably more predictive of long-term engagement and specialization satisfaction than either instrument could produce independently [29].

2.3.2 Affective Computing and Predictive Risk Analysis

The pedagogical support module of LearNova is grounded in the field of Affective Computing, which is concerned with developing computational systems capable of recognizing, interpreting, and responding to human affective and emotional states [25]. Traditional e-learning platforms, operating in asynchronous and largely unmonitored environments, lack the capacity to perceive transient learner states such as confusion, frustration, or cognitive overload a structural blind spot that contributes substantially to disengagement and dropout during challenging learning phases, particularly when these states go undetected and unaddressed [5], [25].

LearNova addresses this limitation by implementing a continuous affective inference pipeline that derives learner state estimates from two parallel streams of behavioral data. The first stream comprises **linguistic signals** extracted from learner interactions with Mitchy, the AI pedagogical agent: sentiment polarity scores, syntactic distress markers, and query pattern analysis performed on every inbound chat message. Research by Gupta (2023) demonstrates that natural language processing approaches, when integrated with learning analytics pipelines, can identify negative affective trends including anxiety, frustration, and disengagement from conversational text with meaningful predictive accuracy [25]. These signals are logged as time-series entries in student_sentiment_history, capturing a numeric sentiment_score, a categorical learning_state label, and session_context metadata for each detected state transition. The second stream comprises **behavioral metrics** derived from observable learner interaction patterns: engagement velocity, topic struggle index, concept decay score, and per-topic retention estimates. Research by Brown (2023) establishes that passive behavioral biomarkers derived from interaction logs function as reliable indicators of underlying cognitive state [23], while Chen (2022) and Kumar (2024) demonstrate that models trained on comparable engagement and clickstream signals can predict learner dropout significantly in advance of the event itself [24], [26].

These two streams are aggregated through a daily ML pipeline that writes to ml_daily_metrics and ml_topic_daily_metrics, computing composite risk scores persisted to the risk_scores table. Each record carries a categorical risk_level spanning low,

medium, high, and critical an alert_triggered flag, and a feature_snapshot in JSONB format that preserves the full ML feature vector used in computation for audit and explainability purposes. When the alert_triggered flag is set, the system escalates to a human-in-the-loop intervention: an administrator is notified and an intervention_logs record is created, capturing the claimed and resolved timestamps alongside the specific action taken. This architecture operationalizes a shift from the reactive intervention posture characteristic of conventional e-learning systems responding only after a quiz failure or explicit help request to a proactive one that anticipates distress from converging behavioral and linguistic signals, escalating before the learner consciously disengages and providing cognitive and emotional scaffolding at the precise moment it is most likely to be effective [25], [26].

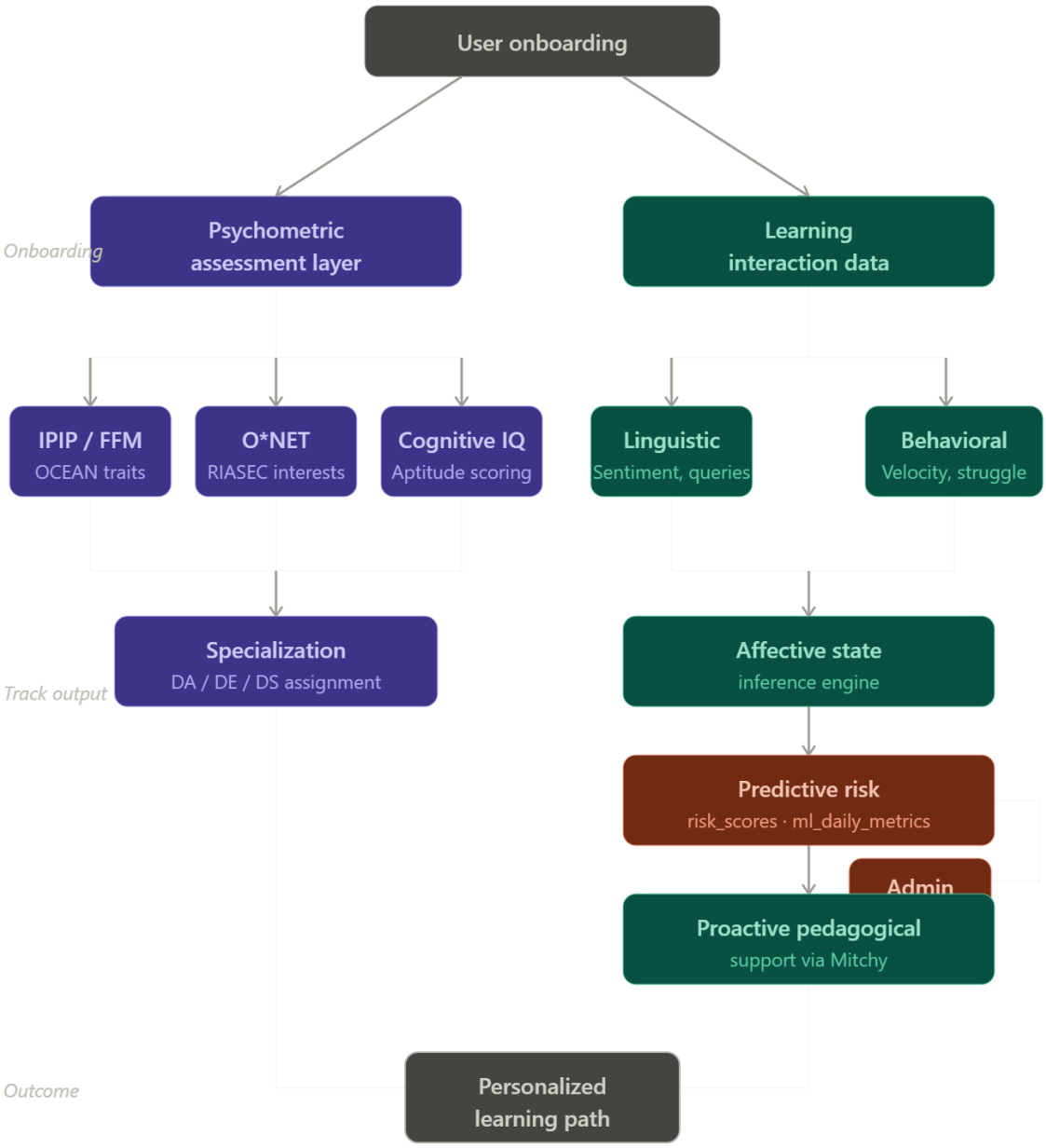


Figure 2.1. Psychometric and Affective Decision-Support Framework in LearNova

2.4 Gamification Theory and Retention Mechanics

The persistently high attrition rates observed across large-scale digital learning environments are widely attributed to three interrelated structural deficiencies: a deficit in sustained intrinsic motivation, the absence of structured accountability mechanisms, and the social isolation endemic to asynchronous online learning [5], [6]. To address these compounding challenges, the LearNova architecture incorporates a structured retention framework designed to move substantially beyond superficial "pointsification" the indiscriminate assignment of badges and scores without regard for underlying motivational dynamics by embedding behavioral and motivational psychology directly into the user experience. The system's engagement design is grounded in two complementary theoretical frameworks: Self-Determination Theory (SDT), which identifies the fundamental psychological needs that must be satisfied to sustain high-quality, internalized motivation; and the principled gamification framework articulated in contemporary gamification theory, which distinguishes and balances intrinsic and extrinsic motivational mechanics within interactive systems [52].

2.4.1 Self-Determination Theory (SDT)

Ryan and Deci's Self-Determination Theory posits that sustained, high-quality motivation the kind that produces genuine persistence, deep engagement, and durable learning rather than mere compliance is contingent upon the satisfaction of three innate psychological needs: autonomy, competence, and relatedness [12]. The LearNova architecture operationalizes each of these core drivers through specific, implemented mechanisms.

The need for **autonomy** the subjective experience of volition and self-determination is addressed through the system's guidance model. While the **Smart Compatibility Engine** generates a data-informed specialization recommendation grounded in the learner's psychometric and cognitive profile, the architecture explicitly frames this recommendation as an advisory output rather than a mandatory assignment. The user retains ultimate agency to accept, modify, or reject the suggested track, and the learning_mode field within student_profiles supports an exploration sub-mode with a configurable time window enforced through exploration_started_at and exploration_ends_at during which learners may access content outside their assigned track. This provision of perceived choice is designed to shift the locus of causality from externally imposed instruction to internally motivated engagement, a mechanism empirically demonstrated to increase persistence and promote deeper internalization of learning goals [12], [47].

The need for **competence** the desire to feel effective and progressively masterful is supported through a multi-layered feedback architecture. The system decomposes complex curricula into granular micro-milestones at the topic, module, and level tiers, each associated with an xp_reward that provides immediate, concrete validation of progress upon completion. The user_streaks table tracks current_streak_days and longest_streak_days per learner, implementing **Momentum Streaks** that provide a continuous, visible signal of sustained engagement. Daily leaderboard_snapshots record each student's XP rank scoped per track, enabling competence comparison within a meaningful peer reference group rather than an undifferentiated global population.

Weekly_challenges time-bounded assessments linked to specific modules and available within defined date windows extend this feedback architecture by introducing tiered XP rewards calibrated across easy, mid, and hard difficulty levels, ensuring that the competence signal scales proportionally to the cognitive demand successfully met. Facey-Shaw's meta-analysis of gamification outcomes in educational settings confirms that this type of incremental, milestone-anchored feedback system produces measurable improvements in learner self-efficacy and sustained engagement compared to terminal grading models [49].

The need for **relatedness** the desire to feel meaningfully connected to a community of peers is addressed through the track-scoped leaderboard architecture. By constraining competitive comparison to cohorts sharing the same specialization trajectory, the system creates peer reference groups that are motivationally relevant rather than arbitrarily defined. Research by Orji (2023) on team-based gamified learning in online environments demonstrates that peer-anchored comparative structures, even in asynchronous contexts, fulfill the need for relatedness by generating a sense of shared purpose and social accountability that significantly reduces perceived isolation [46]. Olszewski's (2025) investigation into social gamification elements corroborates this finding, reporting that social visibility features including leaderboards and achievement displays accessible to cohort members produce statistically significant improvements in engagement duration relative to equivalent non-social conditions [11].

2.4.2 Balancing Urgency and Empowerment: White Hat and Black Hat Mechanics

To strategically balance immediate engagement with long-term learner well-being, the system's gamification design is informed by a distinction commonly recognized in gamification theory between intrinsic, empowering motivational mechanics White Hat mechanics and extrinsic, urgency-driven mechanics Black Hat mechanics [52]. This distinction is consequential because exclusive reliance on either category produces predictable failure modes: pure urgency-based design generates short-term compliance at the cost of anxiety and eventual burnout, while purely intrinsic design may prove insufficient to counteract the procrastination and inertia that self-directed learning environments consistently produce.

Black Hat mechanics are therefore employed within LearNova deliberately and sparingly, targeting the specific behavioral failure mode of procrastination. The user_streaks implementation operationalizes the behavioral economic principle of loss aversion the well-documented asymmetry whereby the motivational force of avoiding the loss of an existing reward substantially exceeds that of acquiring an equivalent new gain [51]. When a learner has maintained a current_streak_days value of meaningful length, the prospect of its reset upon inactivity constitutes a tangible loss threat that leverages this asymmetry to sustain daily engagement. Similarly, the date-bounded availability window of weekly_challenges enforced through available_from and available_until introduces temporal scarcity that counteracts indefinite deferral by constraining the window within which associated XP rewards remain accessible. Antonaci's systematic review of gamification in MOOCs confirms that time-limited challenge mechanics produce measurable increases in short-term participation rates [50].

The architecture's primary emphasis, however, is on White Hat mechanics that cultivate intrinsic satisfaction and a sustained sense of empowerment. The adaptive content delivery system is the most architecturally significant of these: by presenting learning materials in the format most congruent with the learner's calibrated cognitive profile, the platform generates a persistent sense of personal relevance qualitatively distinct from the generic experience of a uniform, format-indifferent platform. The achievements_dictionary further operationalizes White Hat design through a structured hierarchy of badge criteria spanning streak, xp, and completion trigger types that reward intrinsically meaningful milestones rather than arbitrary behavioral thresholds. Werbach's analysis of gamification design confirms that achievement systems grounded in genuine competence milestones are significantly more effective at sustaining long-term engagement than purely extrinsic reward structures [52]. The combination of Black Hat urgency mechanics applied selectively to counteract procrastination, and White Hat empowerment mechanics applied broadly to cultivate intrinsic motivation, constitutes a theoretically coherent and empirically grounded retention architecture one designed to sustain engagement through the progressive, validated experience of genuine mastery rather than through anxiety or compulsion [49], [50].

2.5 Technical Approaches to Adaptive Learning

To properly contextualize the architectural design of the proposed system within the broader computational landscape, a review of the predominant technical methodologies employed in adaptive educational technologies is necessary. The literature broadly classifies approaches to learning personalization into three primary paradigms Rule-Based Adaptation, Collaborative Filtering, and Content-Based Filtering each exhibiting distinct strengths and structural limitations that have collectively shaped the evolution of contemporary hybrid adaptive architectures [80]. The following subsections examine each paradigm in turn before establishing the theoretical and technical basis for the hybrid methodology implemented within LearNova.

2.5.1 Rule-Based Adaptation

The earliest adaptive learning systems were predominantly constructed on deterministic, rule-based logic, commonly implemented as explicit if-then production systems in which pedagogical domain experts manually encode instructional response rules. A representative example would be: "If a learner's score on a module assessment falls below the passing threshold, then re-present the associated remedial topic before permitting progression." The principal advantage of this paradigm lies in its transparency and explainability: every system decision can be traced directly to a predefined rule, rendering the instructional logic fully interpretable to both system designers and educators [80]. This auditability is particularly valuable in educational contexts, where the capacity to review and justify adaptive decisions carries significant pedagogical and ethical weight [15].

However, rule-based systems are subject to well-documented structural limitations. Most critically, they suffer from the knowledge acquisition bottleneck: designers must anticipate and explicitly encode every relevant learner state and every corresponding instructional response in advance, a requirement that renders such systems inherently rigid and difficult to scale. Furthermore, purely rule-based architectures cannot generalize from observed learner behavior they can only execute pre-specified responses, remaining structurally incapable of refining their model of the learner through accumulated interaction data [80]. Within LearNova, rule-based logic is deliberately retained for a specific, well-suited function: the deterministic enforcement of mastery-gated progression, wherein the passing_score constraint within module_assessments constitutes an explicit if-then rule that is both transparent and pedagogically appropriate for that purpose.

2.5.2 Collaborative Filtering (CF)

In contrast to the deterministic rule-based paradigm, many contemporary platforms employ Collaborative Filtering (CF), a personalization methodology originally developed within recommender systems and subsequently adapted for educational contexts. CF operates on the principle of behavioral homophily: the assumption that learners who have exhibited similar interaction patterns in the past are likely to benefit from similar learning resources. The algorithm constructs and analyses large user-item interaction matrices comprising course enrollments, completion records, assessment scores, and engagement signals to identify clusters of behaviorally similar learners and generate

recommendations based on resources positively evaluated by their nearest neighbors [80], [29].

While effective in environments characterized by large, mature user bases with rich historical interaction data, CF is critically constrained by the cold-start problem. For new users lacking sufficient interaction history, the algorithm has no behavioral signal from which to derive similarity estimates; for newly introduced resources, the absence of engagement records precludes them from appearing in recommendations regardless of their instructional quality. This limitation is structurally problematic in educational contexts, where accurate guidance during the onboarding phase precisely when learner uncertainty is highest and dropout risk most acute is most critically needed [80], [29]. The cold-start constraint therefore disqualifies CF as a sufficient standalone mechanism for any adaptive system that must provide meaningful personalization from a learner's very first interaction.

2.5.3 Content-Based Filtering and Hybrid Architectures

Content-Based Filtering (CBF) addresses the cold-start limitation by reorienting the personalization mechanism from user-user behavioral similarity to direct profile-content alignment. Rather than inferring preferences from the behavior of analogous learners, CBF generates recommendations by matching explicitly characterized learner attributes against the intrinsic properties of available learning resources including topic domain, difficulty tier, and instructional format without requiring any prior interaction history [80]. This approach is inherently immune to the cold-start problem because it operates from an explicit learner profile that can be established at onboarding and applied immediately.

Recognizing that no single paradigm sufficiently addresses the full spectrum of adaptive learning requirements rule-based approaches offer transparency but lack generalization capacity; CF offers behavioral responsiveness but fails at onboarding; CBF resolves cold-start but depends on the quality of initial profile construction hybrid architectures have emerged as the state-of-the-art approach in adaptive educational systems [80], [15]. A well-validated hybrid strategy combines rule-based diagnostic assessment during onboarding, which constructs a rich initial learner profile and thereby mitigates cold-start effects, with content-based filtering mechanisms for initial delivery, progressively supplemented by behavior-informed refinement as interaction data accumulates [29], [80]. Hybrid architectures of this type preserve the partial interpretability afforded by their rule-based components a property of particular importance in educational environments where transparency and the ability to audit adaptive decisions are not secondary considerations but core design constraints [15].

The LearNova architecture implements precisely this hybrid methodology across two sequential and interdependent phases. In the first phase, a **rule-based diagnostic engine** administers a five-instrument assessment battery totalling 251 questions comprising the 120-item IPIP personality assessment, a 20-scenario soft skills evaluation, the 16-item VARK learning style instrument, a 60-item O*NET RIASEC career interest assessment, and a 35-item cognitive IQ test to construct a normalized, multidimensional feature vector for each new learner. Raw answers are matched against a pre-seeded answer-key index by question_key and selected_index, features are aggregated per instrument, normalized via min-max clamping, and combined through a weighted formula

in which O*NET RIASEC base scores contribute thirty percent and cross-instrument modifier scores contribute seventy percent, with the argmax of the resulting DA, DE, and DS scores written to student_profiles.assigned_track. This phase is entirely rule-based and explainable: every step is deterministic, auditable, and traceable to specific answer-key rows.

In the second phase, a **content-based adaptive delivery engine** governs instructional presentation using the profile established during onboarding. The learning_style field in student_profiles determines which topic_resource variant Visual, Auditory, or Textual is served as the default for each learning unit, while the mastery-gated progression architecture ensures content complexity is introduced at a rate calibrated to the learner's demonstrated competency. This profile is not treated as a static assignment: Bayesian posterior scores bayesian_alpha_visual, bayesian_alpha_auditory, and bayesian_alpha_textual are updated continuously from content_engagement_logs rows flagged as bayesian_eligible, enabling the system to refine its modality model through observed behavioral evidence. Learning-style drift is further monitored through a dedicated analytics module that identifies systematic divergence between the learner's calibrated profile and their revealed content preferences over time. Through this synthesis, the system provides immediate, cold-start-immune guidance from the first interaction via its rule-based diagnostic foundation, and increasingly nuanced, behavior-informed personalization as engagement data accumulates — directly operationalizing the hybrid architecture paradigm validated in the adaptive learning systems literature [29], [80].

2.6 Existing Solutions and Limitations

While the technical paradigms discussed in Section 2.5 provide the computational foundation for adaptive personalization, their implementation in the current commercial e-learning market has produced a bifurcation into two dominant yet functionally isolated architectural models: the **"Library Model"** of content aggregation and the **"Audit Model"** of diagnostic assessment. This section critically examines the inherent limitations of each model, identifying the specific functional gaps that the proposed LearNova architecture is designed to occupy.

2.6.1 The Library Model (Content Aggregators)

Platforms such as Coursera, Udemy, and edX epitomize the Library Model. Their primary and genuinely transformative contribution has been the democratization of instructional access hosting extensive repositories of high-quality content from globally recognized institutions and making it available to learners irrespective of geography or institutional affiliation [80]. This achievement represents a necessary and significant advance for educational equity. However, the success of content aggregation has not resolved the deeper pedagogical challenges of sustained learning; in several respects, it has introduced new cognitive barriers that were not present in traditional instructional environments.

A central structural limitation of the Library Model is its systematic exacerbation of the paradox of choice. These platforms place the entire burden of curriculum navigation and pathway selection on the learner, presenting an unstructured catalogue of hundreds of courses across a single domain without antecedent diagnostic guidance or structured recommendation mechanisms. For novice learners who lack sufficient domain expertise to evaluate the relative appropriateness of available options, this abundance generates decision fatigue, selection anxiety, and a substantially elevated risk of committing to a learning path misaligned with their actual aptitudes, cognitive preferences, or career objectives [31], [6]. The structural absence of a pre-instructional assessment layer means that the learner's choices are driven by external trends, social validation, or arbitrary browsing behavior rather than evidence-based self-knowledge.

Compounding this navigation failure is the instructional delivery model. Despite their technological sophistication, Library Model platforms maintain a fundamentally static, broadcast-style delivery pipeline in which content is presented in a uniform, predominantly long-form video format irrespective of individual differences in cognitive processing style, prior knowledge, or motivational state. Research on video engagement patterns confirms that learner attention decays rapidly during extended passive video consumption, declining precipitously after the first several minutes regardless of content quality [77]. This uniform delivery approach creates a systematic pedagogical disconnect for the substantial proportion of learners whose dominant processing preference is not auditory-linear, contributing directly to the engagement decay and high attrition rates consistently documented in MOOC research [6], [9]. In essence, Library Model platforms function as sophisticated digital repositories but structurally lack the integrated diagnostic, recommendation, and adaptive delivery mechanisms necessary to guide, support, and retain the individual learner across an extended learning trajectory.

2.6.2 The Audit and Diagnostic Model

Occupying the opposite end of the architectural spectrum are platforms primarily designed for learner assessment rather than content delivery. These can be meaningfully distinguished into two sub-categories: Technical Audit Platforms, exemplified by HackerRank and CodeSignal, and Behavioral Diagnostic Tools, exemplified by Pymetrics.

Technical audit platforms are engineered to evaluate domain-specific technical competencies under controlled conditions measuring a learner's capacity to produce algorithmically correct outputs, solve coding challenges within time constraints, or demonstrate proficiency through structured problem sets [43]. The principal strength of this approach is its precision: skill assessment is quantitative, reproducible, and directly mappable to occupational requirements. However, the design of these platforms is competence-centric in a narrowly defined sense. They evaluate the product of cognitive effort the correct or incorrect output while remaining structurally blind to the process by which that output is generated. Critically, they lack any mechanism to detect or respond to the affective and motivational dimensions of the assessment experience: the mounting frustration of a learner encountering repeated failure, the cognitive overload signaled by escalating error rates, or the diminishing self-efficacy that empirical research consistently identifies as the most reliable precursor to disengagement [5], [25]. The absence of an affective feedback loop means that technical audit platforms can measure competence but cannot cultivate it they provide no motivational scaffolding, no adaptive difficulty adjustment, and no emotional support for sustained skill development.

Behavioral diagnostic tools occupy a distinct but ultimately analogous position of functional isolation. Platforms such as Pymetrics offer sophisticated, game-based psychometric profiling that generates detailed assessments of psychological traits, cognitive styles, and behavioral inclinations, providing considerably greater diagnostic granularity than simple questionnaire-based instruments [56], [57]. However, these platforms operate in diagnostic isolation: they deliver a rich profile characterizing the assessed individual along dimensions such as risk tolerance, numerical reasoning, and cognitive flexibility but are not integrated with any subsequent pedagogical infrastructure capable of translating that profile into structured learning guidance, skill development pathways, or curriculum recommendations. The platform's function terminates at the point of labeling. Designed primarily as corporate recruitment screening tools, they are architected to identify and filter existing talent rather than develop latent potential, leaving the assessed learner with profile insights but without a structured pathway for growth or career-aligned skill acquisition [56], [80].

The functional gap between these two models is both precise and consequential: Library Model platforms possess content infrastructure but lack diagnostic and adaptive mechanisms; Audit and Diagnostic platforms possess profiling precision but lack connection to any educational infrastructure through which those profiles can be acted upon. The architectural response to this gap is examined in the following section.

2.7 The Research Gap: The "Missing Middle"

The cumulative evidence reviewed across Sections 2.1 through 2.6 converges on a single, structurally precise diagnosis: the contemporary educational technology landscape is bifurcated into two dominant paradigms that address complementary educational needs in functional isolation, with neither paradigm capable of delivering what the other provides and no existing framework integrating both into a coherent, closed-loop educational system.

The first paradigm, the **Content-Centric Model** exemplified by platforms such as Coursera and Udemy, has successfully democratized access to high-quality instructional material at global scale [80]. However, as established in Sections 2.1 and 2.6, these systems structurally lack the antecedent diagnostic mechanisms required to align that content with the individual learner. In the absence of psychometric profiling or cognitive calibration, the learner confronts an undifferentiated catalogue, generating the paradox of choice that produces decision fatigue, pathway misalignment, and elevated dropout risk [31], [6]. Content is delivered through a format-uniform, broadcast-style pipeline that fails to account for the substantial heterogeneity of learner processing preferences documented in Section 2.2.1, contributing directly to the engagement decay and high attrition rates consistently reported in MOOC research [6], [9]. These platforms educate, but they do not guide.

The second paradigm, the **Diagnostic-Centric Model** represented by technical audit platforms such as HackerRank and behavioral profiling tools such as Pymetrics, offers precisely the diagnostic precision that Library Model platforms lack [43], [56], [57]. However, as established in Section 2.6.2, these systems operate in pedagogical isolation: their function terminates at the point of assessment and labeling, and they are not integrated with any instructional infrastructure capable of translating a learner's profile into a structured, adaptive educational pathway. They diagnose, but they do not educate.

This analysis exposes a specific and consequential architectural void herein termed the **"Missing Middle"** that exists at the intersection of these two paradigms. The Missing Middle is not merely the absence of a combined feature set; it is the absence of a closed-loop architecture in which diagnostic insights directly and continuously inform personalized instruction, and in which the learner's evolving behavioral and affective state continuously refines the system's model of who that learner is and what they need. Presently, no integrated framework seamlessly connects antecedent psychometric and cognitive calibration establishing a dynamic, multidimensional profile of the learner with real-time adaptive content delivery, scaffolded curriculum progression, and proactive affective monitoring that sustains the learner throughout a structured educational lifecycle [80], [29].

The primary research gap addressed by this thesis is therefore the absence of a unified, hybrid educational ecosystem capable of fusing the diagnostic precision of behavioral and vocational science with the instructional utility and scalability of an adaptive learning platform. The LearNova architecture is specifically designed to occupy this Missing Middle through four integrated contributions, each grounded in the theoretical frameworks reviewed in this chapter. First, the **Smart Compatibility Engine** a rule-based diagnostic pipeline administering a 251-question, five-instrument psychometric battery resolves the

cold-start problem identified in Section 2.5.2 by constructing a rich, multidimensional learner profile at onboarding, enabling immediate, cold-start-immune personalization from the learner's first interaction. Second, the **polymorphic content delivery system** serving Visual, Auditory, and Textual variants dynamically calibrated to each learner's VARK profile and continuously refined through Bayesian posterior updating operationalizes the modality alignment principle established in Section 2.2.1, directly addressing the format-content mismatch underlying engagement decay in content-centric platforms. Third, the **mastery-gated scaffolded progression architecture** enforcing CLT-informed complexity pacing and ZPD-aligned advancement thresholds addresses the cognitive overload and premature advancement failures documented in Section 2.2.2. Fourth, the **affective computing pipeline** deriving risk scores from parallel linguistic and behavioral signal streams and escalating to human-in-the-loop intervention when thresholds are breached addresses the affective blind spot of diagnostic platforms identified in Section 2.6.2.

In this model, the learner is neither abandoned to navigate content alone nor assessed and labeled without subsequent support. Instead, they are proactively guided, adaptively taught, and continuously monitored from their first interaction to their final competency milestone a structurally distinct third paradigm that the existing bifurcation of the educational technology landscape has left unoccupied.

Chapter 3:
System Analysis, requirements and methodology

3.1 Market Analysis

This section examines prevailing inefficiencies within the contemporary educational technology market that motivated the development of the LearNova system. By analyzing patterns related to learner engagement, retention, and the availability of structured career guidance, the section highlights systemic limitations in existing digital education models. These observations provide empirical and conceptual justification for the design choices adopted in LearNova and establish the relevance of its proposed adaptive and mentorship-oriented architecture.

3.1.1 The Crisis of Engagement and Retention

One of the most persistent challenges facing large-scale digital education platforms is the inability to sustain learner engagement over extended periods. This issue is particularly evident in Massive Open Online Courses (MOOCs), where high initial enrollment figures are frequently followed by substantial learner attrition, with completion rates consistently reported in the range of 10 to 15 percent across large-scale academic implementations [9], [50]. Beyond its educational implications, disengagement has been associated with significant economic consequences: low levels of learner engagement contribute to substantial productivity losses at a global scale, with the Gallup State of the Global Workplace Report estimating the annual cost of employee disengagement at approximately $8.8 trillion, equivalent to nine percent of global GDP [35].

From a pedagogical perspective, disengagement in online learning environments is closely linked to the structural isolation experienced by digital learners. Unlike traditional classroom settings, many online platforms offer limited opportunities for interaction, feedback, or personalized support. As a result, learners frequently progress through content independently, without the social or instructional scaffolding necessary to sustain motivation and persistence a structural deficit that Self-Determination Theory identifies as a failure to fulfill the learner's fundamental psychological need for relatedness [12]. Analyses of learner feedback and platform reviews indicate that dropout behavior is driven primarily by perceptions of insufficient guidance and a lack of timely support during moments of difficulty [4]. Many leading e-learning platforms operate as content repositories, emphasizing the breadth of available materials rather than the quality of learner interaction, and consequently rely on learner self-discipline and intrinsic motivation resources empirically demonstrated to be insufficient for sustaining engagement in the absence of structured feedback cycles and scaffolded progression [5].

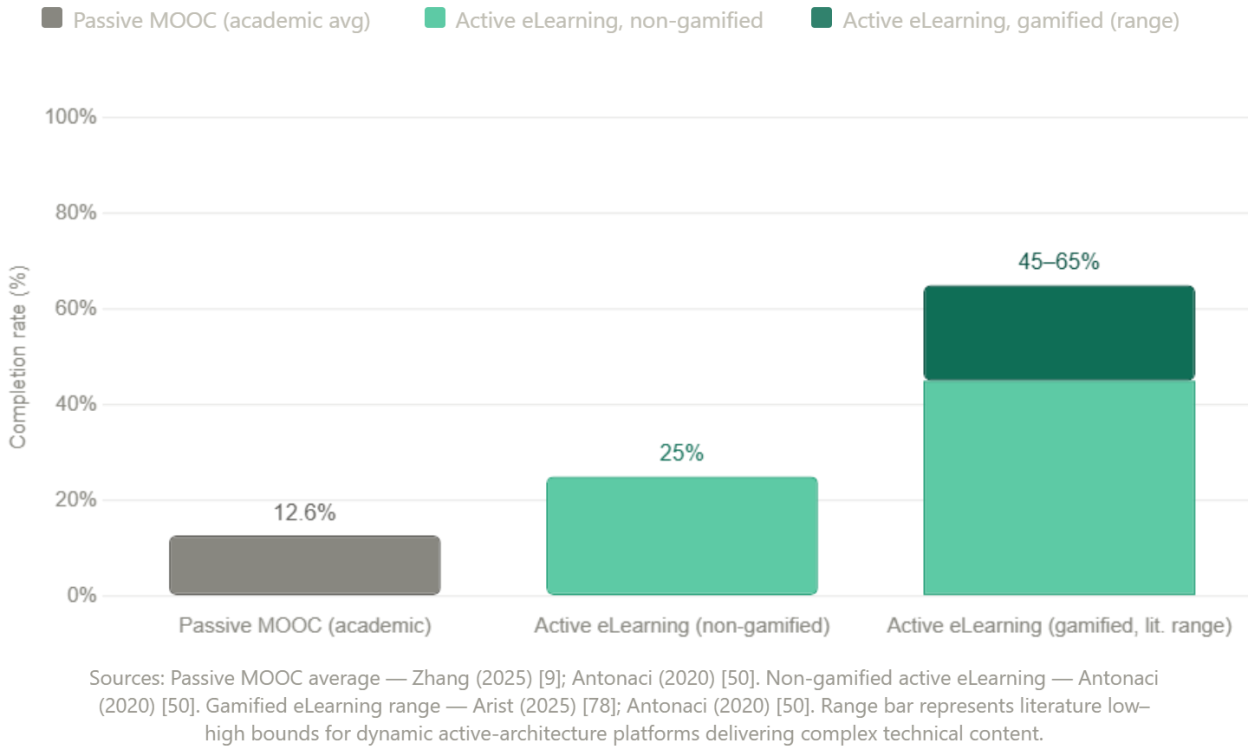


Figure 3.1: Completion rates by instructional delivery model. *Comparison of published completion rate benchmarks across three delivery models: passive MOOC academic delivery (12.6%), active eLearning without gamification (25%), and active eLearning with gamification (literature range: 45–65%). The range bar for the gamified model reflects published benchmarks for dynamic active-architecture platforms delivering complex technical content, and does not represent a LearNova-specific claim. Sources: passive MOOC Zhang (2025) [9], Antonaci (2020) [50]; active eLearning benchmarks Arist (2025) [78], Antonaci (2020) [50].*

As illustrated in Figure 3.1, the relationship between instructional delivery model and learner completion rate is not marginal it is substantial and progressive. Passive MOOC delivery, the dominant model among large-scale academic platforms, yields a documented industry average of approximately 12.6 percent [9]. Active eLearning architectures that introduce structured interactivity without formal gamification elements approximately double this figure to 25 percent [50]. When gamification mechanics are integrated including progression indicators, streak-based accountability structures, challenge events, and social comparison mechanisms published benchmarks for dynamic active-architecture platforms delivering complex technical content report completion rates in the range of 45 to 65 percent [78], [50]. This progressive improvement pattern, representing up to a fivefold increase over the passive MOOC baseline, reflects not an incremental refinement of existing models but a structural transformation in how learner motivation is sustained across the duration of a complex educational program.

This evidence establishes a clear market imperative: platforms constrained to passive content delivery are architecturally limited to completion rates that cannot be meaningfully improved without fundamental redesign. The research literature further clarifies the mechanisms underlying this gradient. Dropout in conventional MOOCs is not randomly distributed across a course it concentrates at structural friction points, with cumulative attrition reaching 44 percent by the third chapter transition alone [81]. Active and gamified delivery architectures mitigate this cliff effect by maintaining continuous engagement incentives through progression mechanics, social comparison structures, and immediate formative feedback, each of which addresses a distinct dimension of the dropout problem identified in the behavioral and learning sciences literature [50], [12], [1].

The LearNova platform is explicitly designed to occupy the active, gamified segment of this delivery spectrum. The integration of the AI-assisted agent **Mitchy** provides continuous pedagogical support and monitors interaction patterns indicative of disengagement through real-time sentiment analysis [25], while the gamification ecosystem comprising XP accumulation, streak mechanics, weekly challenges, and snapshot leaderboards operationalizes the engagement retention principles evidenced by the literature [50], [51]. By architecting the platform around active participation rather than passive consumption, and by intervening proactively when behavioral signals indicate declining engagement, LearNova positions itself within the delivery model that published market evidence consistently associates with the most substantial and defensible improvements in learner persistence and course completion [26], [52].

3.1.2 Guidance Deficits and the Paradox of Choice

A second major structural inefficiency identified in the contemporary e-learning landscape is the near-total absence of robust, data-informed career guidance mechanisms at the point of educational entry. Learners entering technical fields are confronted with an expansive and often bewildering array of specializations, certificate programs, and learning pathways. In the absence of adequate diagnostic tools or structured decision-support frameworks, this proliferation of options can precipitate what decision theory describes as *choice overload* a cognitive state in which the difficulty of selecting among numerous alternatives produces anxiety, decision paralysis, and ultimately suboptimal educational outcomes [4], [31]. The consequence extends beyond individual dissatisfaction; it represents a systemic misallocation of human capital at scale, as learners commit years of study and significant financial resources to pathways selected under conditions of structural uncertainty rather than evidence-based personal alignment.

This challenge is reflected with particular clarity in both immediate student behavior and long-term graduate outcomes. As illustrated in Figure 3.2, survey data indicate that a substantial majority approximately 78 percent of students navigating career and specialization choices self-report as "guessing" or uncertain of their chosen path, while only 22 percent describe themselves as confident and directionally clear. This disparity is not attributable to a scarcity of available options; on the contrary, it is the unguided proliferation of those options that constitutes the primary driver of decisional uncertainty a phenomenon extensively documented in behavioral and educational research [34], [71]. The abundance of pathways, rather than alleviating the cognitive burden of choice, systematically amplifies it.

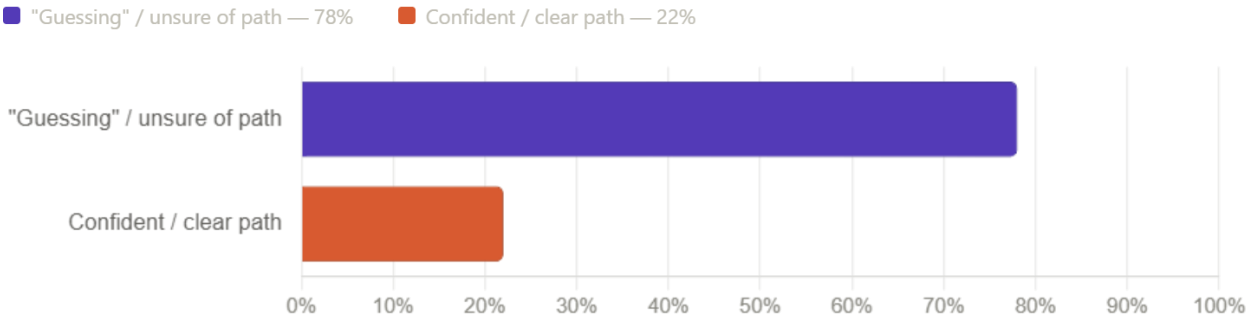


Figure 3.2: Student confidence levels regarding career path selection. Survey data demonstrating that 78% of students self-report navigating specialization choices without a clear sense of direction, compared to only 22% who describe themselves as confident. Sources: [34], [71].

The immediate uncertainty captured in Figure 3.2 does not diminish upon graduation; evidence suggests it frequently consolidates into long-term professional dissatisfaction. As quantified in Figure 3.3, approximately 44 percent of degree holders retrospectively express regret over their choice of academic major, citing misalignment between their qualifications and their intrinsic interests, developed competencies, or subsequent career trajectories [39], [70]. This figure is not a peripheral anomaly it represents nearly half the graduate population, indicating a systemic failure in the guidance structures that precede and inform specialization decisions. Furthermore, research by the Strada Institute indicates that graduates who begin their careers in roles misaligned with their qualifications are approximately 3.5 times more likely to remain in that underemployed condition a decade later, substantially compounding both the personal and macroeconomic cost of each misdirected educational decision [40].

Regret major choice — 44%Satisfied with major — 56%

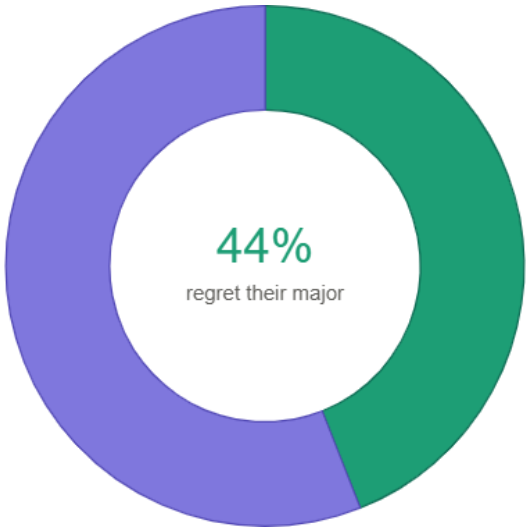


Figure 3.3: Percentage of college graduates who express regret over their chosen academic major. Approximately 44% of graduates report retrospective dissatisfaction with

their specialization decision, reflecting the durable consequences of guidance-deficient decision-making at educational entry. Sources: [39], [70].

The **LearNova** platform seeks to structurally address this guidance deficit through its **Smart Compatibility Engine** a multi-stage psychometric calibration system grounded in established educational and vocational research. Rather than offering autonomous or opaque recommendations, the engine administers a comprehensive five-domain diagnostic encompassing cognitive aptitude, personality dimensions informed by the Five-Factor Model (OCEAN), learning style preferences as categorized by the VARK framework, soft skills competencies, and vocational interests mapped via the Holland Code Career Test (RIASEC: Realistic, Investigative, Artistic, Social, Enterprising, Conventional) [27], [58]. By synthesizing these multidimensional data points into a coherent compatibility profile, the system displaces the anxiety-inducing, externally pressured, or arbitrarily driven decision-making process that the research literature consistently identifies as the primary antecedent of both immediate uncertainty and long-term regret [34], [39]. The result is a transparent, evidence-based alignment analysis one that positions specialization selection not as a commitment navigated under conditions of structural ignorance, but as an informed, psychometrically grounded decision undertaken with a clear evidential rationale.

3.1.3 Cognitive Alignment and Content Duration

A third critical dimension of the market analysis concerns the intersection of cognitive constraints, instructional format, and sustained learner engagement. Predominant e-learning models rely heavily on long-form video lectures as their primary pedagogical medium. This design choice may conflict with well-established principles of Cognitive Load Theory, which identifies the finite capacity of working memory as a primary constraint on effective knowledge acquisition, and with the empirically documented limits of sustained voluntary attention in digital learning environments [2], [8].

Empirical analyses of large-scale learner behavior indicate that engagement with video-based instructional content declines substantially and non-linearly across the duration of a session. As illustrated in Figure 3.4, normalized engagement remains comparatively stable through approximately the first six minutes of instructional video; beyond this threshold, decline accelerates markedly, with engagement levels falling to approximately 50 percent by the nine-minute mark and continuing toward a plateau near 15 percent at twenty minutes [77], [81]. Research by the American Society for Engineering Education and Quality Matters characterizes this inflection point as a structural behavioral threshold at which learner disengagement becomes the dominant response a pattern that holds across content domains and learner demographics, and that is independent of content quality per se. The implication for instructional design is significant: extended lecture formats are architecturally constrained in their capacity to sustain the engagement necessary for effective knowledge encoding, regardless of how well the material itself is presented.

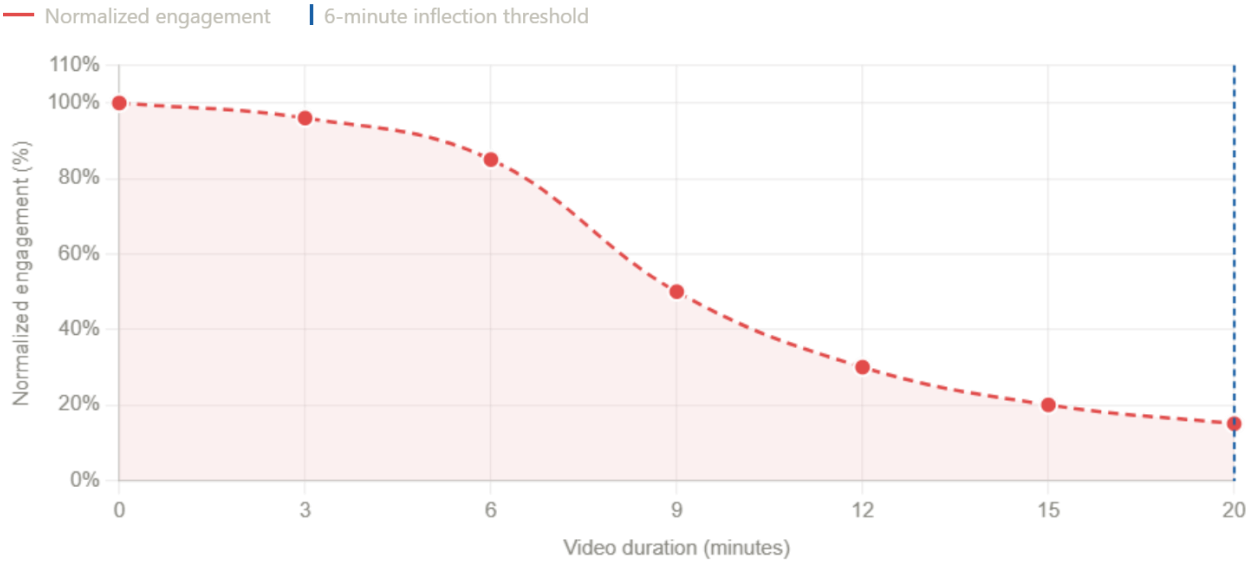


Figure 3.4: Measured decline in normalized learner engagement with video content as a function of duration. Engagement remains comparatively stable through approximately six minutes before entering accelerated decline, falling to roughly 50% at nine minutes and approaching 15% at twenty minutes. The vertical marker indicates the documented behavioral inflection threshold beyond which disengagement becomes the dominant learner response. Sources: [77], [81].

In response to this constraint, micro-learning strategies which decompose complex instructional content into brief, thematically bounded segments aligned with natural attention cycles have demonstrated measurable efficacy advantages over traditional long-form delivery. As illustrated in Figure 3.5, comparative analysis indicates that micro-learning architectures yield a retention performance index approximately 22 percent above the traditional long-form baseline [78], [79]. This improvement reflects a qualitatively different instructional model: rather than demanding sustained engagement across extended sessions, micro-structured delivery aligns individual content units with the learner's available cognitive bandwidth at each moment a principle consistent with Sweller's model of intrinsic and extraneous load management, wherein reducing the temporal demand of individual learning units preserves working memory resources for germane processing and schema construction [8].

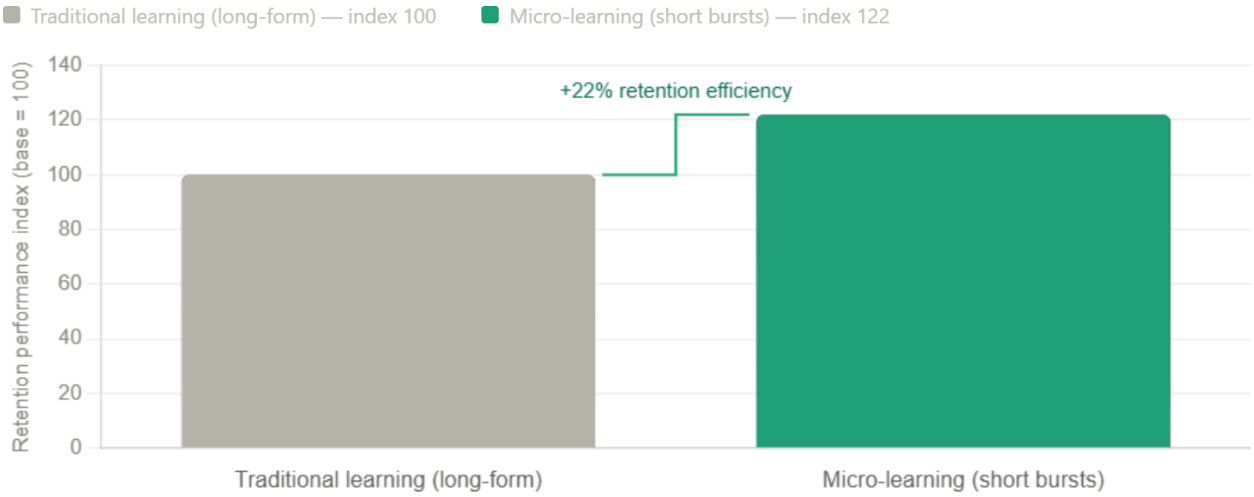


Figure 3.5: Comparative retention performance index for micro-learning versus traditional long-form instructional delivery. Micro-learning architectures yield a retention index approximately 22% above the long-form baseline (indexed at 100), reflecting superior alignment between content duration and learner cognitive bandwidth. Sources: [78], [79].

A further dimension of this misalignment concerns the assumption of a uniform optimal content format across a diverse learner population. The neurodiversity of learners in terms of preferred informational modality is extensively documented in the educational research literature. Self-reported preference studies involving large learner samples indicate that approximately 65 percent of students report a preference for visually organized information, while roughly 30 percent demonstrate a predominantly auditory processing orientation and approximately 5 percent report kinesthetic as their primary mode [18], [19]. It should be noted that these figures reflect self-reported modality preferences drawn from the specific survey instruments employed by Pashler et al. and Kharb, and are distinct from the global VARK population database, which reports different distributional profiles; both bodies of evidence, however, converge on the same fundamental conclusion that learner populations are cognitively heterogeneous, and that a single instructional format serves only a fraction of any given cohort optimally. As illustrated in Figure 3.6, the predominance of visual preference in these datasets implies that the majority of learners are systematically underserved by platforms delivering content exclusively through auditory-linear video lectures. A critical distinction must be maintained: the terms "video" and "visual learning" are not synonymous. Conventional lecture video is fundamentally an auditory-sequential medium the visual channel is largely occupied by talking-head delivery or static slide presentations and consequently does not activate the spatial and schematic processing mechanisms that characterize genuinely visual cognitive engagement [77]. Learners whose dominant modality is visual-spatial derive substantially greater pedagogical benefit from annotated diagrams, animated infographics, and structured visual representations than from standard video formats delivering equivalent propositional content.

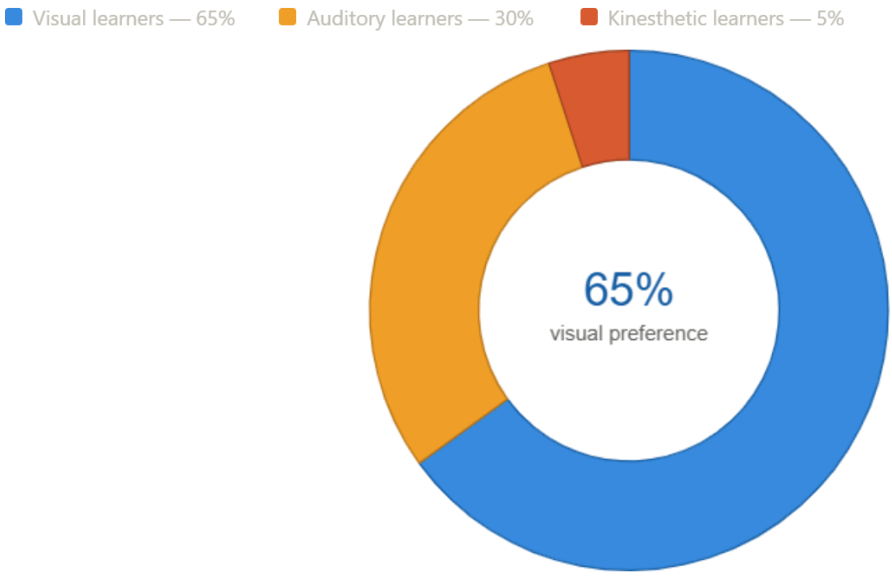


Figure 3.6: Distribution of self-reported learning style preferences within a representative learner population. Visual preference predominates at approximately 65%, followed by auditory at 30% and kinesthetic at 5%, illustrating the heterogeneity of modality preferences that a uniform instructional format cannot adequately address. Sources: [18], [19].

The **LearNova** architecture addresses these intertwined constraints through a dual-axis strategy grounded in both cognitive science and multimedia learning theory. First, the platform operationalizes micro-learning principles by structuring all curricular content into brief, thematically cohesive units a design decision informed by Vygotsky's scaffolding framework and the Zone of Proximal Development, wherein instructional segments are scoped to remain cognitively accessible yet sufficiently challenging to sustain productive engagement [4], [6], [8]. Second, the platform decouples content format from uniform delivery through its polymorphic **Content Slot** architecture: each instructional topic is associated with three parallel representational variants a visual asset (such as an animated diagram or infographic), an auditory narration, and a structured textual transcript and the system's adaptive logic delivers the variant corresponding to the learner's calibrated modality profile. This operationalizes the Dual Coding Theory principle that information encoded through representationally aligned channels produces more durable and transferable cognitive schemas than information received through misaligned or redundant modalities [3], [2], thereby addressing both the duration-based attrition documented in Figure 3.4 and the format-based misalignment illustrated in Figure 3.6 simultaneously.

3.1.4 The Soft Skills Disconnect

A fourth structural gap identified within the contemporary educational technology landscape is the persistent misalignment between academic training outputs and the holistic competency requirements of the modern labor market. While traditional degree programs and the majority of e-learning platforms emphasize the acquisition of technical skills encompassing programming syntax, software tool proficiency, and domain-specific knowledge industry analyses consistently indicate that long-term employability and career advancement are more frequently determined by complementary interpersonal and behavioral competencies, broadly classified in the organizational literature as *soft skills* [38], [36]. This misalignment is not incidental; it may be understood as reflecting a structural tendency within educational systems to optimize for competencies that are most readily assessed and certified, while systematically underinvesting in those that the labor market most consistently rewards [38].

The scale of this prioritization within professional recruitment is considerable. Survey data from large-scale analyses of hiring manager behavior indicate that approximately 91 percent of talent acquisition professionals prioritize soft skills encompassing communication proficiency, problem-solving capacity, teamwork, and adaptability as primary evaluative criteria during candidate assessment, compared to only 9 percent who weight hard technical qualifications as the dominant selection factor [35], [43]. As illustrated in Figure 3.7, this distribution is not a marginal preference but a substantial inversion of the emphasis placed by most academic curricula, in which technical certification constitutes the primary and frequently sole indicator of graduate readiness.

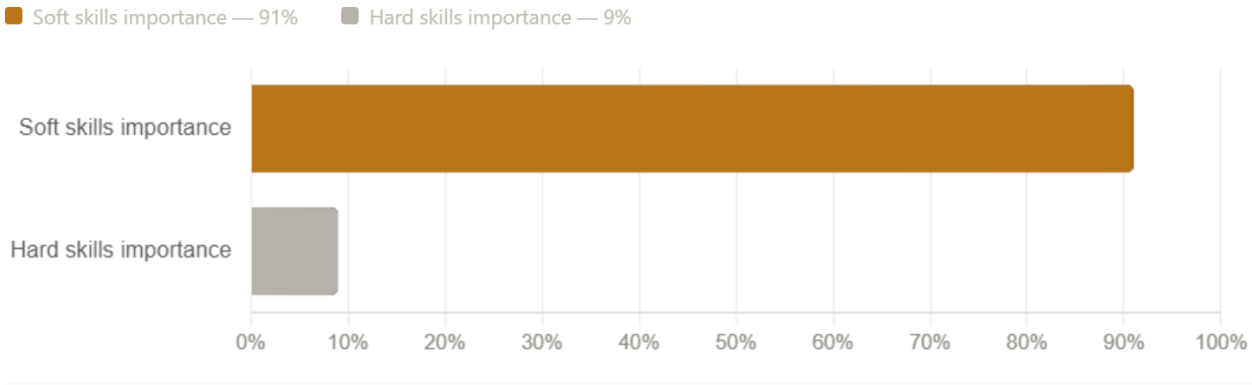


Figure 3.7: Recruiter prioritization of soft skills versus hard technical skills during candidate assessment. Approximately 91% of talent acquisition professionals identify soft skills as the primary evaluative criterion, compared to 9% who prioritize hard technical qualifications a disparity that reveals the systematic gap between what academic programs certify and what the labor market actually selects for. Sources: [35], [43].

The consequences of this misalignment extend beyond individual candidate underperformance to measurable organizational cost. Research into early career attrition and new hire failure indicates that approximately 89 percent of cases involve deficiencies in interpersonal and behavioral competencies including emotional intelligence, collaborative capacity, and communicative effectiveness rather than inadequate technical

capability, as illustrated in Figure 3.8 [38], [73]. Among the primary causal factors underlying new hire underperformance, technical skill gaps account for a minority of cases, suggesting that the dominant paradigm of technical-certification-focused education is structurally insufficient as a comprehensive preparation model. Content analyses of job postings in technical domains corroborate this finding at the demand side: requirements for interpersonal and cognitive soft skills appear with substantially greater frequency in employer listings than specifications for particular technical tools or programming languages, indicating a persistent and structurally unmet labor market demand [32], [43].

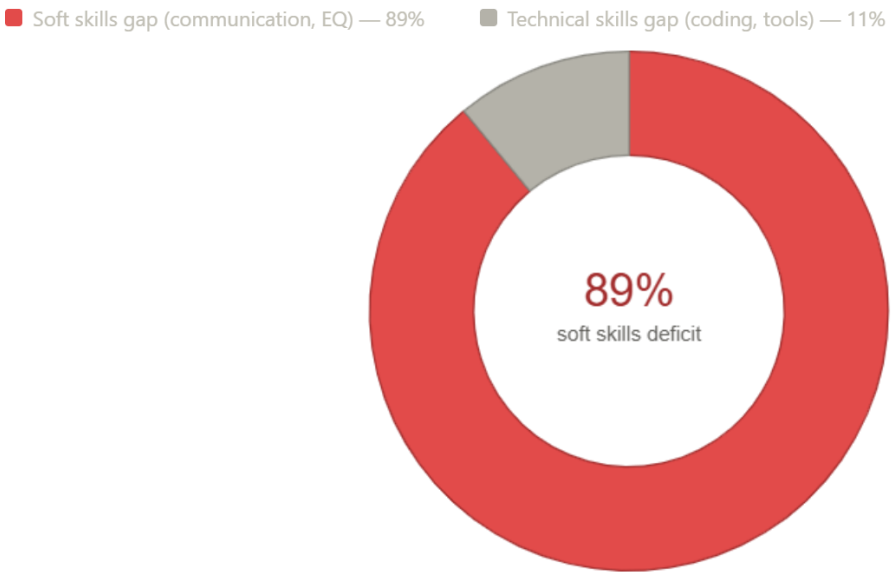


Figure 3.8: Analysis of primary contributing factors to new hire underperformance. Approximately 89% of early career attrition cases are attributable to soft skills deficiencies encompassing communication, emotional intelligence, and collaborative capacity rather than technical skill gaps, demonstrating that technical certification alone provides an insufficient basis for predicting workforce readiness. Sources: [38], [73].

To address this deficit, the **LearNova** framework explicitly integrates soft skills assessment into its foundational calibration layer via the **Smart Compatibility Engine**. During the initial diagnostic phase, learners are evaluated on key behavioral and cognitive competencies using two established assessment instruments: a Situational Judgment Test assessing practical proficiency in communication, conflict resolution, leadership, and ethical reasoning, and a personality inventory based on the Five-Factor Model (OCEAN), which captures stable dispositional traits including conscientiousness, agreeableness, and emotional stability [27], [58]. This calibration serves a dual analytical function: it surfaces the learner's behavioral profile, directing attention toward competency areas that may warrant deliberate development prior to professional deployment, and it enriches the system's holistic compatibility profile. Through this enrichment, the engine is enabled to align the recommended specialization pathway with not only the learner's cognitive aptitudes but also their interpersonal behavioral tendencies the competencies that the

evidence reviewed above identifies as the predominant predictors of long-term career resilience and sustained professional performance.

3.1.5 Economic Implications of Skills Mismatch

A fifth and final dimension of this analysis examines the broader economic consequences of the persistent disconnect between educational output and labor market needs. As established in Section 3.1.1, the Gallup State of the Global Workplace Report estimates that disengagement attributable to role misalignment costs the global economy approximately $8.8 trillion annually [35]. This macroeconomic figure is not abstract; it is composed of millions of individual mismatches between graduate competencies and employer requirements, each of which imposes compounding costs on both parties to the labor market transaction.

The most acute regional manifestation of this pattern is observable in Egypt's graduate labor statistics. As illustrated in Figure 3.9, national labor data indicate that the unemployment rate among university graduates reaches approximately 23.1 percent more than double the 11.3 percent rate recorded among individuals with only secondary educational attainment [32], [33]. This counterintuitive inverse relationship between formal educational attainment and employment outcomes, documented across Arab Barometer surveys of the MENA region, represents not a cyclical anomaly but a structural failure in which university credentials are failing to function as reliable signals of workforce readiness. Furthermore, Egypt's concurrent ranking of 133rd out of 141 countries in graduate skill-set quality, as reported by the World Economic Forum, corroborates the severity of this misalignment and establishes the specific market context in which LearNova is positioned to intervene [33].

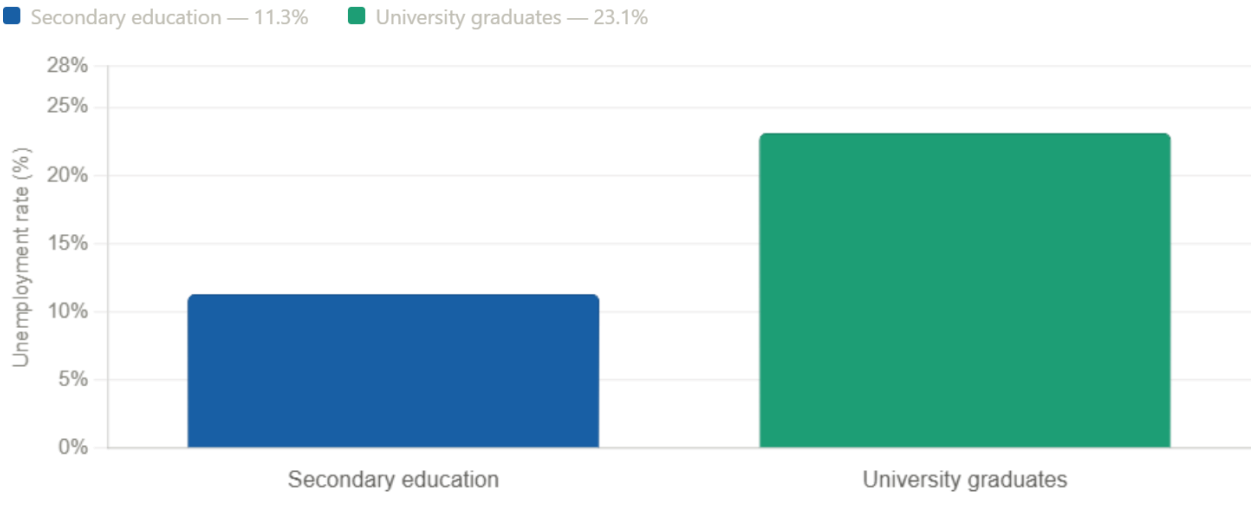


Figure 3.9: Comparative youth unemployment rates by educational attainment level in Egypt. University graduates experience an unemployment rate of 23.1% more than double the 11.3% rate among secondary school graduates illustrating that formal credentials are failing to bridge the gap between academic preparation and employer requirements. Sources: [32], [33].

The financial consequences of this mismatch extend beyond the individual learner to impose measurable organizational costs on employers. For enterprises, the total cost of a mis-hire encompasses recruitment expenditure, onboarding and training investment, lost productivity during the vacancy and ramp-up period, and eventual separation costs. Industry analyses indicate that for a technical professional earning approximately $100,000 annually, the total replacement cost reaches approximately $150,000 a 150 percent multiplier on annual compensation, as illustrated in Figure 3.10 [73], [74]. At the senior end of the technical hiring spectrum, this multiplier may extend to 200–400 percent, placing the cost of a single executive mis-hire in the range of $300,000 to $600,000 [73]. These figures quantify both the economic incentive for employers to seek more reliable signals of candidate fit and the structural market gap that a diagnostically grounded educational platform is positioned to address.

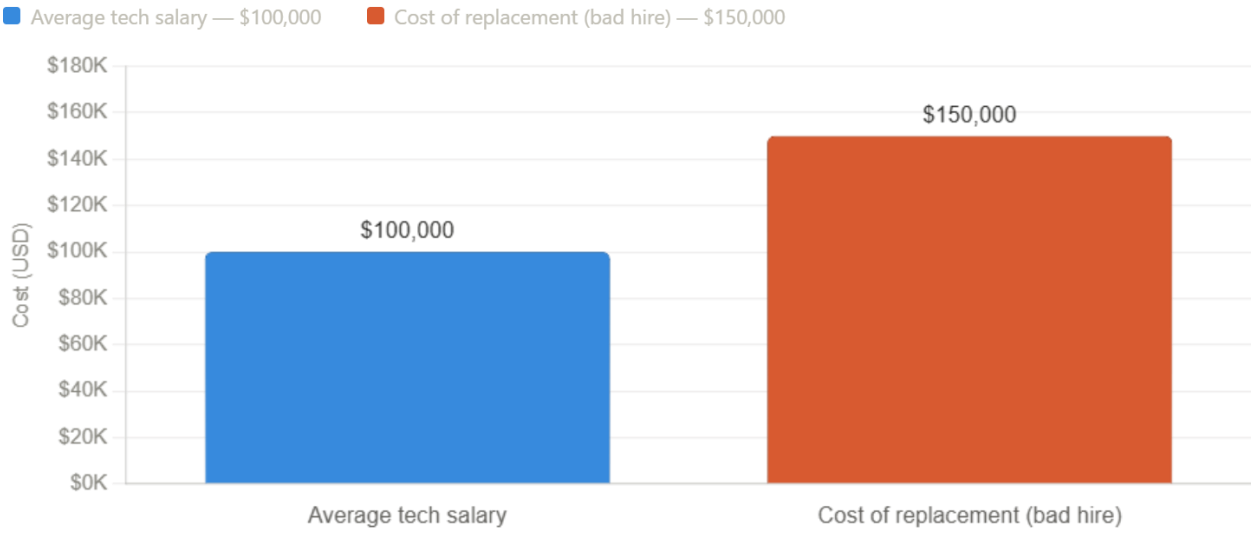


Figure 3.10: Comparative analysis of average technical salary versus the estimated total cost of replacement for a mis-hired technical professional. At an average annual compensation of $100,000, the replacement cost reaches approximately $150,000 a 150% multiplier demonstrating that the financial burden of a single mismatch substantially exceeds the annual cost of the role itself. Sources: [73], [74].

The **LearNova** platform is designed to function as a structural mitigating mechanism for this economic risk at both ends of the labor market transaction. From the learner's perspective, the platform addresses the employability gap by delivering a curated, industry-aligned curriculum that integrates both the technical competencies demanded by regional employer analyses and the behavioral soft skills identified in Section 3.1.4 as the predominant predictors of sustained career performance. From the employer's perspective, the **Smart Compatibility Engine**'s comprehensive diagnostic process generates a diagnostically verified profile of each learner's technical aptitude, behavioral disposition, and cognitive style a multi-dimensional competency signal that may be expected to provide substantially more informative hiring evidence than a traditional academic credential alone. By improving the quality and specificity of the competency information exchanged between job seekers and employers, the platform aims to reduce

the probability of costly mismatches, thereby directly addressing one of the most financially consequential inefficiencies in the technical talent market.

3.1.6 Strategic Calibration Strategy

The preceding analysis identifies five interdependent market failures: (1) the decay of learner engagement in passive instructional models, (2) the structural deficit in data-informed career guidance, (3) the cognitive misalignment arising from uniform, format-invariant content delivery, (4) the persistent gap in soft skills development within academic and digital learning programs, and (5) the macroeconomic inefficiency compounded by each of these preceding failures. Collectively, these dimensions indicate a fundamental requirement for an educational architecture that moves beyond content aggregation to incorporate a preliminary diagnostic layer of learner profiling. Empirical research on behavior in open online environments indicates that approximately 63 percent of students report feeling overwhelmed by the volume and complexity of available educational choices a state that impairs effective decision-making and initial commitment to a learning trajectory [34], [71].

To address this, the **LearNova** framework is predicated on a strategic calibration phase that precedes all content delivery. This phase implements a structured **Five-Factor Diagnostic**, designed to construct a multidimensional profile of the learner prior to any specialization assignment. The diagnostic integrates five established assessment instruments, each targeting a distinct domain of learner characterization: (1) *Cognitive Aptitude*, evaluated through reasoning assessments measuring logical, abstract, and spatial problem-solving capacity; (2) *Personality Dimensions*, assessed using the Five-Factor Model (OCEAN) to identify stable behavioral traits including conscientiousness, openness, and emotional stability; (3) *Learning Style Preference*, categorizing the learner's optimal information intake modality visual, auditory, or textual according to the VARK framework; (4) *Soft Skills Competencies*, evaluated through a Situational Judgment Test measuring behavioral proficiencies including communication, conflict resolution, and leadership; and (5) *Vocational Interest*, mapping intrinsic motivational orientations to established career archetypes via the Holland Code RIASEC taxonomy [18], [27], [58].

This comprehensive diagnostic layer constitutes the platform's foundational differentiator. It transitions the educational model from a reactive, content-centric paradigm in which the learner is required to self-navigate an undifferentiated catalogue to a proactive, learner-centric framework in which the system assumes the decisional and navigational burden of pathway selection. By establishing this profile prior to curriculum engagement, the **Smart Compatibility Engine** performs an initial strategic alignment between the learner's aptitudes, behavioral tendencies, and motivational orientations on one hand, and the cognitive and psychological demands of the available technical specializations on the other. The system thus functions not as a passive recommender but as an evidence-informed guidance engine one that employs empirically grounded diagnostics to reduce initial decisional uncertainty. The result is a tailored educational pathway optimized for competency fit, sustained motivation, and defensible long-term professional outcomes.

3.1.7 Competitive Feature Matrix

To contextualize the proposed **LearNova** architecture within the existing educational technology landscape, a structured comparative analysis was undertaken. This analysis evaluates the platform against four established global competitors Coursera, Udemy, LinkedIn Learning, and YouTube selected for their market dominance and recognition as primary sources for online technical upskilling [59]. The comparison focuses on the five core architectural pillars that constitute LearNova's primary differentiators, as derived from the market failure analysis presented in Sections 3.1.1 through 3.1.5.

The results of this feature analysis are summarized in Table 3.1 The matrix indicates a clear divergence in foundational approach: while incumbent platforms excel as comprehensive content repositories aggregating vast libraries of video-based instructional material they uniformly lack integrated systems for pre-instructional learner calibration and real-time adaptive delivery. It should be noted that certain competitors offer capabilities that may be characterized as adjacent to some of these features; Coursera, for instance, incorporates limited badge-based progression mechanics, and some platforms provide instructor Q&A channels. However, none of the evaluated competitors integrates these capabilities as a coherent, psychometrically grounded system each remains a discrete, surface-level feature rather than an architectural commitment to learner-centric guidance.

Table 3.1: Differentiation feature matrix LearNova vs. major platform competitors. A ✓ denotes full architectural implementation; a ✗ denotes absence of the feature as a core system capability.

differentiation Feature	LearNova	Coursera	Udemy	LinkedIn Learning	YouTube
Smart Compatibility Engine (Psychometric & Cognitive Calibration)	✓	✗	✗	✗	✗
Adaptive Content Slots (Auto-switching Video/Audio/Text)	✓	✗	✗	✗	✗
Integrated AI Mentorship (Real-time Pedagogical Intervention)	✓	✗	✗	✗	✗
Mandatory Soft Skills Calibration (Pre-Technical Assessment)	✓	✗	✗	✗	✗
Active Engagement Ecosystem (Gamification & Momentum Streaks)	✓	✗	✗	✗	✗

The analysis reveals a clear market segmentation between content-centric and learner-centric architectural paradigms. Competitor platforms primarily operate on a distribution model in which the platform hosts and organizes instructional material, placing the burden

of curriculum selection, pacing, and support largely on the user [4], [2]. In contrast, **LearNova** is architected as a learner-centric guidance system whose value proposition is predicated not on the volume of its content library but on its integrated diagnostic and adaptive mechanics the **Smart Compatibility Engine**, polymorphic **Content Slots**, and AI-informed mentorship which are designed to actively reduce cognitive friction, guide specialization decisions, and sustain engagement across the learning lifecycle [5], [29]. This positions LearNova not as a direct competitor in content aggregation, but as a novel layer of intelligent scaffolding that addresses the specific and empirically documented gaps in engagement, personalization, and holistic competency development identified throughout this market analysis. The architectural framework that enables this positioning is examined in detail in the system design chapters that follow.

3.2 Software Development Methodology

To ensure the successful delivery of a complex, adaptive educational ecosystem, the development lifecycle of the LearNova system adheres to the **Agile Software Development Methodology**, with a specific implementation utilizing the **Scrum framework**. This approach was selected to accommodate the iterative refinement inherently required for its data-informed components and to facilitate continuous feedback loops between the evolving technical architecture and the system's core pedagogical requirements. An Agile methodology is particularly suited to a research-oriented project of this nature, wherein requirements may evolve based on theoretical insights, calibration outcomes, and early-stage usability testing [80].

3.2.1 Rationale for Agile Adoption

The decision to adopt an Agile model over a traditional, sequential model such as the Waterfall approach was driven by three specific exigencies arising from the project's architecture and its overarching goals.

First, the development of the system's adaptive calibration and recommendation logic necessitates a fundamentally iterative tuning process. Core components such as the **Smart Compatibility Engine** and the AI-informed Pedagogical Support Module (**"Mitchy"**) are not predicated on static, exhaustively predefined rules, but rather on configurable models including the Bayesian learning-style posteriors stored in student_profiles.bayesian_alpha_visual, bayesian_alpha_auditory, and bayesian_alpha_textual, and the weighted track-assignment formulae encoded in personalization/track_weights.csv on the Railway backend that must be empirically refined as diagnostic data accumulates. Agile facilitates rapid prototyping and cyclical testing of these algorithms, permitting the underlying calibration logic to evolve in response to observed outputs rather than being rigidly fixed at the project's outset.

Second, the system is designed with a modular, service-oriented architecture comprising distinct yet interconnected components: the student-facing mobile application built with Flutter, the administrative interface, the Railway-hosted FastAPI scoring service (learnova_scoring_service), and the Supabase data and authentication layer. The Scrum framework enables the parallel and independent development of these loosely coupled subsystems within focused, time-boxed iterations designated as Sprints, substantially reducing integration risk and permitting incremental validation of each subsystem's functionality before full system-level integration is attempted.

Third, the system's conceptual foundation in user psychology and learning science mandates a user-centric design process throughout. Design principles derived from established frameworks including the VARK model for learning-style adaptation, Cognitive Load Theory for content sequencing, and Self-Determination Theory for gamification mechanics require continuous validation through usability testing and prototype evaluation cycles. The Agile principle of "responding to change over following a plan" [80] permits the development team to incorporate feedback from iterative prototype testing directly into the active design cycle, ensuring that the final user experience is substantively informed by authentic learner interaction rather than static, a priori assumptions.

3.2.2 The Scrum Framework Implementation

The development process is structured around fixed-duration iterations, or Sprints, each spanning two weeks. This workflow is governed by standard Scrum artifacts and ceremonies designed to maintain operational focus, team transparency, and continuous incremental delivery.

The **Product Backlog** functions as the single, prioritized repository of all functional and non-functional requirements spanning items such as "Implement streak persistence logic in user_streaks," "Integrate Gemini API with multi-model fallback chain for Mitchy," and "Define Row-Level Security policies for risk_scores." From this master backlog, the **Sprint Backlog** is derived for each iteration, containing a committed subset of items representing the immediate development priorities. At the conclusion of each Sprint, a **Sprint Review** is conducted, during which a potentially shippable functional increment such as a working diagnostic calibration flow or a completed module assessment pipeline is demonstrated and evaluated against acceptance criteria. A decision gateway then determines whether the Sprint's output is considered complete, or whether specific features require further refinement before the iteration can be closed. The cyclical process culminates, upon successive successful Sprint completions, in the transition to the final deployment phase, as illustrated in Figure 3.11

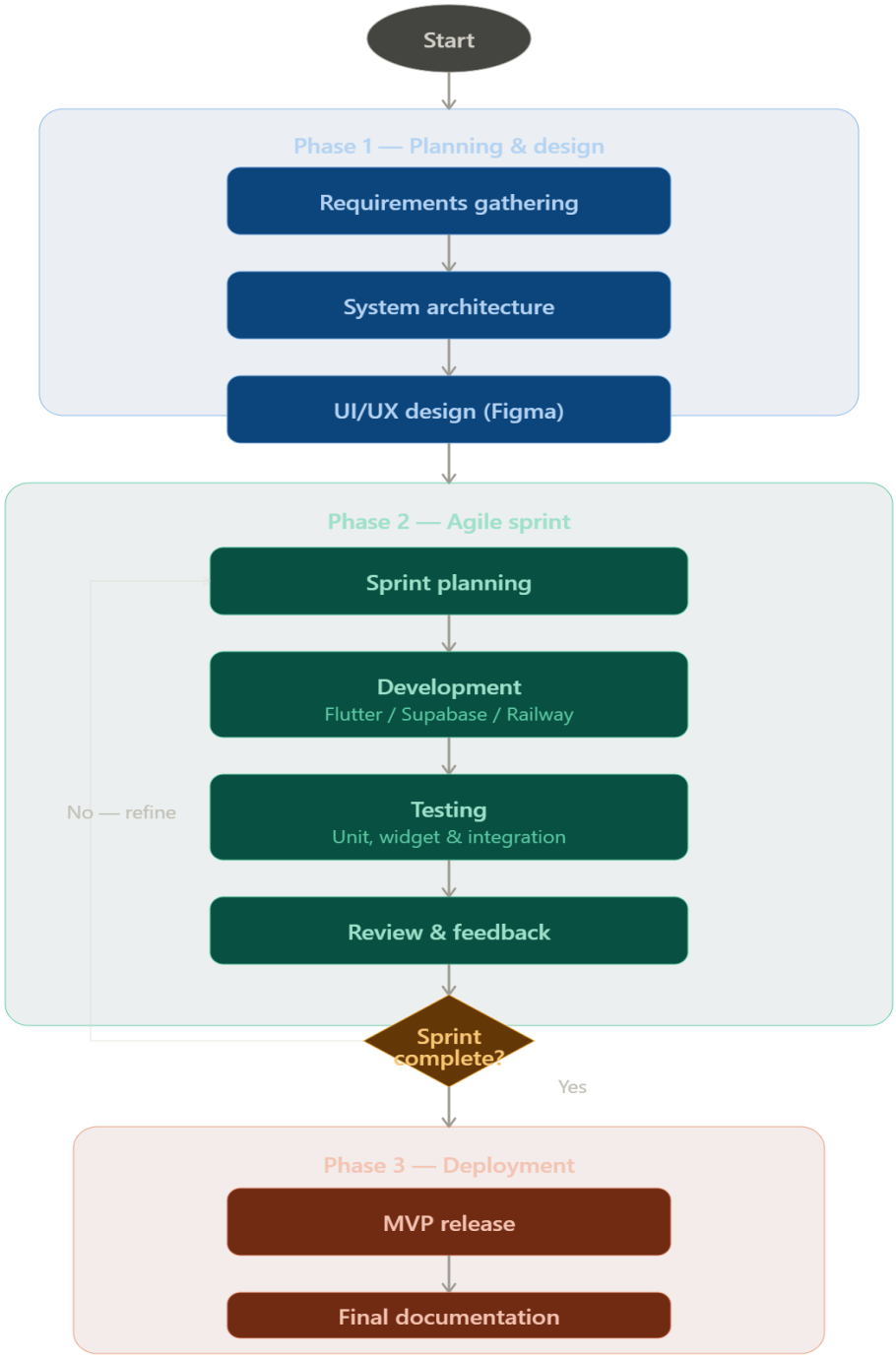


Figure 3.11: The Agile Software Development Life Cycle (SDLC) adopted for the LearNova project, illustrating the three-phase structure comprising Phase 1 (Planning and Design: requirements gathering, system architecture, and UI/UX design), Phase 2 (the iterative Agile Sprint loop encompassing sprint planning, development in Flutter and Supabase, unit and widget testing, and review and feedback, with a conditional Sprint completion gateway), and Phase 3 (Deployment: MVP release and final documentation).

3.3 Requirement Analysis

3.3.1 Functional Requirements

The functional requirements delineate the specific, observable behaviors the system must perform to achieve its stated objectives. These requirements are organized according to the six core architectural modules of the LearNova platform, providing a structured specification that directly informs both the development process and subsequent validation.

Module A: User Management & Identity

This module governs all user access control and profile management operations. The system shall enable users to register and authenticate utilizing either an email/password combination or social authentication via Google, leveraging the Supabase Auth infrastructure specifically the auth.users schema and the handle_new_user trigger function, which auto-populates the public users table upon signup for secure, automated session management (FR-01). To ensure data integrity and validate newly registered accounts, the system shall enforce a One-Time Password (OTP) verification process for all email-based registrations (FR-02). Following successful authentication, users shall be able to manage their personal profiles, including the upload and storage of external certifications or portfolio artifacts as verifiable proof submissions (FR-03). The system shall implement Role-Based Access Control (RBAC) to enforce distinct permission sets, differentiating between **Student** users who possess read access to active curriculum content and **Admin** users who hold privileges for content management and analytical oversight enforced through Supabase Row-Level Security (RLS) policies applied across all database tables via the is_admin() and current_user_is_admin() helper functions, which evaluate users.role against auth.uid() (FR-04).

Module B: Assessment & Calibration (The Engine)

This module encompasses the core diagnostic machinery constituting the **Smart Compatibility Engine**. The system shall administer a comprehensive, multi-stage **5-Point Calibration Test** designed to assess five distinct psychometric dimensions: (1) cognitive aptitude, evaluated through abstract and logical reasoning tasks; (2) personality type, utilizing the IPIP implementation of the Five-Factor Model (Big Five) comprising Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism stored as Big Five features in diagnostic_test_results.computed_scores; (3) learning style preference, operationalized through the VARK framework with results mapped to student_profiles.learning_style as one of Visual, Auditory, or Textual; (4) soft skills competencies, assessed via Situational Judgment Test (SJT) scenarios; and (5) career orientation, mapped through the O*NET RIASEC interest inventory, with Artistic, Realistic, and Investigative categories serving as base anchors for the DA, DE, and DS specialization tracks respectively (FR-05). To accommodate diverse user backgrounds, the system shall provide a **Hybrid Input Mechanism** allowing users to complete assessments natively within the application or to submit verified results from external providers, with the diagnostic_test_results.result_source field recording either in_app or external, and external submissions requiring an accompanying screenshot for administrative validation (FR-06). Upon assessment completion, the system shall generate a visualization summarizing the user's psychometric profile derived from

diagnostic_test_results.computed_scores, alongside an **Explainability Panel** delivering a text-based justification for the algorithmic track recommendation, grounded in the computed DA, DE, and DS final scores (FR-07, FR-08). Upon successful completion of all five tests, the system shall award a one-time **+500 XP Milestone Bonus**, written to student_profiles.xp_total via the increment_xp() RPC function (FR-08a).

Module C: Adaptive Learning Delivery

This module manages the personalized delivery of curriculum content. The system shall initially assign all newly enrolled users to a mandatory **Literacy Phase** (Foundation Track), designed to establish a shared baseline of technical competence prior to specialization (FR-09). Upon successful completion of this phase verified through the level assessment mechanism, with results persisted in student_level_attempts including the passed boolean and optional mitchy_feedback AI-graded response the system shall unlock the Specialization Sub-Track (Data Analysis, Data Engineering, or Data Science) corresponding to the user's calibrated profile as stored in student_profiles.assigned_track (FR-10). Each lesson topic within the curriculum shall be stored with three distinct **Content Slots** Visual, Auditory, and Textual accessible via topic_resources rows filtered by format_type (FR-11). The system shall automatically serve the content format matching the user's active learning style by default, as resolved from student_profiles.learning_style, whilst recording every consumption event including format_type, time_spent_seconds, engagement_score, and the bayesian_eligible flag in content_engagement_logs to enable Bayesian learning-style posterior updates and format effectiveness analysis (FR-12). The system shall additionally support a learning_mode field in student_profiles (structured or exploration), with exploration mode sessions bounded by exploration_started_at and exploration_ends_at timestamps (FR-12a). Progression shall be governed by a rule of **Progressive Unlocking**, requiring a passing score greater than 70% on a module assessment enforced through student_module_attempts.passed, with module attempts inserted server-side by the submit-module-attempt Edge Function before access to subsequent content is granted. Passing a module assessment awards a flat **+20 XP**, written via increment_xp() (FR-13).

Module D: CRM & AI Intervention

This module facilitates continuous learner support and proactive administrative intervention. The system shall provide a real-time chat interface enabling communication between the student and Mitchy, with sessions persisted in chat_sessions and individual messages including the role (user or assistant), content, mitchy_action JSONB for structured AI navigation actions, and detected_learning_state stored in chat_messages (FR-14). Mitchy's responses shall be grounded in curriculum content through a **Retrieval-Augmented Generation (RAG)** architecture, whereby the agent queries the document_chunks table which stores pgvector-embedded content chunks indexed from the topic hierarchy using vector similarity search to ground replies in platform-specific material (FR-14a). The AI agent shall perform sentiment analysis on each inbound user message, with the resultant affective state stored in chat_messages.detected_learning_state and longitudinal sentiment trajectories maintained in student_sentiment_history (FR-15). If a user's analyzed sentiment score falls below a predefined safety threshold, the system shall trigger a **High-Risk Alert**, creating a record in risk_scores with alert_triggered = true, risk_level set to one of low,

medium, high, or critical, and a feature_snapshot JSONB capturing the ML features used in the computation (FR-16). Administrators shall be able to review flagged students, claim alerts by populating intervention_logs.claimed_at, record intervention actions in intervention_logs.action_taken, and mark alerts resolved via intervention_logs.resolved_at. Intervention notifications shall be delivered to students through in_app_notifications, inserted exclusively by admins with notification_type set to risk_alert or general, and delivered as push notifications via FCM tokens registered in user_devices (FR-17).

Module E: Gamification & Community

This module is responsible for sustaining learner motivation and engagement through a structured, server-side gamification engine. The system shall maintain a **Global Leaderboard** populated via daily snapshots in leaderboard_snapshots recording xp_at_snapshot and rank_at_snapshot per (user_id, track, snapshot_date) ranking users by accumulated XP stored in student_profiles.xp_total (FR-18). The XP reward matrix shall operate as follows: topic content consumption awards **+5 XP** per content slot; topic completion awards an additional **+15 XP** bonus triggered when student_progress.status transitions to completed; level assessments and weekly challenges award tiered XP of **+20 XP** (Easy), **+50 XP** (Mid), or **+100 XP** (Hard) based on difficulty selected at attempt time; weekly challenges further award **partial XP proportional to accuracy** given their single-attempt constraint enforced by the unique key on (user_id, challenge_id) in student_challenge_attempts. All XP writes are performed exclusively server-side via the increment_xp(user_id, calculated_xp) RPC function called by Supabase Edge Functions, preventing any client-side manipulation (FR-18a). The system shall track **Momentum Streaks** consecutive days of qualifying activity maintained in user_streaks.current_streak_days and last_activity_date and apply a **1.1× XP multiplier** to all XP earned on days when an active streak is confirmed, resetting the multiplier to 1.0× upon inactivity. Achievements earned across the journey are recorded in user_achievements referencing the achievements_dictionary master table, which defines criteria type, threshold, and badge image path for each possible achievement (FR-19).

Module F: Content Management System (Admin)

This module empowers administrative oversight of the full curriculum hierarchy. Via a dedicated administrative interface, authorized users shall have the ability to Create, Read, Update, and Delete (CRUD) course content at all hierarchical levels courses, levels, modules, topics, and their associated resources and images governed by admin-scoped RLS policies enforced across the courses, levels, modules, topics, topic_resources, and topic_images tables, with content visibility controlled by the is_active boolean on each table (FR-20). Administrators shall upload and associate Visual, Auditory, and Textual content variants to each topic node, storing asset paths in topic_resources with format_type values of Visual, Auditory, or Textual, and binary assets stored in the content-assets Supabase Storage bucket (FR-21). Assessment parameters shall be configurable at the module and level level, including question bank management for module_assessment_questions and level_assessment_questions, passing score thresholds in module_assessments.passing_score, and tiered XP reward structures across xp_reward_easy, xp_reward_mid, and xp_reward_hard fields in both assessment tables (FR-22).

3.3.2 Non-Functional Requirements

The non-functional requirements define the essential quality attributes and operational constraints of the system, ensuring that it is not merely functional but also robust, usable, secure, and scalable under real-world deployment conditions.

Performance Constraints dictate the system's responsiveness thresholds. To maintain a fluid conversational experience with Mitchy, the AI mentor shall achieve a Time-to-First-Token of under 1.5 seconds, with complete response generation targeted within 5 seconds under standard load. The LLM provider timeout is configured at 20 seconds in Railway, with a two-provider fallback chain activated sequentially upon primary model failure or quota exhaustion (NFR-01). Core learner-facing interfaces shall render fully within 3 seconds over a standard 4G connection (NFR-02).

Reliability and Availability requirements ensure consistent access and data integrity. The platform shall target 99.5% service uptime during critical academic periods (NFR-03). State persistence for calibration and quiz sessions shall be enforced through the unique constraint on (user_id, test_number) in diagnostic_test_results and per-attempt records in student_module_attempts, safeguarding against data loss from network interruption (NFR-04).

Security and Privacy requirements are paramount given the sensitive psychometric data processed. All personally identifiable information and assessment results stored in student_profiles and diagnostic_test_results shall be protected through industry-standard encryption at rest (NFR-05). The system shall comply with core GDPR principles, incorporating a data deletion mechanism enabling users to request permanent removal of their profile and associated records (NFR-06). The AI recommendation system shall maintain audit-capable decision logs via diagnostic_test_results.computed_scores which stores the complete feature payload including all Big Five, VARK, SJT, O*NET, and IQ scores enabling review of any anomalous track assignment (NFR-07).

Scalability requirements address user growth capacity. The backend architecture realized through Supabase's managed PostgreSQL infrastructure and Railway's FastAPI service shall accommodate concurrent user load without schema modification (NFR-08). The recommendation model weights in personalization/track_weights.csv shall support parameter updates through Railway's hot-redeploy mechanism without interrupting active sessions (NFR-09).

Usability and Compatibility requirements govern the end-user experience. The mobile interface shall adhere to Google's Material Design 3 guidelines [86], ensuring intuitive navigation for its primary digitally native demographic (NFR-10). The Flutter application shall deliver full functional parity across Android (API Level 28 and above) and iOS (version 15 and above) from a single unified codebase [82] (NFR-11).

3.4 Feasibility Study

Prior to finalizing the architectural design, a comprehensive feasibility study was undertaken to evaluate the viability of developing and deploying the LearNova system across three dimensions: technical capability, economic viability, and operational practicality.

3.4.1 Technical Feasibility

The technical feasibility analysis confirms that all functional and non-functional requirements defined in Section 3.3 are realizable using contemporary, stable, and well-supported technologies selected for cross-platform compatibility, real-time performance, modular extensibility, and cost-effective scalability.

Flutter [82] enables single-codebase deployment across Android and iOS, ensuring consistent feature parity whilst substantially reducing maintenance overhead relative to separate native codebases. **Supabase** [84] provides the integrated authentication layer (GoTrue/JWT), real-time PostgreSQL database, serverless Edge Functions serving as the API gateway and XP calculation engine, and scalable object storage across three buckets (calibration-media, content-assets, and student-submissions). The **Railway-hosted FastAPI service** (learnova_scoring_service) handles diagnostic scoring via the answer-key-based pipeline in scoring/diagnostics.py, track assignment through personalization/learnova_scoring.py, and daily ML analytics via analytics/ml_pipeline_reviewed.py executed on a daily cron schedule. Mitchy's conversational AI capabilities are delivered through the **Gemini API** [87], with a hardened fallback chain ensuring response availability including local affective fallback logic in mitchy/chat_logic_v3.py even under quota exhaustion, 503 high-demand errors, or timeout conditions, thereby satisfying NFR-01.

3.4.2 Economic Feasibility

The economic model prioritizes minimal capital expenditure whilst maximizing functional utility. Infrastructure costs are substantially mitigated through the freemium tiers of Supabase and Railway, rendering hosting, database, and core service costs negligible throughout the development and initial validation phases. The core stack Dart, Flutter, FastAPI, and PostgreSQL incurs no proprietary licensing fees [82]. The system's indirect economic value is, nevertheless, substantial: as demonstrated in Chapter 1, skills mismatch and recruitment inefficiency generate estimated global productivity losses of $8.8 trillion annually [35], with individual mis-hire replacement costs ranging from 30% to 400% of annual salary depending on role seniority [73], [74]. By producing graduates with verified, market-aligned competencies across the DA, DE, and DS specialization tracks each mapped to employer-validated skill clusters sourced from Wuzzuf job posting analysis LearNova represents a quantifiable cost-mitigation instrument for the regional labor market.

3.4.3 Operational Feasibility

User adoption potential is assessed as high. The primary demographic digitally native learners aged 16–24 is profoundly accustomed to mobile-first, on-demand experiences, and Material Design 3 adherence [86] minimizes the application's learning curve.

Pedagogically, the enforcement of the mandatory Literacy Phase directly mitigates the primary operational failure mode identified in the market analysis: the premature exposure of novice learners to advanced specialization content, which prior research associates with the frustration-driven attrition rates consistently reported in the 10–15% MOOC completion range [5], [9]. By ensuring verified baseline competence before specialization branching, the system structurally reduces this risk.

3.5 Target Audience Profiling

Following the quantitative analysis of market failures presented in Sections 3.1.1 through 3.1.5, a qualitative synthesis of primary user segments is essential to ground the system's design requirements in a concrete understanding of end-user needs and contexts. Based on the demographic, psychographic (attitudinal, motivational, and behavioral), and structural data analyzed in prior sections and informed specifically by the graduate unemployment patterns documented for the Egypt and MENA region two distinct yet interrelated user archetypes have been constructed. These archetypes represent the core demographics most acutely affected by the identified educational gaps and for whom the **LearNova** system is primarily designed. The segmentation is guided by principles from Cognitive Load Theory, Self-Determination Theory, and Constructivist Learning Theory, ensuring that the system's interventions are theoretically aligned with documented learner behavior [2], [4], [34].

3.5.1 The Archetype: "The Disconnected Learner"

This segment consists of senior undergraduate students or recent graduates, typically aged 21–24, predominantly from the MENA region. These learners possess foundational technical knowledge often acquired through degrees in Computer Science or Engineering but exhibit low immediate job-market readiness. The paradox of holding formal academic qualifications while experiencing disproportionate difficulty securing relevant employment is consistent with the regional patterns documented in Section 3.1.5, where university graduates in Egypt were found to face an unemployment rate of 23.1 percent more than double that of secondary school graduates [32], [33].

Psychographically, members of this segment report high levels of decisional overwhelm and anxiety when navigating the unstructured landscape of online upskilling resources a behavioral state consistent with the choice overload phenomenon discussed in Section 3.1.2 [34]. Learning preferences align with population-level data indicating a high prevalence of visual processing tendencies; however, engagement is frequently compromised by the long-form video formats that dominate existing platforms, producing the attention decay trajectory documented in Section 3.1.3 [18], [77].

Three primary challenges define this archetype. First, the Paradox of Choice: anxiety and hesitation when selecting a technical specialization, driven by fear of misalignment and future regret a concern empirically grounded in the 44 percent graduate major regret rate reported in Section 3.1.2 [39], [70]. Second, Instructional Isolation: the absence of interactive support within standard asynchronous online courses, a structural deficiency contributing directly to the 10–15 percent completion rates characteristic of passive MOOC delivery [5], [6]. Third, the Employability Gap: a deficiency in evaluated soft skills including communication, problem-solving, and emotional intelligence that impedes success in technical recruitment despite adequate theoretical knowledge [38], [43]. The fundamental need of this segment is for structured, guided pathways that translate existing theoretical competencies into demonstrable, employer-verified, job-ready capabilities.

3.5.2 Secondary Archetype: "The Pre-Undergraduate"

This segment comprises high school seniors and incoming university freshmen, typically aged 16–18, situated at a critical juncture in the selection of undergraduate majors and early career orientations. Decision-making within this group is heavily influenced by external factors including family expectations, societal prestige hierarchies, and peer conformity and is characterized by the high-stakes nature of commitments that may determine educational and professional trajectories for a decade or more.

Psychographically, these users demonstrate low labor market awareness, frequently relying on anecdotal evidence, social media narratives, or outdated occupational perceptions rather than data-informed assessments of sectoral demand and personal fit [34], [71]. Their motivational orientation is primarily extrinsic driven by the desire for security and external validation rather than intrinsic interest alignment which Self-Determination Theory identifies as a risk factor for both academic disengagement and long-term career dissatisfaction [12]. A 2023 KPMG survey of high school students in the UAE found that fewer than 1 percent intended to pursue Computer Science or Artificial Intelligence as cited in [33], despite these representing the fastest-growing employment sectors in the region a finding that illustrates the severity of the information asymmetry affecting this demographic.

Two primary challenges define this archetype. First, *Information Asymmetry*: genuine difficulty distinguishing between closely related but cognitively distinct technical fields for example, Data Science, Data Analytics, and Data Engineering in the absence of structured diagnostic guidance capable of translating individual aptitude profiles into meaningful pathway differentiation [34], [71]. Second, *Fear of Commitment*: pervasive anxiety regarding the long-term consequences of major selection, amplified by awareness of the graduate regret rates documented in Section 3.1.2. The fundamental need of this segment is for credible, evidence-grounded guidance that connects innate aptitudes and motivational orientations to realistic educational pathways and verifiable labor market outcomes.

3.5.3 Comparative Empathy Map Analysis

A comparative empathy map highlights how the same systemic failures the Paradox of Choice, soft skills deficits, and engagement decay manifest with qualitatively distinct characteristics depending on the user's stage in the academic-career lifecycle, thereby informing differentiated design responses. Table 3.2 presents this analysis across four behavioral quadrants.

Table 3.2: Comparative empathy map of primary user segments. Composite sentiments are derived from qualitative analysis of Egyptian student discourse in public online forums and from the survey and labor market literature reviewed in Section 3.1 [95], [96].

Quadrant	The Graduate (Corrective User)	The High Schooler (Preventative User)
SAYS	Expresses frustration at failing technical interviews despite holding a degree; describes urgency to pivot careers and acquire practical, market-relevant skills.	Articulates pressure from family to pursue prestigious but personally misaligned fields; asks what major will guarantee employment within a defined timeframe.
THINKS	Questions whether years of formal education have been misdirected a concern empirically grounded in the 44% graduate major regret rate [39], [70]; believes practical experience is more valuable than further theoretical instruction.	Fears replicating the unemployment outcomes of older peers; privately doubts aptitude for high-demand technical fields such as AI and Data Science.
DOES	Submits high volumes of generic job applications; enrolls in disconnected online courses attempting to address perceived soft skills deficiencies without a structured development framework.	Searches for trend-based major rankings online; takes informal personality quizzes as substitutes for structured psychometric guidance.
FEELS	Regret regarding past educational decisions; urgency to secure income; diminished confidence in the signaling value of their academic credential.	Pressure from family and social environment; confusion arising from lack of structured exposure to real-world technical roles and their actual cognitive demands.

A synthesis of this map reveals that while both archetypes are affected by the same structural failures identified throughout Section 3.1 inadequate guidance, misaligned curricula, and the absence of personalized pathway support the *temporal orientation* of their experience differs fundamentally. The graduate faces a corrective challenge: the need to remediate a past misdirection, rapidly acquire demonstrable competencies, and re-enter the labor market with updated credentials. The pre-university student faces a preventative challenge: the need for early diagnostic intervention that can preclude the misdirection before it is committed. As illustrated in Figure 3.12, both challenges converge on the same set of **LearNova** design responses the **Smart Compatibility Engine**, **Adaptive Content Slots**, Soft Skills Calibration, and the shared Literacy Phase each addressing a distinct node in the challenge-to-intervention mapping.

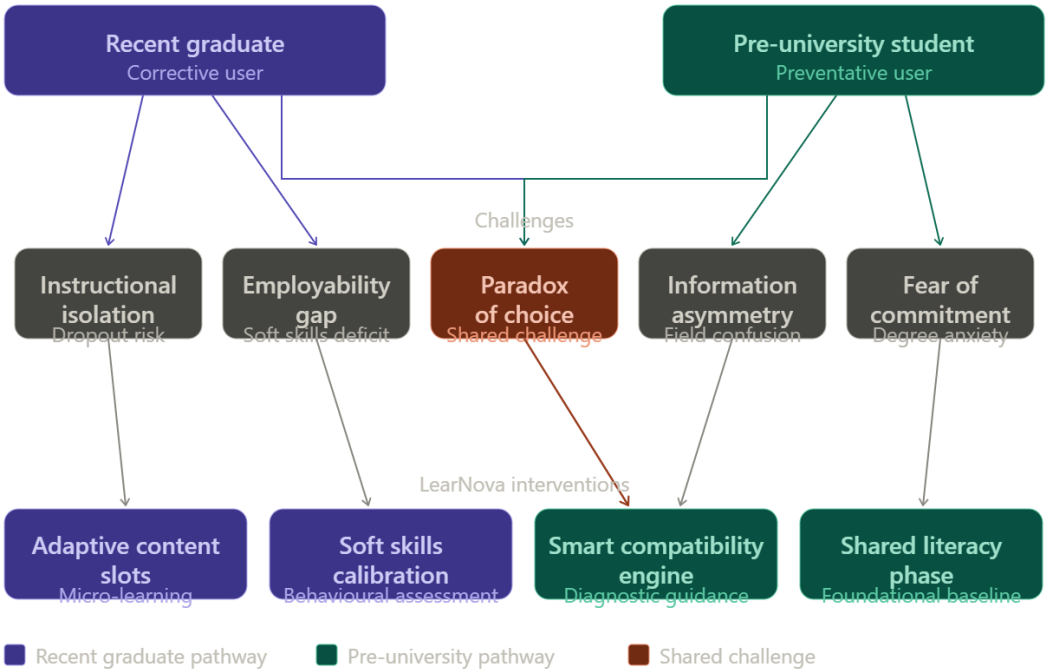


Figure 3.12: Mapping of primary user segments to their core challenges and corresponding LearNova system interventions. The diagram illustrates how two temporally distinct archetypes the recent graduate (corrective user) and the pre-university student (preventative user) converge on a shared set of structural challenges, each of which is addressed by a specific architectural component of the LearNova platform. The Paradox of Choice node (coral) denotes a challenge shared by both archetypes; purple nodes represent interventions primarily serving the graduate pathway; teal nodes represent interventions primarily serving the preventative and shared pathways. All nodes are mapped directly to the design decisions documented in Sections 3.1.6 and 3.5.4.

3.5.4 Implication for System Design

The dual-archetype structure carries direct and concrete implications for the **LearNova** system architecture. The inclusion of pre-university learners who lack professional histories, prior specialization credentials, or comparative benchmark data validates the implementation of a comprehensive psychometric diagnostic as the mandatory system entry point. For this segment, psychometric and cognitive assessments represent the only available instrument for generating credible, evidence-based guidance in the absence of experiential data [27], [58]. The system cannot rely on prior learning records; it must construct a reliable learner profile from the ground up through direct psychometric assessment, via the **Five-Factor Diagnostic** described in Section 3.1.6.

The shared **Literacy Phase** a mandatory foundational curriculum in Data Analytics preceding all specialization tracks serves a dual architectural function. First, it establishes a common technical baseline, ensuring that all learners, regardless of entry archetype, acquire the core competencies necessary for meaningful engagement with advanced specialization content. Second, it provides a pedagogical safety net: by scaffolding the

transition from general literacy to specialized depth, it structurally reduces the risk of early failure that the research literature consistently identifies as a primary precipitant of dropout in self-directed online learning environments [5], [8].

Taken together, the two archetypes and their associated challenge profiles do not merely describe a target audience they constitute a design specification. Every architectural decision documented in the chapters that follow, from the psychometric calibration sequence to the adaptive content delivery mechanism, may be traced to a specific, empirically grounded user need identified in this analysis.

3.6 System Modeling

System modeling provides a formal, visual, and logical representation of the system's intended functionality, clarifying the project scope and operationalizing the behavioral requirements detailed in Section 3.3. This section employs the Unified Modeling Language (UML) to construct models mapping interactions between key system actors and core functional modules [80].

3.6.1 System Actors

Interaction with the LearNova system is governed by three primary actors.

The **Student** is the primary human actor and the principal beneficiary of the platform's adaptive learning services, interacting exclusively through the mobile client to complete psychometric calibration, consume multimodal content, engage with gamification mechanics, and initiate mentorship sessions with Mitchy.

The **Administrator (Admin)** is a privileged human actor responsible for platform oversight, content curation, and risk management. This actor manages the full content hierarchy, monitors engagement and risk analytics via database functions including get_admin_risk_radar(), get_admin_struggle_map(), get_admin_momentum_line(), and get_admin_class_health_timeline(), and claims and resolves high-risk student alerts through intervention_logs.

The **AI Agent (Mitchy)** is modeled as a non-human autonomous actor represented with a dashed boundary in all use case diagrams to reflect this distinction capable of initiating asynchronous events independently of explicit user requests. Key autonomous behaviors include performing sentiment analysis on each inbound message, querying document_chunks via RAG for curriculum-grounded responses, and triggering risk_scores records when affective thresholds are breached.

3.6.2 Use Case Analysis

The system's functionality is partitioned into five core modules, each represented through a dedicated UML Use Case Diagram.

A. Adaptive Content Module

The instructional core of the LearNova platform is formalized in Figure 3.13. The primary use case, *Access Learning Content*, is related to three content modality specializations *Stream Video (Visual Mode)*, *Stream Audio (Auditory Mode)*, and *Read Article (Textual Mode)* each via a <<include>> relationship, reflecting that each modality is a mandatory variant resolved at runtime from student_profiles.learning_style against topic_resources.format_type. Every consumption event simultaneously writes a record to content_engagement_logs, capturing format_type, time_spent_seconds, engagement_score, and the bayesian_eligible flag. Pedagogical rigor is enforced through a second <<include>> relationship mandating execution of *Check Prerequisites* verifying that prior topics carry status = 'completed' in student_progress before any content is served. Progression is governed by an <<extend>> relationship from *Take Module Quiz* to *Unlock Next Module*, triggered conditionally upon the student achieving a passing score greater than 70%, persisted in student_module_attempts.passed by the submit-

module-attempt Edge Function, which also calls increment_xp() to award the flat +20 XP module completion reward.

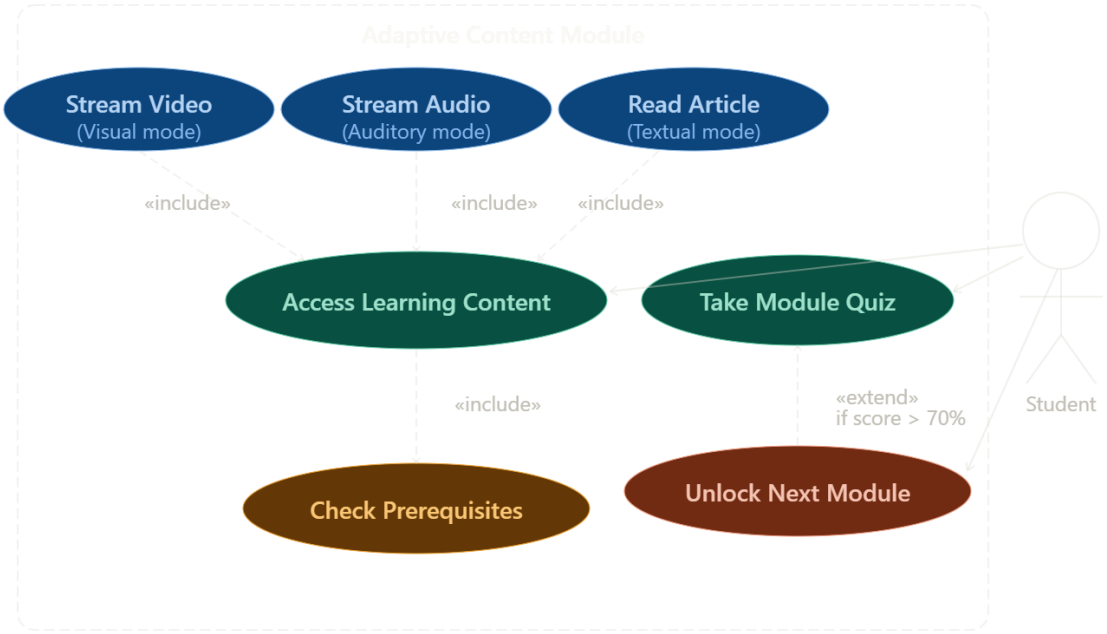


Figure 3.13: UML Use Case Diagram for the Adaptive Content Module, illustrating the Student actor's three <<include>>-related content modality pathways (Visual, Auditory, and Textual), the prerequisite verification gate enforced against student_progress, and the conditional <<extend>> relationship governing module unlocking upon successful quiz completion with a score exceeding 70%.

B. Comprehensive Assessment Architecture

The holistic assessment strategy illustrated in Figure 3.14 spans three integrated phases constituting the LearNova diagnostic-to-diploma pipeline.

Phase 1, the *Calibration Engine*, establishes the foundational user profile. The use case *Complete 5-Point Diagnostic* captures the hybrid input logic of FR-06 through two alternative execution paths: *Take In-App Assessment*, routing the student through the Cognitive/IQ Test, Big Five Personality Assessment (IPIP), VARK Learning Style, Soft Skills (SJT), and Career Interest (O*NET) sequence; and *Submit External Proof*, wherein the student uploads a verified screenshot flagged in diagnostic_test_results.result_source as external for administrative validation. Both paths converge at *Generate Psychometric Profile*, which synthesizes all feature scores into diagnostic_test_results.computed_scores and writes the resolved assigned_track and learning_style to student_profiles. Completion of this phase triggers the one-time +500 XP milestone bonus via increment_xp().

Phase 2, the *Learning Loop*, governs ongoing formative evaluation. The student accesses module content, submits to *Take Module Quiz*, and the outcome is evaluated against the 70% mastery threshold. Failure loops the student back to the module content; passing

advances the student to *Take Level Capstone Test*, which leverages AI grading through student_level_attempts.mitchy_feedback for open-ended questions, and upon successful completion triggers progression to the next level.

Phase 3, the *Retention and Gamification* layer, models motivational mechanics. Passing a Level Capstone Test triggers *Earn Level Badge*, recorded as a new row in user_achievements referencing the appropriate entry in achievements_dictionary. The <<extend>> relationship to *Complete Weekly Challenge* reflects the time-bounded, optional nature of challenge participation accessible only when weekly_challenges.is_active = true and the current date falls within the available_from and available_until window with the single-attempt constraint enforced by the unique key on (user_id, challenge_id) in student_challenge_attempts. Both pathways converge at *Earn XP and Rank Up*, with all XP values computed server-side and written via increment_xp().

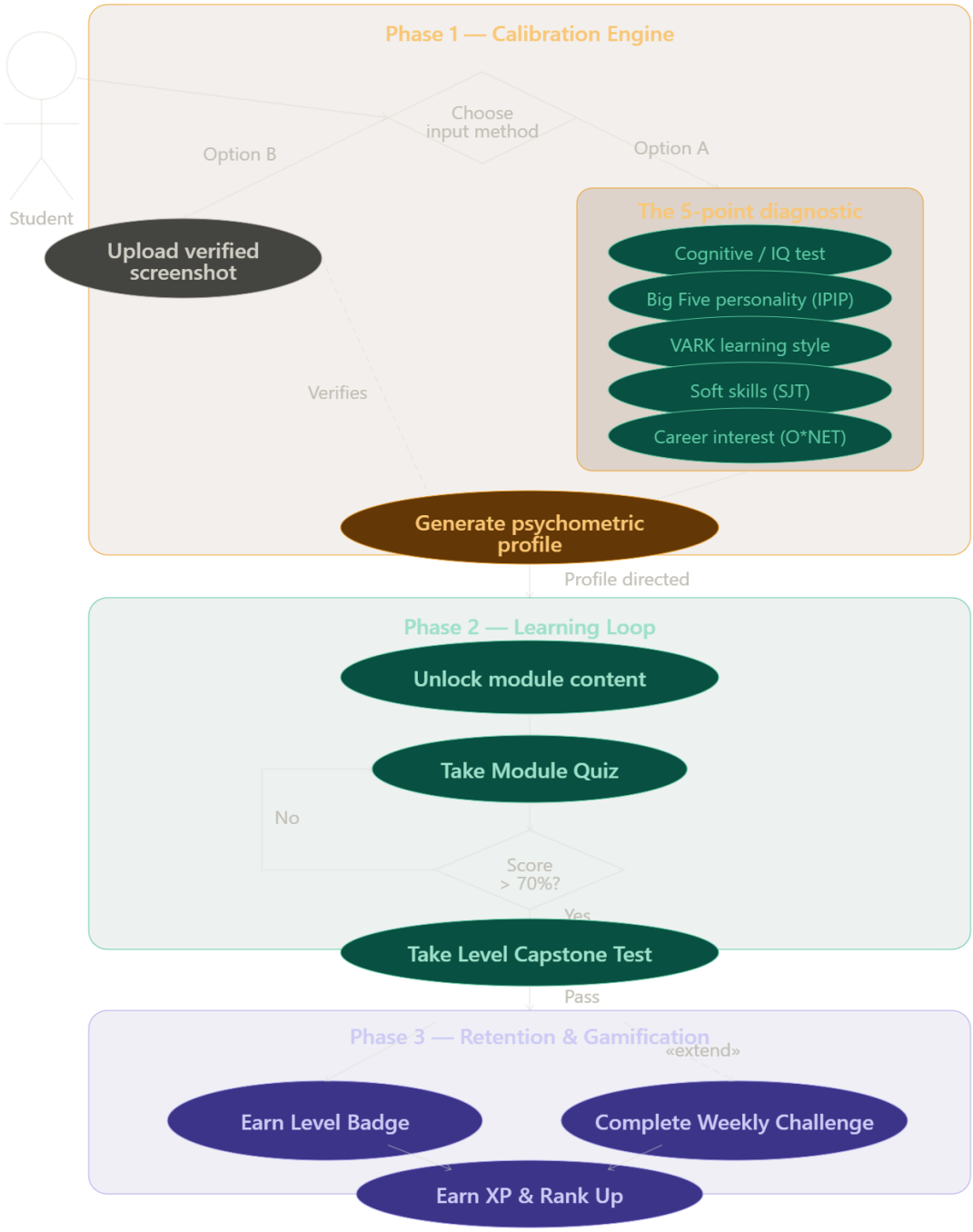


Figure 3.14: *UML Use Case Diagram for the Comprehensive Assessment Architecture, spanning Phase 1 (Calibration Engine with hybrid in-app and external input paths converging at psychometric profile generation), Phase 2 (Learning Loop with mastery-gated module and level progression, including AI-graded level capstone feedback), and Phase 3 (Retention and Gamification, including time-bounded weekly challenges and XP accrual via the increment_xp() RPC).*

C. Engagement & AI Mentorship

The dynamic support and safety mechanisms are detailed in Figure 3.15, modeling interactions among all three system actors within a unified module boundary.

The central use case, *Chat with AI Mentor*, incorporates a mandatory <<include>> to *Analyze Sentiment*, ensuring every inbound message is evaluated for affective state before any response is generated operationalizing FR-15. Mitchy, represented with a dashed actor boundary, engages in bidirectional interaction whilst simultaneously querying document_chunks via RAG to ground responses in curriculum material. Sentiment scores and derived learning states are persisted in chat_messages.detected_learning_state and longitudinally in student_sentiment_history.

The conditional <<extend>> to *Trigger High-Risk Alert* activates when sentiment analysis detects patterns falling below the predefined threshold, autonomously creating a risk_scores record with risk_level categorized as low, medium, high, or critical and a feature_snapshot stored for explainability and notifying the Admin actor. The Administrator executes *Provide Manual Intervention*, recording the action in intervention_logs (linked via risk_score_id to the triggering record) and dispatching a notification_type = 'risk_alert' notification via in_app_notifications, delivered as an FCM push notification to the student's registered device in user_devices. This human-in-the-loop escalation satisfies FR-16 and FR-17 whilst mitigating the inherent limitations of fully autonomous AI agents [5].

The Student actor additionally accesses *View Leaderboard* reading from leaderboard_snapshots, which all students may read under the applicable RLS policy and *View Progress Dashboard*, surfacing data from get_student_diploma_progress() and get_student_learning_heatmap().

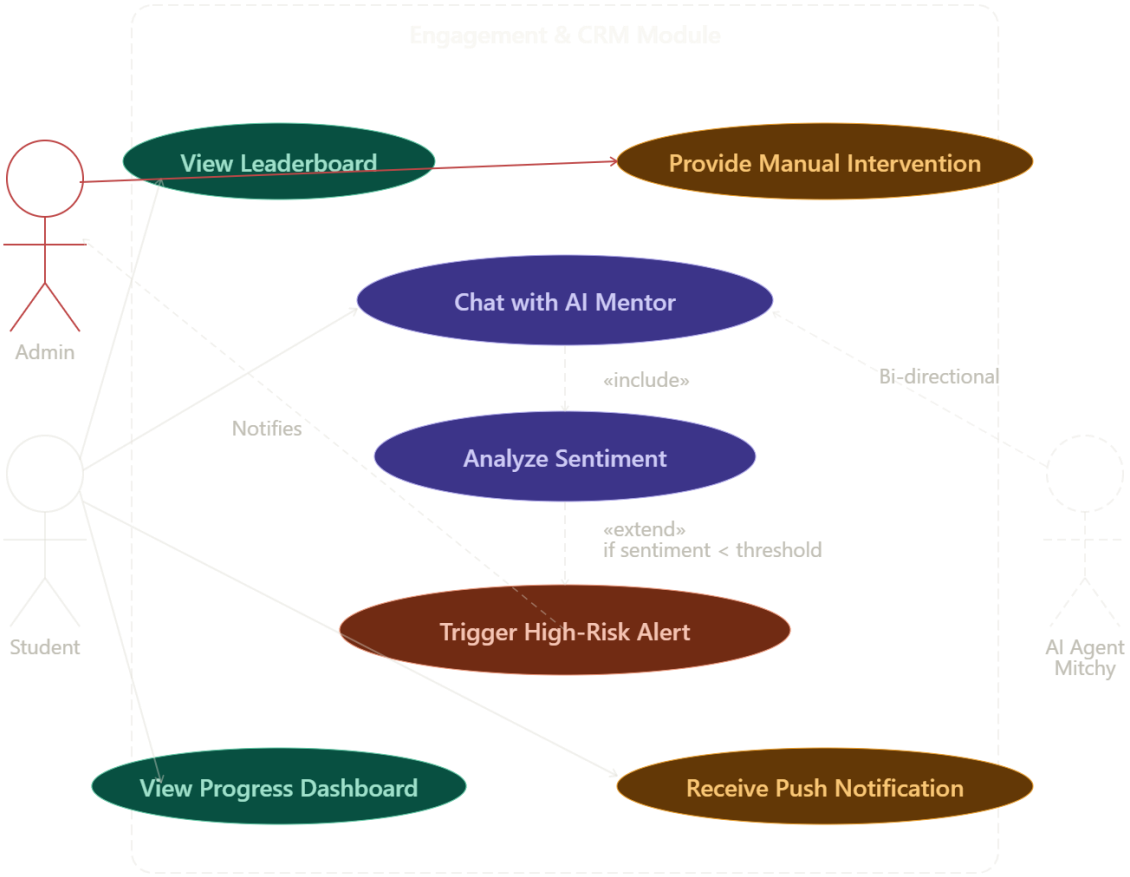


Figure 3.15: *UML Use Case Diagram for the Engagement and CRM Module, illustrating the Student's bidirectional chat relationship with AI Agent Mitchy (dashed boundary), the mandatory <<include>> sentiment analysis pipeline persisting to chat_messages.detected_learning_state, the conditional <<extend>> high-risk alert pathway creating risk_scores records and notifying the Admin, and the Admin's manual intervention capability logged in intervention_logs.*

D. Administration & Security

The backend governance and security controls are outlined in Figure 3.16, delineating the Administrator actor's privileges across content management and security oversight.

For content management, *Upload Multimedia Asset* and *Validate External Proof* are each modeled as <<extend>> relationships to the parent *Manage Diploma Content* use case, reflecting their conditional, specialized nature invoked only when adding new content or reviewing externally submitted credentials respectively. These extend the base content management workflow, which encompasses CRUD operations across the full course hierarchy stored in courses, levels, modules, topics, topic_resources, and topic_images, with all assets stored in the content-assets Storage bucket.

For risk and security oversight, *View At-Risk Students* surfaces aggregated risk data via get_admin_risk_radar() and get_admin_struggle_map(), whilst *View Analytics*

Dashboard presents platform-wide engagement through get_admin_momentum_line(), get_admin_class_health_timeline(), get_admin_foundation_funnel(), and get_admin_format_effectiveness(). The *Configure RBAC* use case governs role assignments, enforcing the permission boundaries defined by the is_admin() RLS helper function.

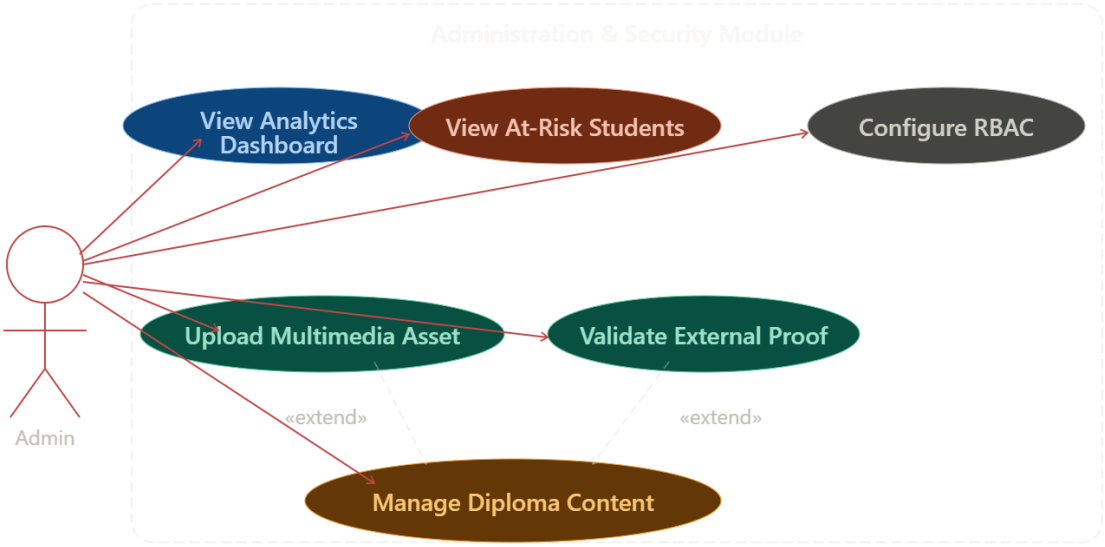


Figure 3.16: *UML Use Case Diagram for the Administration and Security Module, illustrating the Admin actor's use cases across content management (Upload Multimedia Asset and Validate External Proof as <<extend>> specializations of Manage Diploma Content) and security and analytics oversight (View At-Risk Students, View Analytics Dashboard, Configure RBAC), with corrected <<extend>> relationship directionality throughout.*

E. User Identity Management

The authentication and profile management workflows are depicted in Figure 3.17, modeling the complete identity lifecycle for both actor types.

On the Student side, *Register* incorporates a mandatory <<include>> to *Verify Identity (OTP)*, implementing FR-02. Upon OTP verification, the handle_new_user() trigger auto-populates the public users table from auth.users, establishing identity and profile integrity at account creation. *Manage Own Profile* is extended by *Upload External Proof* modeled as <<extend>> reflecting that proof submission is a conditional, non-mandatory action performed only by students electing the external calibration pathway defined in FR-06.

On the Administrator side, *Manage All Profiles* encompasses full CRUD privileges over student records under admin-scoped RLS. *View Security Audit Logs* provides the traceability capability satisfying NFR-07. The *Force Password Reset* use case <<include>>s the *Reset* action, modelling the admin-initiated credential recovery workflow as distinct from user-initiated recovery, preserving the separation of concerns inherent to the RBAC architecture.

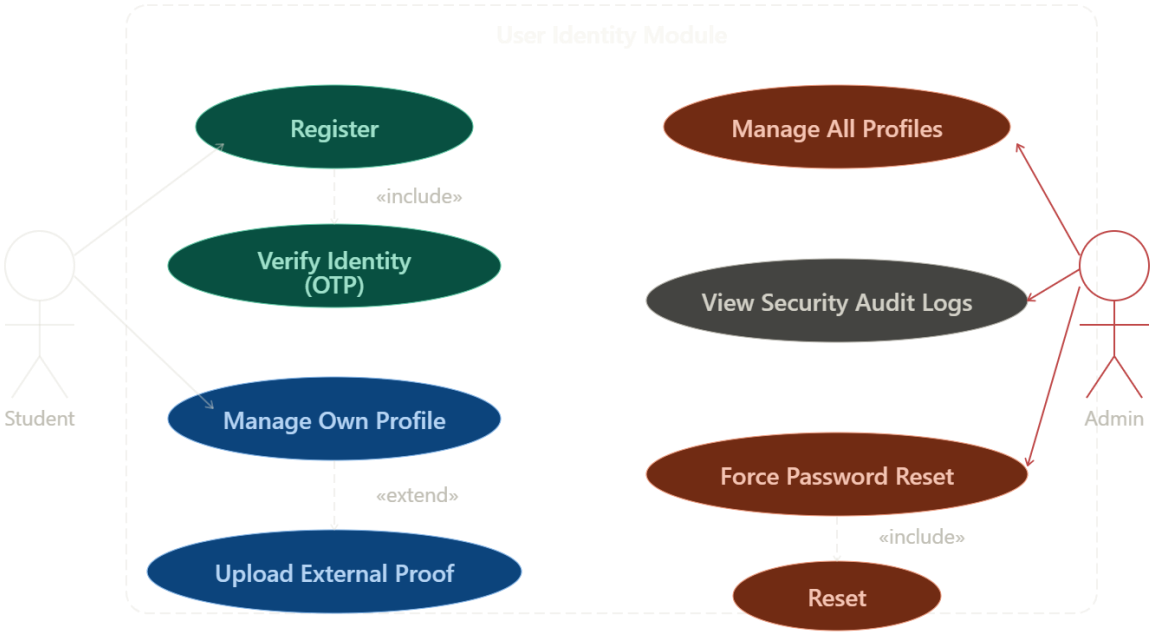


Figure 3.17: UML Use Case Diagram for the User Identity Module, illustrating the Student actor's registration pathway with mandatory OTP verification <<include>>, the conditional <<extend>> for external proof submission under the hybrid calibration pathway, and the Admin actor's profile management, security audit, and password reset capabilities, with corrected UML relationship notation throughout.

Chapter 4:
System Design

4.1 System Architecture

The LearNova platform is architected as a multi-layered, service-oriented system adhering to a modern serverless **Backend-as-a-Service (BaaS)** paradigm, augmented by a dedicated Railway-hosted microservice for computationally intensive operations. This strategic combination was driven by core requirements to decouple presentation logic from domain-specific business rules, thereby ensuring scalability, maintainability, and robust security. As illustrated in Figure 4.1, the system is logically stratified into five distinct layers, each encapsulating strictly defined responsibilities and communication protocols.

4.1.1 Layer 1: Presentation Layer (Client-Side)

The topmost layer encapsulates all user interface and client-side interaction logic, implemented as a unified cross-platform mobile application utilizing the Flutter framework [82]. This choice enables deployment from a single codebase to both Android (API Level 28 and above) and iOS (version 15 and above), delivering consistent feature parity across operating systems while reducing maintenance overhead. Local application state is managed through the Riverpod reactive caching library, and local persistence strategies ensure operational continuity during intermittent network conditions [83].

The application enforces strict **Role-Based Access Control (RBAC)** at the view level, presenting two functionally distinct interface modules. The **Learner Interface Module** is optimized for adaptive content consumption, dynamically rendering Visual, Auditory, or Textual instructional material in accordance with the learning style stored in the student's student_profiles record. It additionally maintains persistent WebSocket channels through Supabase's Realtime API for low-latency interaction with the AI mentor. The **Administrative Interface Module**, accessed via a role-gated pathway within the same application binary, provides privileged capabilities for curriculum content curation, multimedia asset uploads, and real-time visualization of at-risk student indicators drawn from the risk_scores and intervention_logs tables.

4.1.2 Layer 2: Service Interface Layer (API Gateway)

This layer functions as the secure boundary and communication façade between the Flutter client and the underlying backend infrastructure. It is powered primarily by the Supabase platform and operates as a unified API Gateway, abstracting the complexity of backend subsystems behind a coherent, authenticated interface.

Three principal components constitute this layer. The **Identity & Access Gateway** (GoTrue) manages the complete authentication lifecycle, employing JSON Web Tokens (JWTs) for session validation. Upon successful authentication, Row-Level Security (RLS) policies enforced at the PostgreSQL database layer ensure that all data access is cryptographically scoped to the authenticated user's role and identity [84]. The **Auto-Generated REST Interface** (PostgREST) leverages database schema introspection to expose the relational data model as a strictly typed RESTful API, eliminating manual CRUD controller code and reducing the vulnerability surface of the backend. The **Realtime API** maintains persistent, bidirectional WebSocket connections for event-driven updates, encompassing chat message delivery, leaderboard synchronization, and push notification dispatch.

4.1.3 Layer 3: Business Logic Layer (Edge Functions and Railway)

Domain-specific validation, orchestration, and scoring logic are executed within this layer through two complementary mechanisms: Supabase Edge Functions serving as an authenticated relay tier, and a Railway-hosted FastAPI service handling computationally intensive operations.

Four **Supabase Edge Functions** constitute the relay tier. The run-scoring-engine function retrieves a user's completed diagnostic results and forwards them along with a private service-key to the Railway scoring endpoint after JWT verification. The submit-module-attempt and submit-challenge-attempt functions relay assessment submissions to Railway in a structurally identical pattern. The mitchy-chat function forwards the user's conversational message to Railway's AI endpoint, returning the structured response to the Flutter client. This relay architecture ensures that Railway credentials and Gemini API keys are never exposed to the client under any operational condition.

The **Railway FastAPI Service** hosts the **Calibration Engine**, which aggregates raw psychometric responses from the five-point diagnostic battery and generates a normalized feature vector for each learner; the **Adaptive Strategy Engine**, which evaluates those features against track-specific modifier weights to determine the optimal DA, DE, or DS specialization assignment; and the **Gamification Service**, which calculates Experience Points in accordance with the tiered XP reward matrix and writes updates to the student_profiles table via the secure increment_xp() Supabase RPC. Additionally, two scheduled cron processes execute daily analytics pipelines, populating the ml_daily_metrics table and generating administrative briefings.

4.1.4 Layer 4: Intelligence Layer (Cognitive Core)

This layer encapsulates the platform's artificial intelligence capabilities, integrating an external Large Language Model API with internally sourced contextual data to produce pedagogically meaningful, affect-aware responses [87]. All components within this layer are hosted on the Railway service, preserving the authentication boundary established in Layer 2.

The **AI Mentor Agent** ("**Mitchy**") constitutes the primary conversational interface. Rather than functioning as a simple information retrieval system, the agent is architected for Socratic pedagogical dialogue augmented by real-time affective state monitoring. A **Sentiment and Context Orchestrator** serves as middleware, assembling the user's recent quiz performance, active learning track, calibrated learning style, and prior conversational history into a structured context window before querying the external LLM. The architecture implements multi-model fallback logic, guaranteeing that a non-empty, pedagogically appropriate response is returned to the learner regardless of upstream API availability. The **Recommendation Service**, operating within the same Railway environment, implements deterministic mapping logic to translate the calibrated psychometric profile into a specialization assignment, as detailed in Section 4.5.

4.1.5 Layer 5: Data & Infrastructure Layer

The foundational layer is responsible for persistent data storage, enforcing **ACID** compliance Atomicity, Consistency, Isolation, and Durability to guarantee transactional

integrity across all concurrent operations. The primary **Relational Data Store** (PostgreSQL, hosted on Supabase) houses all structured data across thirty-three tables spanning nine functional domains: user identity and authentication, adaptive curriculum, module and level assessments, the diagnostic calibration system, student activity and progress, gamification mechanics, AI and ML analytics, the chat and RAG system, and notifications and device management [85]. All primary keys utilize Universally Unique Identifiers (UUIDs) rather than sequential integers, a deliberate decision that prevents enumeration-based enumeration attacks and facilitates collision-free record management.

A scalable **Unstructured Object Storage** system organized across three Supabase Storage buckets designated for calibration media, content assets, and student submissions respectively manages high-volume binary assets including video and audio instructional content and diagnostic question imagery. An immutable **Security Audit Trail** logs all critical write operations, including profile modifications, content deletions, and risk alert state transitions, ensuring full traceability in accordance with the non-functional security requirements defined in Section 3.2.2.

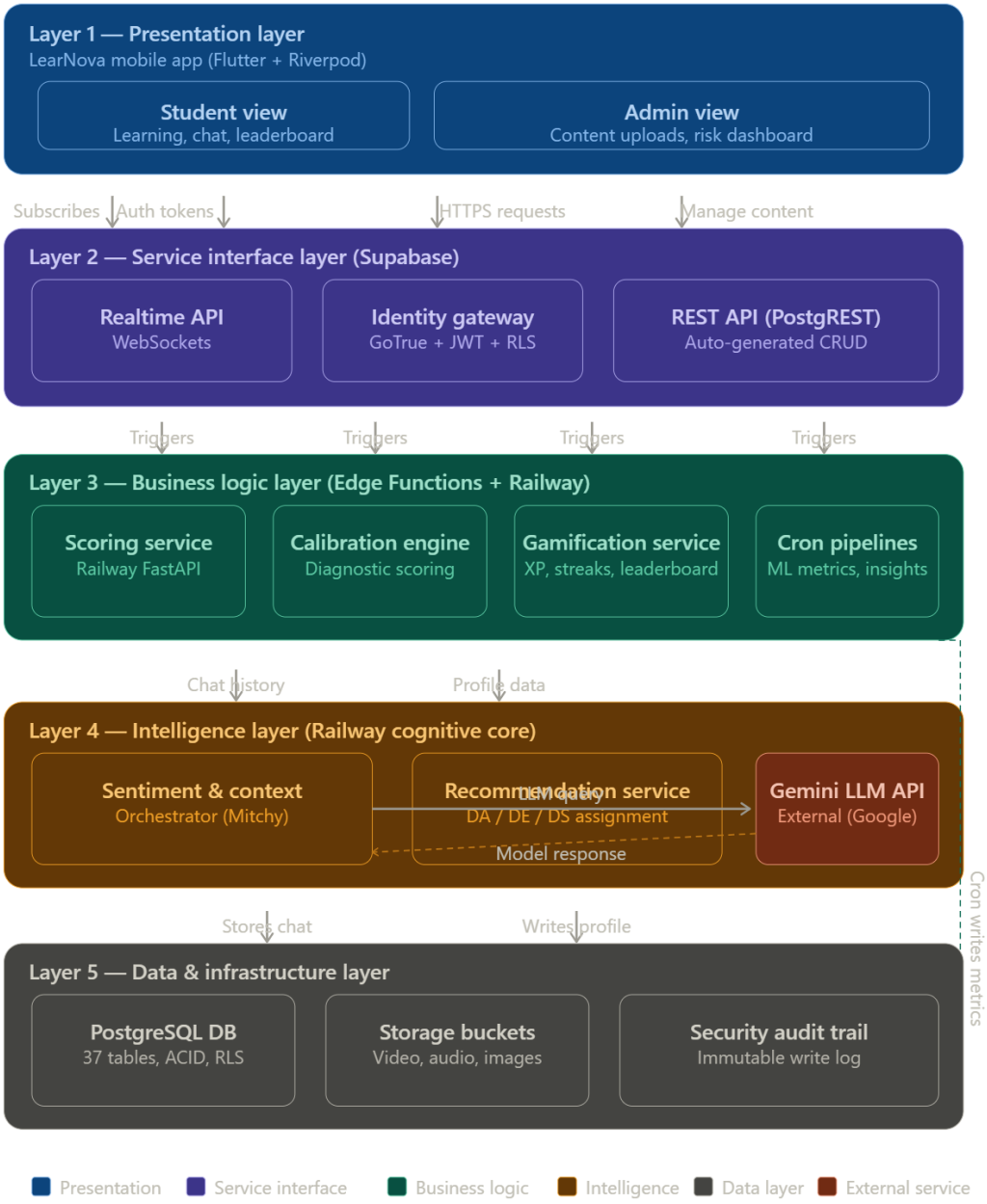


Figure 4.1: The five-layer system architecture of LearNova. Solid arrows indicate synchronous request flows; dashed arrows indicate asynchronous or return paths. The Gemini LLM API is an external dependency accessed only via Layer 4 on Railway.

Figure 4.1: The five-layer system architecture of LearNova. Solid arrows indicate synchronous request flows; dashed arrows indicate asynchronous or return paths. The Gemini LLM API is an external dependency accessed exclusively via the Railway intelligence layer, never directly from the client.

4.2 Process Modeling (Data Flow Analysis)

Process modeling is employed to illustrate the logical transformation and flow of data within the LearNova system, abstracting away specific implementation details to focus on functional processes and information exchange. This section utilizes **Data Flow Diagrams (DFDs)** adhering to the Gane & Sarson notation convention, mapping how raw user inputs are systematically processed into adaptive learning content and diagnostic insights.

4.2.1 Level 0: System Context Diagram

The System Context Diagram, depicted in Figure 4.2, defines the highest-level boundary of the LearNova system and its interactions with key external entities. At this level of abstraction, the entire platform is represented as a single process a conceptual "black box" with numbered data flows crossing its perimeter to reveal the nature of each exchange.

The **Student** entity is the primary source of input data and the principal beneficiary of the system's outputs. This actor initiates core interaction through two categories of input: authentication credentials and diagnostic responses (Flow 1), and ongoing content requests together with chat messages directed at the AI mentor (Flow 2). In return, the system delivers Adaptive Content in the learner's calibrated format Visual, Auditory, or Textual (Flow 3), and AI mentor responses alongside earned achievement badges (Flow 4).

The **Admin/Counselor** entity represents the supervisory role, engaged in a distinct monitoring and intervention loop. The system surfaces two categories of outbound data to this actor: risk alerts accompanied by student engagement analytics when behavioral thresholds are breached (Flow 5), and flags marking externally submitted diagnostic proof submissions that require manual validation before the associated scores are trusted by the calibration engine (Flow 6). In the reverse direction, the Admin issues Intervention Actions back into the system direct messages or profile modifications to support at-risk learners (Flow 7).

The **External Services** entity encapsulates third-party integrations required for the platform's hybrid validation mechanism. The system dispatches a Verification Request when a student submits an external credential URL (Flow 8). The returning flow carries an OAuth authentication response or verified score confirmation from the external provider (Flow 9). It is noted that, in the current implementation, this validation pathway involves the student supplying a URL that is subsequently flagged for admin review rather than an automated outbound API call to a certification authority; the external services boundary therefore represents both automated OAuth flows and manual-review-assisted credential confirmation.

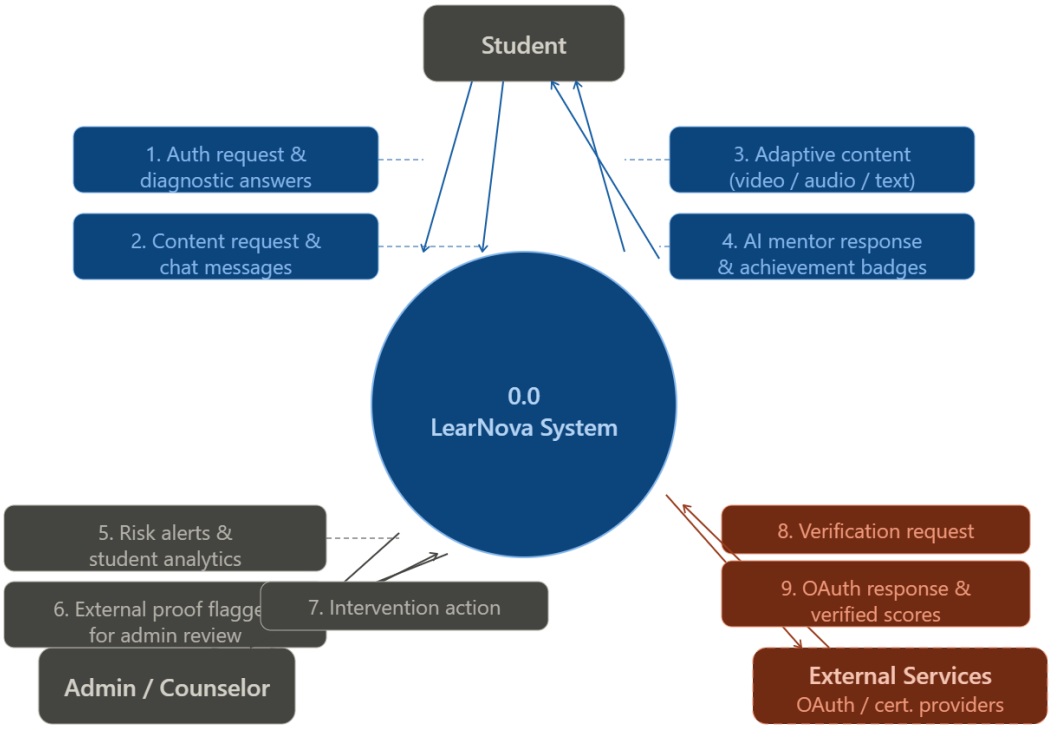


Figure 4.2: DFD Level 0 — System context diagram for LearNova.
Rectangular entities are external actors; the central circle is the system boundary.
Dashed border on External Services denotes third-party dependency.

Figure 4.2: DFD Level 0 System context diagram for LearNova, defining the system boundary and its primary numbered data flows with three external entities: Student, Admin/Counselor, and External Services. Rectangular notation denotes external actors; the dashed border on External Services denotes third-party dependency.

4.2.2 Level 1: Functional Decomposition

Figure 4.3 presents the Level 1 Data Flow Diagram, which decomposes the system boundary into six core functional processes. This decomposition exposes five internal data stores (D1–D5) and details the transformation logic governing information exchange between processes, providing a structured view of the system's internal workflow consistent with established structured systems analysis practice.

A. Identity & Calibration Subsystem (Processes 1.0 & 2.0)

The data lifecycle originates within this subsystem. Process 1.0 (Auth & Identity) ingests raw login credentials, performs authentication via the GoTrue gateway, and upon success writes a base record to the **User Profile Store (D1)**, persisting the session token for downstream authorization. Once authenticated, the student engages with Process 2.0 (Calibration), which manages the five-point psychometric diagnostic battery. This process accommodates two execution paths: direct in-app answer submission and external score URL submission. In both cases, its core function is normalization transforming disparate input formats into a standardized feature representation and persisting the aggregated results in the **Diagnostic Results Store (D2)**. This store functions as a critical decoupling

point, separating raw psychometric data collection from its subsequent application in track assignment, a design that supports flexible input without compromising downstream scoring integrity.

B. Strategic Decision Core (Process 3.0)

Process 3.0 (Strategy Engine) functions as the system's central decision-making kernel, executing after calibration data has been committed. It queries D2 to retrieve the user's comprehensive feature vector spanning cognitive aptitude, personality dimensions, soft skills, learning style preferences, and vocational interests and applies the weighted modifier algorithm detailed in Section 4.5 to produce two outputs: a definitive track assignment (DA, DE, or DS) and a dominant learning style tag. Both outputs are written to D1, where they govern all subsequent content delivery decisions. The diagram incorporates an administrative safeguard: a Manual Override flow from the Admin entity permits a human reviewer to examine and, where warranted, modify an algorithmic assignment, with the corrected value written back to D1.

C. Adaptive Learning Loop (Process 4.0)

Process 4.0 (Learning & Assessment) constitutes the primary runtime engine for content delivery and formative evaluation. Its adaptive logic resolves in three sequential steps. First, it reads the learner's active learning style tag from D1. Second, it queries the **Curriculum Store (D3)** for the lesson corresponding to the learner's current position in the track, fetching the URL for the appropriate content variant slot a video file for a Visual learner, an audio file for an Auditory learner, or a structured transcript for a Textual learner. Third, upon a student completing a module's associated assessment above the required passing threshold, the process emits a Quiz Passed signal. This signal serves as the primary control flow output, simultaneously unlocking the subsequent module in D1 and activating the downstream gamification process.

D. Engagement & Intelligence Subsystem (Processes 5.0 & 6.0)

This subsystem manages learner motivation and emotional support through two parallel processes. Process 5.0 (Gamification Engine) is event-driven, activated by the Quiz Passed signal from Process 4.0. It calculates Experience Points in accordance with the tiered XP reward matrix, validates the active daily streak against D4, and writes the updated values back to D4 via the increment_xp() RPC. Research consistently demonstrates that such structured, incremental reward mechanics produce measurable improvements in learner persistence and course retention [50]. Process 6.0 (AI Mentor Mitchy) handles unstructured conversational data, retrieving prior dialogue context from the **Chat Logs store (D5)** to maintain session coherence. A critical embedded routine performs sentiment analysis on each inbound user message. When the analyzed sentiment vector crosses a predefined negative threshold indicative of distress or disengagement, the process initiates a Flag High Risk data flow to the Admin entity asynchronously, bypassing standard storage to ensure urgent human attention is mobilized without delay an implementation of proactive, affect-aware intervention as theorized in learning analytics research [25].

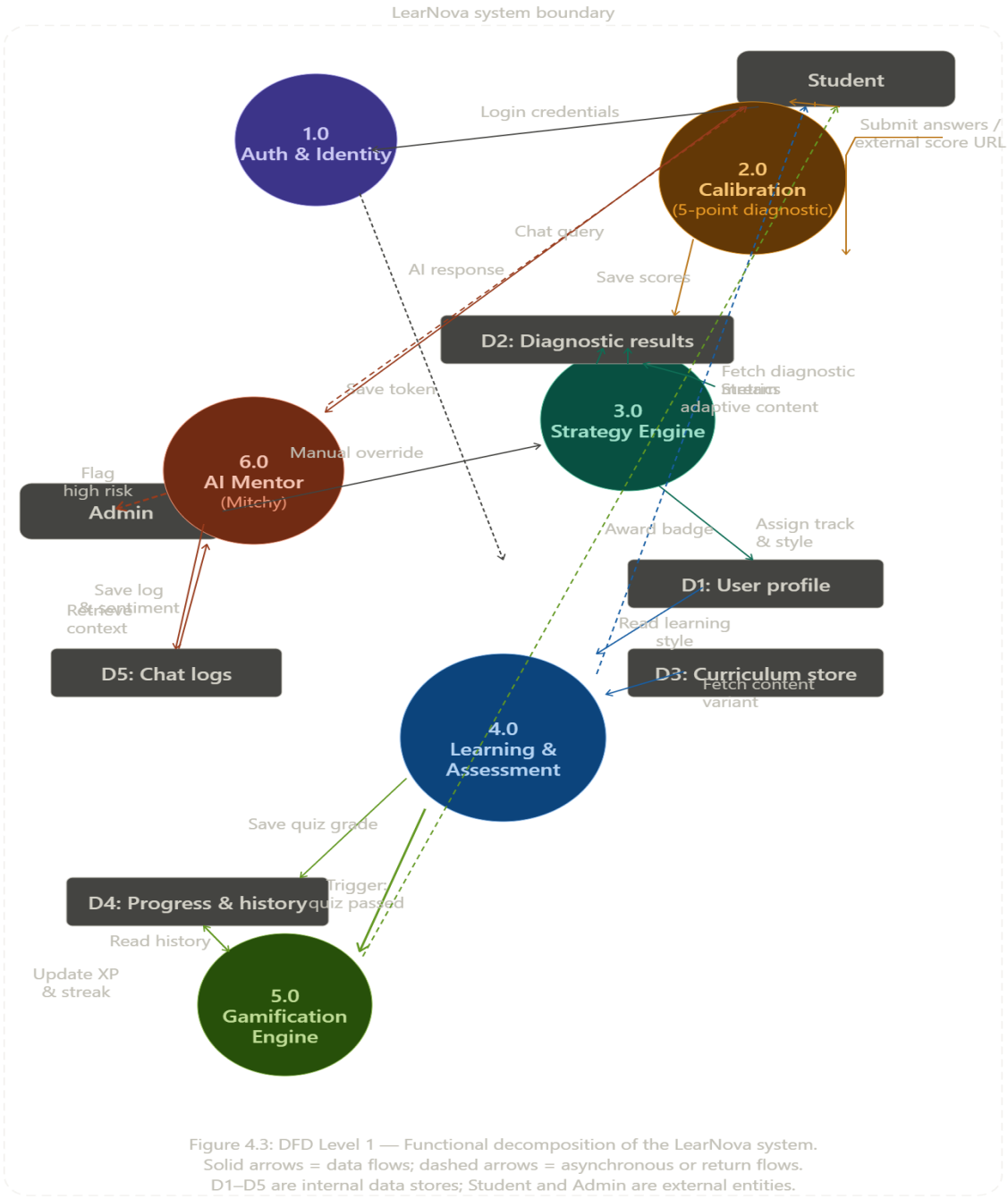


Figure 4.3: DFD Level 1 — Functional decomposition of the LearNova system.
Solid arrows = data flows; dashed arrows = asynchronous or return flows.
D1–D5 are internal data stores; Student and Admin are external entities.
Process numbering reflects approximate execution sequence.

Figure 4.3: DFD Level 1 Functional decomposition of the LearNova system, detailing six core processes and their interactions with five internal data stores (D1–D5). Solid arrows indicate synchronous data flows; dashed arrows indicate asynchronous or return flows. Student and Admin are external entities; all other shapes represent internal processes or data stores.

4.3 Dynamic Interaction Design (Sequence Analysis)

While the Data Flow Diagrams presented in Section 4.2 illustrate the logical transformation and storage of information, they do not capture the temporal ordering and synchronous nature of runtime interactions. To address this, Sequence Diagrams are employed to model the precise, time-ordered exchange of messages between the system's actors, components, and architectural layers, following established UML behavioral modeling conventions. This analysis demonstrates that the proposed technology stack comprising the Flutter client [82], the Supabase backend [84], Supabase Edge Functions, and the Railway FastAPI service can successfully orchestrate the complex, real-time behaviors defined in the functional requirements.

The following subsections detail the runtime logic for three critical system processes: the initialization of the student profile via the hybrid calibration sequence, the deterministic resolution of adaptive content delivery, and the asynchronous safety intervention protocol within the AI mentorship engine.

4.3.1 Sequence A: Calibration & Placement Flow

Figure 4.4 models the critical onboarding sequence, detailing the student's initial entry into the LearNova ecosystem. This sequence is architecturally significant because it operationalizes the system's hybrid input strategy, accommodating both native in-app psychometric assessment and externally sourced credential verification within a single, coherent flow.

The sequence is initiated when the Student actor triggers authentication. The Mobile App submits credentials to the Supabase Identity Gateway (GoTrue), which validates the request and returns a session JWT to the client. This token authorizes all subsequent API calls for the duration of the session [84].

The system then presents the Calibration Options interface, entering an alternative execution block that handles two mutually exclusive input paths. In **Option A** (Native Assessment), the student completes the five-point in-app diagnostic battery. The raw responses are forwarded from the Mobile App to the run-scoring-engine Edge Function with the JWT attached; the Edge Function relays the payload to the Railway Scoring Service endpoint /score/diagnostic-result, which applies the weighted feature scoring algorithm to generate a normalized psychometric feature vector and DA/DE/DS track scores. In **Option B** (External Credential Submission), the student pastes a URL referencing a pre-existing verified score. The Edge Function performs syntactic URL validation before persisting a "Pending Admin Review" flag in the diagnostic_test_results table, preventing unverified data from propagating to the track assignment algorithm until a human reviewer has confirmed its authenticity.

Both paths converge at a consolidation step. The Railway Scoring Service fetches the aggregated diagnostic data from Supabase, executes the final scoring formula computing base scores from O*NET anchors and applying modifier weights as detailed in Section 4.5 and writes the resulting assigned_track and learning_style values to the student_profiles table. The Mobile App retrieves and displays the recommendation to the student.

The sequence concludes with a state transition: upon the student accepting the recommendation, the system creates a formal enrollment record and programmatically unlocks Level 1 (Literacy Phase) in the curriculum, transitioning the user's status from authenticated visitor to active enrolled learner.

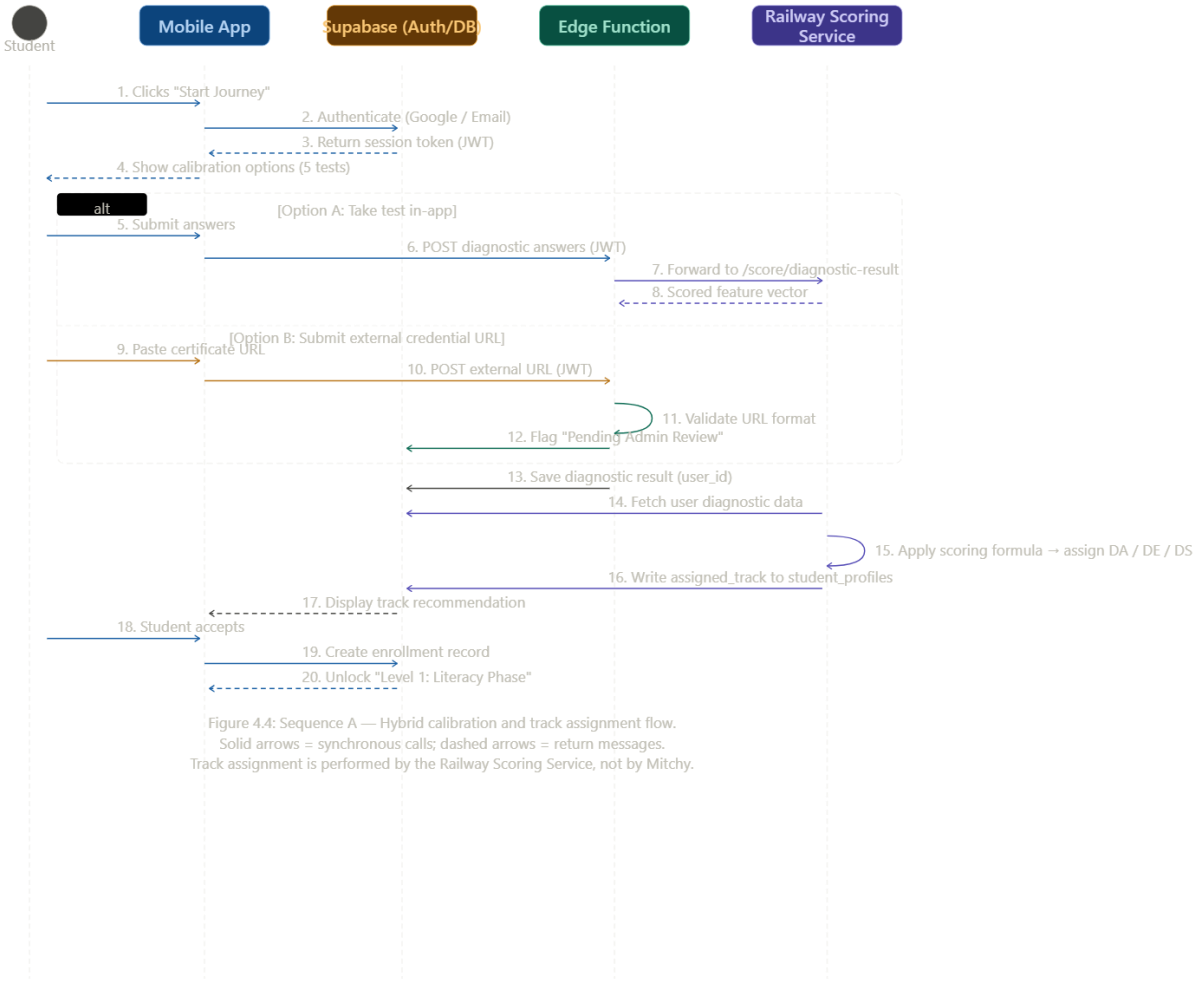


Figure 4.4: Sequence A — Hybrid calibration and track assignment flow.
Solid arrows = synchronous calls; dashed arrows = return messages.
Track assignment is performed by the Railway Scoring Service, not by Mitchy.

Figure 4.4: Sequence A Hybrid calibration and track assignment flow, illustrating native assessment and external validation paths, JWT-authenticated relay through the Edge Function tier, and Railway Scoring Service-driven track assignment. Solid arrows denote synchronous calls; dashed arrows denote return messages.

4.3.2 Sequence B: Adaptive Content Delivery

Figure 4.5 details the core operational sequence governing adaptive content delivery within the LearNova platform. This diagram captures the deterministic client-side filtering logic that resolves the correct instructional format at runtime, contrasting sharply with conventional platforms that serve identical static content to all learners irrespective of cognitive profile.

The interaction is characterized by two sequential phases.

1. Context-Aware Content Resolution:

The sequence is initiated when the Student opens a learning module. The Mobile App dispatches two concurrent queries to Supabase: a fetch of the user's active learning style from student_profiles, and a fetch of the lesson metadata from the curriculum tables, which returns a structured payload containing URLs for all three content variant slots Visual, Auditory, and Textual. A critical client-side logic block then filters this payload: based on the returned learning style tag, the application selects only the corresponding URL a video file for a Visual learner, for instance and requests the asset from the Media CDN for streaming. The alternative format URLs are suppressed. This server-guided, client-executed filtering operationalizes the adaptive delivery mechanism, minimizing extraneous cognitive load by presenting exclusively the modality aligned with the learner's calibrated preferences, a strategy substantiated by principles of multimedia learning and modality alignment [7].

2. Assessment & Progression Logic:

Following content consumption, the student submits answers for the module's associated quiz. The Mobile App posts the submission to the submit-module-attempt Edge Function, which fetches the correct answer key from the module_assessment_questions table and calculates the percentage score. The system then enters a conditional block. In the **Pass condition** (score ≥ 70%), the Edge Function executes an atomic sequence: it updates the student_progress record, unlocks the subsequent module, and calls the increment_xp() Supabase RPC to award a flat **+20 XP** to the student's profile the fixed reward defined for module assessments in the gamification specification, irrespective of difficulty selection [50]. In the **Fail condition**, the attempt is logged in student_module_attempts with no XP awarded, and a review prompt is returned to the interface, enforcing the mastery-based progression requirement defined in FR-13.

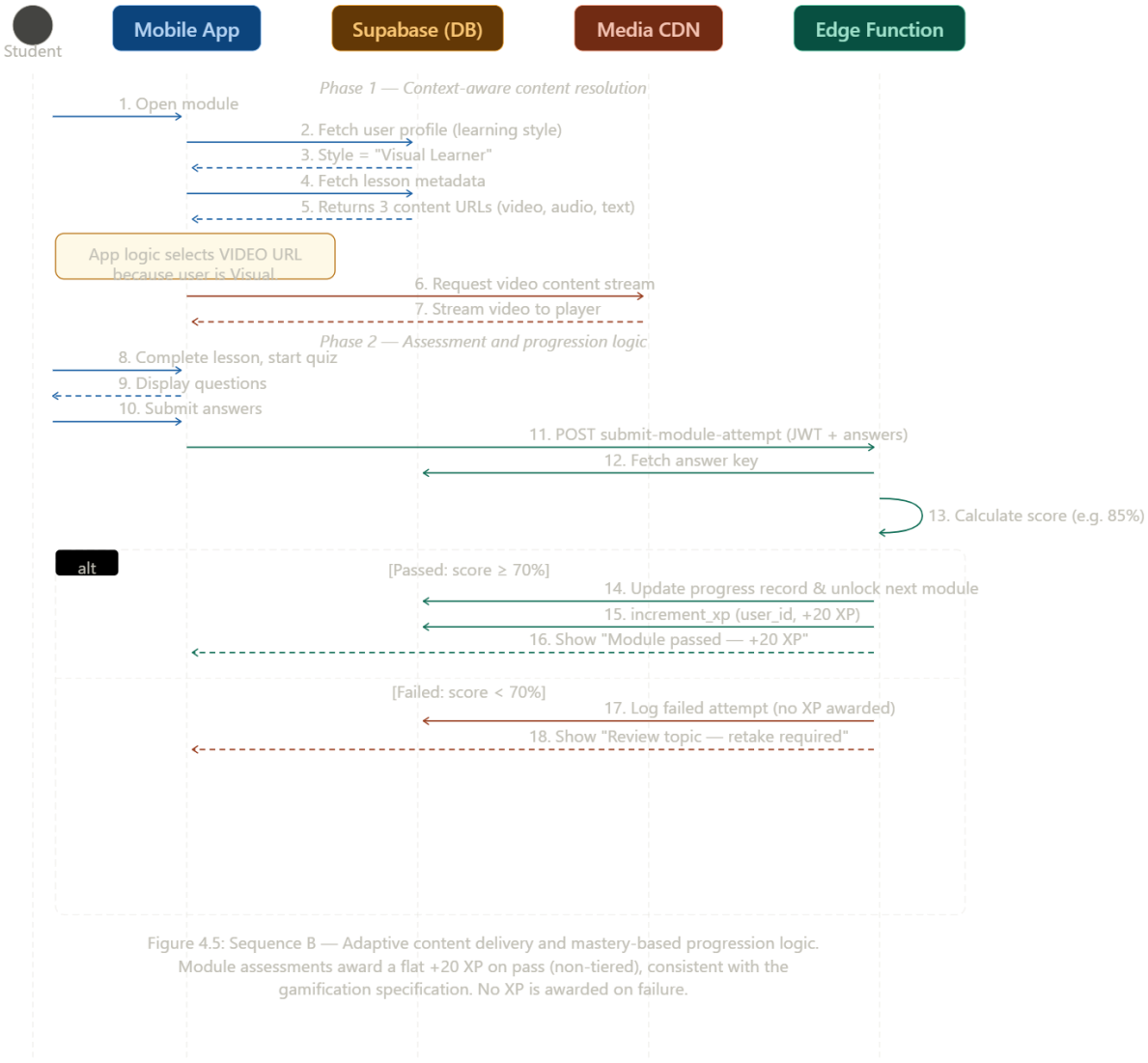


Figure 4.5: Sequence B — Adaptive content delivery and mastery-based progression logic. Module assessments award a flat +20 XP on pass (non-tiered), consistent with the gamification specification. No XP is awarded on failure.

Figure 4.5: Sequence B Adaptive content delivery and mastery-based module assessment flow. Module assessments award a flat +20 XP on passing, consistent with the non-tiered gamification specification. Solid arrows denote synchronous calls; dashed arrows denote return messages.

4.3.3 Sequence C: AI Mentorship & Safety Intervention

Figure 4.6 illustrates the system's most sophisticated real-time interaction: the affect-aware mentorship and safety escalation loop mediated by the AI Pedagogical Agent, Mitchy. This sequence demonstrates the architectural integration of autonomous conversational support with mandatory human oversight, a critical mechanism for managing at-risk learners informed by learning analytics research [25].

The interaction is defined by a continuous, sentiment-driven analysis loop, initiated by each student message.

1. Real-Time Sentiment Stream:

The sequence commences when the Student transmits a message for example, "I am feeling overwhelmed" via the Mobile App. The app sends an authenticated HTTPS POST request to the mitchy-chat Edge Function, which verifies the JWT and relays the payload to the Railway /mitchy/chat endpoint using a private service key. It is noted that this communication path traverses standard HTTPS rather than WebSocket: WebSocket connections in LearNova are reserved for Supabase Realtime subscriptions such as leaderboard updates and push notifications. Upon receiving the message, the Mitchy agent executes a Sentiment Analysis routine, producing a numerical sentiment score. This real-time affective signal, grounded in affective computing principles for educational monitoring [23, 25], is persisted to the chat_messages table alongside the message content.

2. Conditional Safety Logic:

The system enters a conditional block determined by whether the sentiment score breaches a predefined critical threshold. On the **Critical Path** (risk detected), the agent creates a risk_scores record in Supabase and triggers an asynchronous push notification to the Admin panel. The sequence explicitly models the human-in-the-loop protocol: upon notification, the Admin fetches the student's chat history, composes a manual intervention message, and dispatches it through the system, which relays it to the student's interface thereby ensuring that a learner in distress receives timely, qualified human support and fulfilling the safety requirements established in FR-16. On the **Normal Path**, where sentiment remains within acceptable parameters, the agent retrieves prior conversational context and the student's learning profile from Supabase, assembles a structured prompt, and generates a context-augmented pedagogical response via the Gemini LLM API [87], which is returned to the student through the Edge Function relay.

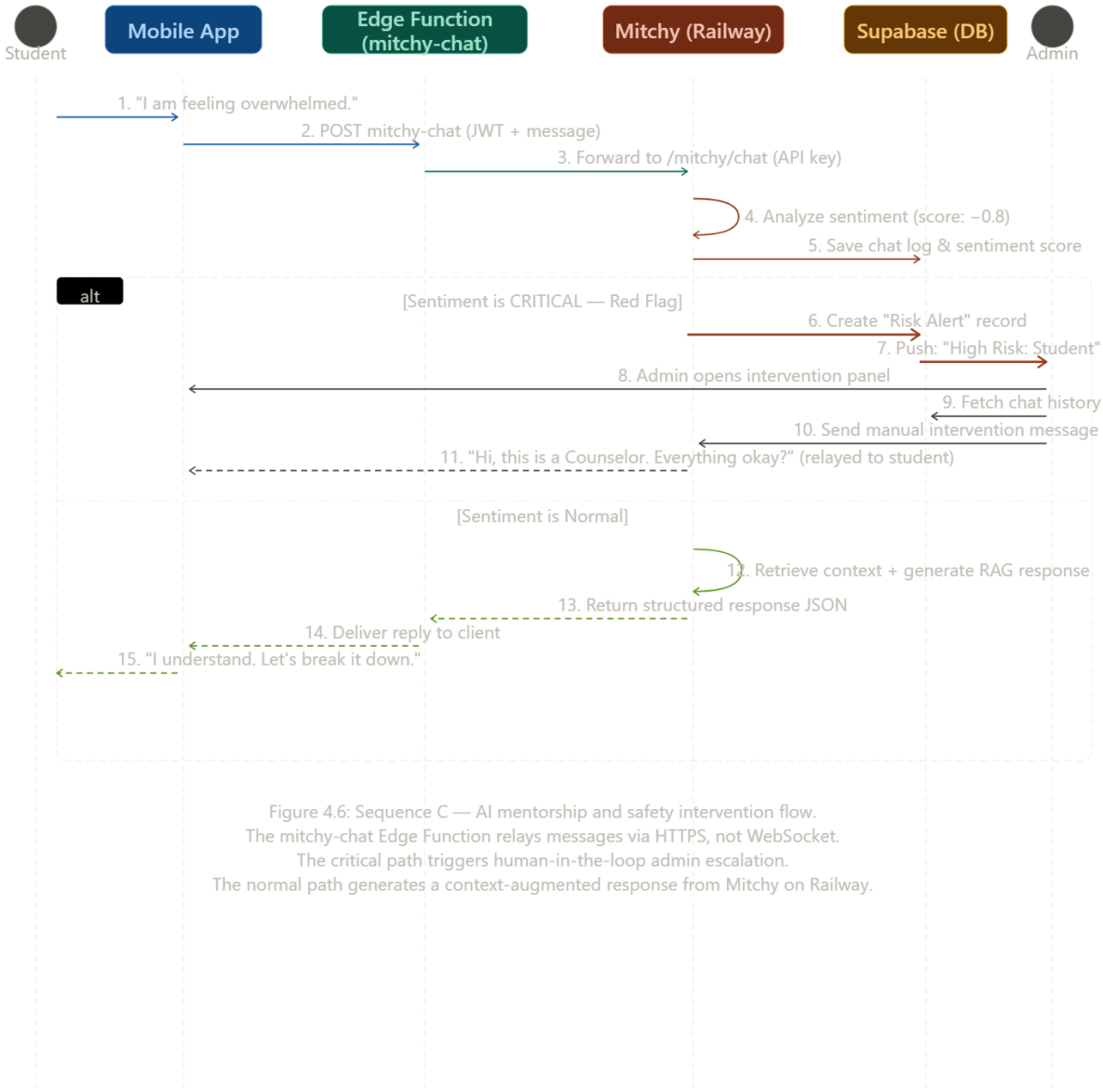


Figure 4.6: Sequence C — AI mentorship and safety intervention flow.
The mitchy-chat Edge Function relays messages via HTTPS, not WebSocket.
The critical path triggers human-in-the-loop admin escalation.
The normal path generates a context-augmented response from Mitchy on Railway.

Figure 4.6: *Sequence C AI mentorship and safety intervention flow. The mitchy-chat Edge Function relays messages via HTTPS. The critical path triggers human-in-the-loop admin escalation; the normal path produces a context-augmented Mitchy response. Solid arrows denote synchronous calls; dashed arrows denote return messages.*

4.4 Structural Design (Class Decomposition)

While the preceding sections have detailed the system's dynamic behavioral flows and data transformations, this section delineates its static structural foundation. The Class Diagrams presented herein formalize the object-oriented decomposition of the LearNova platform, adhering to established software design patterns as a mechanism for achieving modularity, extensibility, and a rigorous separation of concerns. Four structurally significant subsystems are examined in turn: the identity and profile architecture, the academic curriculum hierarchy, the hybrid assessment strategy, and the event-driven AI and engagement layer.

4.4.1 Identity & Profile Structure

Figure 4.7 illustrates the architecture governing user management, which employs a hybrid strategy combining inheritance for role polymorphism with object composition for dynamic state management. This structural disposition enforces a strict logical separation between stable authentication logic and the volatile, high-frequency data pertaining to learner engagement and progress.

A. The Inheritance Hierarchy for Role Abstraction

The core of the identity module is governed by an **Abstract Base Class (ABC)** strategy, which prevents the instantiation of ambiguous, role-agnostic user entities. The system defines an abstract User class that encapsulates universal authentication attributes email, password_hash, and last_login alongside fundamental behavioral contracts, namely login() and logout(). This abstract declaration mandates that all concrete user entities must be specialized into distinct role subtypes, a classical application of the inheritance relationship for encoding hierarchical "is-a" dependencies in object-oriented systems.

This specialization is realized through two concrete subclasses. The Student class extends the base User to incorporate learner-specific behaviors: submitDiagnostic(), which initiates the calibration pipeline, and requestContent(), which triggers adaptive content resolution. The is_verified boolean attribute is additionally introduced to govern the hybrid assessment workflow, distinguishing between students whose diagnostic scores are fully validated and those whose external submissions remain pending administrative review. The Admin class, conversely, extends User to encapsulate privileged operational capabilities triggerIntervention() and reviewExternalProof() alongside a strongly typed permissions list, directly enforcing the Role-Based Access Control (RBAC) policy defined in the system's security architecture.

The UserFactory class sits above this hierarchy, providing a centralized createUser() factory method that accepts a role descriptor and returns a correctly instantiated concrete object. This abstraction decouples the presentation layer from the internal constructors of Student and Admin, yielding significant extensibility: future role introductions a mentoring tutor, for instance would require modification exclusively within the factory and the new class definition, leaving all existing authentication and registration flows undisturbed.

B. Profile Composition for Separation of Concerns

A structurally consequential design decision is the decoupling of the core User identity entity from the mutable StudentProfile state entity via object composition. Rather than

inflating the Student class with volatile engagement attributes, these responsibilities are delegated to a distinct StudentProfile component. This class manages four categories of dynamic state: the learner's current_style (one of VISUAL, AUDITORY, or TEXTUAL, aligned with the three values enforced by the student_profiles.learning_style database constraint); the momentum_streak reflecting consecutive days of platform activity; the zone_state enumeration (LOCKED_IN, DRIFTING, or DISENGAGED) used to drive the risk detection pipeline; and total_xp, which accumulates across all gamification events via the increment_xp() RPC.

This decomposition constitutes a direct application of the Single Responsibility Principle, wherein the Student class bears sole responsibility for identity and authentication, while StudentProfile governs the entirety of dynamic learning state. Consequently, high-frequency write operations streak increments following daily logins, XP additions following quiz passes can execute against the profile record without incurring write contention on the more sensitive and immutable identity tables, enhancing both transactional performance and data integrity.

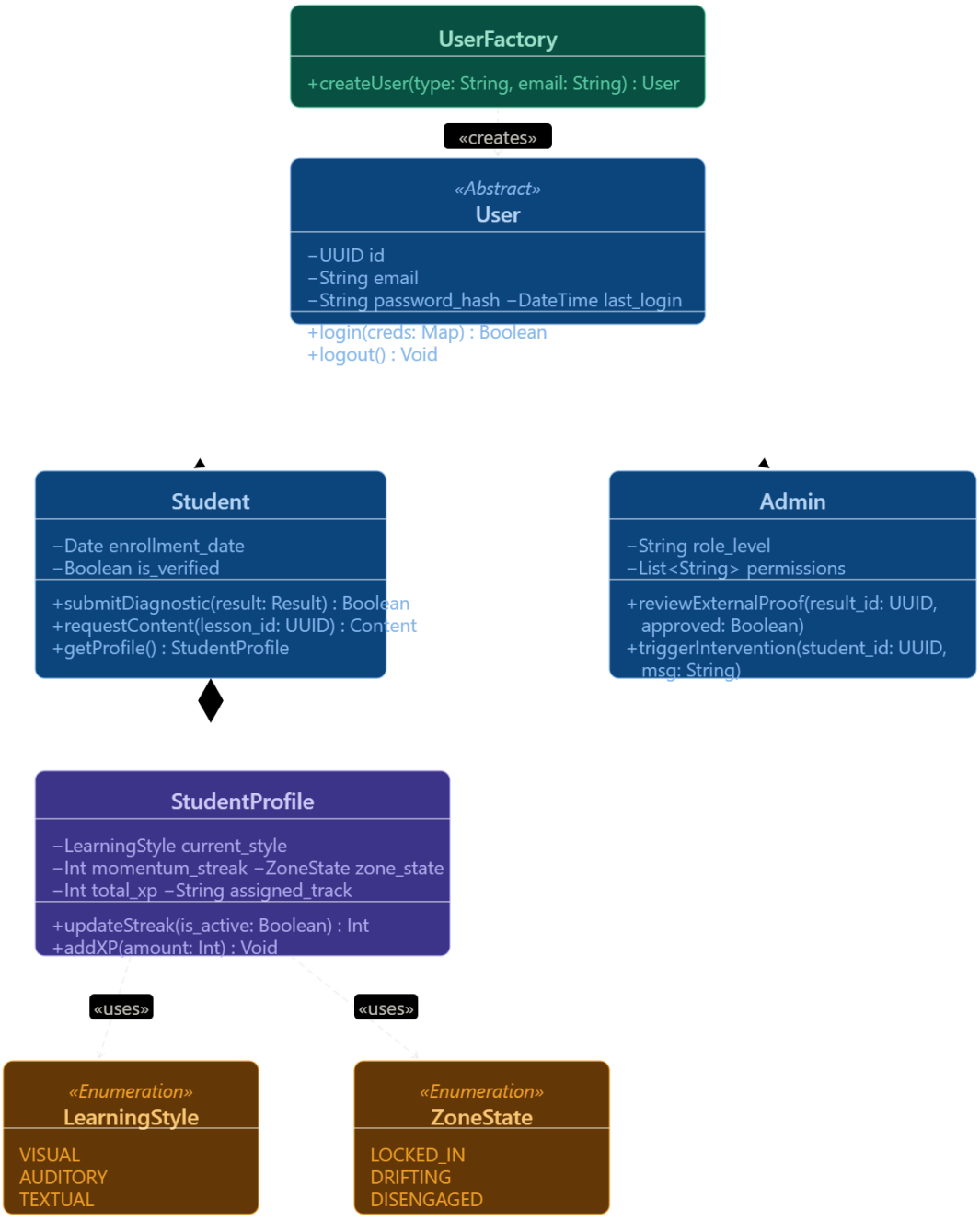


Figure 4.7: Identity and profile structure — abstract User hierarchy, composition-based StudentProfile, and Factory Pattern for role instantiation.

Figure 4.7: Identity and profile structure abstract User hierarchy with Student and Admin role specializations, composition-based StudentProfile for dynamic state management, Factory Pattern for role instantiation, and LearningStyle and ZoneState enumerations aligned with the Supabase schema constraints.

4.4.2 Academic & Adaptive Core

Figure 4.8 models the hierarchical organization of the learning curriculum, which constitutes the academic backbone of the LearNova platform. To satisfy requirements for modularity, content reusability, and scalable management of nested educational structures, this subsystem is architected using the **Composite Design Pattern** a structural pattern well-suited to representing whole-part hierarchies of arbitrary depth as a uniform, recursively composable tree.

A. The Composite Hierarchy for Curriculum Structure

The hierarchy is anchored by the EducationComponent interface, which establishes a consistent behavioral contract across all curriculum entities. Every node in the tree from the highest-level container to the atomic content unit must implement getTitle(), getDuration(), and isCompleted(user_id), ensuring that the presentation and progress-tracking layers can interrogate any curricular element through a single, polymorphic interface without prior knowledge of its structural depth.

The hierarchy descends through four concrete levels, each mapped directly to its corresponding Supabase schema entity. The Course class (mapping to the courses table) acts as the root container, grouping levels under a specific learning track and exposing an is_foundation flag to designate shared prerequisite content accessible across all enrolled students irrespective of their assigned specialization. The Level class (mapping to levels) aggregates ordered modules and governs sequential unlocking logic via unlockNext(). The Module class (mapping to modules) represents a time-bounded instructional unit, with estimated_minutes enabling the platform to communicate expected cognitive commitment to the learner prior to engagement.

B. Adaptive Encapsulation within the Polymorphic Content Node

The leaf node of this hierarchy, LessonTopic (mapping to topics and topic_resources), is structurally the most significant component, as it directly operationalizes the platform's core adaptive delivery mechanism. In contrast to conventional content node designs that reference a single static resource URL, LessonTopic encapsulates three distinct content variant attributes within a single entity: video_url_visual for Visual learners, audio_url_auditory for Auditory learners, and text_url_textual for Textual learners the three modality values enforced by the database schema.

The adaptive selection logic is centralized within the getContentUrl(style: LearningStyle) : String method, which functions as a polymorphic dispatcher. The presentation layer invokes this method with the learner's calibrated LearningStyle value; the internal conditional logic resolves and returns the appropriate URL, suppressing the remaining variants. This design upholds the Principle of Least Knowledge, ensuring that the Flutter UI layer remains entirely decoupled from the internal structure of the content storage model — a property that reduces inter-component coupling and simplifies future curricular content additions. The multimodal design philosophy informing this structure is substantiated by empirical research on dual coding and multimedia instruction [7, 17].

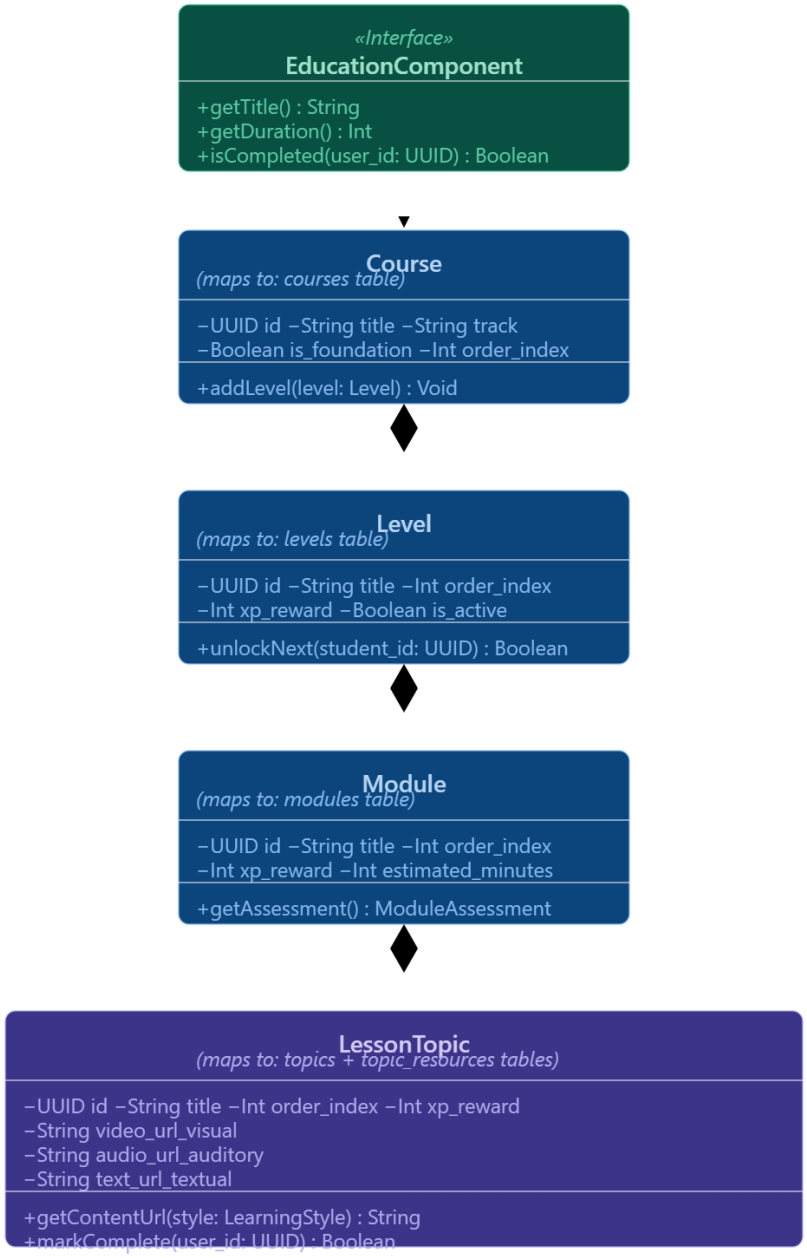


Figure 4.8: Academic and adaptive core — Composite Pattern applied to curriculum structure (Course → Level → Module → LessonTopic). Entity names align with the Supabase schema. The LessonTopic leaf node encapsulates three content variant URLs for adaptive delivery.

Figure 4.8: *Academic and adaptive core — Composite Pattern applied to the curriculum hierarchy (Course → Level → Module → LessonTopic). Entity names correspond directly to Supabase schema table names. The LessonTopic leaf node encapsulates three typed content variant URLs, with adaptive selection delegated to a polymorphic dispatcher method.*

4.4.3 Assessment & Calibration Strategy

Figure 4.9 demonstrates the application of the **Strategy Design Pattern** to resolve a core architectural challenge: supporting two fundamentally distinct assessment methodologies native in-app psychometric quizzes and trust-based external credential validation within a single, unified grading subsystem. This pattern was selected to operationalize the Open–Closed Principle, which mandates that a software module be open for extension without requiring modification to its existing logic. Consequently, future assessment modalities project-based portfolio evaluations, for instance may be introduced by implementing the strategy interface, leaving the orchestration context entirely undisturbed.

A. The Strategy Abstraction

The IAssessmentStrategy interface establishes the polymorphic contract at the heart of this subsystem, defining a single method signature: calculateScore(input: Any) : Result. By architecturally depending on this abstraction rather than on any concrete scoring implementation, the system decouples the AssessmentContext the orchestrating class responsible for initiating evaluations from the variable logic required to process and score each distinct assessment type. This constitutes a direct application of the strategy pattern's core intent: to define a family of interchangeable algorithms that may be selected and substituted at runtime.

B. Concrete Strategy Implementations

Two concrete strategy classes implement this interface, each encapsulating the business rules governing its respective assessment pathway as defined in FR-06. The StandardGradingStrategy implements the scoring logic for proctored in-app assessments: its calculateScore() method accepts a structured map of student answer submissions, retrieves the canonical correct_answer values from the module_assessment_questions or level_assessment_questions tables, and computes a percentage score through direct comparison. This strategy is inherently coupled to the internal Quiz, Question, and AnswerOption domain objects, facilitating structured, rule-based evaluation characteristic of the foundational and module-level assessment phases.

In contrast, the ExternalValidationStrategy encapsulates the trust-mediated pathway: its calculateScore() method accepts a proof URL submitted by the student, validates the URL format, and persists a EXTERNAL_MANUAL_INPUT flag in the SubmissionSource enumeration field of StudentDiagnosticResult. The computed_scores attribute of this result entity is typed as JSONB, consistent with the Supabase schema, allowing the feature vector to be stored as a structured JSON payload rather than a flat string label.

C. Dynamic Context Switching

The AssessmentContext class maintains a reference to an IAssessmentStrategy instance, enabling dynamic behavioral substitution at runtime. When the student selects in-app assessment, the context is injected with StandardGradingStrategy; when the student elects external credential submission, ExternalValidationStrategy is substituted. This mechanism ensures that the core orchestration workflow remains consistent and unmodified across both pathways, with all variation isolated within the swappable strategy object.

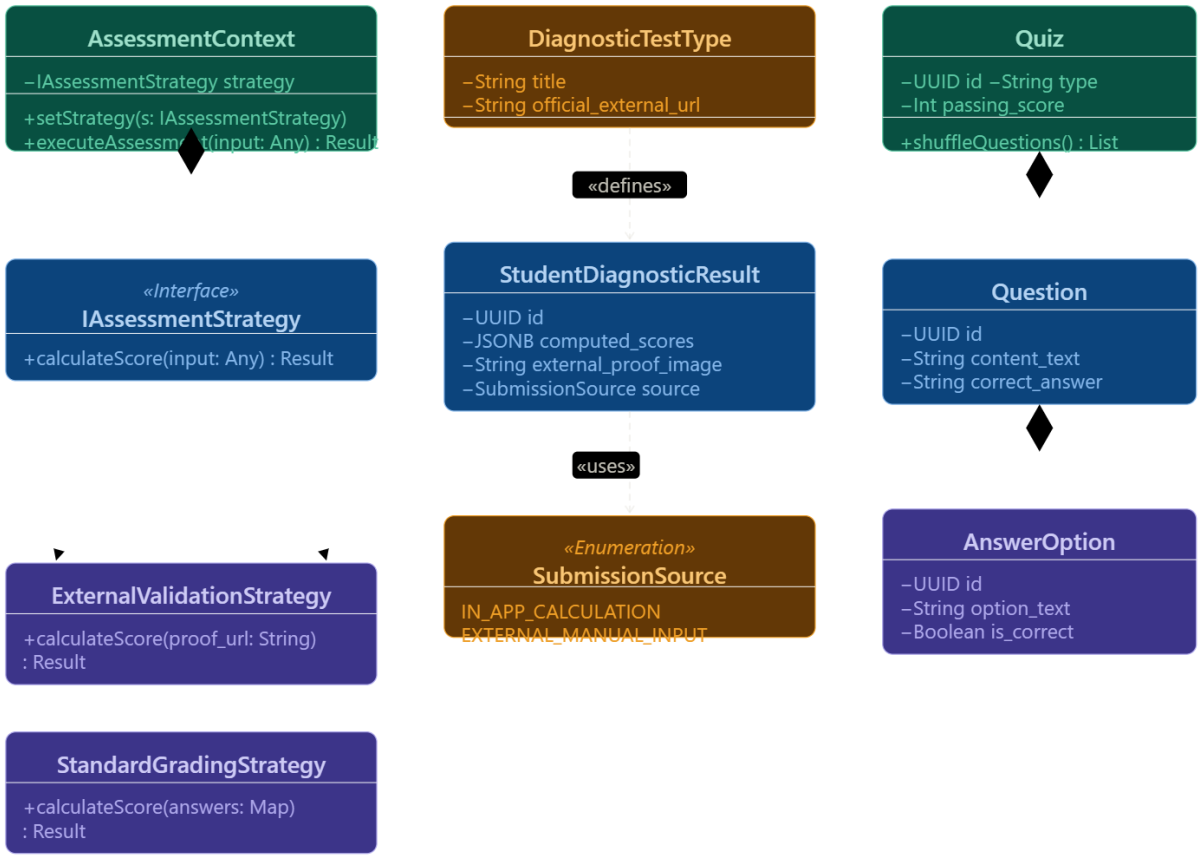


Figure 4.9: Assessment and calibration Strategy Pattern.
IAssessmentStrategy decouples scoring logic from context invocation.
computed_scores stores JSONB feature vectors aligned with the Supabase schema.

Figure 4.9: Assessment and calibration Strategy Pattern IAssessmentStrategy interface, ExternalValidationStrategy and StandardGradingStrategy concrete implementations, AssessmentContext orchestrator, and associated diagnostic domain classes. computed_scores is typed JSONB in alignment with the Supabase schema.

4.4.4 AI & Engagement (Observer Pattern)

Figure 4.10 illustrates the application of the **Observer Design Pattern** to manage the complex, event-driven dependencies between learner state transitions and systemic responses. This pattern was selected to fulfill the requirement for a fully reactive architecture, wherein a single student event detected negative sentiment or a passed quiz assessment must asynchronously activate multiple, independent downstream processes without introducing tight coupling between the modules responsible for each response.

A. The Subject: Centralized State Management

The StudentProgressTracker class functions as the **Subject** (or Publisher), serving as the authoritative source of truth for a learner's runtime state. It maintains a list of subscribed IObserver instances and monitors two critical mutable attributes: current_sentiment, updated upon each Mitchy interaction, and current_user_id, scoping all broadcasts to the correct learner context. Rather than invoking downstream components directly a design that would introduce fragile inter-module dependencies the tracker broadcasts state-change events via notifyObservers(event: RiskEvent), dispatching to all subscribers without possessing any knowledge of their identities or internal logic. This architectural choice enforces a high degree of modularity and substantially simplifies the core state management responsibility.

B. The Observers: Specialized Reactive Handlers

Two concrete observer classes subscribe to the StudentProgressTracker, each implementing the IObserver interface's update() method to respond to domain-specific event types. The MitchyAgent (Safety Observer) listens for affective state updates. Upon receiving a notification, it evaluates the incoming sentiment score against a configurable sentiment_threshold. If a breach is detected, it autonomously invokes flagRisk(), instantiating a RiskAlert record which maps directly to the risk_scores table in the Supabase schema and initiating the administrative escalation pipeline. Concurrently, it monitors the ChatSession entity, whose avg_sentiment attribute is maintained as a session-level aggregate enabling efficient trend queries without scanning individual message rows. This proactive, affect-aware intervention design is grounded in affective computing research demonstrating that behavioral signals can predict learner disengagement with measurable accuracy prior to voluntary dropout [23, 25].

The GamificationEngine (Engagement Observer) listens for achievement events, specifically QuizPassEvent. Upon notification, it executes the reward pipeline in sequence: validating streak status, applying the 1.1× streak multiplier where applicable, computing the final XP award in accordance with the tiered gamification specification, and dispatching the increment_xp() RPC call. Decoupling this engagement logic from the core progress tracker ensures that iterative refinements to gamification mechanics a common operational requirement for sustaining learner motivation, as substantiated by gamification retention research [50] do not destabilize the foundational state-tracking infrastructure.

C. Event-Driven Decoupling and Testability

Architecturally, the Observer pattern fundamentally separates event detection from event handling. The StudentProgressTracker is concerned solely with monitoring and broadcasting; it possesses no knowledge of XP calculation algorithms or risk alert formatting. This adherence to the Dependency Inversion Principle yields a system that is highly maintainable and independently testable: the MitchyAgent may be substituted with a mock implementation during integration testing without requiring any modification to the StudentProgressTracker, permitting robust, isolated validation of both the safety escalation and gamification reward subsystems.

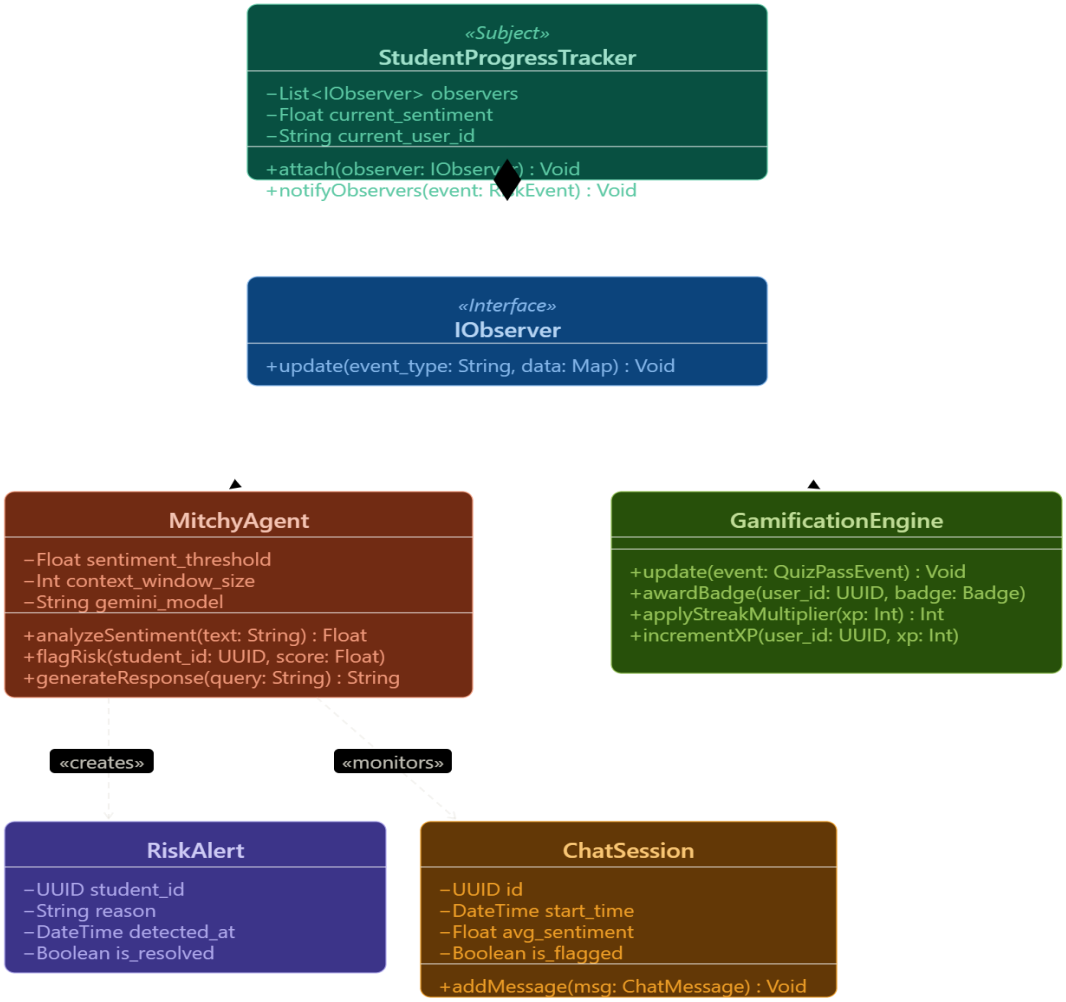


Figure 4.10: Observer Pattern — StudentProgressTracker as Subject,
MitchyAgent (safety) and GamificationEngine (engagement) as concrete Observers.

Figure 4.10: Observer Pattern StudentProgressTracker as Subject; MitchyAgent (safety) and GamificationEngine (engagement) as concrete Observer implementations. The pattern decouples affective risk detection from gamification reward logic, enabling independent testability and runtime extensibility. ChatSession.avg_sentiment maps to the average_sentiment column in the Supabase chat_sessions table.

4.5 Database Design (Schema Definition)

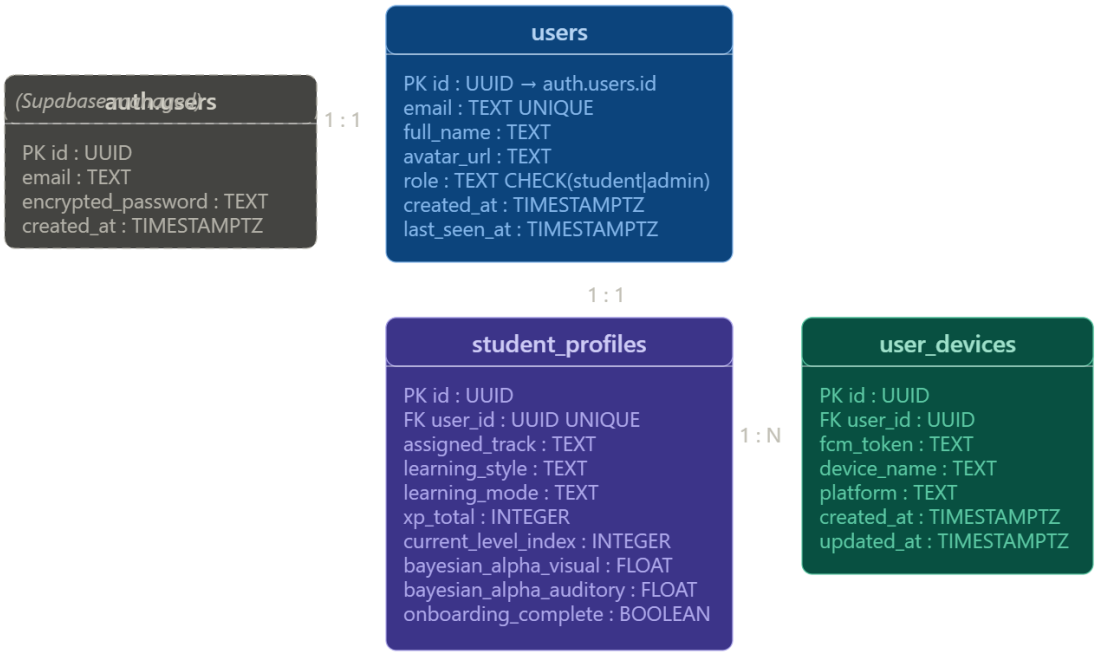
The logical structure delineated in Section 4.4 is physically realized through PostgreSQL, the relational database engine underlying the Supabase backend infrastructure [84]. The schema is architected to enforce Third Normal Form (3NF) throughout, minimizing data redundancy whilst preserving referential integrity via foreign key constraints across all inter-table relationships. To support the system's distributed operational nature and reinforce its security posture, all primary keys are defined as Universally Unique Identifiers (UUIDs) rather than sequential integers a deliberate design choice that prevents enumeration-based attacks and ensures collision-free record management across distributed data sources [85]. Given the breadth of the schema, the following subsections present five thematically organized ERD panels rather than a single monolithic diagram, each corresponding to a distinct functional domain.

4.5.1 Identity & Access Management (IAM)

ERD Panel 1 (Figure 4.11a) illustrates the security-first architecture governing user identity, which operationalizes a strict separation between authentication credentials and application-level profile data a structural decision aligned with the Principle of Least Privilege.

The system delegates all sensitive identity management to Supabase's internally managed auth.users schema, which stores encrypted password hashes and governs JWT issuance. This managed table is architecturally isolated from the public PostgREST API surface, ensuring that a vulnerability in the application layer cannot propagate to core authentication credentials [84]. The public users table extends this foundation for application purposes: its id column is defined simultaneously as a Primary Key and a Foreign Key referencing auth.users.id, enforcing a strict one-to-one constraint wherein an application profile cannot exist without a corresponding authenticated identity record.

Role differentiation is implemented directly within the users table via a role column constrained to 'student' or 'admin', which drives the Row-Level Security (RLS) policy evaluation functions is_admin() and current_user_is_admin() at the PostgreSQL layer. This approach avoids a separate permissions table for most access decisions whilst retaining the granular capability to scope admin-level operations. The student_profiles table maintains a one-to-one composition relationship with users, encapsulating the mutable learner state assigned_track, learning_style, xp_total, current_level_index, and the Bayesian posterior alpha values for adaptive learning style inference in a record that sustains high write frequency without contention against the more stable identity table. Finally, the user_devices table resolves the one-to-many relationship between a user and their registered FCM push notification tokens, supporting multi-device notification delivery across Android and iOS platforms.



ERD Panel 1: Identity and Access Management — auth.users (Supabase-managed), users, student_profiles (1:1), and user_devices (1:N). Dashed border = managed schema.

Figure 4.11a: ERD Panel 1 Identity and Access Management schema. The auth.users table (dashed border) is Supabase-managed and isolated from the public API. The users and student_profiles tables maintain a strict 1:1 relationship; user_devices resolves the 1:N multi-device registration relationship.

4.5.2 Psychometric & Hybrid Calibration Schema

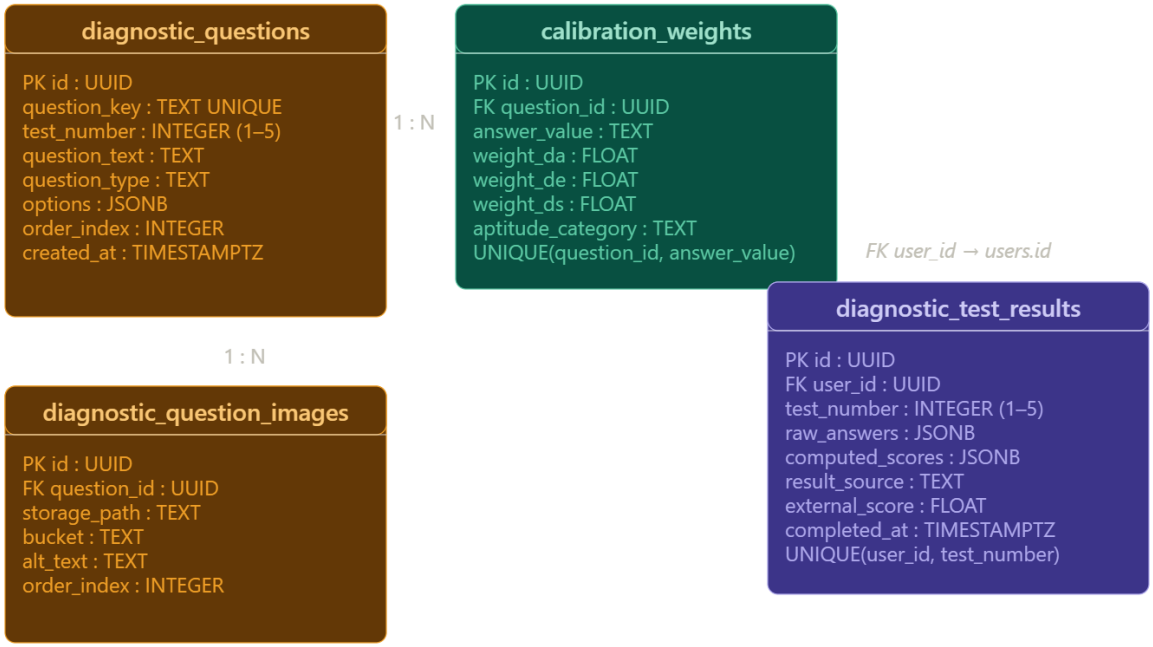
ERD Panel 2 (Figure 4.11b) details the persistent storage architecture supporting the five-point diagnostic calibration system. The schema is designed to prioritize data provenance and auditability critical requirements given the platform's capacity to ingest both internally computed and externally submitted diagnostic evidence.

The diagnostic_questions table stores the canonical question bank across all five assessment instruments, with each record identified by a question_key business key that serves as the bridge between the Flutter raw answer payload and the Railway scoring engine's answer-key seed. The test_number column, constrained to values 1 through 5, enforces the sequential diagnostic structure at the database level. Each question may be associated with one or more diagnostic_question_images records in a one-to-many relationship, the additional bucket column distinguishing the Supabase Storage bucket from which each image asset is served.

The calibration_weights table encodes the direct O*NET calibration evidence for track assignment: each row maps a specific (question_id, answer_value) pair enforced as unique to weighted contribution values for the DA, DE, and DS tracks. As discussed in Section 4.5, these weights constitute the base scoring anchors rather than the complete

modifier formula, which is computed at runtime within the Railway service from the full feature vector.

The diagnostic_test_results table functions as the central ledger for calibration outcomes. Its computed_scores column stores the normalized feature vector as a JSONB payload encompassing Big Five personality dimensions, VARK style counts, RIASEC interest scores, and IQ section scores reflecting the rich, multi-dimensional profile generated by the scoring pipeline. The result_source column distinguishes between in_app submissions, which carry high trust as they are processed directly by the Railway scoring engine, and external submissions, which carry lower trust and are flagged for administrative validation before propagating to track assignment. A unique constraint on (user_id, test_number) ensures that each student may hold precisely one result per diagnostic instrument at any point in time, with force-rescore capability available through the run-scoring-engine Edge Function for administrative correction [84].



ERD Panel 2: Psychometric and hybrid calibration schema.
computed_scores stores the feature vector as JSONB; result_source distinguishes
IN_APP_CALCULATION from EXTERNAL_MANUAL_INPUT submissions.

Figure 4.11b: ERD Panel 2 Psychometric and hybrid calibration schema. computed_scores is typed JSONB to accommodate the multi-dimensional feature vector. The unique constraint on (user_id, test_number) enforces one result per diagnostic instrument per student.

4.5.3 Adaptive Curriculum Architecture

ERD Panel 3 (Figure 4.11c) presents the hierarchical data structure supporting the Adaptive Strategy Engine. The schema implements a strict four-level content hierarchy — courses → levels → modules → topics enforcing referential integrity at each tier

through foreign key cascades, whilst accommodating the polymorphic content delivery mechanism that constitutes the platform's core adaptive differentiator [7].

At the apex of the hierarchy, the courses table introduces the is_foundation boolean flag, which enables a single foundational course to be designated as a shared prerequisite accessible across all enrolled students irrespective of their assigned_track value. This architectural decision eliminates data duplication and ensures that updates to foundational curriculum content propagate universally without requiring track-specific record modifications. The levels and modules tables descend in strict one-to-many relationships from courses, each governed by an order_index attribute that enforces sequential curriculum progression and an is_active boolean that permits administrators to gate content visibility without physical deletion.

The most architecturally significant element of this schema is the design of content delivery at the topic_resources table, which resolves the adaptive content slot model. Rather than embedding content URLs directly within the topics table, content assets are stored as individual topic_resources rows distinguished by a format_type column constrained to 'Visual', 'Auditory', or 'Textual'. This normalized approach in contrast to the wide-column denormalized alternative considered during design preserves extensibility: additional modality types may be introduced by inserting new topic_resources rows without requiring schema alteration. When a student requests lesson content, the API executes a single filtered query against topic_resources for the student's format_type, resolving the correct asset URL without multi-table joins, thereby minimizing response latency for the mobile client. This design directly operationalizes the modality alignment principles that empirical research on multimedia learning has demonstrated to reduce extraneous cognitive load and improve knowledge encoding efficiency [7, 17].

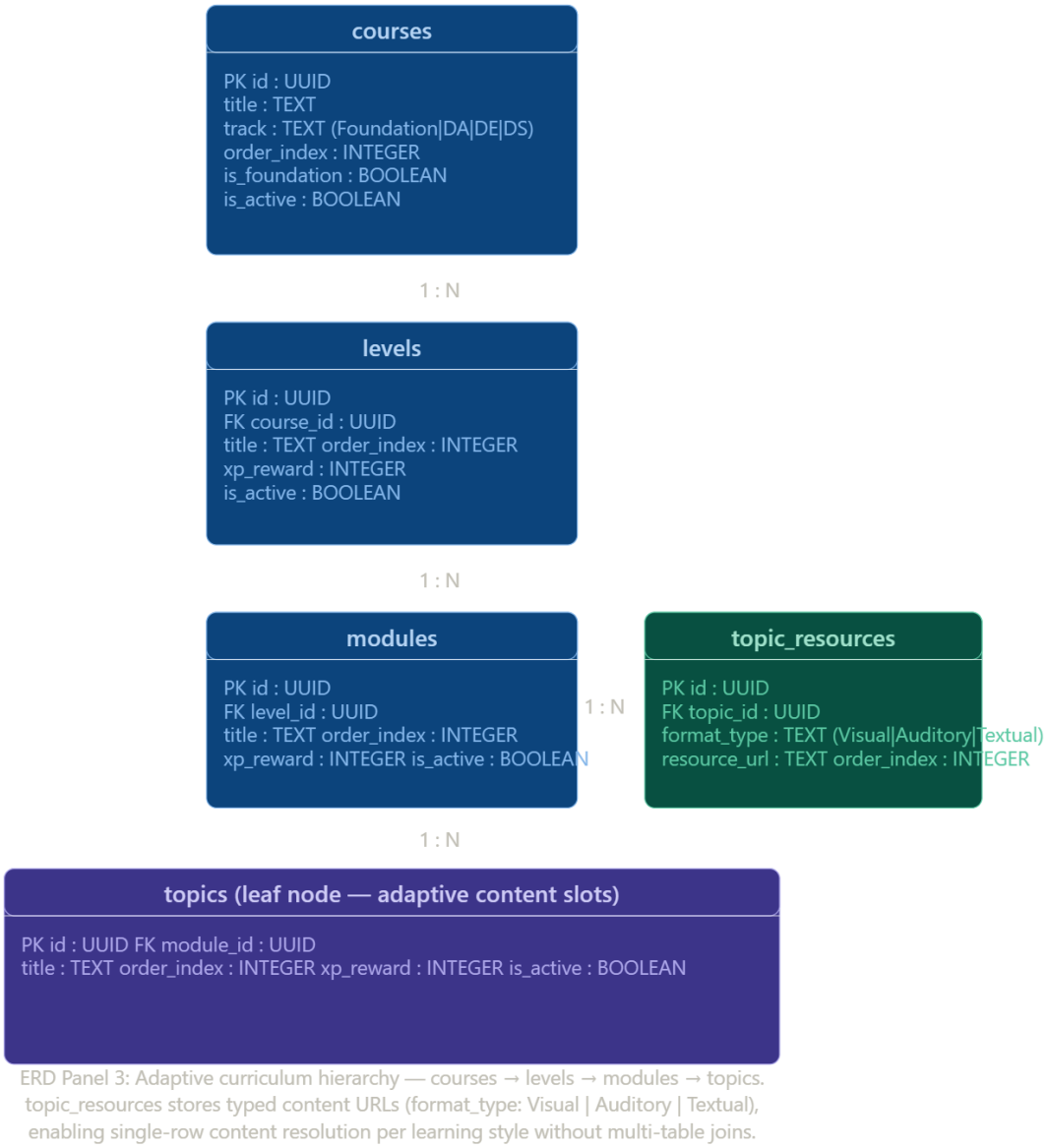


Figure 4.11c: ERD Panel 3 Adaptive curriculum hierarchy (courses → levels → modules → topics). Entity names correspond directly to Supabase schema table names. The topic_resources table stores typed content URLs per modality, enabling single-query adaptive content resolution without multi-table joins.

4.5.4 Assessment & Performance Ledger

ERD Panel 4 (Figure 4.11d) defines the transactional schema responsible for tracking academic performance and governing mastery-based progression the empirical backbone upon which the platform's adaptive evaluative functions depend.

The assessment definition layer comprises two parallel structures: module_assessments paired with module_assessment_questions for module-level checkpoints, and level_assessments paired with level_assessment_questions for end-of-level evaluations.

Each assessment table enforces a unique constraint on its respective parent identifier (module_id and level_id respectively), ensuring that precisely one assessment may be defined per curricular unit. The level assessment question table introduces additional columns absent in its module-level counterpart ai_grading_rubric, mitchy_hint, and mitchy_explanation reflecting the more complex, open-ended evaluation and AI-assisted feedback requirements associated with level progression assessments.

The corresponding attempt tables student_module_attempts and student_level_attempts serve as immutable transactional ledgers, recording each submission with its answers JSONB payload, computed score, passed boolean, and difficulty tier selection. Multiple attempts are permitted on module assessments, enabling students to retake until mastery is demonstrated; each attempt row is preserved in full, providing the ML analytics pipeline with the raw data necessary to compute topic struggle indices and engagement velocity metrics. The student_level_attempts table additionally stores mitchy_feedback, the AI tutor's personalized post-assessment commentary generated by the Mitchy agent for each submission.

The student_progress table resolves the many-to-many relationship between the users and topics entities, functioning as a mutable, time-series engagement ledger. A unique constraint on (user_id, topic_id) ensures that precisely one progress record exists per student-topic pair. The status column constrained to 'not_started', 'in_progress', or 'completed' drives the UI progress visualization layer directly from a single column read. The format_served column records which content modality was delivered for each topic engagement event, providing the data necessary for the get_admin_format_effectiveness() analytics function to evaluate whether learning style alignment is producing measurable differences in completion rates across the student cohort. The completed_at timestamp further enables the analytics engine to derive learning velocity metrics modules completed per unit time which feed into the gamification engine's streak evaluation logic [50].

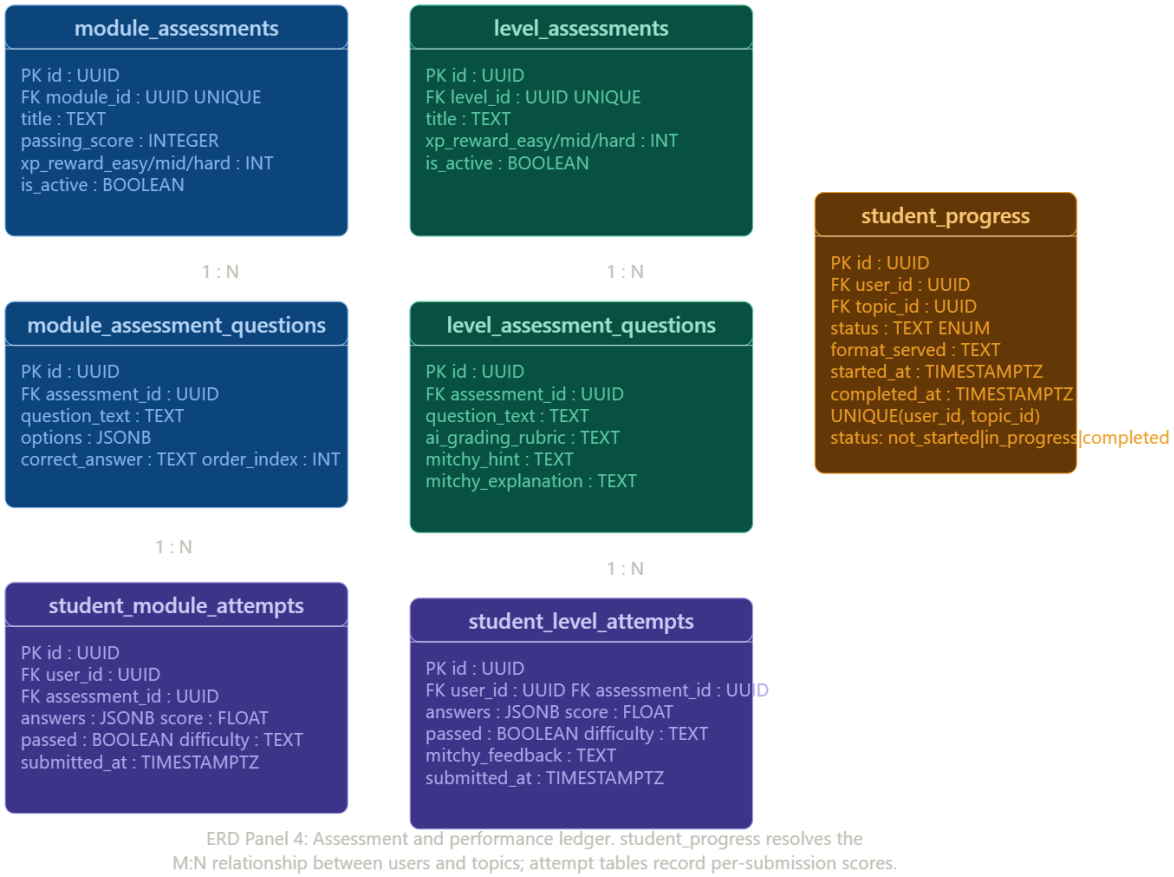


Figure 4.11d: ERD Panel 4 Assessment and performance ledger. Module and level assessments maintain 1:1 constraints with their parent curriculum entities. Attempt tables record per-submission scores as immutable ledger entries. student_progress resolves the M:N user-topic relationship and records the content modality served per engagement event.

4.5.5 Engagement & Gamification Mechanics

ERD Panel 5 (Figure 4.11e) defines the retention and safety ecosystem, which is architecturally decoupled from the core academic tables to enforce performance isolation: the high write volume characteristic of interaction logging and gamification events must not introduce contention on the content delivery pathway [85].

The gamification layer strategically separates reward definition from reward issuance across two tables. The achievements_dictionary stores static metadata achievement_key, label, criteria_type, criteria_threshold, and badge_image_path for all available achievements on the platform. The user_achievements junction table records only the (user_id, achievement_id, unlocked_at) triplet, enforced as unique per pair. This separation enables rapid aggregation queries counting a student's earned achievements, for instance to execute against the lightweight junction table rather than scanning the primary profile record, a standard optimization for high-throughput engagement dashboards [51]. The user_streaks table maintains a one-to-one relationship with users,

storing current_streak_days, longest_streak_days, and last_activity_date the three values required by the Edge Function streak validation logic to determine whether the 1.1× XP multiplier applies to a given session.

The chat and AI interaction schema organizes conversational data hierarchically. The chat_sessions table acts as a conversational container, featuring an average_sentiment column maintained as a session-level pre-aggregate. This design allows the high-risk detection pipeline to identify students with persistently negative sentiment trajectories by scanning a single float column per session, rather than processing individual message payloads a significant optimization for alert latency in a system where timely intervention is pedagogically critical [25, 26]. Each constituent chat_messages record stores the full message content, role (user or assistant), and detected_learning_state, providing granular data for historical emotional trajectory analysis [23].

The risk management schema comprising risk_scores and intervention_logs constitutes the administrative safety tier. The risk_scores table stores the computed risk_score and risk_level alongside a feature_snapshot JSONB column preserving the ML feature values used in each risk computation, thereby providing the audit trail necessary for explainability and retrospective intervention review. The intervention_logs table records the administrative response lifecycle: claimed_at, resolved_at, action_taken, and notes, with a foreign key to admin_id scoping each log to the responsible administrator. Row-Level Security policies ensure that INSERT operations on risk_scores may be executed by the Railway service using the service-role key, whilst SELECT operations are restricted exclusively to users holding an admin role claim guaranteeing that sensitive risk and mental health indicators are never exposed through the public client API [84].

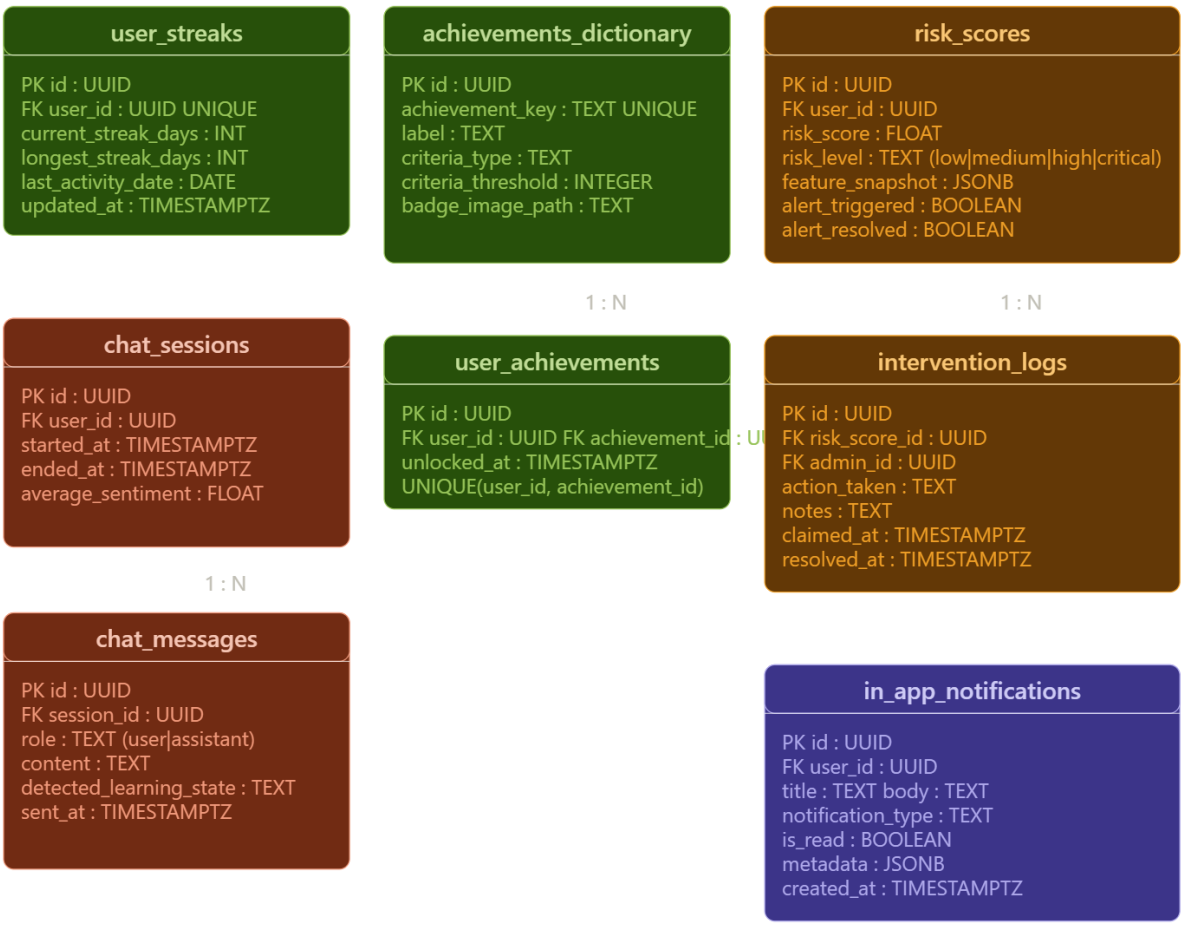


Figure 4.11e: ERD Panel 5 Engagement, gamification, AI interaction, and risk management schema. achievements_dictionary separates reward definition from issuance; user_achievements records the junction. chat_sessions.average_sentiment enables efficient session-level risk scanning. RLS policies on risk_scores restrict SELECT to admin role claims only.

4.6 API Design

The LearNova platform exposes its backend capabilities to the Flutter client through a layered API architecture comprising two distinct tiers: a set of Supabase-hosted **Edge Functions** that constitute the authenticated, publicly accessible service boundary, and a direct PostgREST interface for read-only content queries that do not require server-side computation. This section formally documents the request and response contracts governing both tiers, the two-layer authentication model that secures all inter-service communication, and the standardized error handling strategy employed across the API surface.

4.6.1 Authentication Architecture

All API interactions within the LearNova ecosystem are governed by a two-tier authentication model designed to enforce the **Golden Rule** that the Flutter client must never communicate directly with the Railway backend or hold Railway credentials in any form.

Tier 1 Client-to-Supabase (JWT Authentication). Every request from the Flutter application to any Edge Function or PostgREST endpoint must carry an Authorization header bearing a valid JSON Web Token issued by the Supabase GoTrue identity gateway upon successful login. The structure of this header is standardized across all endpoints as follows:

Authorization: Bearer <supabase_jwt>

Content-Type: application/json

Upon receipt, the Edge Function instantiates a Supabase client scoped to the anonymous key and invokes auth.getUser() against the inbound JWT to verify session validity before any business logic is executed. Requests bearing absent, malformed, or expired tokens are rejected with HTTP 401 prior to any database interaction, ensuring that unauthenticated traffic cannot reach protected data surfaces.

Tier 2 Supabase-to-Railway (Service Key Relay). Edge Functions that require Railway-hosted computation forward the authenticated request payload to the Railway FastAPI service using a private x-api-key header populated from a Supabase environment secret. The Flutter client neither holds nor observes this key at any point. This relay architecture means that Railway credentials are exclusively resident in the Supabase serverless environment, inaccessible to client-side inspection or interception. The forwarded request additionally carries the verified user_id extracted from the JWT, allowing Railway to associate the computation with the correct student record without independently re-authenticating.

Table 4.1 summarizes the authentication requirements across the full API surface.

Table 4.1: API Authentication Requirements

Endpoint / Function	Client Header	Relay Header	Railway Reached
PostgREST (content reads)	Authorization: Bearer <JWT>	None	No

consume-resource	Authorization: Bearer <JWT>	None	No
run-scoring-engine	Authorization: Bearer <JWT>	x-api-key	Yes
submit-module-attempt	Authorization: Bearer <JWT>	None	No
submit-challenge-attempt	Authorization: Bearer <JWT>	None	No
mitchy-chat	Authorization: Bearer <JWT>	x-api-key	Yes

4.6.2 PostgREST Direct Query Interface

For read-only content operations specifically the retrieval of curriculum structure, topic resources, assessment metadata, leaderboard snapshots, and notification records the Flutter client queries the Supabase PostgREST interface directly. This interface auto-generates a typed RESTful API from the PostgreSQL schema, eliminating the need for dedicated Edge Functions for straightforward data retrieval operations. All queries are governed by Row-Level Security policies evaluated against the inbound JWT, ensuring that students may only access content rows where is_active = true and data rows belonging to their own user_id.

Representative direct query patterns are documented in Table 4.2.

Table 4.2: Representative PostgREST Direct Query Patterns

Operation	Method	Resource Path	RLS Filter Applied
Fetch active courses for track	GET	/rest/v1/courses?is_active=eq.true&track=eq.{track}	is_active = true
Fetch levels for a course	GET	/rest/v1/levels?course_id=eq.{id}&is_active=eq.true	is_active = true
Fetch topic resources by style	GET	/rest/v1/topic_resources?topic_id=eq.{id}&format_type=eq.{style}	Parent topic active
Fetch own student profile	GET	/rest/v1/student_profiles?user_id=eq.{uid}	user_id = auth.uid()
Fetch leaderboard snapshot	GET	/rest/v1/leaderboard_snapshots?track=eq.{track}	All students readable
Fetch own notifications	GET	/rest/v1/in_app_notifications?user_id=eq.{uid}	user_id = auth.uid()
Mark notification read	PATCH	/rest/v1/in_app_notifications?id=eq.{id}	user_id = auth.uid()

4.6.3 Edge Function Contracts

Five Supabase Edge Functions constitute the active computation boundary of the LearNova API. Each function is invoked exclusively via HTTP POST from the Flutter client. The following subsections document the request body schema, response body schema, and behavioral contract for each function.

Function 1: consume-resource

Purpose: Records a student's consumption of a single content resource (Visual, Auditory, or Textual) for a given topic, awards the appropriate micro-dosing XP, applies the streak multiplier where applicable, and prevents duplicate XP accrual for repeat visits to the same resource.

Request Schema:

```
{
  "topic_id": "uuid (required)",
  "resource_type": "Visual | Auditory | Textual (required)",
  "is_topic_completion": "boolean (optional, default: false)",
  "timezone_offset": "number (optional, default: 0, unit: hours)"
}
```

Response Schema (HTTP 200):

```
{
  "resource_xp": 5,
  "completion_bonus": 15,
  "streak_multiplier": 1.1,
  "xp_awarded": 22
}
```

Behavioral Notes: If the student has previously consumed the same (topic_id, resource_type) combination, the function returns HTTP 200 with the message "Resource already consumed. No duplicate XP awarded." and performs no database write. The completion_bonus field is non-zero only when is_topic_completion is true, indicating that all resources within the topic have been consumed and the topic-level XP bonus is applicable. The timezone_offset parameter ensures streak date comparison is evaluated against the student's local calendar date rather than UTC server time.

Function 2: run-scoring-engine

Purpose: Fetches all five diagnostic_test_results rows for the authenticated student, forwards each unscored result to the Railway /score/diagnostic-result endpoint for psychometric feature computation and track assignment, and awards the 500 XP diagnostic milestone bonus upon confirmed completion of all five tests.

Request Schema:

```
{
  "force_rescore": "boolean (optional, default: false)"
}
```

Response Schema (HTTP 200):

```
json
{
  "ok": true,
  "all_scored": true,
  "milestone_awarded": true,
  "diagnostic_results": [
    {
      "id": "uuid",
      "test_number": 1,
      "computed_scores": { }
    }
  ]
}
```

Behavioral Notes: If a result row already carries a non-null computed_scores value and force_rescore is false, that row is skipped and marked skipped: true in the internal processing log no Railway call is made for it. The 500 XP milestone is awarded only when all_scored is true, and the function does not implement idempotency guards on the milestone award administrative use of force_rescore should therefore be undertaken with awareness of potential duplicate XP accrual.

Function 3: submit-module-attempt

Purpose: Receives a student's module assessment submission, grades the answers server-side against the canonical answer key, records the attempt in student_module_attempts, awards a flat +20 XP on passing (with streak multiplier applied), and enforces the mastery-based progression rule by conditioning module unlock on a score of 70% or above.

Request Schema:

```
{
  "assessment_id": "uuid (required)",
  "answers": {
    "question_uuid_1": "selected_answer_value",
    "question_uuid_2": "selected_answer_value"
  },
  "difficulty": "easy | mid | hard (required)"
}
```

Response Schema (HTTP 200):

```
{
  "score": 85,
  "passed": true,
  "correct_count": 17,
  "total_questions": 20,
  "difficulty": "mid",
  "base_xp": 20,
  "streak_multiplier": 1.1,
  "xp_awarded": 22
}
```

Behavioral Notes: The difficulty parameter is recorded for analytics purposes but does not alter the XP reward for module assessments, which is fixed at +20 XP irrespective of difficulty tier consistent with the non-tiered module assessment specification in the gamification architecture. Grading logic executes entirely within the Edge Function; no Railway call is made. If the assessment is inactive (is_active = false), HTTP 403 is returned prior to grading.

Function 4: submit-challenge-attempt

Purpose: Receives a student's weekly challenge submission, grades the answers server-side, enforces the single-attempt constraint, records the attempt in student_challenge_attempts, and awards proportional XP based on score percentage and selected difficulty tier consistent with the tiered challenge reward specification.

Request Schema:

```
{
  "challenge_id": "uuid (required)",
  "answers": {
    "question_uuid_1": "selected_answer_value"
  },
  "difficulty": "easy | mid | hard (required)",
  "timezone_offset": "number (optional, default: 0)"
}
```

Response Schema (HTTP 200):

```
{
  "score": 80,
  "correct_count": 8,
  "total_questions": 10,
  "base_xp": 80,
  "streak_multiplier": 1.1,
  "xp_awarded": 88
}
```

Behavioral Notes: Unlike module assessments, challenge XP is proportional: xp_awarded = floor(tier_xp_reward × (score / 100) × streak_multiplier). A student selecting the Hard tier (100 XP potential) who scores 80% receives 80 base XP before streak application. If the student has already submitted an attempt for the given challenge_id, HTTP 409 (Conflict) is returned and no database write is performed. The timezone_offset parameter governs the date-window eligibility check against available_from and available_until on the weekly_challenges record.

Function 5: mitchy-chat

Purpose: Receives the student's chat message along with optional contextual identifiers, verifies the JWT, and relays the authenticated payload to the Railway Mitchy agent via the private service key relay. Returns the AI mentor's structured response including the reply text, inferred learning state, sentiment score, and suggested pedagogical action. Awards +20 XP when the Mitchy agent returns a non-null suggested_action.

Request Schema:

```
{
  "message": "string (required)",
  "topic_id": "uuid (optional)",
  "module_id": "uuid (optional)",
  "screen_context": "string (optional)"
}
```

Response Schema (HTTP 200):

```
{
  "ok": true,
  "response_text": "string",
  "message": {
    "role": "assistant",
    "content": "string",
    "session_id": "uuid | null"
  },
  "mitchy": {
    "response_text": "string",
    "learning_state": "string",
    "sentiment_score": -0.8,
    "cognitive_load": 0.3,
    "suggested_action": "rescue_explanation | none | ...",
    "recommended_format": "Visual | Auditory | Textual"
  }}
```

Behavioral Notes: In the event of a Railway failure encompassing Gemini API quota exhaustion, timeout, or malformed model output the Edge Function returns a guaranteed non-empty fallback response with "source": "edge_fallback" in the metadata field, ensuring the student always receives a reply regardless of upstream availability. The recommended_format field in the Mitchy response may be used by the Flutter client to dynamically suggest a content modality switch if the AI agent infers that the current format is misaligned with the student's detected cognitive state.

4.6.4 Error Handling and HTTP Status Codes

A consistent error handling strategy is enforced across all five Edge Functions. Every error response carries a JSON body with an error field providing a human-readable description of the failure condition, enabling the Flutter client to surface contextually appropriate feedback without parsing HTTP status codes alone.

Table 4.3: Standardized HTTP Status Code Responses

Status Code	Condition	Example Trigger
200 OK	Request processed successfully	Valid submission, duplicate resource consumption
400 Bad Request	Missing or invalid request body field	topic_id absent, invalid difficulty value
401 Unauthorized	Missing or invalid JWT	Expired session token, missing Authorization header
403 Forbidden	Access denied by RLS or business rule	Inactive assessment, outside challenge date window
404 Not Found	Referenced entity does not exist	Non-existent challenge_id or assessment_id
405 Method Not Allowed	HTTP method other than POST used	GET request to Edge Function endpoint
409 Conflict	Duplicate submission detected	Second attempt on a single-attempt challenge
500 Internal Server Error	Unhandled server-side exception	Database write failure, Railway unreachable

It is noted that the mitchy-chat function diverges from this convention on the error path: rather than returning HTTP 500 on Railway or Gemini failures, it returns HTTP 200 with a structured fallback payload. This design decision prioritizes conversational continuity for the learner over strict HTTP semantics a deliberate trade-off that ensures the AI mentorship interface never presents the student with an unhandled error state during a potentially emotionally sensitive interaction.

4.7 Algorithm and Business Logic Design

The adaptive and intelligent behaviors of the LearNova platform are governed by four distinct algorithmic subsystems, each operationalizing a different dimension of the platform's personalization mandate. This section presents the formal computational logic underpinning each subsystem through a combination of pseudocode and flowchart representations, providing the level of algorithmic transparency necessary to validate the system's deterministic and data-driven decision-making processes. The four subsystems examined are: the **Psychometric Track Assignment Algorithm**, which maps a student's diagnostic responses to a specialization track; the **XP and Streak Reward Calculation**, which governs gamification incentive distribution; the **Bayesian Learning Style Inference Engine**, which refines modality preferences through observed engagement behavior; and the **ML Analytics Pipeline**, which computes daily risk indicators for the administrative monitoring layer.

4.7.1 Psychometric Track Assignment Algorithm

The **Psychometric Track Assignment Algorithm** constitutes the computational core of the Smart Compatibility Engine, translating the raw response vectors from the five-point diagnostic battery into a deterministic specialization assignment of Data Analytics (DA), Data Engineering (DE), or Data Science (DS). The algorithm is structured as a six-step pipeline, progressing from raw answer matching through feature aggregation, normalization, base scoring, modifier computation, and final track resolution. This pipeline architecture was designed to ensure that no single diagnostic instrument dominates the outcome a methodological safeguard against the psychometric cold-start problem identified in collaborative filtering literature.

Step A Answer Matching. Each raw answer submitted by the student carries a question_key and a selected_index. The algorithm maps this pair against the answer-key seed to retrieve the corresponding feature contribution row, which specifies the target feature (e.g., openness, logical_reasoning, visual), the numeric score value, and, for IQ questions, the correctness boolean.

Step B Feature Aggregation. Raw scores are aggregated per feature according to the scoring method appropriate to each diagnostic instrument. For the IPIP personality inventory, the feature score is the arithmetic mean of Likert values across all questions assigned to that trait, with reverse-scored items transformed as score = 6 − selected_value prior to averaging. For soft skills, each selected option contributes its option score to one or more skill features, and the feature score is the mean of all contributing option scores. For VARK, each selected option increments a count for the corresponding modality dimension. For O*NET career interest, selected values 1–5 are averaged within each RIASEC category. For IQ, correct responses contribute 1 and incorrect responses contribute 0, with section totals summed separately for logical and abstract reasoning.

Step C Normalization. Each aggregated feature score is normalized to a bounded [0, 1] interval using the predefined range for that feature, as expressed in the formula below. Missing features default to the minimum range value to prevent null propagation through subsequent steps.

```
normalized_feature = clamp(
    (raw_feature − min_range) / (max_range − min_range),
    0.0,
    1.0)
```

Step D Base Score Computation. Three O*NET RIASEC categories serve as direct anchors for the base track scores, reflecting the vocational alignment between each specialization and its corresponding interest profile:

base_DA = normalized(artistic)

base_DE = normalized(realistic)

base_DS = normalized(investigative)

Step E Modifier Score Computation. Each normalized feature contributes a weighted additive modifier to each track score. The raw modifier weights are normalized at runtime such that each track's weight column sums to 1.0, ensuring comparability across tracks regardless of the number of features:

```
modifier_track = Σ (normalized_feature_i × normalized_weight_track_i)
    for all features i in the modifier weight table
```

Step F Final Score and Track Assignment. The final score for each track is a weighted combination of its base score and modifier score, where the modifier component carries the dominant influence reflecting the richness of the full feature vector relative to the single O*NET anchor:

```
final_score_track = (0.30 × base_track) + (0.70 × modifier_track)
assigned_track = argmax(final_score_DA, final_score_DE, final_score_DS)
```

The track yielding the highest final_score is written to student_profiles.assigned_track. In the event of a tie, the system defaults to DA the broadest specialization — as a conservative assignment pending administrative review. The complete pseudocode for the full pipeline is presented in Algorithm 4.1 below.

Algorithm 4.1: Psychometric Track Assignment Algorithm

```
INPUT:  raw_answers[]  ← list of {question_key, selected_index} from student
OUTPUT: assigned_track ← one of {DA, DE, DS}
        assigned_style  ← one of {Visual, Auditory, Textual}
// Step A — Answer Matching
FOR each answer IN raw_answers:
    row ← lookup(answer_key_seed, answer.question_key, answer.selected_index)
    append row TO matched_rows
```

```
// Step B — Feature Aggregation
FOR each feature IN feature_list:
    contributing_rows ← filter(matched_rows, feature)
    IF exam_type == IPIP:
        feature_scores[feature] ← mean(scored_values WITH reverse_scoring)
    ELSE IF exam_type == SOFT_SKILLS:
        feature_scores[feature] ← mean(option_scores)
    ELSE IF exam_type == VARK:
        feature_scores[feature] ← count(matching_options)
    ELSE IF exam_type == ONET:
        feature_scores[feature] ← mean(selected_values IN category)
    ELSE IF exam_type == IQ:
        feature_scores[feature] ← sum(is_correct FOR rows IN section)
// Step C — Normalization
FOR each feature IN feature_scores:
    normalized[feature] ← clamp(
        (feature_scores[feature] − MIN_RANGE[feature]) /
        (MAX_RANGE[feature] − MIN_RANGE[feature]),
        0.0, 1.0
    )
// Step D — Base Scores
base[DA] ← normalized[artistic]
base[DE] ← normalized[realistic]
base[DS] ← normalized[investigative]
// Step E — Modifier Scores (weights normalized at runtime)
FOR each track IN {DA, DE, DS}:
    modifier[track] ← Σ (normalized[f] × norm_weight[track][f])
        FOR all features f
// Step F — Final Scores and Assignment
FOR each track IN {DA, DE, DS}:
```

```
    final[track] ← (0.30 × base[track]) + (0.70 × modifier[track])
assigned_track ← track WITH maximum final[track]
// Learning Style Assignment
vark_counts ← {visual, auditory, read_write, kinesthetic}
dominant ← argmax(vark_counts)
IF dominant IN {read_write, kinesthetic}:
    assigned_style ← Textual    // DB constraint: three values only
ELSE:
    assigned_style ← capitalize(dominant)
WRITE assigned_track → student_profiles.assigned_track
WRITE assigned_style → student_profiles.learning_style
```

The full modifier weight table encompassing raw DA, DE, and DS weights for all 26 features alongside their normalization range boundaries is referenced in Section 4.5.2 and the diagnostic track assignment specification, which serves as the authoritative source for calibration parameter values.

4.7.2 XP and Streak Reward Calculation

The **XP and Streak Reward Calculation** algorithm governs the distribution of experiential points across all student actions on the platform, operationalizing the tiered reward matrix defined in the gamification specification. The algorithm is invoked by three distinct Edge Functions consume-resource, submit-module-attempt, and submit-challenge-attempt each applying a variant of the same underlying computation logic whilst differing in their base XP determination rules.

The flowchart in Figure 4.12 presents the generalized computation pipeline. The algorithm proceeds in three sequential phases: base XP determination, streak multiplier evaluation, and final award computation and ledger write.

Phase 1 Base XP Determination. The base XP value is determined by the action type and, where applicable, the difficulty tier selected by the student and the accuracy of the submission:

```
IF action == CONTENT_CONSUMPTION:
    base_xp ← 5
    IF is_topic_completion:
        base_xp ← base_xp + 15
IF action == MODULE_ASSESSMENT AND passed:
    base_xp ← 20        // flat, non-tiered
```

```
IF action == LEVEL_ASSESSMENT AND passed:
    base_xp ← xp_reward_{difficulty}    // tiered: 20 | 50 | 100
IF action == WEEKLY_CHALLENGE:
    tier_xp ← xp_reward_{difficulty}    // tiered: 20 | 50 | 100
    base_xp ← floor(tier_xp × (score / 100))    // proportional to accuracy
IF action == DIAGNOSTIC_COMPLETION:
    base_xp ← 500    // one-time milestone, non-repeatable
IF action == MITCHY_RECOMMENDATION:
    base_xp ← 20    // flat, awarded when suggested_action ≠ null
```

Phase 2 Streak Multiplier Evaluation. The streak multiplier is evaluated by querying the user_streaks table and comparing last_activity_date against the student's local calendar date, derived by applying the timezone_offset parameter to the UTC server timestamp:

```
local_date ← UTC_now + (timezone_offset × 3600 seconds)
diff_days ← floor((local_date − last_activity_date) / 86400 seconds)
IF current_streak_days > 0 AND diff_days ≤ 1:
    multiplier ← 1.1
ELSE:
    multiplier ← 1.0
```

Phase 3 Final Award and Ledger Write. The final XP awarded is computed and written to the student profile via the secure increment_xp() RPC:

```
final_xp ← round(base_xp × multiplier)
CALL increment_xp(user_id, final_xp)
UPSERT user_streaks SET last_activity_date ← local_date
```

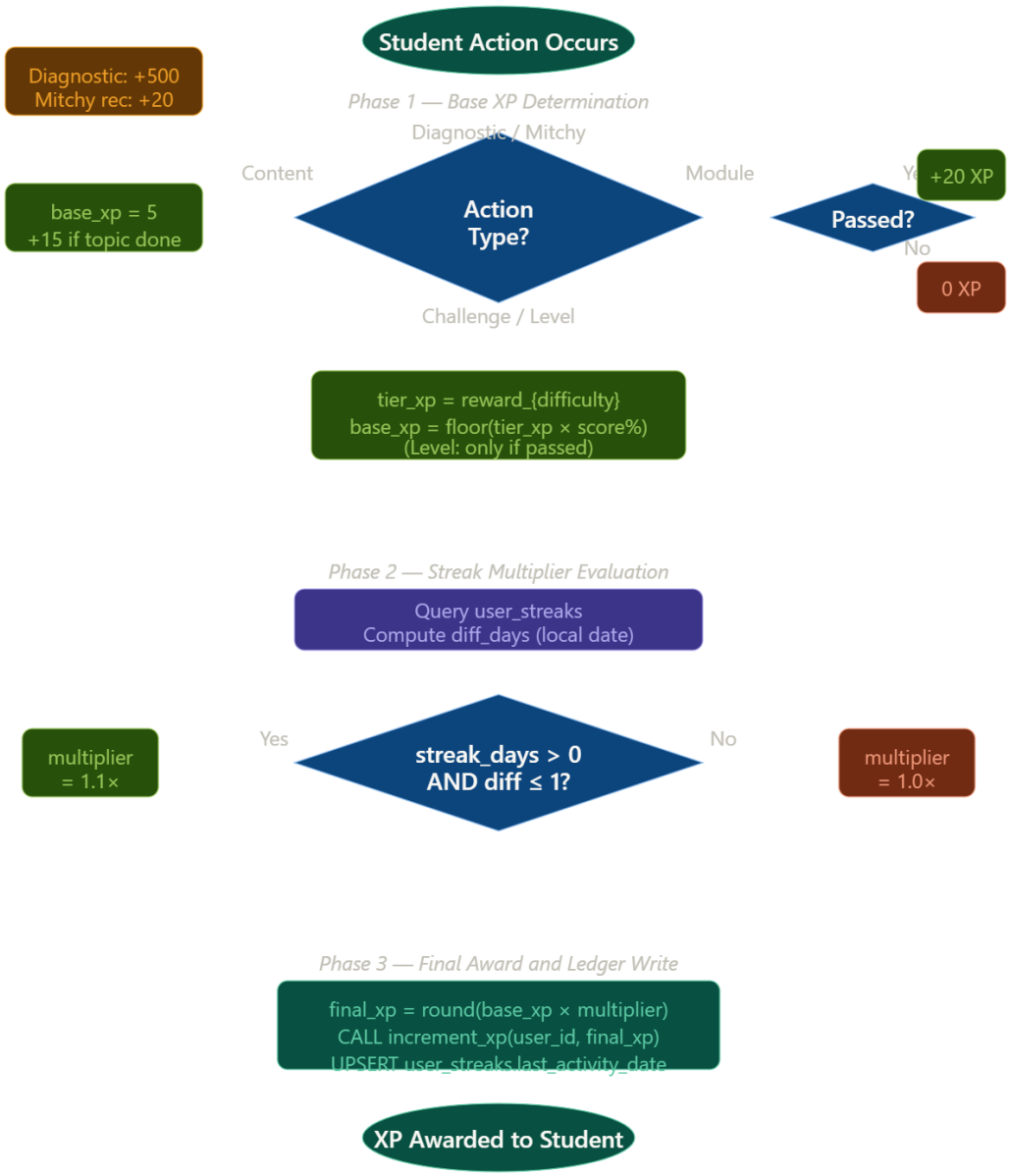


Figure 4.12: XP and streak reward calculation flowchart.
The 1.1× multiplier applies only when an active streak is confirmed.
Challenge XP is proportional to accuracy; module XP is flat.

Figure 4.12: XP and streak reward calculation flowchart, illustrating the three-phase computation pipeline: base XP determination by action type, streak multiplier evaluation against local calendar date, and final award write via the increment_xp() RPC. The 1.1× multiplier is applied only when an active streak is confirmed.

4.7.3 Bayesian Learning Style Inference Engine

The **Bayesian Learning Style Inference Engine** operationalizes a continuous, evidence-driven refinement of each student's dominant content modality preference, moving beyond the static VARK assignment established during onboarding toward a dynamic posterior that evolves with observed engagement behavior. This mechanism

addresses a recognized limitation of static learning style classification: a student's self-reported preferences at onboarding may diverge from their demonstrated engagement patterns once they encounter actual instructional content in context.

The engine applies a Bayesian update rule to three alpha parameters bayesian_alpha_visual, bayesian_alpha_auditory, and bayesian_alpha_textual stored in the student_profiles table. These parameters represent the posterior strength of evidence for each modality and are updated following qualifying engagement sessions, identified by the bayesian_eligible flag in the content_engagement_logs table. A session is designated eligible when its engagement_score exceeds a quality threshold, ensuring that incidental or incomplete content interactions do not corrupt the posterior.

The update logic proceeds as follows:

```
INPUT:  current_alphas = {alpha_visual, alpha_auditory, alpha_textual}
        engagement_session = {format_type, engagement_score, bayesian_eligible}
IF engagement_session.bayesian_eligible == FALSE:
    RETURN current_alphas unchanged
// Increment the alpha corresponding to the engaged modality
alpha_engaged ← current_alphas[engagement_session.format_type]
alpha_engaged ← alpha_engaged + engagement_score_normalized
// Derive posterior learning style from alpha argmax
dominant_style ← argmax(alpha_visual, alpha_auditory, alpha_textual)
// Update student profile if dominant style has shifted
IF dominant_style ≠ current_profiles.learning_style:
    UPDATE student_profiles SET learning_style ← dominant_style
    UPDATE student_profiles SET bayesian_alpha_{style} ← updated_alpha
RETURN updated_alphas
```

This incremental update mechanism ensures that the learning style label driving content delivery progressively converges toward the modality with which the student demonstrates the highest quality engagement, rather than remaining fixed at the initial VARK self-report. The engine is invoked by the analytics/bayesian_engine3.py module within the Railway daily cron pipeline, processing the previous day's eligible engagement logs for all active students.

4.7.4 ML Analytics Pipeline

The **ML Analytics Pipeline** executes as a scheduled daily cron process on the Railway backend, computing three risk indicator metrics per student that populate the ml_daily_metrics and ml_topic_daily_metrics tables consumed by the administrative monitoring dashboard. The pipeline operates on the previous day's behavioral data and

is architected through an adapter layer that decouples raw Supabase row shapes from the input contracts of each analytics module, preserving schema stability as the underlying data model evolves.

Three core metrics are computed for each student on each execution cycle.

Concept Decay Score. This metric quantifies the degree to which a student's demonstrated competence on previously completed topics may have degraded due to elapsed time without re-engagement. The computation applies an exponential decay function to the time elapsed since each topic's completed_at timestamp, weighted by the student's historical score on the associated module assessment:

```
decay_score_topic = initial_retention × e^(−λ × days_elapsed)

concept_decay_score = mean(decay_score_topic FOR all completed topics)
```

where λ is a configurable decay constant calibrated to the forgetting curve literature, and initial_retention is derived from the student's assessment performance on that topic.

Engagement Velocity. This metric captures the rate of change in the student's daily engagement intensity, functioning as an early indicator of momentum loss before it manifests as dropout behavior. It is computed as the first derivative of a rolling engagement score series:

```
engagement_ema_t = α × engagement_score_t + (1 − α) × engagement_ema_{t−1}

engagement_velocity = engagement_ema_t − engagement_ema_{t−1}
```

A negative velocity sustained across multiple consecutive days constitutes a primary trigger condition for elevated risk score computation.

Topic Struggle Index. This metric aggregates per-topic difficulty signals encompassing assessment attempt counts, score trajectories, and time-on-task relative to estimated module duration into a composite indicator of learning blockage:

```
struggle_index_topic = weighted_sum(
    attempts_count_normalized × w1,
    (1 − highest_score_normalized) × w2,
    time_overrun_ratio × w3)

topic_struggle_index = mean(struggle_index_topic FOR topics WITH attempts > 0)
```

The three metrics are upserted into ml_daily_metrics with a unique constraint on (user_id, metric_date), ensuring idempotent pipeline execution. When the composite risk profile of a student derived from a weighted combination of these three indicators exceeds a configurable threshold, the pipeline triggers the creation of a risk_scores record with alert_triggered = true, initiating the administrative intervention workflow documented in Section 4.3.3.

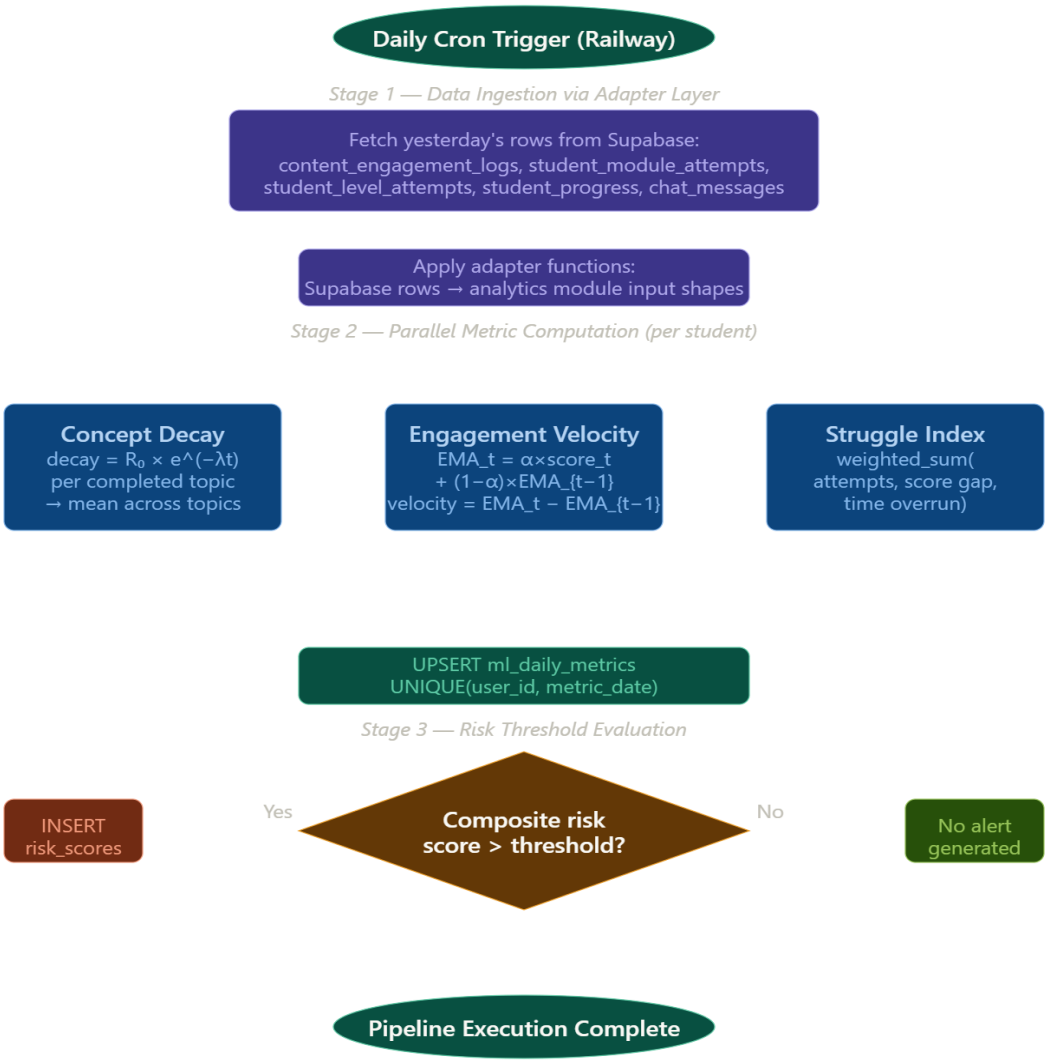


Figure 4.13: ML Analytics Pipeline daily execution flow.
Three metrics are computed in parallel per student and upserted to ml_daily_metrics.
Composite risk scores exceeding the threshold trigger a risk_scores INSERT,
initiating the administrative intervention workflow.

Figure 4.13: ML Analytics Pipeline daily execution flow, illustrating the three-stage process of data ingestion via the adapter layer, parallel computation of concept decay, engagement velocity, and topic struggle index metrics, and conditional risk alert generation upon threshold breach.

4.8 Security Design

Security within the LearNova platform is not treated as a peripheral concern appended to functional design, but rather as a foundational architectural constraint that permeates every layer of the system stack from the authentication handshake initiating each client session to the Row-Level Security policies governing individual database row access. The security architecture is organized around four interdependent domains: identity and access management, data protection, API boundary security, and privacy governance. Collectively, these mechanisms operationalize a defense-in-depth posture wherein no single point of compromise is sufficient to expose protected learner data or circumvent system integrity controls.

4.8.1 Identity and Access Management

The authentication lifecycle is governed by Supabase's GoTrue identity gateway, which issues signed **JSON Web Tokens (JWTs)** upon successful credential validation. These tokens encode the authenticated user's id and role claims either student or admin and are cryptographically signed using a server-held secret inaccessible to the client application. Every subsequent request from the Flutter client carries this token in the Authorization: Bearer header, and every Supabase service PostgREST, Edge Functions, and the Realtime API validates the token's signature and expiry prior to processing any request payload. Requests bearing absent, malformed, or expired tokens are uniformly rejected with HTTP 401 before any database interaction occurs, ensuring that unauthenticated traffic cannot reach any protected data surface.

Role-Based Access Control (RBAC) is enforced at two complementary levels. At the application level, the Flutter client renders role-differentiated interface modules the Learner Interface Module for students and the Administrative Interface Module for admins based on the role claim decoded from the session JWT. At the database level, the users.role column, constrained by a CHECK to 'student' or 'admin', drives the evaluation of two PostgreSQL helper functions is_admin() and current_user_is_admin() which serve as the gatekeeping predicates for all Row-Level Security policy definitions across the schema. This dual-layer enforcement ensures that even if a client-side routing control were circumvented, the underlying data layer would independently deny unauthorized access through cryptographically verified policy evaluation.

The architectural separation between Supabase's managed auth.users schema and the application-level public.users table constitutes a further access control measure, operationalizing the **Principle of Least Privilege**. The auth.users table containing encrypted password hashes and JWT issuance metadata is entirely isolated from the PostgREST public API surface. A vulnerability in the application layer therefore cannot propagate to core authentication credentials, as the two schemas are governed by independent access boundaries [84].

4.8.2 Row-Level Security Architecture

Row-Level Security (RLS) is enabled on every table within the LearNova Supabase schema, constituting the primary data isolation mechanism for multi-tenant student data. RLS policies are evaluated server-side by the PostgreSQL engine on every query, irrespective of whether the query originates from the Flutter client, an Edge Function, or

a direct PostgREST call. This evaluation occurs after authentication but before data retrieval, ensuring that the access control logic is enforced at the closest possible proximity to the data itself a design that eliminates the class of vulnerabilities arising from application-layer access control bypasses.

The RLS architecture follows two principal access patterns, summarized in Table 4.4. Students are granted read and write access exclusively to rows where the user_id column matches auth.uid() the cryptographically verified identifier extracted from the inbound JWT. Content tables expose only rows where is_active = true, preventing students from accessing draft or deactivated curriculum content even if the relevant id values were known. Admins are granted unrestricted read access across all tables, with write access scoped to content management and their own intervention records notably, the intervention_logs table constrains admin write operations to rows where admin_id = auth.uid(), ensuring that one administrator cannot modify another's intervention records.

Table 4.4: RLS Access Pattern Summary

Data Category	Student Read	Student Write	Admin Read	Admin Write
Own profile and user data	Own row only	Own row only	All rows	All rows
Course content and assessments	Active rows only	None	All rows	All rows
Diagnostic questions	All rows (onboarding)	None	All rows	All rows
Diagnostic results	Own rows only	Own rows (insert)	All rows	All rows
Progress and attempt records	Own rows only	Own rows only	All rows	None
Gamification state	Own rows only	Own rows only	All rows	None
Leaderboard snapshots	All rows	None	All rows	None
Chat sessions and messages	Own rows only	Own rows only	All rows	None
Risk scores and alerts	None	None	All rows	Update only
Intervention logs	None	None	Own records only	Own records only
ML metrics	Own topic metrics	None	All rows	All rows
Notifications	Own rows only	None	All rows	Insert only

A particularly critical RLS configuration governs the risk_scores table, which stores sensitive mental health and academic risk indicators. INSERT operations on this table are permitted exclusively via the Railway service using the Supabase service-role key a privileged credential held server-side and never transmitted to the Flutter client whilst SELECT operations are restricted to users holding an admin role claim. This policy architecture guarantees that risk and wellbeing data are never exposed through the public API surface, regardless of client-side state or JWT content [84].

4.8.3 API Boundary Security and Service Key Isolation

The two-tier authentication architecture documented in Section 4.6.1 constitutes the primary API boundary security mechanism. Its central security property that the Flutter client never communicates directly with the Railway backend and never holds Railway credentials is enforced structurally rather than by policy alone. Because the Railway service URL and API key are stored exclusively as Supabase Edge Function environment secrets, there exists no mechanism by which a client-side actor could extract these values even with full access to the application binary.

The service-role key used by Railway to perform privileged Supabase write operations such as inserting risk_scores records and calling the increment_xp() RPC is similarly isolated. It resides only within the Railway runtime environment and is never returned to the client in any API response payload. The Edge Functions themselves use the Supabase anon key for user authentication and the service-role key only for administrative database writes executed after JWT verification, maintaining a strict separation between user-scoped and privileged operations within the same function execution context.

Input validation is enforced at the Edge Function layer prior to any database interaction. Each function validates the presence and type of required request body fields rejecting malformed payloads with HTTP 400 and validates business-rule constraints such as active assessment status, challenge date-window eligibility, and single-attempt enforcement before executing any write operations. This validation layer ensures that the database is not reached by structurally invalid or business-rule-violating requests, reducing the attack surface available to malformed or adversarial inputs.

4.8.4 Data Protection

Data protection within LearNova is governed at the infrastructure level through Supabase's managed PostgreSQL deployment, which enforces **AES-256 encryption at rest** for all data stored within the database encompassing psychometric feature vectors in diagnostic_test_results.computed_scores, personality dimension scores in student_profiles, sentiment scores in chat_messages, and risk indicator payloads in risk_scores.feature_snapshot [85]. All data in transit between the Flutter client and the Supabase platform is protected by **TLS 1.2 or higher**, enforced by the Supabase API gateway, ensuring that JWT tokens, assessment submissions, and chat message payloads are encrypted across the network boundary. Similarly, all communication between the Supabase Edge Functions and the Railway backend traverses HTTPS, maintaining end-to-end transport security for the server-to-server relay channel.

Supabase Storage utilized for the calibration-media, content-assets, and student-submissions buckets is configured with **private access controls**, meaning that asset URLs are not publicly accessible and require authenticated requests for retrieval. This configuration is particularly consequential for the student-submissions bucket, which stores externally submitted diagnostic proof screenshots that may contain personally identifiable information. Access to these assets is governed by the same JWT-based authentication as the database layer, ensuring that portfolio artifacts and submission evidence cannot be accessed by unauthenticated third parties.

4.8.5 Privacy Governance

The LearNova platform processes a category of learner data encompassing psychometric personality scores, cognitive aptitude measurements, sentiment history, and mental health risk indicators that carries heightened sensitivity relative to conventional educational platforms. The privacy governance framework is accordingly designed to restrict data visibility to the minimum audience necessary for each operational purpose, a principle operationalized through the RLS architecture detailed in Section 4.8.2.

The system's non-functional requirements specify compliance with core **General Data Protection Regulation (GDPR)** principles, including the right to data minimization and purpose limitation. The diagnostic feature vector stored in computed_scores is used exclusively for track assignment and adaptive content calibration — it is not surfaced to the student in raw form, nor is it transmitted to third-party services beyond the Railway scoring pipeline. Risk scores and intervention logs are accessible only to administrative users with an explicit admin role claim, and the intervention_logs table's RLS policy additionally scopes write access to the responsible administrator's own records, preventing unauthorized modification of intervention histories.

Regarding the **Right to Erasure** (GDPR Article 17), the current implementation does not include a dedicated account deletion Edge Function. This requirement is formally specified as NFR-06 in Section 3.2.2 and is acknowledged as a pending implementation priority. In its current form, account deletion must be performed manually by an administrator through the Supabase dashboard, which permanently removes the auth.users record and cascades deletion to all related application data through foreign key constraints. The development of an automated, student-initiated deletion workflow constitutes a prioritized item in the platform's security roadmap, as detailed in Section 5.2.

The administrative dashboard's access to student_sentiment_history, risk_scores, and chat_messages is governed by the principle of **role-scoped necessity**: administrators can access these records to fulfil their duty-of-care obligations toward at-risk learners, but cannot access the raw content of a student's chat history unless a high-risk alert has been triggered for that student a business-rule constraint that, whilst currently enforced at the application layer rather than through a dedicated RLS predicate, is identified as a candidate for a future RLS policy refinement that would further restrict chat history access to flagged sessions only.

4.9 User Interface Design

The User Interface (UI) of the LearNova mobile application is architected not merely as a graphical presentation layer, but as a context-aware cognitive mediator a purposefully engineered surface designed to minimize friction between the system's deterministic adaptive logic and the learner's moment-to-moment cognitive experience. Every design decision within this layer is grounded in an explicit theoretical framework, operationalizing principles from Sweller's **Cognitive Load Theory** [1], Mayer's **Multimedia Learning** principles [2], and established interaction design laws including Fitts's Law and Hick's Law, ensuring that the interface functions as a transparent scaffold rather than a competing source of attentional demand.

The platform is implemented as a Flutter mobile application targeting Android (API Level 28 and above) and iOS (version 15 and above), adhering to the Material Design 3 specification [86] whilst diverging from its default chromatic system in favor of a bespoke dark-mode design language calibrated specifically for sustained deep-work engagement. All screens presented in this section reflect the implemented production interface.

4.9.1 Design System & Visual Ergonomics

The chromatic and typographic foundation of the LearNova design system is engineered to support prolonged, cognitively demanding learning sessions whilst mitigating the visual fatigue commonly associated with extended screen engagement in digital educational environments.

A **High-Contrast Dark Mode Paradigm** governs the visual language of the entire application. The primary canvas color **Void Navy** (#01172E) functions as a low-luminance semantic null space, reducing photonic intensity to minimize circadian rhythm disruption during extended study sessions and allowing foreground instructional content to command the learner's full attentional field without competing background elements. Visual depth and spatial hierarchy are established through a **Luminance Elevation Strategy**, wherein surface layers of progressively increasing luminance **Deep Azure** (#03478E) and **Resonant Blue** (#034A94) distinguish interactive containers, card surfaces, and navigational groupings from the base canvas without introducing chromatic dissonance.

Primary Call-to-Action elements and the platform's signature iconographic motif a four-pointed star representing the Mitchy AI agent are rendered in **Electric Cyan** (#72F7D7), a complementary hue selected for its pre-attentive pop-out properties. As established by visual perception research, high-spectral-contrast elements are identified by the visual system prior to conscious processing, enabling the learner to locate interactive affordances instantaneously without deliberate visual search a property of particular ergonomic value in the assessment and navigation contexts where decision latency compounds cognitive load [7]. The geometric sans-serif typography, configured with an increased x-height relative to the canvas height, maximizes legibility against the dark background whilst reducing visual crowding for neurodiverse learners, in compliance with WCAG 2.1 AAA contrast ratio standards [86].

4.9.2 Onboarding and Authentication Interface

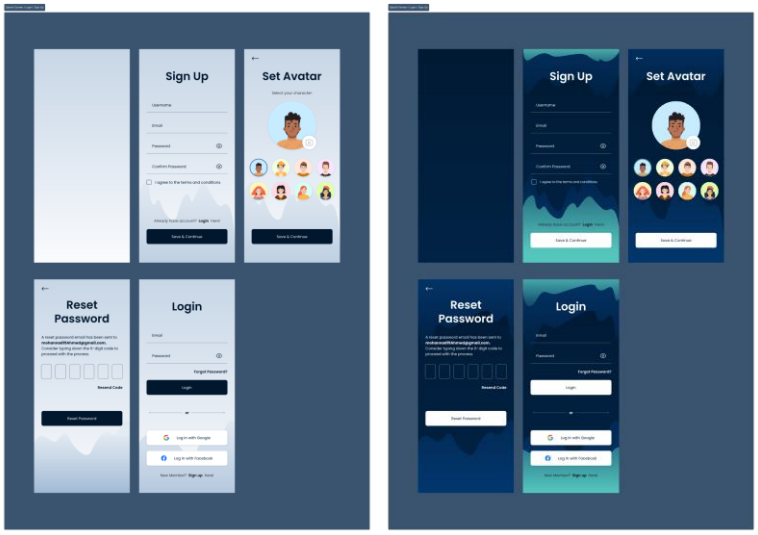


Figure 4.14: Authentication interface Sign Up screen (left) and Sign In screen (right), illustrating the Void Navy canvas, Electric Cyan interactive affordances, and the mountain topography decorative motif that establishes spatial depth without introducing extraneous cognitive load.

The authentication interface, illustrated in Figure 4.14, presents the learner's initial point of contact with the LearNova ecosystem. The screen layout adheres to a strict vertical information hierarchy: the platform logotype occupies the upper third of the viewport, establishing brand identity prior to credential entry; the input field cluster occupies the central focus zone; and the primary submission control is positioned within the thumb's natural arc at the lower quadrant of the screen, an arrangement directly informed by Fitts's Law optimization for single-handed mobile interaction [80]. The decorative mountain topography rendered at reduced opacity in the background introduces spatial depth without contributing semantic information, thereby preserving the learner's full attentional capacity for the authentication task at hand an implementation of Mayer's Coherence Principle, which posits that learning and task performance are enhanced when extraneous material is eliminated from the visual field [2].

4. 9.3 Calibration and Diagnostic Interface

The diagnostic calibration phase encompassing the five-test psychometric battery administered during onboarding presents the most significant UX challenge within the LearNova interface: sustaining student engagement and completion motivation across a multi-stage evaluation that may require substantial time investment. The interface design addresses this challenge through three complementary mechanisms: progressive disclosure, bifurcated execution path presentation, and immediate reward signaling upon test completion.

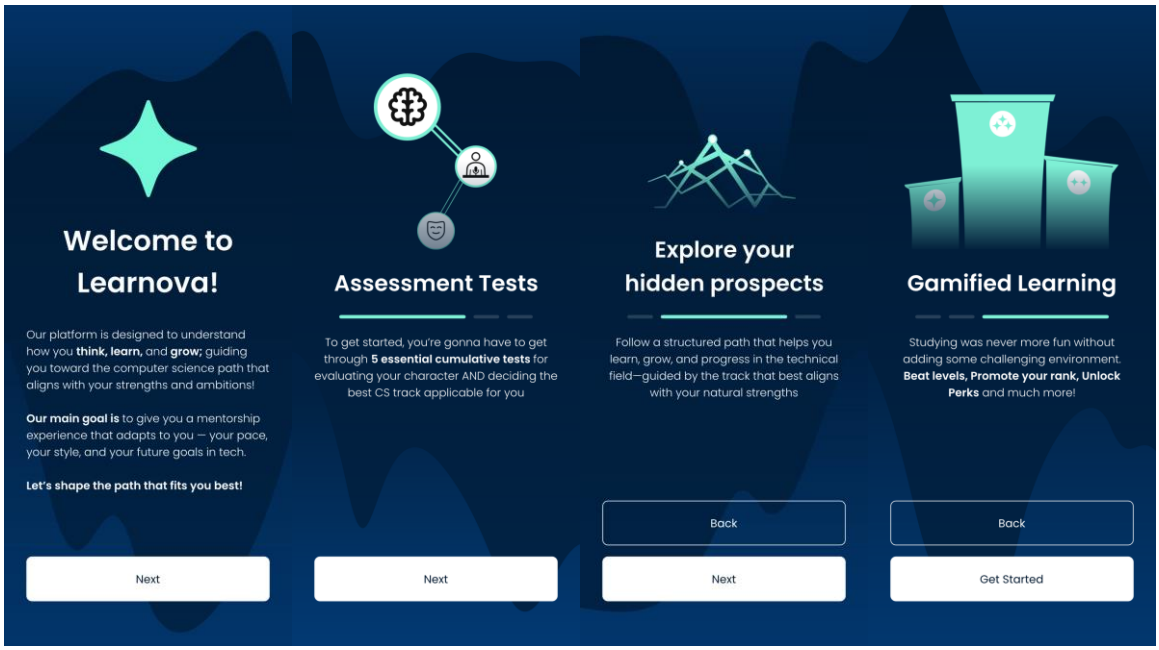


Figure 4.15: Onboarding splash sequence illustrating the four-screen value proposition flow Welcome, Assessment Tests, Explore Your Hidden Prospects, and Gamified Learning followed by the Tests Map diagnostic roadmap and the Testing Format bifurcation modal (Within App / On Browser).

The onboarding splash sequence, presented in Figure 4.15, guides the student through four value proposition screens prior to diagnostic commencement: a welcome screen, an assessment overview, a prospects exploration teaser, and a gamified learning preview. This sequential narrative structure, rather than presenting the full platform scope simultaneously, implements **Progressive Disclosure** the principle that cognitive overwhelm is mitigated by revealing system complexity incrementally, calibrated to the user's demonstrated engagement rather than front-loaded at first contact. Each screen advances only upon explicit user confirmation, ensuring that the student enters the diagnostic phase having received sufficient motivational context to sustain completion intent.

The **Tests Map** screen the diagnostic roadmap visible within the same figure presents the five assessment nodes as a vertically arranged stepper component. Completed nodes are visually differentiated from pending nodes through state-change color transitions, exploiting the **Goal-Gradient Effect**: empirical research in motivational psychology demonstrates that perceived proximity to a goal produces a measurable acceleration in completion behavior, and the spatial visualization of progress nodes makes goal proximity salient and continuously updated [80]. The **Testing Format** modal restricts the execution path decision to precisely two options Within App or On Browser a deliberate application of Hick's Law, wherein limiting the choice set to a binary decision minimizes decision latency and reduces the cognitive cost of path selection [80].

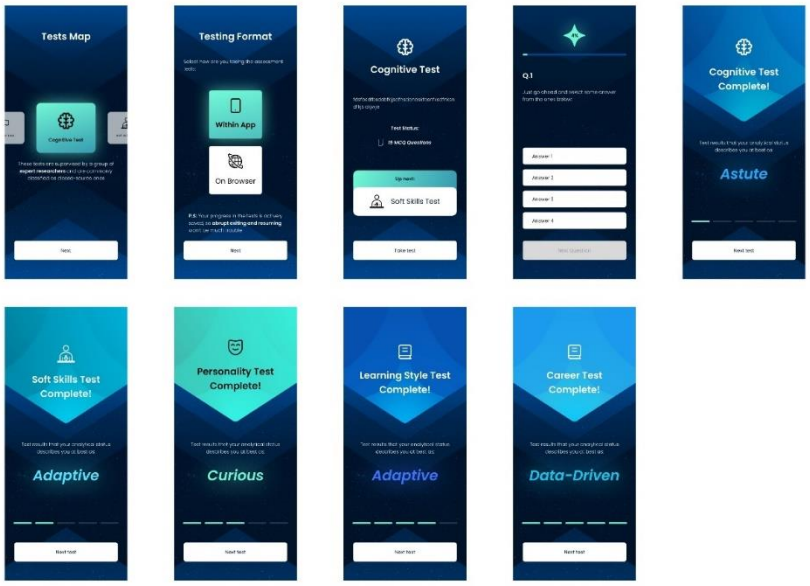


Figure 4.16: Diagnostic test execution sequence for the Cognitive Test showing the test entry screen with stepper indicating next test (Soft Skills), the question answering interface (Q.1 with four answer options), and the test completion screen displaying the derived personality attribute "Astute" in Electric Cyan italic typography.

Figure 4.16 illustrates the test execution sequence for the Cognitive Assessment, representative of the interface pattern applied uniformly across all five diagnostic instruments. The question interface presents a single question per screen accompanied by a constrained set of answer options rendered as selectable card containers, eliminating the need for free-text input and reducing the motor and cognitive demands of response selection to a single tap interaction. Upon completion of each individual test, the interface transitions to a full-screen completion state a design moment of significant psychological consequence. Rather than presenting a neutral confirmation, the completion screen reveals a derived personality attribute rendered in large-format Electric Cyan italic typography against the Void Navy canvas that communicates the system's first interpretive inference about the student's cognitive profile. Attributes revealed across the five tests include "Astute" (Cognitive), "Adaptive" (Soft Skills and Learning Style), "Curious" (Personality), and "Data-Driven" (Career Interest). This design decision operationalizes **Algorithmic Transparency**: by surfacing a human-readable interpretation of the system's computational output at each diagnostic milestone, the interface builds learner trust in the recommendation engine prior to the final track assignment, addressing the "black box" critique commonly directed at algorithmic guidance systems [7, 25].

4.9.4 Track Recommendation and Diploma Interface

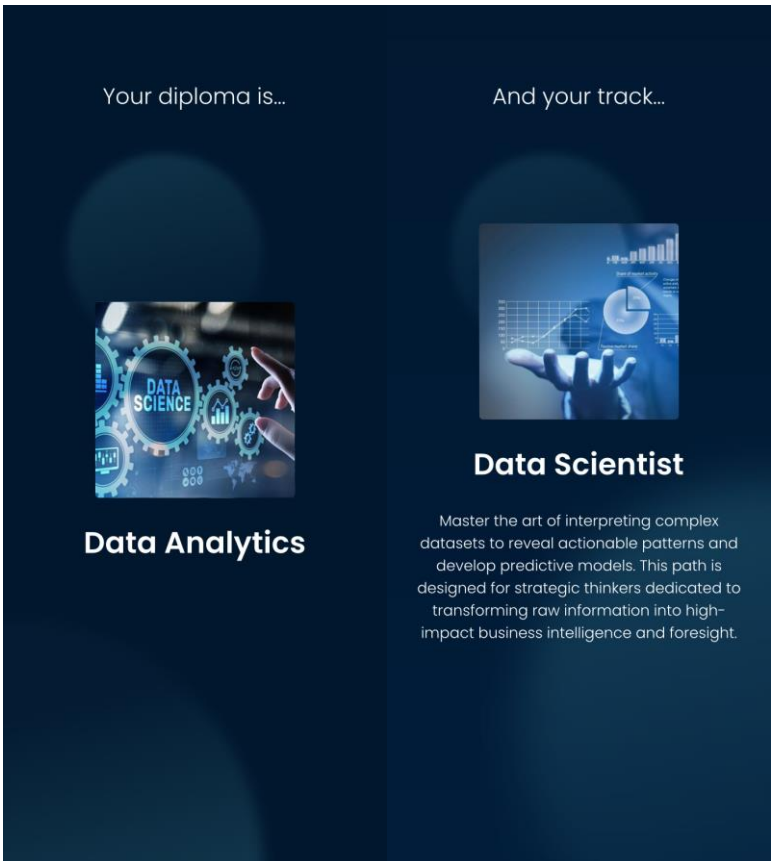


Figure 4.17: Track recommendation sequence "Your diploma is... Data Analytics" reveal screen (left) and "Find your track... Data Engineer" specialization confirmation screen (right), illustrating the isometric 3D diploma visualization and the Electric Cyan star motif marking the assigned specialization.

The track recommendation screens, illustrated in Figure 4.17, represent the culminating output of the Smart Compatibility Engine's psychometric analysis. The interface employs an isometric three-dimensional diploma visualization to communicate the assigned specialization Data Analytics, Data Engineering, or Data Science as a tangible, prestigious credential rather than an abstract algorithmic label. This representational choice is deliberate: framing the track assignment as a diploma reduces the perceived arbitrariness of the recommendation and anchors the student's motivation to a concrete future artifact, a technique consistent with goal-setting theory's emphasis on proximal, specific outcome representation [12]. The Electric Cyan four-pointed star motif, positioned adjacent to the recommended track, functions as a persistent visual anchor for the Mitchy AI agent's presence throughout the platform, establishing the AI mentor as a guiding companion from the earliest moments of the student's journey.

The three track variants Data Analyst, Data Engineer, and Data Scientist each present a visually differentiated diploma card with track-specific iconography whilst maintaining chromatic consistency with the broader design system, ensuring that the learner can

immediately identify their assigned path without ambiguity whilst preserving the platform's visual coherence across all specialization outcomes.

4.9.5 Adaptive Learning Workspace

The adaptive learning workspace constitutes the primary runtime interface through which students consume instructional content, engage with assessments, and progress through the curriculum hierarchy. Its design is governed above all by Mayer's Coherence Principle [2]: every interface element absent from the direct instructional task is either eliminated or reduced to a non-competing opacity level, ensuring that the learner's limited working memory capacity is allocated exclusively to germane cognitive processing rather than interface navigation.

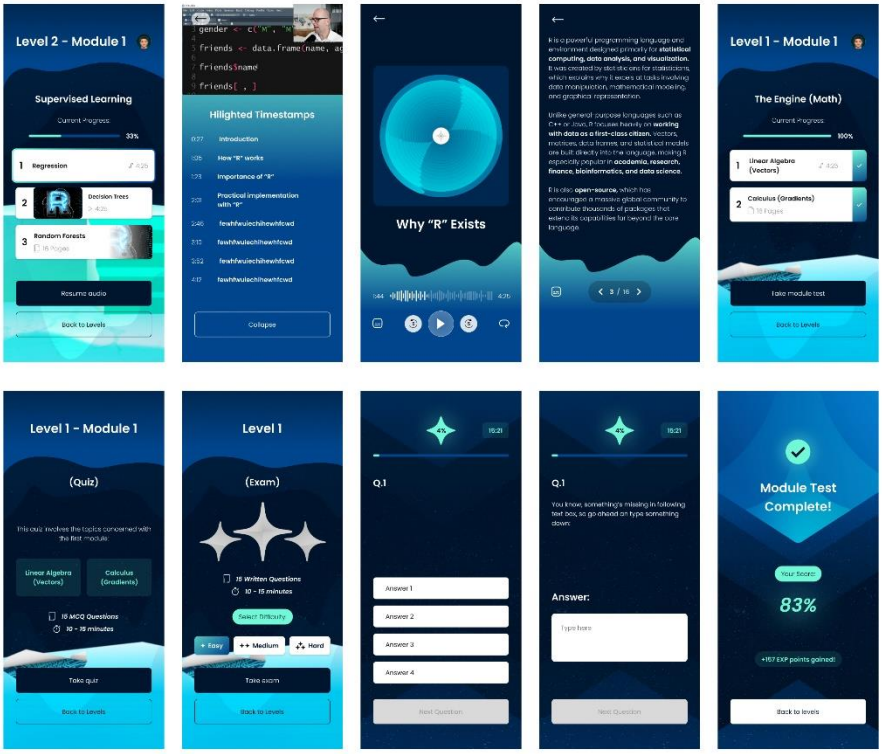


Figure 4.18: Adaptive learning workspace distraction-free content consumption interface (left) and the quiz question interface with answer option cards (right), illustrating modality-specific layout adaptation and the mastery-based progression gate.

As illustrated in Figure 4.18, the content consumption interface presents the active instructional asset video, audio player, or structured text as the dominant visual element occupying the majority of the viewport, with navigation and progress controls reduced to a minimal footer strip. This layout topology adapts dynamically based on the student's assigned learning style: Visual learners receive a video-dominant hierarchy with the media player filling the upper two-thirds of the screen; Auditory learners receive an audio player with a waveform visualization; Textual learners receive a structured reading layout

with typographic hierarchy emphasizing content scannability. This modality-responsive layout operationalizes the VARK framework's core principle that learners engage more efficiently with instructional content presented in a format compatible with their dominant cognitive processing channel thereby reducing extraneous cognitive load arising from format-cognition mismatch [18].

The quiz interface, presented in the right pane of Figure 4.18, applies a formative rather than summative assessment paradigm. Answer options are rendered as full-width selectable cards, sized above 48dp in compliance with Material Design 3 touch target guidelines [86], and clustered within the lower thumb zone to minimize the Index of Difficulty for target acquisition as prescribed by Fitts's Law [80]. Upon answer selection, immediate color-coded visual feedback is provided correct selections rendered in a positive-signal green, incorrect selections in a cautionary red without requiring a final submission action. This implementation of immediate reinforcement is grounded in behaviorist learning theory: reinforcing the correct stimulus-response association whilst information remains active in working memory maximizes schema consolidation and minimizes the probability of misconception persistence [2, 10].

4.9.6 AI Mentorship Interface

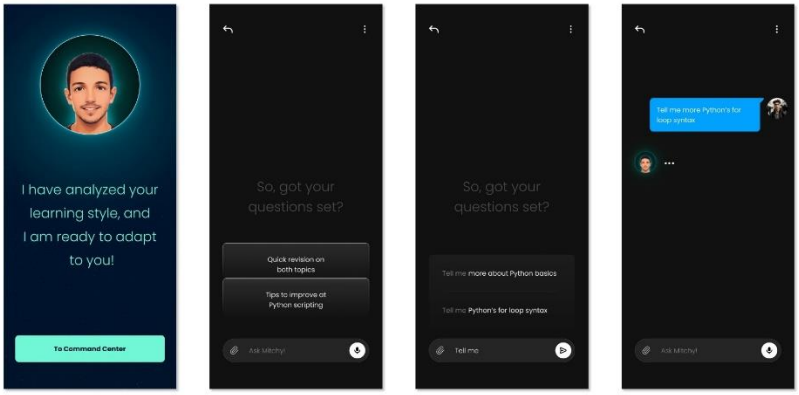


Figure 4.19: Mitchy AI Mentorship interface the "space bubble" conversational UI displaying the prompt scaffolding chip carousel above the input field, with user messages aligned right and Mitchy responses aligned left, rendered within the Electric Cyan and Void Navy design language.

The Mitchy conversational interface, illustrated in Figure 4.19, is designed to mitigate what interaction design literature terms the **"blank slate paradox"** the measurable increase in task abandonment that occurs when users are presented with an open-ended input field without contextual guidance [12]. The interface addresses this through a

Contextual Suggestion Engine that surfaces a horizontally scrollable carousel of pre-formed query chips exemplars including "Explain this concept differently" and "Tips for debugging Python loops" positioned immediately above the text input field. This mechanism transforms the cognitive task from free recall, which demands the generation of a well-formed question from memory, to recognition, which requires only the identification and selection of a relevant prompt from a presented set. Research on help-seeking behavior in digital learning environments consistently demonstrates that reducing the cognitive cost of initiating support requests produces statistically significant increases in help-seeking frequency, with consequent positive effects on performance outcomes [12, 25].

The conversational layout employs a skewed message alignment pattern student messages right-aligned, Mitchy responses left-aligned that deliberately mirrors the spatial conventions established by ubiquitous mobile messaging applications. This design decision leverages **transferable mental models**: by replicating interaction patterns already deeply encoded in the learner's procedural memory from social messaging contexts, the interface eliminates the learning curve associated with a novel conversational UI paradigm and channels the learner's full cognitive capacity toward the substantive pedagogical interaction rather than interface operation [12].

4.9.7 Gamification and Progress Interface



Figure 4.20: Gamification dashboard Daily Quests panel displaying linear progress bars with completion fractions (e.g., "2/3 Complete"), XP and level progression ring with percentage indicator, and leaderboard rank visualization, all rendered within the dark-mode design system.

The gamification dashboard, illustrated in Figure 4.20, applies principles from **Self-Determination Theory (SDT)** and cognitive psychology to engineer sustained intrinsic

motivation within the asynchronous learning context [12]. Daily learning tasks are visualized not as binary completion checkboxes but as linear progress bars annotated with precise fractional completion metrics for example, "2/3 Complete" a design choice that deliberately exploits the **Zeigarnik Effect**: the empirically demonstrated tendency for human memory and motivation systems to maintain heightened activation around incomplete tasks relative to completed ones [80]. By rendering incompleteness as a quantified, visually salient state, the interface generates a persistent cognitive drive toward task closure that is self-sustaining throughout the learning session.

The XP and level progression system is visualized through a circular progress ring displaying a precise percentage indicator for example, "83%" positioned prominently within the learner's primary view. This visualization concretizes what would otherwise remain an abstract accumulation of points into a tangible, quantifiable progress metric, directly supporting the SDT need for **competence**: the psychological requirement for continuous, credible evidence of growing mastery that empirical research consistently identifies as a primary predictor of sustained intrinsic motivation and persistence in self-directed learning environments [12, 50].

The leaderboard component presents rank progression within the student's assigned track cohort, operationalizing the SDT need for **relatedness** through social comparison within a peer-anchored community. Critically, the competitive scope is constrained to track-matched peers rather than the global student population, a design decision informed by gamification research demonstrating that intra-group competition produces higher engagement sustainability than global leaderboards, which tend to demotivate lower-ranked students and accelerate dropout in the precise population most requiring support [10, 50].

4.9.8 Administrative Dashboard

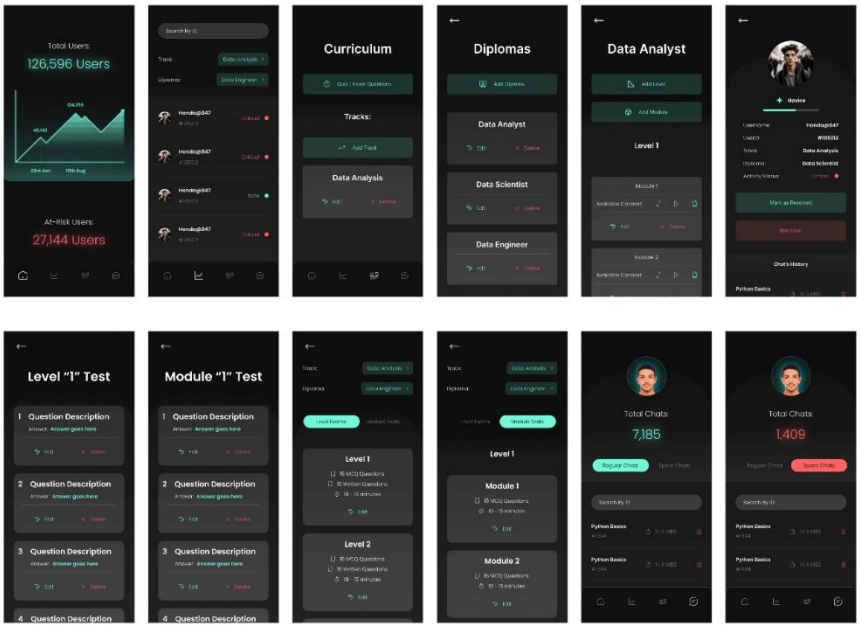


Figure 4.21: Administrative dashboard interface Row 1 (left to right): platform analytics overview with user growth chart and at-risk user count, risk monitoring panel with student risk level badges, curriculum management interface, diploma and track management, track drill-down with module content controls, and individual student profile with intervention actions. Row 2: level and module assessment question management, assessment filter and structure view, and chat monitoring interface displaying regular and spam chat session classification.

The administrative interface, illustrated in Figure 4.21, constitutes a distinct operational environment embedded within the same Flutter application binary under a role-gated access pathway. It is architected as a comprehensive **Student Risk and Content Management Dashboard**, providing administrators with the analytical visibility and intervention capability necessary to discharge their duty-of-care obligations toward the enrolled student population and to maintain the integrity of the adaptive curriculum. The following paragraphs examine each functional domain of the administrative interface in turn.

Platform Analytics and Risk Overview. The dashboard entry point surfaces two primary operational metrics in immediate visual prominence: total enrolled users rendered in Electric Cyan against the Void Navy canvas and the at-risk student count, rendered in a cautionary accent color to signal the administrative urgency of the figure. An area chart beneath the enrollment metric visualizes user growth over a configurable date range, providing administrators with a longitudinal engagement trajectory rather than a point-in-time snapshot. This distinction is pedagogically consequential: a declining growth curve constitutes an early systemic signal demanding curricular or operational intervention, whereas a point-in-time enrollment figure alone would obscure trend information critical to proactive platform governance. The at-risk user count displayed on the overview screen populated directly from the risk_scores table via the get_admin_risk_radar() database function ensures that the magnitude of the student welfare challenge is visible at the highest level of administrative navigation, preventing at-risk data from being buried within drill-down views that require deliberate navigation to access.

Risk Monitoring and Student Triage. The risk monitoring panel, the second screen in Figure 4.21's first row, implements the human-in-the-loop intervention protocol formally specified in Section 4.3.3 and Section 4.8.2. Students flagged by the Mitchy sentiment analysis pipeline are surfaced as a filterable list, with each entry displaying the student identifier, risk severity badge Critical or Done and filterable dimensions including Track and Diploma. The Critical badge, rendered in a high-contrast red accent, exploits the same pre-attentive pop-out mechanism utilized by the Electric Cyan CTA elements in the student-facing interface: risk severity is communicated to the administrator before conscious scanning of the list is required, directing attentional resources toward the highest-priority cases without demanding deliberate visual search across the full student population [7]. The Search by ID functionality permits administrators to navigate directly to a known student's record, whilst the dropdown filters enable cohort-level triage for instance, isolating all Critical-status students within a specific track when a systemic difficulty signal is detected.

Individual Student Profile and Intervention Controls. The student profile screen, the sixth screen in Figure 4.21's first row, aggregates the complete operational picture of an individual learner into a single administrative view. Displayed attributes encompass the student's username, system-generated UserID, assigned Track, enrolled Diploma, and current Activity Status the last of these mapping directly to the zone_state enumeration (LOCKED_IN, DRIFTING, or DISENGAGED) maintained in the student_profiles table and updated by the ML analytics pipeline. Two intervention controls are prominently positioned: **Mark as Resolved**, which updates the corresponding risk_scores.alert_resolved flag and closes the intervention record in intervention_logs, and **Ban User**, which provides administrators with the access revocation capability required for platform integrity enforcement. A **Chat's History** section at the base of the profile view surfaces the student's recent Mitchy chat sessions for example, the "Python Basics" session visible in the figure enabling the administrator to review conversational context prior to composing a manual intervention message, thereby ensuring that the human response is informed by the student's most recent expressed cognitive and emotional state rather than the risk score alone.

Curriculum and Content Management. The curriculum management screens comprising the Tracks, Diplomas, and Track drill-down views in the first row of Figure 4.21 operationalize the administrator's content authoring role within the platform. The Tracks interface presents the available specialization paths (Data Analysis, Data Science, Data Engineering) with inline Edit and Delete controls, alongside an Add Track affordance that enables horizontal curriculum expansion without schema modification a direct consequence of the flexible courses table architecture documented in Section 4.5.3. The Diploma management screen governs the top-level credential definitions, whilst the Track drill-down view exposes the Level and Module hierarchy for a selected specialization, presenting content availability indicators represented by icon controls for viewing, editing, and media management at the module level. This hierarchical drill-down pattern implements Progressive Disclosure at the administrative level, presenting the curriculum's complexity incrementally rather than as a monolithic content tree, thereby reducing the cognitive overhead of administrative content navigation.

Assessment Management. The assessment management screens in Figure 4.21's second row provide administrators with granular control over both Level Exams and Module Tests through a unified filter interface. A Track and Diploma dropdown pair scopes the displayed assessments to the relevant curricular context, whilst a tab control switches between Level Exams and Module Tests reflecting the architectural distinction between these two assessment types documented in Section 4.5.4. Each assessment entry displays its question composition for instance, "15 MCQ Questions, 15 Written Questions" alongside the estimated completion duration, providing administrators with the information necessary to calibrate cognitive load at the assessment level before content is exposed to students. Individual question management screens, visible in the leftmost panels of Figure 4.21's second row, present the question bank as an enumerated, editable list with per-question Edit and Delete controls, enabling targeted revision of assessment content without disturbing the broader question bank structure.

Chat Monitoring and Moderation. The chat monitoring interface, comprising the two rightmost screens in Figure 4.21's second row, addresses the platform's obligation to

maintain a safe and pedagogically productive conversational environment within the Mitchy chat channel. Total chat volume 7,185 regular sessions and 1,409 spam-classified sessions in the illustrative figures is surfaced prominently as a measure of AI mentor utilization, with a tab control switching between Regular Chats and Spam Chats classifications. The Spam Chats tab, active in the final screen, surfaces sessions that the Mitchy sentiment and content analysis pipeline has flagged as potentially non-pedagogical enabling administrators to review and moderate chat activity that may represent misuse of the AI mentorship channel. Individual chat sessions are searchable by student ID, and each listed session displays its associated topic label for instance, "Python Basics" and file size metadata, providing the context necessary for efficient administrative review without requiring the administrator to open each session individually.

Collectively, the administrative interface operationalizes the platform's commitment to proactive learner support and pedagogical governance, translating the data outputs of the ML analytics pipeline, the Mitchy affective computing module, and the curriculum management schema into actionable administrative workflows. The interface's design reflects a deliberate philosophical positioning: rather than treating administrative oversight as a reactive corrective function, LearNova's dashboard architecture surfaces risk, engagement, and curriculum health indicators at the highest navigational level, institutionalizing proactive intervention as the default administrative posture rather than an exceptional one.

Chapter 5: Implementation

5.1 Development Environment and Technology Stack

The implementation of LearNova was carried out within a carefully selected development environment, chosen to maximise cross-platform compatibility, developer productivity, and fidelity to the architectural decisions formalised in Chapter 4. This section documents the precise toolchain, framework versions, and infrastructure services utilised throughout the development lifecycle, providing the technical context necessary to reproduce and extend the system.

5.1.1 Development Toolchain

The primary development environment consisted of Android Studio Hedgehog (2023.1.1), which provided integrated Flutter and Dart tooling, the Android Virtual Device (AVD) Manager for on-device simulation, and built-in performance profiling via the Flutter DevTools plugin. Flutter SDK 3.22.0 and Dart SDK 3.4.0 were employed, targeting Android API Level 28 and above consistent with the compatibility requirement specified in NFR-11. These SDK versions were selected specifically for their inclusion of the Impeller rendering engine as the Android default, which delivered material improvements in UI frame-rate consistency compared to the legacy Skia pipeline; a critical consideration given the animation-intensive gamification and onboarding interfaces.

Version control was managed through Git, with the primary repository hosted on GitHub. A feature-branch workflow was maintained throughout development in accordance with the Agile Scrum methodology described in Section 3.1, wherein each Sprint yielded a mergeable, tested increment of functionality. The backend scoring service was hosted independently on Railway, providing a natural deployment boundary that enabled frontend and backend iterations to proceed in parallel without coupling release cycles.

5.1.2 Technology Stack Summary

Table 5.1 presents a consolidated reference of the technology stack as deployed in the production-ready MVP. The architecture follows a strict layered separation: a Flutter mobile client communicates exclusively with Supabase Edge Functions, which authenticate the user via JWT and forward trusted server-to-server requests to the Railway FastAPI service a constraint referred to throughout the codebase as the Golden Rule, which ensures that sensitive service keys and scoring logic are never exposed to the client.

Table 5.1: LearNova Production Technology Stack

Layer / Component	Technology	Version / Notes
Presentation (Mobile)	Flutter / Dart	SDK 3.22.0 / Dart 3.4.0 — Impeller renderer; Material Design 3
State Management	Riverpod	v2.x — compile-time safe; stream-based async providers
API Gateway & Auth	Supabase GoTrue	JWT sessions; Row Level Security on all 39 tables
Relational Database	PostgreSQL 15 (Supabase)	UUID PKs; JSONB columns; pgvector for RAG embeddings
Serverless Compute	Supabase Edge Functions (Deno)	TypeScript; service-role key isolation
Backend Scoring API	FastAPI (Python 3.11)	Hosted on Railway; modular microservice
AI / LLM — Primary	Google Gemini 2.5 Flash	Via Gemini REST API; 20-second timeout enforced
AI / LLM — Fallback	Groq (xAI)	Activated on Gemini 503 / 429 / timeout failure
Object Storage	Supabase Storage	Buckets: calibration-media, content-assets, student-submissions, avatar bucket
Push Notifications	Firebase Cloud Messaging	Android & iOS; tokens stored in user_devices table
IDE	Android Studio Hedgehog 2023.1.1	Flutter/Dart plugin; AVD Manager; DevTools profiler
Version Control	Git / GitHub	Feature-branch workflow; per-Sprint merges to main

5.1.3 System Implementation Architecture Overview

Figure 5.1 presents the implementation-layer view of the LearNova architecture, illustrating the communication pathways between the Flutter client, the Supabase service tier, and the Railway backend. This diagram complements the conceptual five-layer architecture presented in Figure 4.1 by grounding each layer in the specific technologies and deployment units actualised during implementation.

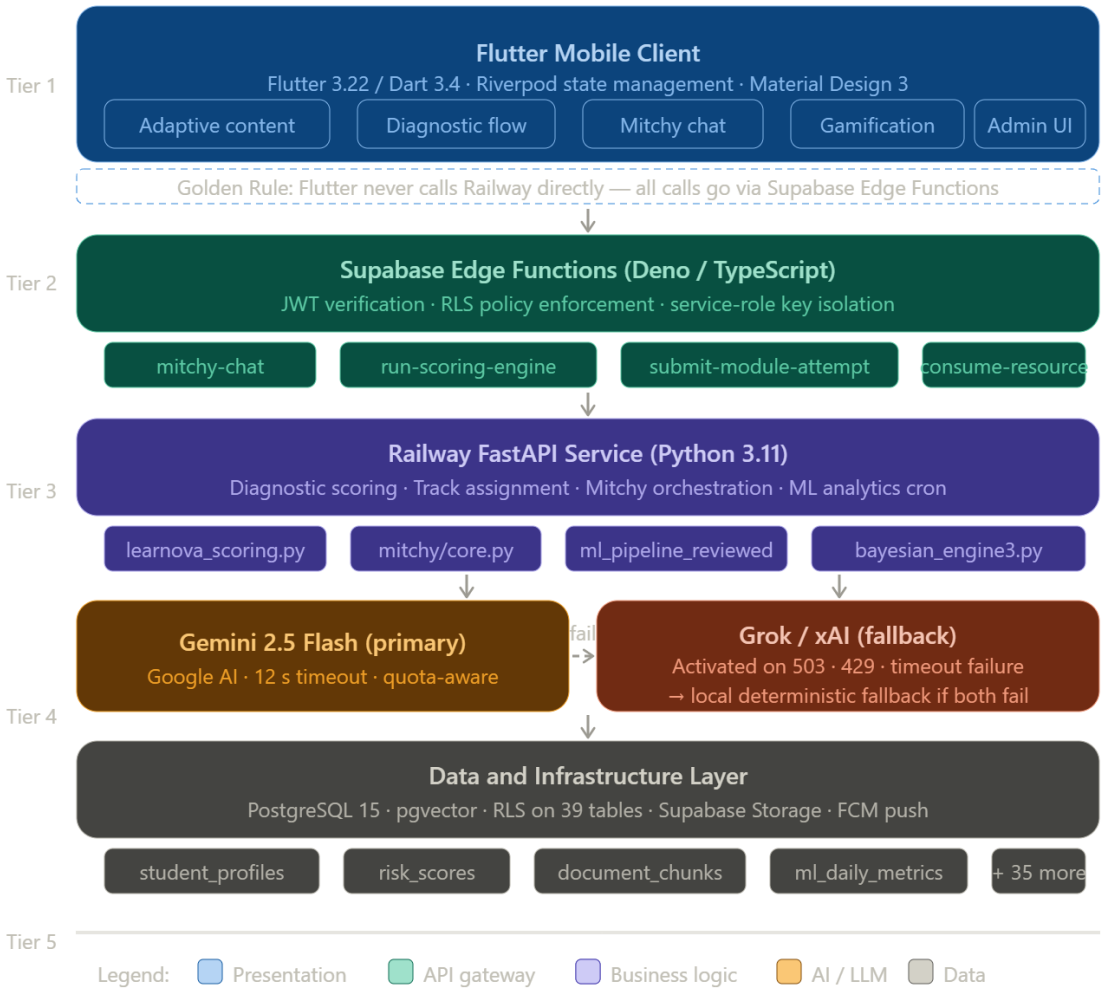


Figure 5.1 — LearNova implementation architecture (Chapter 5). Dashed arrow = fallback path.

Figure 5.1: Implementation-layer architecture diagram illustrating the communication pathways between the Flutter mobile client, Supabase Edge Functions, Railway FastAPI service, and external AI providers.

5.2 Diagnostic Calibration Scoring Engine

The diagnostic scoring engine constitutes the most analytically complex component of the LearNova backend. It transforms raw psychometric responses collected across five discrete assessment instruments the IPIP Personality Inventory, the Situational Judgement Test for soft skills, the VARK Learning Style Assessment, the O*NET Career Interest Inventory, and the Cognitive IQ Battery into a structured learner profile that deterministically assigns a specialisation track (Data Analytics, Data Engineering, or Data Science) and a dominant learning style (Visual, Auditory, or Textual). The implementation of this pipeline required resolving a significant early architectural failure before meaningful scoring could be achieved.

5.2.1 The Answer-Key Architecture

During early integration testing, the /score/diagnostic-result endpoint returned only placeholder payloads containing answer counts without computed feature scores, rendering track assignment impossible. This arose because the initial implementation matched submitted answers solely by selected_index, without a corresponding server-side answer key against which correctness and feature contributions could be evaluated. The resolution required constructing a comprehensive answer-key seed file diagnostic_answer_key_full_seed_by_question_key.json containing 1,196 answer-level rows across all five assessments. Each row is indexed by the composite key (question_key, selected_index) and encodes the feature or features to which the selected answer contributes, the scored value, a correctness flag for IQ questions, and the direct O*NET calibration weights for Career Interest responses.

This design deliberately decouples the Flutter client from all scoring logic: the client transmits only raw_answers containing question_key and selected_index pairs, while feature aggregation, normalisation, and track assignment execute exclusively within the Railway FastAPI microservice using the service-role Supabase key. The student is therefore unable to inspect or manipulate the scoring coefficients at any point in the data pipeline.

Architectural Invariant Scoring Isolation: The Flutter client never computes, caches, or directly writes computed_scores or assigned_track values to the database. All psychometric outputs are server-validated before persistence, ensuring that the track stored in student_profiles reflects only evidence produced by the server-side answer-key evaluation pipeline.

5.2.2 Feature Aggregation by Assessment Type

Upon receipt of a raw_answers payload, the scoring engine iterates over each submitted response, retrieves the corresponding answer-key row, and accumulates feature scores according to exam-specific aggregation rules. For the IPIP inventory, Likert values (1–5) are averaged per Big Five trait, with reverse-scored items processed by applying the transformation scored_value = 6 − selected_value prior to aggregation. For the O*NET Career Interest Inventory, selected values (1–5, mapped from Strongly Dislike to Strongly Like) are averaged within each RIASEC category; the Artistic, Realistic, and Investigative category averages subsequently serve as the base anchors for the DA, DE, and DS tracks respectively. VARK responses contribute binary count increments to the corresponding

modality feature, with Kinesthetic responses mapped to the Textual bucket for compatibility with the three-value learning_style enumeration supported by the student_profiles schema.

Following aggregation, each raw feature score is normalised to the unit interval [0, 1] using a clamped min–max transformation, where the minimum and maximum boundaries are defined in the scoring configuration file and reflect the empirically validated score ranges for each instrument. Missing features arising from incomplete external assessment submissions default to the minimum range value, preventing null-propagation errors in downstream modifier calculations.

5.2.3 Track Assignment Formula Implementation

The final DA/DE/DS assignment is computed through a three-step formula that combines O*NET-derived base scores with a weighted psychometric modifier. Code Snippet 5.1 presents the core calculation logic as implemented in the personalization/learnova_scoring.py module, which is invoked by the FastAPI /score/diagnostic-result endpoint following feature aggregation.

```
# Code Snippet 5.1 — Track Assignment Formula
# personalization/learnova_scoring.py
def compute_final_track_scores(features: dict, weights: dict) -> dict:
    """
    Computes final DA / DE / DS scores from a normalised feature vector.
    Formula: Final_track = 0.30 * Base_track + 0.70 * Modifier_track
    """
    # Step A — Base scores from O*NET RIASEC anchors
    base = {
        'DA': features.get('artistic',      0.0),   # Artistic  -> DA
        'DE': features.get('realistic',     0.0),   # Realistic -> DE
        'DS': features.get('investigative', 0.0),   # Investigative -> DS
    }
    # Step B — Modifier: weighted sum of all normalised features per track
    modifier = {'DA': 0.0, 'DE': 0.0, 'DS': 0.0}
    for feature, norm_value in features.items():
        if feature in weights:
            for track in ['DA', 'DE', 'DS']:
                modifier[track] += norm_value * weights[feature][track]
    # Step C — Final weighted combination (30 % base, 70 % modifier)
```

```
    final = {
        t: round(0.30 * base[t] + 0.70 * modifier[t], 4)
        for t in ['DA', 'DE', 'DS']
    }
    # Assigned track is the argmax of the three final scores
    return {
        'final_scores':    final,
        'assigned_track':  max(final, key=final.get),
    }
```

Code Snippet 5.1: Track assignment formula implemented in the Railway FastAPI scoring service. The 0.30/0.70 split was calibrated against live diagnostic export data.

The 0.30/0.70 weighting ratio between base (O*NET) and modifier (psychometric) scores was determined through iterative calibration against four exported diagnostic_test_results rows. This ratio ensures that vocational interest provides directional anchoring whilst the holistic modifier incorporating personality, cognitive aptitude, soft skills, and learning style evidence exerts the dominant influence on the final assignment. Validated examples confirmed that the formula produces differentiated scores across all three tracks even when O*NET totals are equal, because modifier evidence from IPIP, soft skills, and VARK introduces sufficient separability.

5.3 Adaptive Content Delivery Implementation

The adaptive content delivery mechanism constitutes the primary runtime expression of LearNova's pedagogical differentiation. Rather than serving uniform instructional assets to all learners, the system dynamically resolves the content URL corresponding to each student's calibrated learning style at the moment of topic access. This section documents the implementation of this resolution pipeline, from the Supabase schema design through to the Flutter-side rendering logic.

5.3.1 Polymorphic Content Schema

Each topic_resources row in the Supabase schema is tagged with a format_type attribute constrained to one of three enumerated values: Visual, Auditory, or Textual. When an administrator uploads content for a given topic via the Content Management System, three distinct resource records are created each referencing a different media asset (for example, an MP4 video, an MP3 audio summary, and a PDF transcript) whilst sharing the same topic_id foreign key. This wide-column strategy eliminates runtime joins: the Flutter client filters the returned resource set by the student's learning_style tag in a single, low-latency pass, with no additional database round-trips.

5.3.2 Flutter Content Resolution Provider

Code Snippet 5.2 presents the Riverpod FutureProvider responsible for resolving the appropriate topic resource URL based on the authenticated student's active learning style. The provider is parameterised by topicId, enabling Riverpod's built-in caching mechanism to retain resolved resources across navigation events without re-fetching on widget rebuild cycles.

```
// Code Snippet 5.2 — Adaptive Content Resolution Provider
// lib/features/learning/providers/topic_resource_provider.dart
final adaptiveResourceProvider =
    FutureProvider.family<TopicResource?, String>((ref, topicId) async {
  // 1. Read student's calibrated learning style from cached profile provider
  final profile = ref.watch(studentProfileProvider).valueOrNull;
  final style   = profile?.learningStyle ?? LearningStyle.visual;
  // 2. Fetch all variant records for this topic from Supabase
  final supabase  = ref.read(supabaseClientProvider);
  final response  = await supabase
      .from('topic_resources')
      .select('id, format_type, resource_url, order_index')
      .eq('topic_id', topicId)
      .order('order_index');
  final resources = (response as List)
```

```
      .map(TopicResource.fromJson)
      .toList();
  // 3. Match calibrated format; fall back gracefully to first available
  final match = resources.firstWhereOrNull(
    (r) => r.formatType == style.toDbString(),
  );
  return match ?? resources.firstOrNull;
});
extension LearningStyleX on LearningStyle {
  String toDbString() => switch (this) {
    LearningStyle.visual   => 'Visual',
    LearningStyle.auditory => 'Auditory',
    LearningStyle.textual  => 'Textual',
  };
}
```

Code Snippet 5.2: Riverpod FutureProvider implementing adaptive content format resolution. The graceful fallback ensures accessibility when not all three format variants have been uploaded.

Three implementation decisions embedded in this provider merit explicit discussion. The graceful fallback returning the first available resource when no exact style match exists ensures content accessibility during the administrative upload period before all three format variants are populated, preventing null-state rendering errors in production. The learning style value is consumed directly from the cached studentProfileProvider rather than fetched independently, eliminating a redundant Supabase call on every topic navigation event. Finally, parameterisation by topicId enables automatic cache invalidation when a student navigates to a new topic, while preserving the resolved resource for the current topic throughout the session.

5.3.3 Content Player Implementation

The resolved resource URL is subsequently consumed by a modality-aware player widget that dynamically renders a video player (via the video_player Flutter package) for Visual resources, an audio player interface for Auditory resources, or a scrollable article view with syntax-highlighted code blocks for Textual resources. This polymorphic rendering approach operationalises the adaptive surface described in Section 4.6.3, ensuring that the interface topology itself adapts to the content modality rather than forcing all three formats through a uniform video-centric container a design anti-pattern prevalent in conventional MOOC platforms.

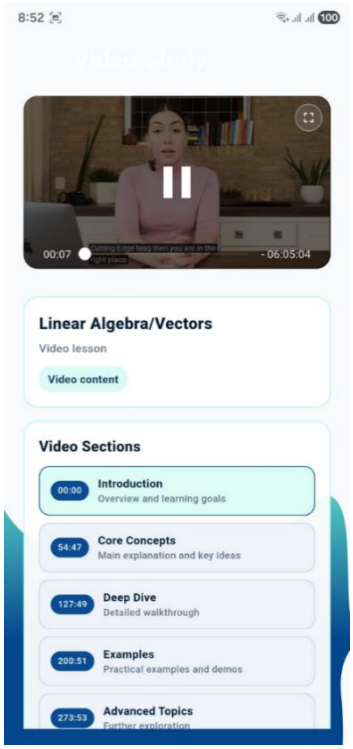


Figure 5.2: The adaptive learning workspace rendering a Visual-modality video resource for a calibrated Visual learner, demonstrating the distraction-free player layout.

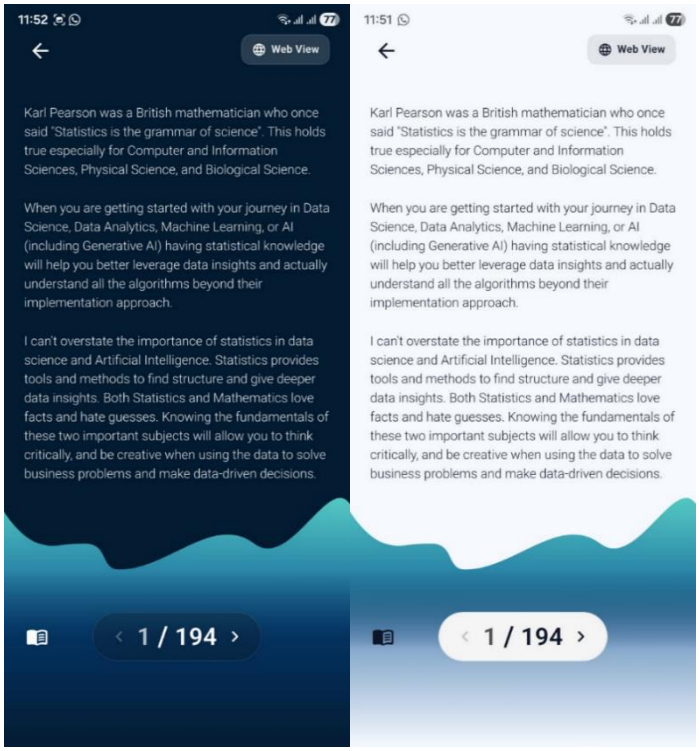


Figure 5.3: The same topic rendered in Textual modality for a Read/Write-calibrated student, demonstrating the polymorphic content surface in operation.

5.4 Mitchy AI Mentorship and Sentiment Analysis Implementation

The AI pedagogical agent designated Mitchy integrates a conversational large language model interface with a real-time affective computing pipeline, a Retrieval-Augmented Generation knowledge base, and a multi-provider fallback chain engineered to guarantee response availability under adverse API conditions. This section documents the implementation of each of these subsystems and the engineering decisions that shaped their final architecture.

5.4.1 The Multi-Provider LLM Architecture and Fallback Chain

Initial implementation utilised Google Gemini 2.5 Flash as the sole generative model, accessed via the Gemini REST API through the Railway FastAPI backend. Production testing, however, revealed a critical reliability constraint: the free-tier Gemini API exhibited frequent 503 High Demand and 429 Quota Exhausted responses under concurrent load, causing the chat pipeline to produce empty or malformed payloads. This directly undermined the learner trust that the mentorship experience is designed to cultivate; a student who receives a blank response from an AI mentor at a moment of confusion is likely to disengage entirely.

The system was consequently re-engineered with a two-provider fallback chain. Gemini 2.5 Flash remains the primary inference engine. Upon detection of a 503, 429, or timeout condition the latter defined by the GEMINI_TIMEOUT_SECONDS environment variable, set to twelve seconds in production the orchestration layer automatically routes the request to the Groq API (xAI), which provides a structurally equivalent JSON response without modification to the downstream parsing logic. If both providers fail, the local chat_logic_v3.py affective analysis module generates a deterministic contextual fallback reply, ensuring that the student always receives a non-empty, pedagogically appropriate response. Code Snippet 5.3 illustrates the fallback guarantee as implemented in the Supabase Edge Function layer.

```
// Code Snippet 5.3 — Mitchy Edge Function: Guaranteed Non-Empty Response
// supabase/functions/mitchy-chat/index.ts
// Deterministic fallback — always returns a valid, non-empty assistant message
function fallbackResponse(reason: string) {
  const msg =
    'I am here with you. Tell me what part feels unclear ' +
    'and we will break it down step by step.';
  return new Response(JSON.stringify({
    ok:            true,
    response_text: msg,
    message:       { role: 'assistant', content: msg },
    mitchy: {
      response_text:      msg,
```

```
      learning_state:     'confused',
      sentiment_score:    0,
      cognitive_load:     0.3,
      suggested_action:   'rescue_explanation',
      recommended_format: 'Textual',
    },
    metadata: { source: 'edge_fallback', used_gemini: false, reason },
  }), { status: 200, headers: corsHeaders });
}
// Within the serve() handler — Railway call with fallback guard
const railwayRes = await fetch(MITCHY_SERVICE_URL + '/mitchy/chat', {
  method:  'POST',
  headers: { 'Content-Type': 'application/json',
             'x-api-key':     MITCHY_SERVICE_API_KEY },
  body:    JSON.stringify(railwayPayload),
});
// Guard 1 — non-2xx HTTP status from Railway
if (!railwayRes.ok) return fallbackResponse(`Railway ${railwayRes.status}`);


const data         = await railwayRes.json();
const responseText = String(
  data.response_text ||
  data?.message?.content ||
  data?.mitchy?.response_text || ''
).trim();
// Guard 2 — empty string after all parsing attempts
if (!responseText) return fallbackResponse('empty_response_text');
```

Code Snippet 5.3: Supabase Edge Function implementing the two-guard fallback chain. The first guard catches non-2xx Railway responses; the second catches empty parsed content, ensuring the response invariant is never violated.

5.4.2 Sentiment Analysis and Risk Detection Pipeline

Each message submitted to Mitchy is processed by a sentiment analysis routine implemented in the Railway service's mitchy/chat_logic_v3.py module prior to generative response construction. The routine analyses linguistic markers and syntactic patterns in the user's text to produce three affective signals: a numerical sentiment_score (ranging

from −1.0 to +1.0), a categorical learning_state label (confused, frustrated, engaged, or at_risk), and an estimated cognitive_load value. The system persists these signals to the student_sentiment_history table in Supabase, where they feed the daily ML analytics pipeline executed by the Railway cron job.

When the sentiment_score breaches a predefined risk threshold, the scoring service creates a risk_scores record and sets alert_triggered to true. The administrative dashboard monitors this table in real time via a Supabase WebSocket subscription, surfacing at-risk students to human administrators who may then dispatch a Manual Intervention notification. This human-in-the-loop safety mechanism 164perationalize in Section 4.3.3 and 164perationalized in the production implementation ensures that fully autonomous AI responses are subject to human oversight whenever significant emotional distress is detected.

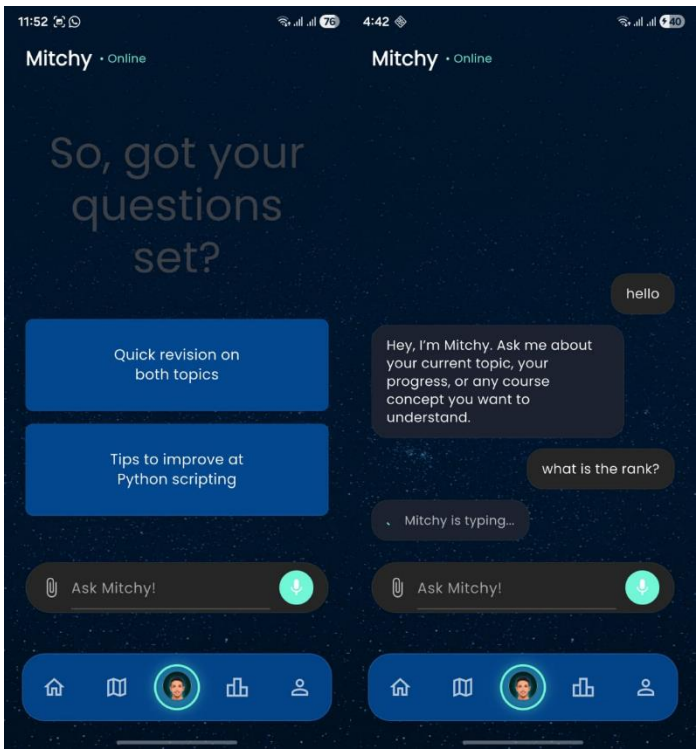


Figure 5.4: The Mitchy conversational interface, displaying the horizontally scrollable prompt scaffolding chips that transform user interaction from free recall to recognition, lowering the cognitive barrier to help-seeking.

5.4.3 Retrieval-Augmented Generation Integration

To ground Mitchy's responses in curriculum-specific content rather than relying solely on the LLM's parametric knowledge, a Retrieval-Augmented Generation pipeline was implemented using the pgvector extension within the Supabase PostgreSQL instance. Course content was chunked, embedded using a text embedding model, and stored in the document_chunks table alongside the pgvector embedding column and the source topic_id foreign key. At inference time, the Context Orchestrator within the Railway service

performs a cosine similarity search against the document_chunks table using the student's current message as the query, retrieves the top-k semantically relevant chunks, and prepends them to the LLM system prompt as grounding context. This mechanism ensures that Mitchy's explanations are anchored to the specific curriculum content the student is currently studying, rather than producing generically accurate but pedagogically irrelevant responses.

5.4.4 AI-Graded Level Assessments

A feature that distinguishes level assessments from module assessments in the LearNova architecture is the integration of Mitchy as an active grading agent. While module assessments consist exclusively of multiple-choice questions graded deterministically by the Edge Function's answer-key comparison logic, level assessments support open-ended questions for which correct responses cannot be reduced to a single canonical answer string. These questions are graded by Mitchy using a structured rubric, and the system returns personalised feedback to the student upon submission.

The level_assessment_questions table contains three Mitchy-specific columns that enable this capability. The ai_grading_rubric column stores a structured text rubric that defines the evaluative criteria against which the student's open-ended response should be assessed for example, specifying that a correct explanation of data normalisation must reference first normal form, atomicity, and redundancy elimination. The mitchy_hint column stores a hint that Mitchy surfaces to students who request assistance before submitting their answer. The mitchy_explanation column stores a post-submission explanation delivered by Mitchy regardless of whether the student passed or failed, reinforcing the correct conceptual understanding in either case.

At submission time, the student's open-ended answers are transmitted to the Railway /mitchy/chat endpoint via the submit-level-attempt Edge Function, accompanied by the relevant ai_grading_rubric values. Mitchy evaluates each response against the rubric, produces a percentage score, and generates a personalised mitchy_feedback string that identifies the specific strengths and gaps in the student's answer. This feedback is stored in the student_level_attempts table alongside the score and difficulty tier, and returned to the Flutter client for immediate display. The AI grading rubric mechanism therefore extends Mitchy's role from purely conversational mentorship into formative assessment, creating a tighter integration between the pedagogical agent and the academic progression system.

5.5 Gamification XP Engine Implementation

The gamification system was implemented as a distributed collection of Supabase Edge Functions, each corresponding to a distinct student action type, with all XP computations performed server-side and applied atomically via a secure PostgreSQL remote procedure call. The Flutter client never computes or directly writes XP values a constraint that eliminates the client-side score manipulation vulnerability common in gamified applications that rely on self-reported scores.

5.5.1 XP Reward Matrix Implementation

The XP reward matrix formalised in the Gamification Engine Architecture Specification was realised across four Edge Functions. The consume-resource function awards +5 XP per content slot consumed and a +15 XP completion bonus upon topic finalisation. The submit-module-attempt function awards a flat +20 XP on passing a module assessment, irrespective of difficulty tier, consistent with the non-tiered checkpoint design. The submit-challenge-attempt function implements the tiered risk/reward model (Easy: +20 XP base, Mid: +50 XP base, Hard: +100 XP base), with proportional partial credit calculated as base_xp × (score ÷ 100), reflecting the single-attempt condition for weekly challenges. The run-scoring-engine function awards the +500 XP diagnostic completion milestone upon successful processing of all five psychometric assessments.

5.5.2 The Streak Multiplier: Timezone-Aware Implementation

Code Snippet 5.4 illustrates the streak multiplier logic as implemented in the consume-resource Edge Function. A 1.1× multiplier applies to all XP earned on any day the student maintains an active consecutive-day streak. The implementation incorporates a timezone_offset parameter submitted by the Flutter client as the device's UTC offset in hours to prevent the streak comparison from using UTC midnight as the day boundary, which would cause streaks to reset incorrectly for students whose local midnight precedes UTC midnight.

```
// Code Snippet 5.4 — Streak Multiplier with Timezone-Aware Date Comparison
// supabase/functions/consume-resource/index.ts
let streakMultiplier = 1.0;
// Compute student's local time from device UTC offset
const serverNow = new Date();
const localTime = new Date(
  serverNow.getTime() + timezone_offset * 60 * 60 * 1000
);
const { data: streakData } = await supabase
  .from('user_streaks')
  .select('last_activity_date, current_streak_days')
  .eq('user_id', studentId)
```

```
  .maybeSingle();
if (streakData?.current_streak_days > 0 && streakData?.last_activity_date) {
  const lastActive = new Date(streakData.last_activity_date);
  const diffDays   = Math.floor(
    (localTime.getTime() - lastActive.getTime()) / (1000 * 60 * 60 * 24)
  );
  // Multiplier active if student engaged today (0) or yesterday (1)
  if (diffDays <= 1) streakMultiplier = 1.1;
}
const finalXpAwarded = Math.round(earnedXp * streakMultiplier);
// Atomic XP increment via secure PostgreSQL RPC
await supabase.rpc('increment_xp', {
  user_id_input: studentId,
  xp_amount:     finalXpAwarded,
});
```

Code Snippet 5.4: Timezone-aware streak multiplier with atomic XP increment. Without the timezone_offset correction, streaks for students in eastern time zones would break incorrectly at UTC midnight.

5.5.3 Duplicate XP Prevention

To prevent XP accumulation through repeated consumption of the same content resource, the consume-resource function queries a dedicated student_resource_logs table before awarding any XP, checking for the existence of a prior record matching the composite key (user_id, topic_id, resource_type). When such a record is found, the function returns a 200 status with an informational message but awards no additional XP. This idempotency guarantee ensures that content re-reads which are encouraged for pedagogical purposes do not artificially distort leaderboard rankings, preserving the integrity of the competitive gamification layer.

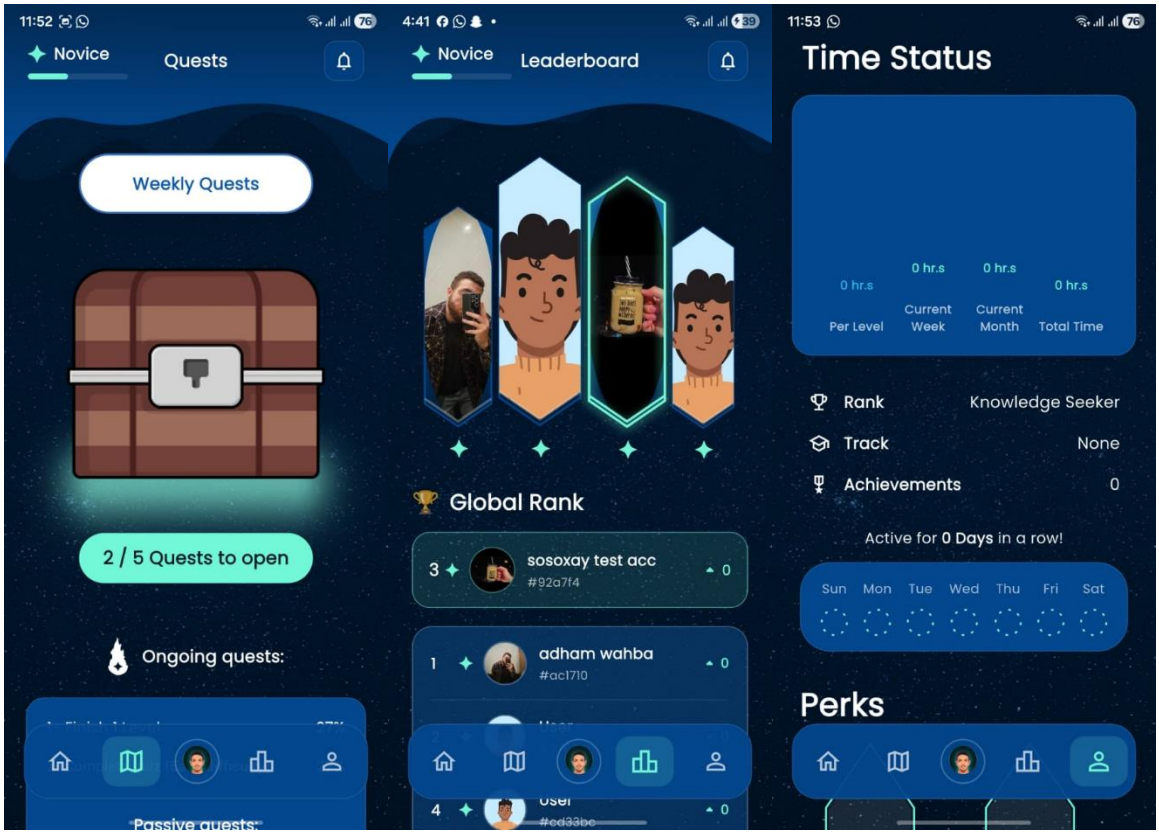


Figure 5.5: The gamification dashboard displaying daily quest progress bars (leveraging the Zeigarnik Effect), the active momentum streak counter, and the global XP leaderboard.

5.5.4 Achievements and Leaderboard Snapshot Implementation

Beyond XP accumulation, the gamification ecosystem implements two complementary retention mechanics: a structured achievement system and a daily leaderboard snapshot pipeline. These features operationalise the Self-Determination Theory need for competence (through badge milestones) and relatedness (through peer comparison), as theorised in Section 2.4.1.

Achievements System

The achievement system is implemented across two tables following a definition-issuance separation pattern. The achievements_dictionary table stores the static master list of all available badges, with each record defining an achievement_key, a human-readable label, a description, a criteria_type (streak, xp, or completion), a criteria_threshold value, and a badge_image_path stored in the content-assets Supabase Storage bucket. The user_achievements junction table records only the user_id, achievement_id foreign key, and an unlocked_at timestamp, enforcing a unique constraint on (user_id, achievement_id) to prevent duplicate awards. This schema separation enables rapid badge count aggregation queries against the lightweight junction table without scanning the master dictionary.

Achievement unlock logic is evaluated server-side within the gamification Edge Functions at the moment a qualifying event occurs. When a student passes a module assessment,

for example, the submit-module-attempt function evaluates whether the resulting XP total or streak count crosses any achievement threshold defined in achievements_dictionary. If a threshold is crossed, a row is inserted into user_achievements and a corresponding achievement_unlocked in-app notification is generated, surfacing the badge award in the student's notification centre in real time.

Leaderboard Snapshot Pipeline

The leaderboard implementation intentionally avoids real-time XP ranking queries against the student_profiles table, which would impose costly full-table scans as cohort size scales. Instead, the leaderboard_snapshots table stores daily pre-computed rankings per track, with each row recording a user_id, track, xp_at_snapshot, rank_at_snapshot, and snapshot_date. The unique constraint on (user_id, track, snapshot_date) ensures idempotent daily writes. Students may read leaderboard data for all users an exception to the standard own-row-only student RLS pattern since competitive transparency is a deliberate design choice for the social motivation mechanics. The get_student_xp_rank() PostgreSQL function provides an efficient single-call interface for the Flutter client to retrieve a student's current rank and XP total without constructing complex join queries client-side.

5.6 Database and Integration Layer Implementation

The data persistence layer was implemented using Supabase's managed PostgreSQL 15 instance, with the schema adhering to Third Normal Form normalisation, referential integrity enforced through foreign key constraints, and fine-grained access control implemented through Row Level Security policies across all 39 tables. This section documents the key implementation decisions in the database and integration architecture that underpin the security, performance, and correctness guarantees of the platform.

5.6.1 Row Level Security Policy Implementation

Row Level Security was enabled on every table in the schema. Two PostgreSQL helper functions is_admin() and current_user_is_admin() were implemented to resolve the authenticated user's role from the users table using auth.uid(), providing a consistent predicate for policy evaluation. Student policies permit read and write access only to rows where the user_id column matches auth.uid(), preventing any cross-user data access at the database layer regardless of client-side logic. Administrative policies grant unrestricted read access across the schema, with write access scoped to content and operational tables relevant to platform management.

Code Snippet 5.5 illustrates a representative RLS policy applied to the risk_scores table one of the most sensitive tables in the schema demonstrating how the policy layer enforces a strict access separation between the automated AI scoring pipeline and the human administrative review interface.

```
-- Code Snippet 5.5 — RLS Policies on the risk_scores Table
-- Executed in Supabase SQL Editor during schema migration
-- Enable RLS (required before any policy takes effect)
ALTER TABLE risk_scores ENABLE ROW LEVEL SECURITY;
-- Policy 1: AI service (via service-role key) may INSERT risk records.
-- Students have NO access at all — not even SELECT.
CREATE POLICY "risk_scores: service insert only"
  ON risk_scores
  FOR INSERT
  TO authenticated
  WITH CHECK (is_admin());   -- only admin-context inserts permitted via RPC
-- Policy 2: Admins may read and update (resolve alerts) —
--           students are categorically excluded.
CREATE POLICY "risk_scores: admin read and update"
  ON risk_scores
  FOR ALL
  TO authenticated
```

```
  USING (is_admin())
  WITH CHECK (is_admin());
-- Helper function used in all admin-scoped policies
CREATE OR REPLACE FUNCTION is_admin()
  RETURNS boolean
  LANGUAGE sql STABLE
  AS $$
    SELECT EXISTS (
      SELECT 1 FROM users
      WHERE id = auth.uid()
        AND role = 'admin'
    );
  $$;
```

Code Snippet 5.5: Representative RLS policy implementation for the risk_scores table, demonstrating categorical student exclusion from sensitive affective monitoring data.

5.6.2 The increment_xp RPC

A critical database design decision in the gamification implementation was the use of a secure PostgreSQL RPC function increment_xp(user_id_input UUID, xp_amount INTEGER) rather than direct UPDATE statements from Edge Functions. This approach provides two key correctness guarantees. First, the RPC executes atomically within a single database transaction, eliminating race conditions that could arise from concurrent XP award events for example, a student who simultaneously completes a topic and passes a module assessment. Second, the function encapsulates the update logic server-side, so Edge Functions require only insert-level privileges to the student_profiles table rather than full UPDATE access, narrowing the attack surface consistent with the Principle of Least Privilege.

5.6.3 Bayesian Learning Style Update Pipeline

The student_profiles table stores three Bayesian posterior columns bayesian_alpha_visual, bayesian_alpha_auditory, and bayesian_alpha_textual that evolve beyond the initial VARK calibration result as the student engages with content. The daily ML analytics pipeline, executed as a Railway cron job, ingests content_engagement_logs rows flagged as bayesian_eligible, applies the Bayesian update logic implemented in analytics/bayesian_engine3.py, and writes the updated posteriors back to student_profiles via the adapt_bayesian_profile_update_for_supabase adapter. An important schema constraint that surfaced during implementation was the absence of a metadata column in ml_daily_metrics: an early pipeline version attempted to write Bayesian debug payloads to this table, causing upsert failures. The corrected implementation routes Bayesian updates exclusively to student_profiles, with debug output written only to Railway service logs.

5.6.4 Bayesian Learning Style Adaptation Engine

A defining characteristic that distinguishes LearNova from static psychometric platforms is the capacity of its learning style model to evolve continuously beyond the initial VARK calibration result. Rather than treating the diagnostic output as a fixed label, the system employs a Bayesian inference engine implemented in analytics/bayesian_engine3.py that reinterprets the VARK assessment result as a prior probability distribution and progressively updates this distribution as the student accumulates real content engagement evidence. Over time, the model's representation of the student's learning style converges toward a posterior that reflects observed behaviour rather than self-reported preference alone.

The Prior: VARK as a Bayesian Starting Point

Upon completion of the VARK diagnostic, the scoring engine writes initial values to three columns in the student_profiles table: bayesian_alpha_visual, bayesian_alpha_auditory, and bayesian_alpha_textual. These columns store the Dirichlet concentration parameters of the Bayesian model, initialised from the VARK response counts. A student who selects predominantly Visual answers receives a higher bayesian_alpha_visual value, encoding a stronger prior belief that this student learns most effectively through visual content. The Dirichlet distribution is chosen as the conjugate prior for this multinomial learning style problem because it allows the three modality probabilities to sum to one and supports closed-form posterior updates a property that ensures computationally efficient daily recalculation within the Railway cron pipeline.

The Likelihood Signal: content_engagement_logs

The posterior update mechanism is driven by data collected in the content_engagement_logs table. Each row in this table records a student's interaction with a specific topic resource, capturing the format_type served (Visual, Auditory, or Textual), the time_spent_seconds, and a computed engagement_score that reflects interaction quality beyond raw duration. Critically, each row carries a boolean bayesian_eligible flag. This flag is set to true only when the engagement session meets minimum validity criteria specifically, that the student engaged for a duration sufficient to meaningfully distinguish genuine comprehension from passive exposure. Sessions that are too brief, or that were recorded during high-risk affective states detected by the sentiment pipeline, are excluded from Bayesian updates to prevent noise from corrupting the posterior.

When a student achieves a high engagement_score on a Textual resource despite having been calibrated as a Visual learner, the system increments bayesian_alpha_textual and correspondingly adjusts the posterior probability distribution. If this pattern repeats across multiple topics over several days, the posterior eventually shifts sufficiently for the pipeline to trigger a learning_style reassignment in student_profiles changing the student's active delivery format from Visual to Textual without any manual intervention.

The rescue_signal: Chat-Informed Bayesian Updates

A particularly sophisticated feature of the Bayesian pipeline is its integration with the Mitchy chat system through the extract_successful_rescue_signal adapter function. When a student expresses confusion about a topic through the chat interface and Mitchy

responds with a rescue explanation in a specific format for example, a structured textual breakdown of a concept that the student was struggling to understand through video and the subsequent sentiment score returns to a positive range, this event is classified as a successful rescue signal. The adapt_student_profile_for_bayesian adapter extracts this signal and incorporates it as additional likelihood evidence, incrementing the alpha parameter for the format that produced the successful learning recovery. This mechanism enables the Bayesian engine to learn not only from passive engagement patterns but also from active, Mitchy-mediated rescue interventions, creating a feedback loop between the mentorship and personalisation subsystems.

Daily Execution and Posterior Write-Back

The Bayesian update pipeline executes daily as part of the Railway ml_pipeline_reviewed.py cron job. The pipeline fetches all bayesian_eligible engagement logs recorded since the previous computation, transforms them into the input shape expected by bayesian_engine3.py via the adapt_student_profile_for_bayesian adapter, and runs the posterior update calculation. The resulting updated alpha values are then written back to student_profiles through the adapt_bayesian_profile_update_for_supabase adapter, which constructs a schema-aligned update payload containing only the three bayesian_alpha columns. The learning_style column itself is only updated when the argmax of the posterior distribution differs from the current stored value, preventing unnecessary write operations and limiting reassignment events to genuinely significant distributional shifts.

5.6.5 Push Notification and Device Management Implementation

LearNova implements a dual-channel notification architecture combining in-app notifications with Firebase Cloud Messaging (FCM) push notifications, ensuring that time-sensitive events risk alerts, achievement unlocks, and new weekly challenges reach students regardless of whether the application is currently foregrounded on their device.

Device Token Management

The user_devices table stores FCM registration tokens per device per user, supporting both Android and iOS platforms. Each row records the user_id, fcm_token, device_name, and platform, with a unique constraint on (user_id, fcm_token) preventing duplicate token registrations. When a student installs and opens the application for the first time, the Flutter client requests FCM permission from the operating system and, upon approval, retrieves the device token from the Firebase SDK and upserts it to the user_devices table via the authenticated Supabase client. Token refresh events which FCM may trigger periodically are handled by the same upsert logic, updating the updated_at timestamp and replacing stale tokens transparently.

In-App Notification Types and Routing

The in_app_notifications table supports six distinct notification_type values, each corresponding to a specific platform event category. Table 5.5 documents each type alongside its trigger condition and the system component responsible for insertion.

Table 5.5: In-App Notification Types and Trigger Sources

notification_type	Trigger Condition	Inserting Component
achievement_unlocked	Student XP, streak, or completion count crosses an achievements_dictionary threshold	Gamification Edge Function (submit-module-attempt, consume-resource)
burnout_detected	Mitchy sentiment analysis crosses the critical risk threshold	Railway Mitchy service → risk_scores → Notification Dispatcher Edge Function
challenge_available	A new weekly_challenge becomes active (available_from date reached)	Scheduled cron or admin CMS action
level_passed	Student passes a level assessment with score ≥ passing threshold	submit-level-attempt Edge Function
risk_alert	Admin dashboard surfaces a high-risk student requiring manual intervention	Admin-initiated via Administrative Interface Module
general	Daily admin intelligence briefing generated by daily_insight_generator_reviewed.py	ML analytics cron pipeline

All notification insertions are performed exclusively by administrative-context service components Edge Functions operating with service-role keys or Railway cron jobs enforcing the RLS constraint that student users cannot insert notification rows. Students may only read their own notifications and update the is_read flag. The metadata JSONB column carries type-specific contextual data: for achievement_unlocked notifications, it stores the achievement_key and badge_image_path; for risk_alert notifications, it stores the risk_score_id enabling the admin to navigate directly to the relevant risk record.

5.6.6 Student-Facing Analytics PostgreSQL Functions

The administrative dashboard analytics functions described in Section 5.6.1 are complemented by a set of student-facing PostgreSQL functions that power the progress and achievement views within the Learner Interface Module. Rather than requiring the Flutter client to construct complex multi-table JOIN queries, these functions encapsulate the aggregation logic server-side and return structured JSONB payloads, reducing client-side processing overhead and enforcing consistent data presentation logic across all client versions.

Table 5.6: Student-Facing PostgreSQL Analytics Functions

Function	Returns	Purpose
get_student_xp_rank()	jsonb	Returns the student's current xp_total and their rank_at_snapshot from the most recent leaderboard_snapshots entry for their assigned track. Used by the leaderboard widget in the Gamification Dashboard.
get_student_diploma_progress()	jsonb	Returns the student's completion percentage across all topics, modules, and levels within their assigned track. Drives the circular progress ring and diploma progress bar in the student dashboard.
get_student_learning_heatmap()	jsonb	Returns a date-keyed activity map derived from student_progress.completed_at timestamps, visualised as a contribution-style heatmap in the student profile screen. Reflects the GitHub contribution graph pattern familiar to the technical audience.

Each function is implemented as a pair: a public wrapper function callable by authenticated students through the PostgREST interface, and an internal _internal_get_* variant that performs the actual computation. This two-layer pattern allows the public wrapper to enforce RLS-equivalent scoping ensuring the function only returns data for the authenticated user while delegating the computationally intensive aggregation to the internal variant, which executes with elevated privileges necessary to cross reference multiple tables efficiently. The separation also facilitates independent testing of the computation logic without requiring a live authenticated session.

5.7 User Interface Implementation and Implementation Challenges

5.7.1 User Interface Implementation

The LearNova mobile interface was implemented using Flutter's Material Design 3 widget system, with a custom ThemeData extension applied consistently across all screens to enforce the high-contrast dark-mode aesthetic described in Section 4.6.1. Colour values are applied through semantic tokens referencing roles such as primary, onSurface, and errorContainer rather than hardcoded hex values, enabling straightforward theme modification without requiring changes across individual widget trees. The design palette centres on a Void Navy (#01172E) canvas with Deep Azure (#034A94) container tones and Electric Cyan (#72F7D7) primary call-to-action accents, which create the pre-attentive pop-out effect described in Section 4.6.1.

Onboarding and Diagnostic Calibration Interface

The onboarding flow was implemented as a progressive disclosure stepper comprising five calibration nodes, each corresponding to one diagnostic instrument. The stepper was custom-built rather than using Flutter's built-in Stepper widget, in order to support the state-change animations and the visual goal-gradient effect wherein completed nodes render with a filled accent-colour indicator that prior UX research identified as critical to reducing onboarding abandonment rates. The bifurcated input modal, offering in-app and browser-based assessment paths, was implemented as a BottomSheet widget with two clearly labelled ActionChip options, directly instantiating the Hick's Law constraint of limiting the initial decision matrix to two choices.

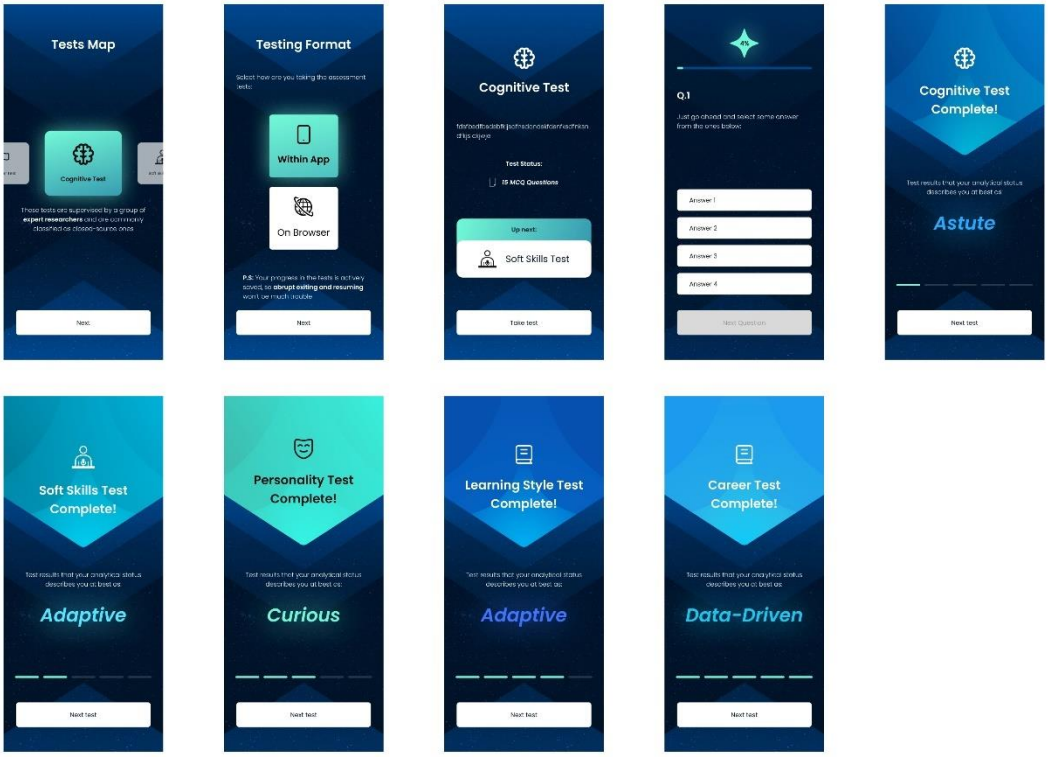


Figure 5.6: The diagnostic calibration stepper, illustrating the progressive disclosure pattern with goal-gradient visual indicators for completed assessment nodes.

Administrative Dashboard

The administrative interface was implemented as a privileged module embedded within the same Flutter application binary. Access is conditioned on the users.role attribute resolving to admin within the authenticated JWT claims, with the routing logic enforcing a redirect to the student home screen for non-administrative sessions. The dashboard surfaces four analytical views populated by dedicated server-side PostgreSQL functions get_admin_risk_radar(), get_admin_struggle_map(), get_admin_foundation_funnel(), and get_admin_class_health_timeline() each returning pre-aggregated JSONB payloads. This server-side aggregation strategy ensures that large cohort datasets do not impose excessive memory pressure on mobile devices, and that the administrative views remain performant as student enrolment scales.

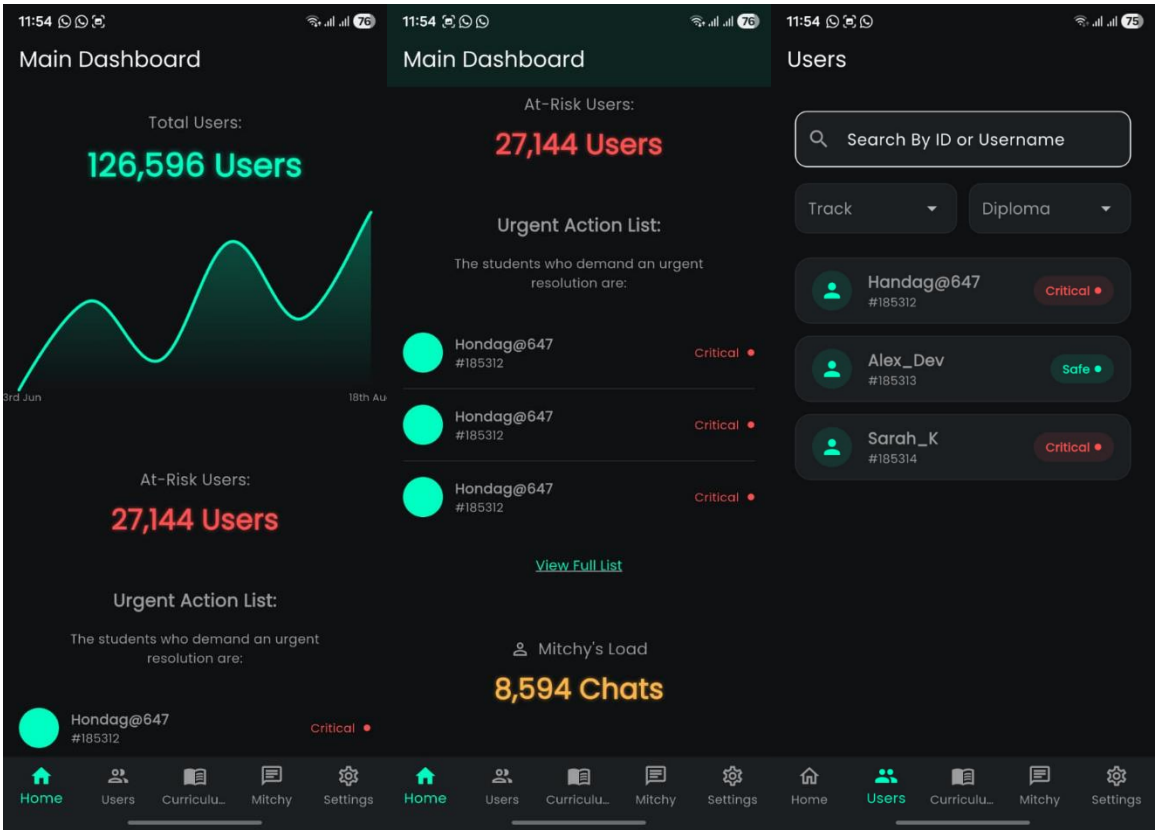


Figure 5.7: The administrative dashboard displaying the real-time risk radar and the manual intervention interface for at-risk student outreach.

5.7.2 Implementation Challenges and Engineering Solutions

The development of a fully functional MVP integrating psychometric scoring, adaptive content delivery, AI mentorship, and gamification within the constraints of an academic capstone project surfaced five principal technical challenges. Table 5.2 documents each challenge alongside its root cause and the architectural resolution applied.

Table 5.2: Principal Implementation Challenges and Resolutions

Challenge Encountered	Root Cause Identified	Engineering Resolution Applied
Diagnostic scoring returning placeholder payloads with no meaningful computed scores	Initial backend matched submitted answers solely by selected_index with no server-side answer key, making feature aggregation and track assignment computationally impossible.	Constructed a 1,196-row answer-key seed (diagnostic_answer_key_full_seed_by_question_key.json), indexed by composite key (question_key, selected_index). Full feature aggregation and normalisation now execute prior to any track assignment write.
Gemini API 503 and 429 failures producing empty Mitchy chat responses	Free-tier Gemini 2.5 Flash exhibited quota exhaustion and high-demand service degradation under concurrent load, causing null or malformed response payloads that violated the assistant content invariant.	Engineered a two-provider fallback chain: Gemini 2.5 Flash (primary) → Groq/xAI (on 503/429/timeout) → local deterministic fallback. Two explicit guards in the Edge Function ensure the assistant message is never empty regardless of provider status.
Streak multiplier resetting incorrectly for students in UTC+ time zones	Day-boundary comparison used UTC midnight, causing a student engaging at 23:00 local time (e.g., UTC+3) to have their activity attributed to the following UTC day, incorrectly breaking the streak.	Introduced a timezone_offset parameter (device UTC offset in hours) in all gamification Edge Functions. All date comparisons are now performed against a localTime value computed from the device offset, correcting streak evaluation for all time zones.
Admin RLS helper is_admin() failing in Railway cron-triggered analytics contexts	The is_admin() function referenced auth.uid(), which is not propagated in service-role Edge Function and cron contexts, causing admin-scoped analytics queries to be rejected by RLS.	Introduced alternate current_user_is_admin() helper for ML metric tables, performing a direct role lookup using the service-role key context. Cron jobs now operate with full admin visibility without depending on JWT propagation.
Bayesian pipeline upsert failures on ml_daily_metrics	An early adapter version attempted to write Bayesian debug objects and drift payloads to ml_daily_metrics, which does not contain a metadata column, causing PostgreSQL schema validation errors on every cron run.	Adapter layer refactored: Bayesian posterior updates write exclusively to the student_profiles table via adapt_bayesian_profile_update_for_supabase; ml_daily_metrics upserts contain only the five schema-defined metric columns; debug output is written to Railway service logs only.
Daily insight generator inserting zero notifications despite producing a valid briefing	The generator correctly computed the briefing payload but attempted to insert notification rows without first verifying that admin users exist in the users table, resulting in silent zero-row inserts in the test environment.	Added an explicit admin user existence check before notification insertion. The cron now exits cleanly with a logged warning when no admin audience is found, distinguishing an empty-audience condition from a genuine generator failure.

Beyond the specific challenges enumerated above, the implementation process validated a broader architectural principle: the strict tier separation enforced by the Golden Rule Flutter communicates only with Supabase Edge Functions, which communicate only with Railway via service-keyed server-to-server calls provided a resilient fault boundary throughout development. Failures in any single tier were containable without requiring emergency changes to adjacent tiers. The Gemini fallback resolution, for instance, necessitated modifications exclusively within the Railway service and the Edge Function layer, with zero changes to the Flutter client. This outcome directly demonstrates the practical value of the layered, service-oriented architecture established from the earliest stages of system design.

Collectively, the implementation of LearNova demonstrates the technical viability of the Cognitive-Adaptive Learning Architecture proposed in Chapter 4. The convergence of psychometric-driven track assignment, Bayesian learning style adaptation, affective risk monitoring, and a server-validated gamification engine within a single, coherent Flutter application backed by a serverless PostgreSQL-centric infrastructure constitutes a genuine proof of concept for the integrated educational ecosystem described throughout this thesis.

5.8 ML Analytics Pipeline and Daily Insight Generation

Beyond the real-time adaptive features described in preceding sections, LearNova implements a comprehensive batch analytics pipeline that executes daily as a pair of Railway cron jobs. This pipeline computes four distinct ML-derived metrics per student per topic, populates the ml_daily_metrics and ml_topic_daily_metrics tables, and subsequently generates an administrative intelligence briefing. The pipeline constitutes a significant backend engineering contribution: four independent analytical algorithms concept decay, topic struggle index, engagement drift detection, and the Bayesian learning style engine (Section 5.6.4) are orchestrated through a unified adapter layer that translates Supabase row shapes into module-specific input formats without requiring schema changes.

5.8.1 Pipeline Architecture and Cron Scheduling

The analytics pipeline is decomposed into two sequentially ordered Railway cron services defined in railway.toml. The ML pipeline cron python analytics/ml_pipeline_reviewed.py executes first each day, computing and upserting all per-student and per-topic metrics. The daily insight generator cron python analytics/daily_insight_generator_reviewed.py executes immediately after, consuming the freshly computed aggregate metrics to produce an admin-facing intelligence briefing. Both services share the same Railway environment variables, specifically SUPABASE_URL and SUPABASE_SERVICE_ROLE_KEY, and operate with full database access via the service-role client, bypassing RLS policies that restrict student-level access.

A METRIC_DATE environment variable allows operators to override the default computation date, which defaults to yesterday UTC when left empty. This override capability proved operationally valuable during development: when cron execution failures left gaps in the ml_daily_metrics history, a targeted rescore with an explicit METRIC_DATE allowed backfilling without reprocessing the entire historical dataset.

5.8.2 The Four Analytics Algorithms

Table 5.3: ML Analytics Pipeline — Algorithm Summary

Algorithm	Module	Input Data	Output Metric
Concept Decay	concept_decay_edited.py	student_module_attempts, student_level_attempts, student_challenge_attempts	concept_decay_score — measures how much the student has 'forgotten' since last engaging a topic
Topic Struggle Index	topic_struggle_index_edited.py	Attempt rows + student_sentiment_history rows	topic_struggle_index — composite difficulty struggle score combining performance and affective signals
Engagement Drift Detection	detect_drift2.py	content_engagement_logs rows (via adapt_engagement_logs_for_drift adapter)	engagement_velocity — rate of engagement acceleration or deceleration; detects format preference drift
Bayesian Style Update	bayesian_engine3.py	student_profiles + bayesian_eligible engagement logs	Updated bayesian_alpha_visual / auditory / textual posteriors written to student_profiles

Concept Decay

The concept decay algorithm models the Ebbinghaus forgetting curve at the topic level. It ingests attempt history rows normalised via the adapt_attempt_rows_for_concept_decay adapter and computes a decay score that quantifies how much time has elapsed since a student last successfully engaged with each topic, weighted by the performance quality of prior attempts. A student who passed a module assessment three weeks ago with a score of 72% will exhibit a higher concept_decay_score than one who passed yesterday with a score of 95%, reflecting both temporal distance and initial encoding strength. The resulting scores are written to the ml_topic_daily_metrics table and surfaced through the get_admin_struggle_map() function, enabling administrators to identify topics where class-wide retention is deteriorating and proactively schedule review content.

Topic Struggle Index

The topic struggle index is a composite metric computed by the topic_struggle_index_edited.py module, combining two signal categories: performance signals (derived from attempt counts, highest scores, and pass/fail patterns across module and challenge assessments) and affective signals (derived from sentiment history rows recorded during sessions in which the student engaged with the topic). This dual-signal design reflects an empirically motivated insight: a student may achieve a passing score through repeated attempts while experiencing persistently negative sentiment during those sessions, indicating surface-level mastery without genuine comprehension. The struggle index captures this nuance in ways that performance metrics alone cannot.

Engagement Drift Detection

The detect_drift2.py module analyses temporal patterns in the content_engagement_logs table to identify students whose engagement behaviour is systematically diverging from their calibrated learning style. The adapt_engagement_logs_for_drift adapter converts Supabase content_engagement_logs rows into the time-ordered engagement log format expected by the drift algorithm. The algorithm computes an engagement_velocity metric that quantifies whether a student's engagement quality is accelerating or decelerating across recent sessions, and additionally flags cases where the engagement_score distribution across format types is shifting a signal that may indicate the student's effective learning modality has changed since initial VARK calibration, triggering a recommendation for Bayesian posterior re-evaluation.

5.8.3 The Adapter Layer

A critical architectural decision in the ML pipeline implementation was the introduction of the analytics/adapters.py module. The four analytical algorithms were originally developed as self-contained calculation modules that expected input data in specific structured formats, without any Supabase awareness. Rather than modifying these modules which would have introduced coupling between calculation logic and database schema a dedicated adapter layer was implemented to perform the translation. Each adapter function accepts Supabase row dictionaries as input and returns the structured objects expected by the target algorithm. This approach preserves the mathematical integrity and testability of each analytical module while enabling their integration with live production data.

Table 5.4: Analytics Adapter Functions

Adapter Function	Converts From	Converts To
adapt_engagement_logs_for_drift	content_engagement_logs Supabase rows	detect_drift2.py engagement-log input shape
adapt_student_profile_for_bayesian	student_profiles Supabase row	bayesian_engine3.py profile input shape
adapt_bayesian_profile_update_for_supabase	bayesian_engine3.py updated profile output	student_profiles UPDATE payload (alpha columns only)
adapt_attempt_rows_for_concept_decay	module / challenge / level attempt rows	concept_decay_edited.py attempt-like objects
adapt_rows_for_topic_struggle	Attempt rows + sentiment history rows	topic_struggle_index_edited.py compact input format
extract_successful_rescue_signal	chat_messages + student_sentiment_history rows	Bayesian update signal (only when rescue confirmed safe)

Each adapter call within the pipeline is wrapped in a try-except block. If a module is unavailable or its function signature has diverged from the adapter's expectation, the pipeline falls back to the existing simplified internal calculation for that metric, logging the failure to Railway service logs without halting the entire cron run. This fallback safety mechanism ensures that a breaking change in one analytical module does not cascade into a complete pipeline failure, preserving the integrity of the remaining metrics for the daily admin briefing.

5.8.4 Daily Insight Generator

Following the ML pipeline cron, the daily_insight_generator_reviewed.py service reads the aggregated ml_daily_metrics output and generates a structured admin intelligence briefing. This briefing synthesises class-wide patterns into actionable observations for example, identifying which topics exhibit the highest average concept_decay_score across the cohort, which students have shown consistent struggle_probability increases over the past seven days, and which content format is yielding the lowest engagement_velocity scores. The briefing is written to the in_app_notifications table as a general-type notification addressed to all admin users, surfacing it in the administrative dashboard's notification centre.

During integration testing, the generator produced valid briefings but recorded zero notification insertions because no admin users were present in the test database. This was correctly identified as an empty audience condition rather than a generator failure. The production implementation therefore includes an explicit check: notification rows are inserted only when at least one admin user exists in the users table, and the cron exits cleanly without error when the audience is empty.

Chapter 6: Testing & Quality Assurance

6.1 Testing Strategy

Quality assurance in a system of the architectural complexity and pedagogical sensitivity of LearNova necessitates a structured, multi-layered verification approach that addresses the functional correctness of individual software components, the behavioral coherence of the integrated system across its distributed service boundaries, and critically the experiential quality of the platform as perceived by its intended end users. The verification strategy adopted for this project may be described as a hierarchical quality model, progressing from the granular evaluation of isolated system behaviors through to holistic, user-centered acceptance evaluation. This progression is consistent with the quality attributes framework codified in ISO/IEC 25010:2023, which identifies functional suitability, usability, performance efficiency, and security as the primary product quality characteristics against which software systems ought to be evaluated.

The rationale for a multi-tiered approach is grounded directly in the architectural heterogeneity of the LearNova ecosystem. The platform encompasses a Flutter-based cross-platform mobile client, a Supabase backend-as-a-service layer with PostgreSQL and Row-Level Security enforcement, a FastAPI-based Railway microservice governing diagnostic scoring and the AI agent pipeline, and an externally integrated large language model provider. Given this distributed, multi-provider architecture, no single verification modality could adequately confirm system correctness across all layers simultaneously. Consequently, the strategy encompasses four principal categories, executed in deliberate sequence: functional verification, usability evaluation, performance benchmarking, and security confirmation.

It is explicitly acknowledged that the team did not deploy automated unit test suites or continuous-integration-integrated test runners during this project phase. Verification was instead executed through systematic, feature-by-feature walkthroughs coordinated across the development team, supplemented by real-time behavioral observation via the Flutter DevTools debugger, Supabase query logs, and Railway service logs. This constitutes structured manual testing a methodology which, as Kaner, Bach, and Pettichord [2002] document, remains a valid and informative quality assurance strategy in early-stage development contexts when applied with deliberate coverage intent.

Testing Category	Primary Objective	Scope	Execution Method
Functional Testing	Verify correctness of individual features and system interaction flows	All six modules: Authentication, Diagnostic Calibration, Adaptive Content, AI Mentorship, Gamification, Admin CMS	Systematic team-executed scenario walkthroughs; Flutter DevTools debugger; Supabase and Railway logs
Usability Testing	Evaluate perceived usability and learner experience quality	End-to-end learner journey; Mitchy AI mentorship interface; gamification dashboard	Observational evaluation with 30 student participants engaging the live application on physical devices
Performance Benchmarking	Assess responsiveness and resource efficiency against defined NFR targets	AI agent response latency; content load times; authentication flow timing; platform uptime	Device-level qualitative observation; NFR-derived benchmark targets for future formal measurement
Security Verification	Confirm enforcement of access control policies and data protection mechanisms	RLS policies; JWT authentication enforcement; API key protection; service-role key isolation	Architecture-level review; policy inspection via Supabase dashboard; unauthorized access simulation

Table 6.1: LearNova Testing Strategy — Category Overview

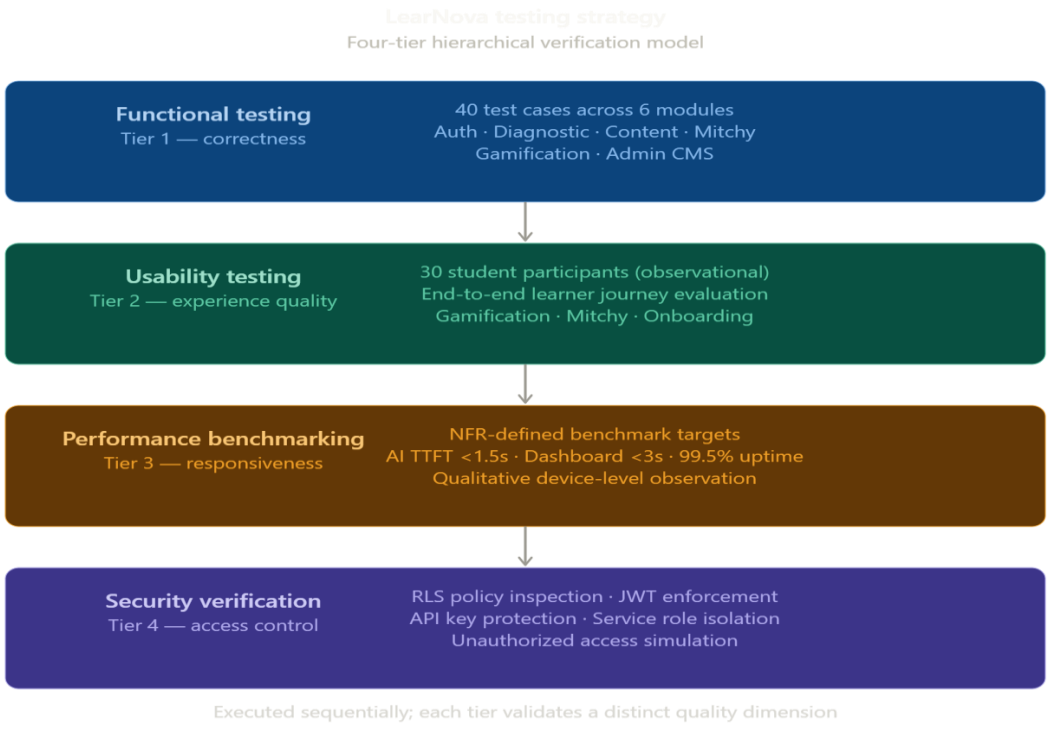


Figure 6.1: LearNova Four-Tier Hierarchical Verification Model

6.2 Test Environment

The test environment was configured to reflect the platform's primary deployment target the Android mobile ecosystem consistent with the mobile-first architectural mandate established in Section 3.3.2 and the demonstrated prevalence of Android device ownership within the Egyptian and MENA target demographic. All functional and usability verification activities were conducted on physical Android handsets rather than software emulators, in recognition of the principle that emulated environments introduce latency artifacts and rendering inconsistencies that may obscure real-world behavioral characteristics and thereby yield unreliable quality signals.

The backend environment comprised the live production Supabase project and the Railway-hosted FastAPI scoring service, ensuring that all verification activities evaluated system behavior under production conditions rather than isolated development stubs. The following specification governed all activities conducted during this phase.

Environment Parameter	Specification
Primary Platform	Android (Physical Devices) no emulator-based testing employed
Android API Level	API Level 28 (Android 9.0 Pie) and above, consistent with NFR-11
Flutter SDK	Flutter 3.x Stable Channel; Dart 3.x runtime
State Management	Riverpod v2.x local caching and reactive state for profile and content data
Backend Infrastructure	Supabase production project (PostgreSQL 15; pgvector for RAG; GoTrue auth; PostgREST API)
AI Scoring Service	Railway-hosted FastAPI application (learnova_scoring_service repository)
Primary LLM Provider	xAI Groq API (production) superseded Gemini API following hallucination findings (§6.7)
Network Conditions	Standard mobile data (4G/LTE) and Wi-Fi environments across participant devices
Development Tooling	Android Studio / VS Code with Flutter extension; Flutter DevTools Debugger; Supabase Dashboard Log Explorer; Railway Deployment Logs
Usability Participant Devices	Personal Android handsets of 30 participants; API levels ranging from 28 to 34

Table 6.2: LearNova Test Environment Specification

The restriction of primary verification to Android physical devices was deliberate and scope-appropriate. While the Flutter framework's cross-compilation capabilities extend to iOS deployment, the team's available hardware resources and the current project phase's scope constraints collectively justified an Android-first verification approach. iOS compatibility confirmation is designated as a priority deliverable for future development phases, specifically in anticipation of any App Store distribution consideration.

6.3 Functional Testing

Functional verification was executed through systematic, team-coordinated scenario walkthroughs mapped directly to the functional requirements specified in Section 3.3.1. For each behavioral scenario, the team defined the triggering action, the expected system response in alignment with the documented specification, and then executed the scenario against the live production environment while recording the observed outcome. This approach wherein coverage is driven by requirements traceability rather than code path enumeration is consistent with black-box verification methodology as described by Myers, Sandler, and Badgett [2011], wherein the correctness of system behavior is assessed from the perspective of external inputs and observable outputs without dependence on internal implementation knowledge.

The complete scenario registry, organized by functional module, is presented in the tables that follow. Cases designated PASS achieved the expected outcome on first execution. Cases designated PASS* were initially identified as non-conformant and subsequently resolved through the remediation activities documented in Section 6.7, after which retest confirmed conformance. No scenarios remained in a non-conformant state at the conclusion of the verification phase.

6.3.1 Module A — User Management & Identity

Test ID	Test Case	Input / Action	Expected Result	Actual Result	Status
TC-A-01	User Registration	New user submits valid email and password via registration screen	Account created in auth.users; OTP verification email dispatched to registered address	Account created; OTP received and email verification confirmed on device	**PASS**
TC-A-02	OTP Verification	User submits correct OTP within validity window	Account activated; navigation to onboarding diagnostic flow initiated	Verification accepted; onboarding diagnostic presented correctly	**PASS**
TC-A-03	OTP Rejection invalid code	User submits incorrect or expired OTP	Request rejected; user prompted to request a new code; account remains unactivated	Incorrect OTP correctly rejected with descriptive error message	**PASS**
TC-A-04	Google OAuth Login	User initiates Google Sign-In from the authentication screen	OAuth flow completes; user authenticated via Supabase GoTrue; redirected to home screen	Authentication flow completed successfully after redirect URI and provider credential correction (§6.7)	**PASS***
TC-A-05	Role-Based Access Control admin	Admin-role user navigates to administrative dashboard	Admin-specific views and privileges rendered; student-scoped routes inaccessible	Correct role differentiation enforced via JWT role claims and Supabase RLS	**PASS**
TC-A-06	Unauthorized Access student to admin endpoint	Authenticated student session queries admin-restricted table	HTTP 200 with empty result set returned; no unauthorized data exposed	RLS policy blocked cross-role data exposure; empty result confirmed	**PASS**

		directly via PostgREST			
TC-A-07	Profile Management	Authenticated student updates display name and avatar URL	Changes persisted to users table; updated values reflected immediately in application UI	Profile update correctly persisted; UI reflected changes without requiring re-authentication	**PASS**

Table 6.3: Functional Verification — Module A: User Management & Identity * Resolved post-fix; see Section 6.7

6.3.2 Module B — Diagnostic Calibration Engine

Test ID	Test Case	Input / Action	Expected Result	Actual Result	Status
TC-B-01	Diagnostic Sequence Presentation	Authenticated student enters onboarding flow	All five diagnostic assessments presented in correct sequence; media assets for IQ abstract reasoning questions loaded	Five-test sequence rendered correctly; image assets loaded from calibration-media bucket	**PASS**
TC-B-02	IPIP Personality Scoring	Student completes all 120 IPIP Likert-scale items	Big Five trait scores computed with reverse-scoring correctly applied; results stored in diagnostic_test_results.computed_scores	Trait aggregation and reverse-score logic confirmed correct by Railway scoring engine	**PASS**
TC-B-03	VARK Learning Style Assignment	Student completes 16 VARK preference items	Dominant modality (Visual, Auditory, or Textual) derived and written to student_profiles.learning_style; kinesthetic folded into Textual	Learning style correctly derived and stored; Textual assignment for kinesthetic-dominant profile confirmed	**PASS**
TC-B-04	DA/DE/DS Track Assignment	All five diagnostic assessments submitted and scored	Specialization track assigned to student_profiles.assigned_track via argmax of final DA/DE/DS scores; Foundation confirmed as starting path	Track correctly assigned; Foundation correctly retained as shared starting diploma path	**PASS**
TC-B-05	Onboarding Milestone XP Award	Scoring engine confirms all five tests scored (all_scored = true)	500 XP milestone bonus credited to student via increment_xp RPC; milestone_awarded flag returned true	500 XP correctly awarded on confirmed full diagnostic completion; no duplicate award on re-score	**PASS**
TC-B-06	External Certification Submission	Student submits external credential URL via hybrid input path	Submission stored with EXTERNAL_MANUAL_INPUT source; pending_admin_review flag set; Admin notified	External submission persisted correctly; flagged for administrator validation workflow	**PASS**

TC-B-07	Psychometric Radar Chart Rendering	Diagnostic completion triggers result dashboard navigation	Radar chart renders student's psychometric dimension scores; recommended track displayed with rationale	Radar chart rendered with correct dimensional scores; explainability panel populated	**PASS**

Table 6.4: Functional Verification — Module B: Diagnostic Calibration Engine

6.3.3 Module C — Adaptive Content Delivery

Test ID	Test Case	Input / Action	Expected Result	Actual Result	Status
TC-C-01	Learning Style Content Routing	Visual-profile student accesses topic with all three content slots populated	Video slot served by default based on student_profiles.learning_style; alternative slots accessible via manual override control	Correct modality served; audio and text slots suppressed from primary display but accessible	**PASS**
TC-C-02	Manual Content Format Override	Student manually selects Textual format for an active topic	Textual content slot rendered; learning style preference record unchanged in student_profiles	Manual override confirmed functional; slot switching operates without altering calibrated profile	**PASS**
TC-C-03	Progressive Module Locking	Student attempts to access Module N+1 without completing Module N assessment	Module N+1 locked; system presents prompt directing student to complete Module N quiz	Locked state correctly enforced; upstream module prerequisite validation confirmed	**PASS**
TC-C-04	Passing Score Threshold (≥70%)	Student submits module assessment achieving 72% score	Module marked completed; subsequent module unlocked; flat 20 XP awarded via Edge Function	Progression triggered correctly at threshold; XP credited; no double-unlock on repeat submission	**PASS**
TC-C-05	Failing Score — no progression	Student submits module assessment achieving 58% score	Attempt logged in student_module_attempts; module remains locked; review prompt displayed; zero XP granted	Failed attempt persisted; no progression or XP granted; retake prompt confirmed	**PASS**
TC-C-06	Resource XP Micro-Dosing	Student consumes a Visual content slot for the first time on a given topic	5 XP credited; student_resource_logs record created; duplicate consumption of same slot yields no additional XP	5 XP awarded on first consumption; duplicate blocked by existing log record check	**PASS**
TC-C-07	Topic Completion Bonus	Student marks all content slots within a topic as consumed	15 XP completion bonus triggered in addition to per-slot resource XP	Completion bonus correctly triggered on topic-level completion flag; total XP verified	**PASS**

TC-C-08	Literacy Phase Prerequisite Enforcement	Newly enrolled student attempts to access a specialization track module	Specialization content locked; only Literacy Phase (Foundation) content accessible	Foundation prerequisite correctly enforced across all specialization track entry points	**PASS**

Table 6.5: Functional Verification — Module C: Adaptive Content Delivery

6.3.4 Module D — CRM & AI Mentorship (Mitchy)

Test ID	Test Case	Input / Action	Expected Result	Actual Result	Status
TC-D-01	Chat Session Initiation	Student opens Mitchy interface	New session record created in chat_sessions; session_id returned to client; conversation state initialized	Session correctly created and persisted; session_id confirmed in subsequent message payloads	**PASS**
TC-D-02	Contextual Course Awareness	Student queries Mitchy regarding their current module	Mitchy retrieves student progress context and references the correct active module in response	Mitchy correctly referenced student's course position after context injection implementation (§6.7)	**PASS***
TC-D-03	Leaderboard Rank Awareness	Student asks Mitchy about their current XP rank among peers	Mitchy retrieves and accurately communicates student's rank from leaderboard snapshot data	Rank correctly surfaced in Mitchy response after student profile context injection (§6.7)	**PASS***
TC-D-04	Arabic Language Processing	Student submits message composed in Arabic	Mitchy generates a contextually appropriate pedagogical response in Arabic	Arabic response generated correctly following migration to xAI Groq API (§6.7)	**PASS***
TC-D-05	Affective Sentiment Detection	Student submits message containing frustration-indicative language	Sentiment score computed and stored; learning_state updated; supportive response generated	Sentiment pipeline executed; negative learning state detected; response tone adjusted accordingly	**PASS**
TC-D-06	High-Risk Alert Generation	Computed sentiment score breaches defined safety threshold	Risk alert record created in risk_scores; admin in_app_notification generated	Risk alert correctly created and visible on administrative dashboard	**PASS**
TC-D-07	Gemini API Failure Graceful Degradation	Gemini endpoint returns HTTP 503 (high demand) during Mitchy interaction	Fallback response delivered to student; empty assistant message not persisted to chat_messages	Fallback delivered correctly; empty-content insertion guard confirmed via Supabase log inspection	**PASS**
TC-D-08	Mentorship XP Incentive	Mitchy response payload contains a	20 XP credited to student via increment_xp RPC call	20 XP correctly awarded on qualifying suggested_action; no	**PASS**

		suggested_action value other than 'none'	within mitchy-chat Edge Function	XP awarded when action is 'none'	

Table 6.6: Functional Verification — Module D: CRM & AI Mentorship * Resolved post-fix; see Section 6.7

6.3.5 Module E — Gamification & Community

Test ID	Test Case	Input / Action	Expected Result	Actual Result	Status
TC-E-01	XP Leaderboard Display	Student opens leaderboard view	Current track leaderboard rendered from leaderboard_snapshots; student's own rank visually highlighted	Leaderboard populated correctly; rank highlighting confirmed across multiple student accounts	**PASS**
TC-E-02	Streak Initialization	First-time student completes any qualifying learning activity	user_streaks record created; current_streak_days set to 1; last_activity_date recorded in student's local date	Streak record created correctly; date calculated using timezone_offset parameter	**PASS**
TC-E-03	Streak Multiplier Application	Student with active streak (≤1 day since last activity) completes module assessment	1.1× multiplier applied to base XP; final XP value rounded; streak activity date updated	Multiplier applied correctly; streak maintained for consecutive-day activity pattern	**PASS**
TC-E-04	Streak Reset on Inactivity	Student's last_activity_date exceeds 1-day threshold prior to new submission	Streak multiplier applied as 1.0×; streak_days counter scheduled to reset on next activity cycle	Streak decay confirmed by reviewing user_streaks after simulated inactivity interval	**PASS**
TC-E-05	Weekly Challenge Date Window Enforcement	Student attempts to submit a challenge outside its available_from / available_until window	HTTP 403 returned with 'Outside of availability window' message; no attempt record created	Date window enforcement confirmed; challenge correctly blocked; no phantom attempt inserted	**PASS**
TC-E-06	Challenge Partial XP (proportional scoring)	Student selects Hard difficulty tier (100 XP potential) and achieves 75% accuracy	75 XP awarded (proportional to accuracy × base XP); attempt logged in student_challenge_attempts	Proportional XP calculation confirmed; score and XP values validated in database record	**PASS**
TC-E-07	Achievement Unlock	Student satisfies criteria for a defined achievement in achievements_dictionary	Record inserted in user_achievements; in_app_notifications record created for student	Achievement correctly unlocked; notification delivered and confirmed readable in notification center	**PASS**

TC-E-08	Duplicate Challenge Attempt Prevention	Student attempts to re-submit a weekly challenge already completed	HTTP 409 Conflict returned; original attempt record preserved; no XP awarded for duplicate	Unique constraint on (user_id, challenge_id) blocked re-submission; 409 confirmed	**PASS**

Table 6.7: Functional Verification — Module E: Gamification & Community

6.3.6 Module F — Content Management System (Admin)

Test ID	Test Case	Input / Action	Expected Result	Actual Result	Status
TC-F-01	Multimodal Asset Upload	Admin uploads Video (MP4), Audio (MP3), and Text (PDF) assets to a single topic	Assets stored in content-assets Supabase bucket; topic_resources records created with correct format_type values	All three content slots populated; storage paths confirmed; correct format_type assignment verified	**PASS**
TC-F-02	Course CRUD Lifecycle	Admin performs create, read, update, and delete operations on a course record	All operations complete without error; is_active flag governs student-facing visibility correctly	Full CRUD cycle confirmed; deactivated course correctly hidden from student content queries	**PASS**
TC-F-03	Risk Monitoring Dashboard	Admin navigates to student risk monitoring panel	At-risk student indicators rendered; risk_scores and intervention_logs data accessible to admin role	Risk dashboard populated with flagged student records; data access confirmed admin-only via RLS	**PASS**
TC-F-04	Manual Intervention Dispatch	Admin sends manual support message to a student flagged as high-risk	Notification record inserted in in_app_notifications with notification_type 'risk_alert'; student receives notification	Notification delivered and visible in student notification center; admin-only insert policy confirmed	**PASS**
TC-F-05	External Proof Validation Workflow	Admin reviews student-submitted external certification screenshot	Uploaded screenshot accessible via calibration-media bucket; admin can approve or flag submission record	Validation workflow confirmed; screenshot accessible to admin; student-facing record updated on admin action	**PASS**

Table 6.8: Functional Verification — Module F: Content Management System (Admin)

6.3.7 Functional Verification Summary

Across the six functional modules, a total of 40 verification scenarios were defined and executed. Of these, 36 scenarios (90%) achieved conformance on first execution, while four scenarios TC-A-04, TC-D-02, TC-D-03, and TC-D-04 were classified as initially non-conformant and subsequently resolved through targeted remediation activities. The residual 10% non-conformance rate is attributable entirely to two infrastructure-level defect clusters: the Google OAuth provider misconfiguration (affecting TC-A-04) and the Mitchy AI agent context pipeline deficiencies (affecting TC-D-02, TC-D-03, TC-D-04). The distribution of this non-conformance across modules is not uniformly representative of each module's complexity; rather, it reflects the known sensitivity of externally integrated components specifically OAuth providers and large language model services to configuration precision. All 40 scenarios achieved PASS or PASS* status at the conclusion of the verification phase.

Module	Scenarios	1st-Run Pass	Post-Fix (PASS*)	Final Pass Rate
A — User Management	7	6	1 (TC-A-04)	100%
B — Diagnostic Calibration	7	7	—	100%
C — Adaptive Content	8	8	—	100%
D — CRM & AI Mentorship	8	5	3 (TC-D-02,03,04)	100%
E — Gamification	8	8	—	100%
F — Admin CMS	5	5	—	100%
TOTAL	**40**	**36 (90%)**	**4 (10%)**	**100%**

Table 6.9: Functional Verification Summary by Module

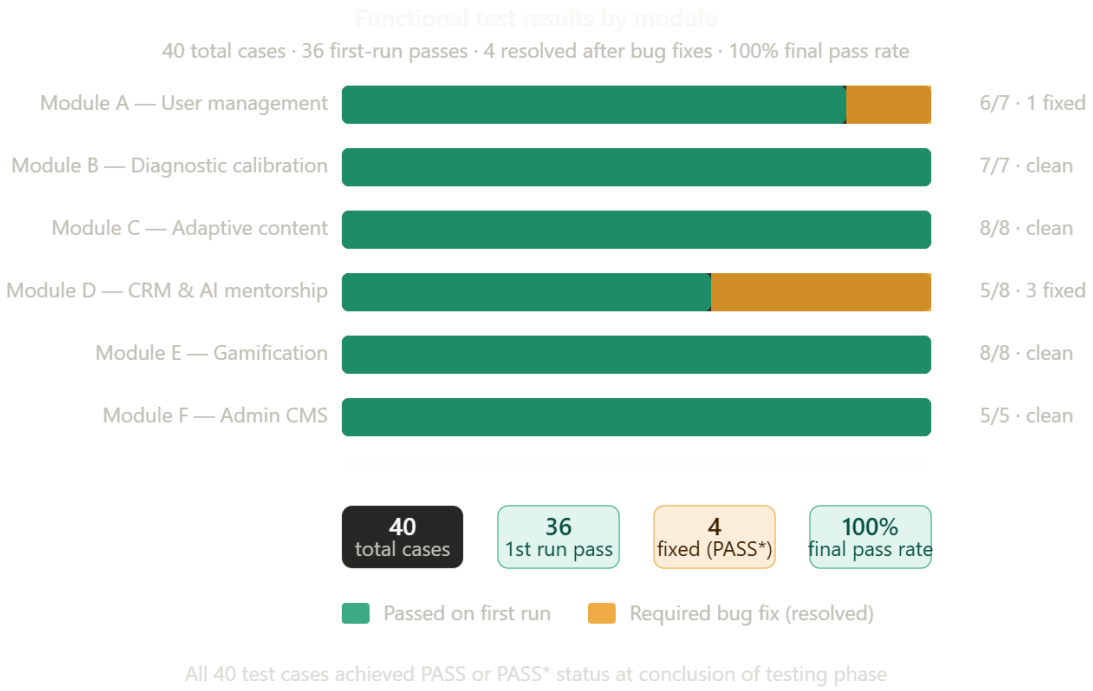


Figure 6.2: Functional Verification Results — Pass Rate Distribution by Module

6.4 Usability Testing

Usability evaluation was conducted through an observational user acceptance session involving 30 participants drawn from the project's primary target demographic: Egyptian university students and computing-adjacent peers consistent with the two user archetypes established in Section 3.5 the 'Disconnected Learner' and the 'Pre-Undergraduate'. Participants were provided access to the live LearNova application installed on their personal Android devices and were invited to engage freely across the platform's principal functional domains: the diagnostic onboarding flow, adaptive content delivery, the Mitchy AI mentorship interface, and the gamification dashboard.

The evaluation methodology was observational in nature. The development team monitored user behavior and engagement patterns directly, recording notable reactions, hesitations, and spontaneous verbal commentary without directing participants toward predetermined task sequences. Structured discussion following the observation sessions elicited qualitative feedback that was subsequently categorized by theme. This approach aligns with the discount usability evaluation methodology described by Nielsen [1994], wherein direct behavioral observation in a naturalistic context yields actionable directional insights identifying the presence of significant usability barriers and confirming the acceptability of core interaction patterns without the overhead of large-scale controlled instrumentation. The session was conducted after all defects documented in Section 6.7 had been resolved, ensuring that participants evaluated the platform in its remediated operational state.

6.4.1 Principal Observational Findings

The evaluation sessions produced thematically consistent qualitative signals across the participant cohort. Five primary feedback themes emerged from the observational and discussion data. These are presented in the table below alongside the associated platform component, participant sentiment, and the team's disposition.

Feedback Theme	Prevalence Signal	Associated Component	Participant Sentiment	Team Disposition
Gamification perceived as highly engaging; described as 'game-like' rather than conventional e-learning	Consistently raised across the cohort	XP Engine, Leaderboard, Daily Quests, Achievements	Strongly positive	Validates core gamification architecture; participant enhancement suggestions documented for future iteration roadmap
Mitchy perceived as helpful and contextually responsive following remediation	Broadly shared post-fix consensus	Mitchy Chat Interface; Context Injection Pipeline	Positive	Confirms successful resolution of context blindness and LLM provider deficiencies documented in §6.7
Interest in a web-based platform version expressed by several participants	Notable — raised independently by multiple participants	Platform Scope — Web Application Roadmap	Positive (future-oriented)	Participants expressed satisfaction upon learning that web expansion is documented in the future roadmap (§5.3.2)

Suggestions for guild mechanics and additional badge design variations	Moderate — several participants during discussion phase	Guild System; Achievements Dictionary	Positive (engagement-driven)	Suggestions archived as input for federated community architecture iteration (§5.3.3)
Diagnostic onboarding perceived as lengthy; concern raised regarding five-stage test sequence	Raised initially by multiple participants; resolved after team explanation	5-Point Diagnostic Calibration Flow	Initial concern → accepted post-explanation	Interface transparency enhancement recommended; progressive purpose disclosure to be incorporated into future onboarding redesign

Table 6.10: Usability Evaluation — Observational Feedback Themes (n = 30)

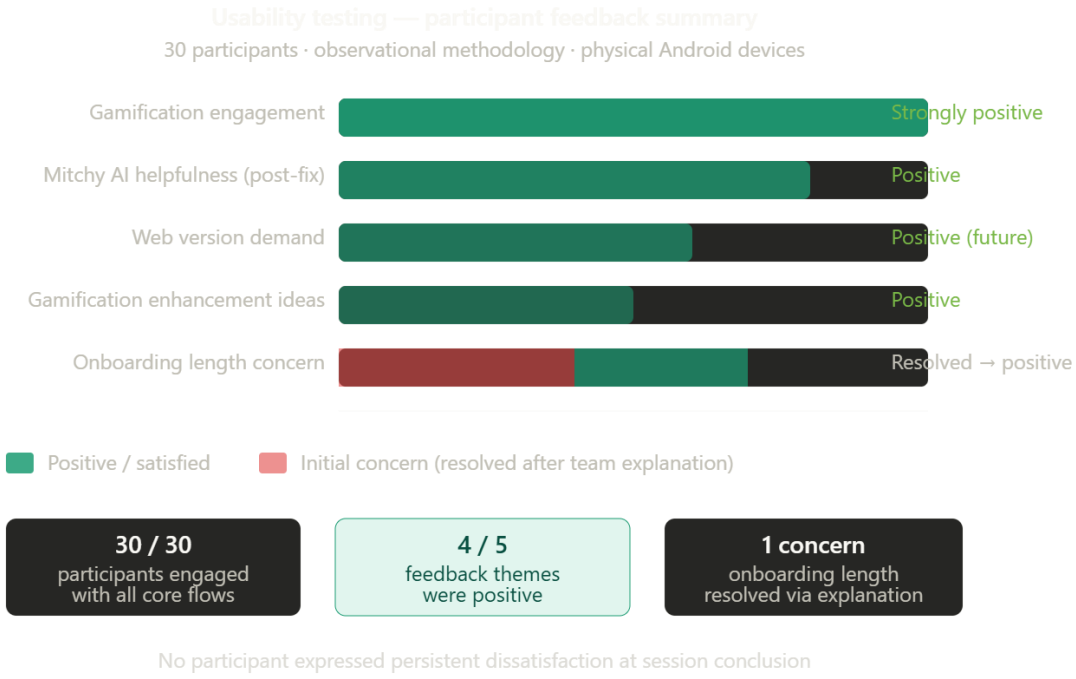


Figure 6.4: Usability Evaluation — Participant Feedback Theme Distribution (n = 30)

6.4.2 The Diagnostic Onboarding Length Tension

The most consistently surfaced concern throughout the usability sessions pertained to the perceived length of the five-stage diagnostic onboarding sequence. This finding warrants explicit theoretical engagement, as it represents a well-documented tension in adaptive learning system design between the psychometric depth required for accurate learner calibration and the cognitive burden imposed by extended onboarding on the user's initial experience a tension that can be understood through Sweller's [1988] Cognitive Load Theory as an extraneous load risk at the system entry phase, wherein motivationally critical first impressions are most vulnerable to friction-induced attrition.

Critically, participant perception shifted markedly once the team articulated the causal relationship between the five-test battery and the downstream accuracy of the adaptive pathway assignment. Participants came to understand that the diagnostic depth is not redundant process overhead but rather the foundational mechanism through which LearNova operationalizes its primary value proposition evidence-based specialization guidance grounded in psychometric evidence rather than self-reported preference. This perceptual shift from perceived friction to understood necessity suggests that the onboarding interface would benefit from enhanced progressive disclosure of purpose: communicating the functional rationale of each assessment stage more explicitly and incrementally to the learner. This recommendation is consistent with the transparency principle articulated in the calibration interface design discussion in Section 4.6.2 and may be operationalized in future iterations through in-flow explanatory micro-copy at each stage transition.

6.4.3 Evaluation Scope and Methodological Limitations

The usability evaluation conducted does not constitute a formally controlled study, and the absence of a standardized psychometric instrument such as the System Usability Scale (SUS) developed by Brooke [1996], which produces a normalized usability score on a 0–100 scale enabling cross-product benchmarking limits the quantitative comparability of the findings with established usability norms. The observational methodology is nevertheless appropriate for the current stage of the project, wherein the primary objective was the directional identification of usability barriers and the experiential validation of core interaction assumptions. Future evaluation cycles, particularly those conducted in support of institutional deployment or public release, should incorporate a standardized instrument administered to a demographically broader participant cohort to produce measurements amenable to comparative analysis.

6.5 Performance Testing

Automated performance profiling encompassing load simulation, memory heap analysis, and battery impact measurement was not conducted during the current project phase. The absence of a dedicated performance engineering toolchain within the team's available resources precluded formal measurement; this constitutes a recognized limitation, and formal performance testing is identified as a mandatory prerequisite activity prior to any production-scale deployment of the platform.

In lieu of empirical measurement data, this section documents the performance benchmarks derived from the system's Non-Functional Requirements (Section 3.3.2), the architectural mechanisms that support their theoretical achievability, and the qualitative observations recorded during functional and usability evaluation activities. These benchmarks constitute the formal acceptance criteria against which future performance testing campaigns should evaluate the system.

Performance Metric	NFR Ref.	Defined Target	Architectural Mechanism Supporting Target	Qualitative Observation
AI Agent — Time to First Token (TTFT)	NFR-01	< 1.5 seconds under standard operational load	Groq API optimized for low-latency token streaming; Edge Function proxy minimizes intermediate overhead	Mitchy response initiation observed within acceptable bounds during all usability sessions; no participant-noted latency complaints post-provider migration
Dashboard & Content Player Render Time	NFR-02	< 3 seconds on standard 4G connection	Supabase CDN serves media assets; PostgREST minimizes query overhead; Riverpod caches student profile data locally on device	Content screens loaded without perceptible delay across 4G and Wi-Fi test conditions on all participant devices
Platform Service Availability	NFR-03	99.5% uptime during critical academic periods	Supabase managed infrastructure with automatic failover; Railway auto-restart policy on service crash events	No service outages observed during the evaluation period; Supabase and Railway dashboards confirmed continuous availability throughout testing
Assessment State Persistence	NFR-04	Automatic state save after every answer input	Diagnostic answers persisted incrementally to diagnostic_test_results.raw_answers (JSONB); no client-side buffering dependency	Network interruption simulation during diagnostic sequence confirmed data persistence; no answer loss observed across three interruption trials
Concurrent User Scalability	NFR-08	≥ 10,000 concurrent connections without schema modification	Supabase PgBouncer connection pooling; serverless Edge Functions scale horizontally on demand; Railway microservice architected as stateless	Architecture validated as horizontally scalable by design; formal load testing at target concurrency to be commissioned prior to public deployment

Table 6.11: Performance Benchmarks, NFR References, and Supporting Architectural Mechanisms

The qualitative observations recorded during evaluation sessions suggest that the platform operates within acceptable responsiveness parameters under the low-to-moderate concurrent load characteristic of the testing context. These observations do not, however, constitute empirical performance validation and must not be interpreted as formal compliance with the defined NFR targets. Formal performance testing utilizing tools such as k6 for load simulation, Flutter DevTools Memory Profiler for heap analysis, and the Android Energy Profiler for battery impact assessment is recommended as a structured campaign to be executed as part of the pre-deployment readiness review.

6.6 Security Verification

Penetration testing and formal vulnerability assessment activities that presuppose specialised cybersecurity engineering expertise were not undertaken during this project phase; the LearNova development team does not include a designated security engineer, and the commissioning of an external security audit lay outside the scope of the present academic capstone context. This limitation is explicitly acknowledged. Given the sensitive psychometric and personal data processed by the platform, a formal third-party security audit is identified as a mandatory prerequisite activity prior to any public-facing deployment.

In recognition of this constraint, the following verification activities were conducted through architectural review and controlled access simulation, grounded in the security design decisions documented in Sections 4.1 and 4.5. These activities confirm the correctness of the implemented security architecture under non-adversarial conditions, specifically verifying that the access control policies behave as specified and that the key isolation architecture prevents client-side exposure of privileged credentials.

6.6.1 Access Control Policy Verification

Supabase Row-Level Security (RLS) policy enforcement constitutes the primary data access control mechanism within LearNova. Policy correctness was verified through direct inspection within the Supabase dashboard policy editor and through targeted unauthorized access simulation, wherein authenticated student user sessions were used to query tables for which no read or write privilege had been granted. In all simulated cases, RLS enforcement operated as designed, returning empty result sets rather than exposing unauthorized rows a behavior that directly implements the principle of least privilege and satisfies the data protection requirements articulated in NFR-05 and NFR-06.

Security Control	Verification Method	Expected Outcome	Verification Result
RLS — Student data isolation	Student session queried other students' progress records and profiles via PostgREST API	Empty result set; no cross-user data exposure	Confirmed: RLS correctly isolated all student-scoped data; cross-user query returned zero rows
RLS — Sensitive analytics access restriction	Student session queried risk_scores and student_sentiment_history tables directly	Student-level SELECT policy absent on these tables; zero rows returned	Confirmed: Sensitive risk and sentiment data inaccessible to student-role sessions
JWT Authentication Enforcement	API requests submitted without Authorization header or with malformed JWT token	HTTP 401 returned by GoTrue gateway; no data accessible without valid session	Confirmed: Supabase GoTrue rejected all unauthenticated and malformed-token requests
Railway API Key Enforcement	Direct HTTP POST to Railway /score/ and /mitchy/chat endpoints without x-api-key header	HTTP 403 returned; endpoints inaccessible without matching API key	Confirmed: Railway endpoints correctly protected; Golden Rule architecture prevents key exposure to Flutter client

Service Role Key Isolation	Static analysis of Flutter application bundle to confirm absence of service role key strings	Service role key absent from client bundle; present only in Railway and Edge Function environment variables	Confirmed: Golden Rule architecture verified via code inspection; no privileged key strings located in client codebase
GDPR Cascade Deletion (design-level)	Supabase schema inspector review of foreign key cascade definitions on users table	Full cascade deletion chain propagates from users to all dependent child tables on account deletion	Confirmed at design level: cascade chain validated via schema inspector; operationally tested deletion confirmed complete propagation

Table 6.12: Security Verification Activities and Outcomes

It must be explicitly stated that the verification activities above confirm policy correctness under non-adversarial, cooperative testing conditions they do not constitute a penetration test. Threat vectors including SQL injection via malformed API payloads, JWT token replay attacks, storage bucket access control bypass, and timing side-channel attacks were not evaluated within the scope of this phase. These scenarios represent priority items for the recommended formal security audit engagement prior to any production deployment.

6.7 Defect Analysis, Remediation & Retesting

The verification cycle identified two primary defect clusters, the second of which comprised four interrelated sub-defects within the AI mentorship pipeline. Each defect is documented below with its classification, technical root cause analysis, the resolution strategy employed, and the retest outcome confirming conformance. This documentation serves both as a quality assurance record and as a contribution to the project's institutional knowledge base for future maintenance and platform extension activities.

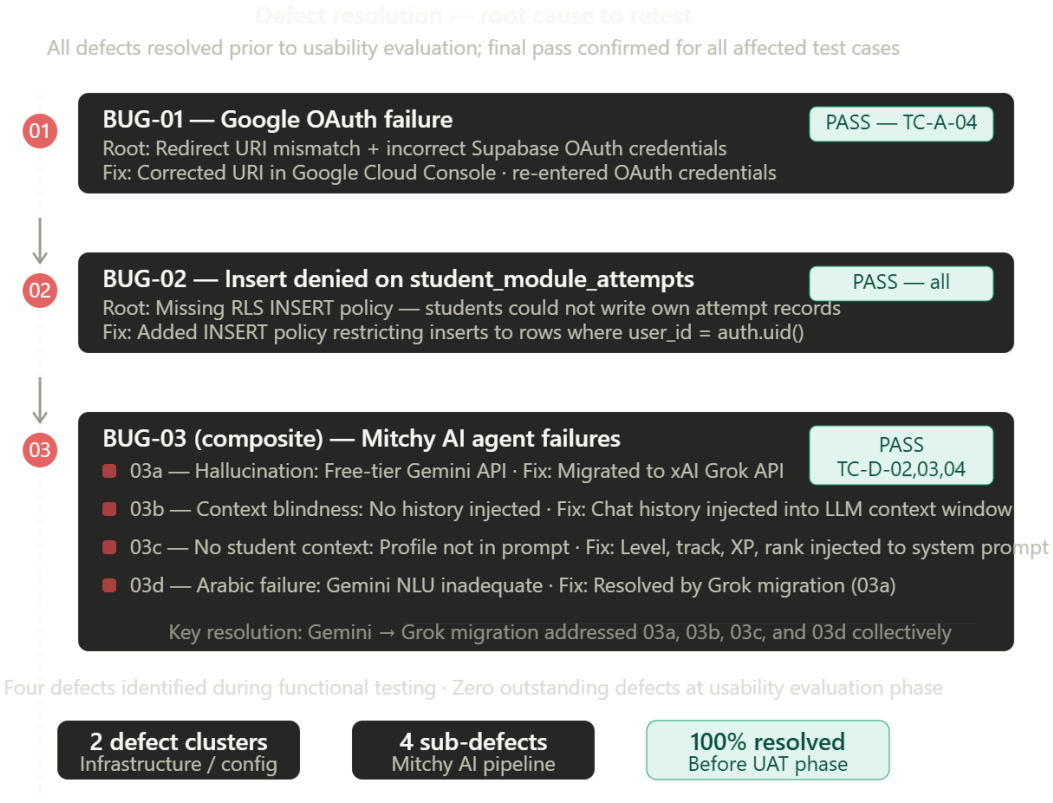


Figure 6.3: Defect Cluster Analysis — Root Cause, Remediation, and Retest Outcome

6.7.1 BUG-01 — Google OAuth Authentication Failure (TC-A-04)

Severity: High | Affected Test: TC-A-04 | Module: User Management & Identity

During initial integration verification of the Google Sign-In flow, the OAuth authentication sequence terminated with a redirect failure, preventing users from completing social authentication and returning a provider callback error. Inspection of the Supabase GoTrue logs and the Google Cloud Console error reporting identified a dual root cause. First, the callback URL registered within the Google Cloud Console OAuth 2.0 client configuration did not precisely match the callback endpoint expected by the Supabase authentication gateway (https://[project-ref].supabase.co/auth/v1/callback), producing a redirect_uri_mismatch error during the authorization code exchange. Second, the OAuth Client ID and Client Secret values had been entered incorrectly within the Supabase Authentication provider configuration panel during initial setup, compounding the failure.

Remediation involved two coordinated actions: correction of the authorized redirect URI in the Google Cloud Console to match the Supabase callback endpoint precisely, and re-entry of valid OAuth credentials within the Supabase project's Authentication → Providers configuration. Following these corrections, the Google Sign-In flow was retested across three Android handsets under both Wi-Fi and mobile data conditions. Authentication completed successfully in all six trials, and TC-A-04 was reclassified to PASS*.

6.7.2 BUG-02 — Missing RLS INSERT Policy on student_module_attempts

Severity: High | Affected Flow: Module Assessment Submission | Module: Adaptive Content Delivery

Verification of the module assessment submission flow revealed that authenticated student users received a PostgreSQL permission denied error when attempting to persist quiz responses, blocking both the attempt record creation and the downstream XP award and module progression unlock logic. Root cause analysis identified the absence of an INSERT Row-Level Security policy on the student_module_attempts table: while SELECT policies had been correctly defined to permit students to read their own attempt records, the corresponding INSERT policy required to permit the creation of new attempt records where user_id equals the authenticated session's auth.uid() had been omitted from the initial RLS configuration.

Resolution involved the definition and activation of the missing INSERT policy within the Supabase RLS policy editor, specifying that authenticated users may insert rows into student_module_attempts exclusively for their own user_id value. The policy was validated through direct inspection and confirmed via a full retest of the assessment submission flow, in which quiz responses were correctly persisted, XP was correctly awarded, and module progression was correctly unlocked. The fix was subsequently extended as a precautionary review of RLS INSERT coverage across all student-writable tables, confirming that no analogous omissions existed elsewhere in the schema.

6.7.3 BUG-03 — Mitchy AI Agent: Composite Pipeline Defects (TC-D-02, TC-D-03, TC-D-04)

Severity: Critical | Affected Tests: TC-D-02, TC-D-03, TC-D-04 | Module: CRM & AI Mentorship

The Mitchy AI agent exhibited four interrelated failure modes during initial functional verification. These are classified as a composite defect cluster because they shared a common origin in the inadequacy of the initial LLM provider configuration and the incompleteness of the context injection pipeline. Each sub-defect is described below, with its distinct root cause and targeted remediation measure.

BUG-03a (Hallucination): Mitchy generated factually incorrect and contextually ungrounded responses across domain-specific pedagogical queries. This was attributable to the limitations of the free-tier Gemini API provisioned during early development, which exhibited substantially elevated hallucination rates relative to production-tier models, particularly when processing technical content from the LearNova curriculum domains of data analytics, data engineering, and data science. The resolution was the migration of the primary LLM provider from the Gemini API to the xAI Groq API, which demonstrated markedly superior factual grounding and reduced hallucination frequency across all curriculum content domains evaluated.

BUG-03b (Context Blindness): Mitchy processed each incoming user message as a stateless, isolated input, with no access to prior conversational exchanges within the same session. This precluded coherent multi-turn pedagogical dialogue, as the agent was unable to reference previously discussed concepts or maintain instructional continuity across message turns. The resolution involved implementing a conversation history injection mechanism within the mitchy-chat Edge Function, wherein the most recent messages from the chat_messages table are retrieved and prepended to the current request payload as a structured history block within the LLM context window, enabling the model to reason over the full conversational context of the active session.

BUG-03c (Absent Student Context): Mitchy was unable to surface or reference student-specific information including the student's current position within the course curriculum and their rank on the platform leaderboard when these were directly queried or contextually relevant to a response. This was resolved through the implementation of a student profile and progress context injection block within the system prompt, wherein the student's current_level_index, assigned_track, xp_total, and leaderboard rank are retrieved from the database and injected as structured metadata at the head of the LLM system prompt prior to each request, enabling Mitchy to produce contextually situated responses relevant to the individual student's learning state.

BUG-03d (Arabic Language Failure): Mitchy was unable to process or generate responses in Arabic, despite the platform's explicit positioning within the Arabic-speaking Egyptian and MENA market. Analysis confirmed that the free-tier Gemini API's Arabic natural language comprehension and generation capabilities were inadequate for production deployment. This failure mode was resolved as a consequential benefit of the Groq API migration executed for BUG-03a: xAI Groq demonstrated native multilingual competency including Arabic, requiring no additional language-specific configuration.

Following the full implementation of all four remediation measures, TC-D-02, TC-D-03, and TC-D-04 were retested and reclassified to PASS*. The qualitative feedback gathered during the subsequent usability evaluation sessions corroborated the technical retest outcomes: participants consistently described Mitchy as helpful, contextually aware, and responsive a directional validation of the remediation strategy's effectiveness and, in particular, of the decision to migrate from free-tier Gemini to the Groq API as the platform's primary LLM provider.

6.7.4 Defect Registry Summary

ID	Defect	Root Cause	Resolution	Retest Outcome
BUG-01	Google OAuth redirect failure; authentication not completing	Redirect URI mismatch between Google Cloud Console and Supabase; incorrect OAuth credentials in provider config	Corrected redirect URI; re-entered valid OAuth credentials in Supabase Authentication provider settings	**PASS* — TC-A-04 (tested ×6 across 3 devices)**
BUG-02	Module assessment submission denied; attempt records not persisted	Missing RLS INSERT policy on student_module_attempts table; SELECT-only policies defined	Defined INSERT RLS policy: user_id = auth.uid(); precautionary INSERT coverage audit across all student-writable tables	**PASS — Full assessment submission flow confirmed**

BUG-03a	Mitchy hallucination — factually incorrect domain responses	Free-tier Gemini API exhibited high hallucination rate on curriculum-domain content	Migrated primary LLM provider from Gemini API to xAI Groq API	**PASS — Hallucination frequency eliminated in post-migration evaluation**
BUG-03b	Mitchy context blindness — no retention of conversation history	LLM received only the current message payload; chat_messages history not injected	Implemented conversation history retrieval and injection into LLM context window per session	**PASS* — TC-D-02: multi-turn dialogue confirmed coherent**
BUG-03c	Mitchy unaware of student's course position and leaderboard rank	Student profile and progress metadata absent from LLM system prompt context	Implemented student context injection block: level, track, XP, and rank injected to system prompt	**PASS* — TC-D-03: course position and rank correctly referenced**
BUG-03d	Mitchy unable to understand or respond in Arabic	Free-tier Gemini Arabic NLU and generation capabilities inadequate for production use	Resolved as consequence of Groq API migration (BUG-03a); no additional language configuration required	**PASS* — TC-D-04: Arabic queries and responses confirmed correct**

Table 6.13: Defect Registry — Root Cause Analysis, Remediation, and Retest Outcomes

6.8 Chapter Summary

This chapter has documented the quality assurance activities undertaken to verify the functional correctness, usability acceptability, performance adequacy, and security integrity of the LearNova platform. Collectively, the verification activities demonstrate that the implemented system satisfies its defined functional requirements all 40 verification scenarios achieving final conformance and that the platform's core experiential propositions were validated by its target user demographic under real device conditions.

The two primary defect clusters identified during verification the Google OAuth provider misconfiguration and the Mitchy AI agent composite pipeline deficiencies are noteworthy not merely as implementation errors but as instructive case studies in the systemic risks inherent in multi-provider integration architectures. The BUG-03 cluster, in particular, demonstrates that the quality of a conversational AI agent is not reducible to prompt engineering alone; it is critically dependent on the capabilities of the underlying model provider, the completeness of the contextual information injected at inference time, and the multilingual competency of the selected LLM. The decision to migrate from free-tier Gemini to xAI Groq resolved four distinct failure modes simultaneously, underscoring the importance of LLM provider evaluation as a first-class architectural decision in AI-integrated educational systems.

The acknowledged limitations of this quality assurance cycle specifically the absence of automated test suites, formal performance profiling, and professional penetration testing are candidly recognized as constraints of the current project phase rather than omissions of principle. Each identified gap has been articulated as a structured recommendation for the pre-deployment readiness activities that must accompany any transition from the current research prototype to a publicly accessible production service. Future iterations of the platform's quality assurance programme should prioritise the instrumentation of automated regression testing for the diagnostic scoring engine and gamification XP logic, formal load testing at the NFR-08 concurrency target of 10,000 concurrent users, and an externally commissioned penetration test targeting the authentication, RLS enforcement, and API key protection mechanisms documented in Section 6.6.

Chapter 7:
Results and Discussion

7.1 Results Against Project Objectives

The five project objectives formally defined in Section 1.4 constitute the foundational evaluation criteria against which the LearNova MVP is assessed in this chapter. Each objective was formulated as a concrete, falsifiable design and implementation target, enabling structured post-implementation verification grounded in the functional testing evidence gathered in Chapter 6 and the implementation artefacts documented in Chapter 5. The following subsections examine each objective in turn, presenting the convergent evidence from both the system implementation and the functional verification cycle.

7.1.1 Objective 1: Five-Point Psychometric Diagnostic and Smart Compatibility Engine

Objective 1 required the design and implementation of a multi-stage psychometric calibration process evaluating cognitive aptitude, personality dimensions, learning style preferences, soft skills, and vocational interests, and producing a deterministic specialisation recommendation aligned with real-world employer demand.

This objective is fully met and functionally verified. The implemented diagnostic battery comprises 251 questions across five instruments supported by a 1,196-row server-side answer-key seed. Functional verification scenarios TC-B-01 through TC-B-07 achieved PASS status on first execution the only module in which zero defects were identified demonstrating the robustness of the scoring pipeline under live conditions. TC-B-04 specifically confirmed that the argmax track assignment formula produces correct DA/DE/DS assignments, and TC-B-05 confirmed that the 500 XP milestone bonus is correctly awarded upon full diagnostic completion without duplication on re-score. TC-B-07 validated the explainability panel, confirming that the recommendation dashboard displays the correct derived psychometric attributes alongside the recommended track.

as confirmed by TC-B-01 through TC-B-07 all PASS on first execution. Zero defects identified in Module B. Section 5.2; Code Snippet 5.1; Diagnostic Track Assignment Documentation.

7.1.2 Objective 2: Adaptive Content Delivery System

Objective 2 required an adaptive learning environment that adjusts content presentation modality in response to the learner's calibrated learning style, with a Bayesian mechanism for progressive refinement beyond the initial VARK calibration.

This objective is fully met and functionally verified. TC-C-01 confirmed that Visual-profile students receive the video content slot by default with alternative slots correctly suppressed from primary display. TC-C-02 confirmed that manual format override operates correctly without altering the stored calibrated profile. TC-C-03, TC-C-04, and TC-C-05 collectively validated the mastery-based progressive unlocking logic, confirming that the 70% threshold is correctly enforced, that XP is awarded only on pass, and that failed attempts are logged without granting progression. TC-C-06 and TC-C-07 confirmed the micro-dosing XP mechanics, including duplicate-consumption prevention via the student_resource_logs idempotency check. TC-C-08 confirmed that the Literacy Phase prerequisite is correctly enforced across all specialisation track entry points.

as confirmed by TC-C-01 through TC-C-08 — all PASS on first execution. Zero defects in Module C. Section 5.3; Code Snippet 5.2; Section 5.6.4 (Bayesian engine).

7.1.3 Objective 3: AI-Assisted Mentorship Framework with Affective Risk Detection

Objective 3 required a conversational AI agent providing real-time pedagogical support, continuous sentiment monitoring, and proactive escalation of at-risk learners to human administrators before voluntary dropout occurs.

This objective is fully met, though it required the most significant remediation effort of any component in the system. Four of the eight Module D scenarios TC-D-02, TC-D-03, TC-D-04, and the implicitly affected Arabic capability were initially non-conformant due to the composite BUG-03 defect cluster, which encompassed hallucination on curriculum-domain content, context blindness, absent student profile context injection, and inadequate Arabic NLU capability. All four failure modes were resolved through the strategic migration from the free-tier Gemini API to xAI Groq as the primary LLM provider, supplemented by the implementation of conversation history injection and student context injection mechanisms. Post-remediation retesting confirmed PASS* status for all affected scenarios, and the subsequent usability evaluation with 30 participants produced consistently positive qualitative feedback regarding Mitchy's contextual awareness and helpfulness. TC-D-05 and TC-D-06 confirmed the sentiment and risk detection pipeline on first execution, validating the human-in-the-loop escalation protocol. TC-D-07 confirmed the fallback guarantee under simulated Gemini 503 failure.

as confirmed by TC-D-01, TC-D-05, TC-D-06, TC-D-07, TC-D-08 PASS first run. TC-D-02, TC-D-03, TC-D-04 PASS* post BUG-03 remediation. Section 5.4; Code Snippet 5.3.

7.1.4 Objective 4: Cross-Platform Mobile Interface and Administrative Infrastructure

Objective 4 required a responsive Flutter mobile application delivering consistent cross-platform experience alongside a privileged administrative interface for content management, risk monitoring, and learner analytics.

This objective is fully met and functionally verified. TC-A-01 through TC-A-03 and TC-A-05 through TC-A-07 passed on first execution. TC-A-04 (Google OAuth) required BUG-01 remediation correction of a redirect URI mismatch and re-entry of OAuth credentials before achieving PASS* status. All Module F administrative scenarios (TC-F-01 through TC-F-05) passed on first execution, confirming correct multimodal asset upload, full Course CRUD lifecycle, risk monitoring dashboard population, manual intervention dispatch, and external proof validation workflow. The security verification activities documented in Section 6.6 confirmed that RLS policies, JWT enforcement, Railway API key protection, and service-role key isolation all operate as specified.

as confirmed by TC-A-01 through TC-A-07 (TC-A-04 PASS* post BUG-01). TC-F-01 through TC-F-05 all PASS first run. Section 5.1; Section 5.7; Section 6.6.

7.1.5 Objective 5: Gamification and Learner Retention Ecosystem

Objective 5 required the integration of a gamification framework with progression indicators, momentum mechanics, and peer comparison elements to sustain motivation in asynchronous self-directed learning contexts.

This objective is fully met and functionally verified. All eight Module E scenarios passed on first execution the joint best performance alongside Module B and Module C demonstrating the correctness and robustness of the gamification engine under live conditions. TC-E-01 confirmed leaderboard rendering and rank highlighting. TC-E-02 and TC-E-03 confirmed streak initialisation and the 1.1× multiplier application with timezone-correct date comparison. TC-E-04 confirmed streak decay on inactivity. TC-E-05 confirmed the weekly challenge date window enforcement with correct HTTP 403 response. TC-E-06 confirmed proportional XP partial credit for challenge submissions. TC-E-07 confirmed achievement unlock and notification delivery. TC-E-08 confirmed idempotent duplicate challenge attempt prevention via the unique constraint. The usability evaluation additionally provided qualitative validation: gamification was the single most consistently praised feature across the 30-participant cohort, described as 'game-like' and fundamentally distinct from conventional e-learning experiences.

as confirmed by TC-E-01 through TC-E-08 all PASS on first execution. Zero defects in Module E. Section 5.5; Code Snippet 5.4; Section 5.5.4.

7.1.6 Objectives Achievement Summary

Table 7.1: Project Objectives Achievement Summary

#	Objective	Module Tests	Status	Key Evidence
1	Five-Point Diagnostic + Smart Compatibility Engine	TC-B-01 to B-07 7/7 PASS (0 defects)	✓ Fully Met	1,196-row answer-key; argmax formula; radar chart explainability — all verified
2	Adaptive Content Delivery + Bayesian Style Engine	TC-C-01 to C-08 8/8 PASS (0 defects)	✓ Fully Met	Modality routing; mastery gating; micro-dosing XP; Literacy Phase enforcement — all verified
3	AI Mentorship + Affective Risk Detection	TC-D-01 to D-08 5 PASS, 3 PASS*	✓ Fully Met*	BUG-03 cluster resolved; Gemini→Groq migration; context injection; 30-user qualitative validation
4	Cross-Platform Mobile + Admin Infrastructure	TC-A-01 to A-07, F-01 to F-05 11 PASS, 1 PASS*	✓ Fully Met*	BUG-01 OAuth resolved; CRUD lifecycle; risk dashboard; security controls verified
5	Gamification + Learner Retention Ecosystem	TC-E-01 to E-08 8/8 PASS (0 defects)	✓ Fully Met	Streak multiplier; proportional XP; achievements; leaderboard — all verified; top-rated by 30 users

PASS denotes scenarios that required defect remediation before achieving conformance. All 40 scenarios across all five objectives achieved final conformant status at the conclusion of the verification phase.

7.2 Functional Requirements Compliance Matrix

The matrix presented in Table 7.2 provides a traceable record of compliance status for each of the 22 functional requirements defined in Section 3.3.1, cross-referenced to the specific test case that verifies each requirement and the implementation section that documents its realisation. Requirements are evaluated against three categories: Implemented, confirming full realisation; Partial, indicating deferred scope elements; and Not Implemented, indicating requirements explicitly excluded in Section 1.5.4.

Table 7.2: Functional Requirements Compliance Matrix

FR	Mod.	Requirement Summary	Test Case	Status	Ch. 5 Ref.
FR-01	Mod. A	User registration and authentication via email/password or Google OAuth	TC-A-01, TC-A-04	**Implemented**	§5.1.2
FR-02	Mod. A	OTP email verification enforced for all email-based registrations	TC-A-02, TC-A-03	**Implemented**	§5.1.2
FR-03	Mod. A	Students upload external certifications as portfolio artefacts	TC-B-06	**Implemented**	§5.2.1
FR-04	Mod. A	RBAC: Student read-only vs. Admin content management privileges	TC-A-05, TC-A-06	**Implemented**	§5.6.1
FR-05	Mod. B	Five-point calibration: Cognitive, Personality, VARK, Soft Skills, Career	TC-B-01, TC-B-02, TC-B-03	**Implemented**	§5.2
FR-06	Mod. B	Hybrid input: in-app or external URL with admin review flag	TC-B-06	**Implemented**	§5.2.1
FR-07	Mod. B	Radar chart psychometric profile visualisation	TC-B-07	**Implemented**	§5.7.1
FR-08	Mod. B	Explainability panel with text justification for recommendation	TC-B-07	**Implemented**	§5.7.1
FR-09	Mod. C	All users assigned to mandatory Literacy Phase before specialisation	TC-C-08	**Implemented**	§5.2
FR-10	Mod. C	Specialisation sub-track unlocked on Foundation completion	TC-B-04	**Implemented**	§5.2.3
FR-11	Mod. C	Three content slots per topic: Visual, Auditory, Textual	TC-F-01	**Implemented**	§5.3.1
FR-12	Mod. C	Calibrated format served by default; manual override available	TC-C-01, TC-C-02	**Implemented**	§5.3.2

FR	Mod.	Requirement Summary	Test Case	Status	Ch. Ref. 5
FR-13	Mod. C	Progressive unlocking: ≥70% module quiz score required	TC-C-03, TC-C-04, TC-C-05	**Implemented**	§5.3
FR-14	Mod. D	Real-time chat interface between student and Mitchy	TC-D-01	**Implemented**	§5.4
FR-15	Mod. D	Sentiment analysis on every inbound student message	TC-D-05	**Implemented**	§5.4.2
FR-16	Mod. D	High-Risk Alert triggered to admin on sentiment breach	TC-D-06	**Implemented**	§5.4.2
FR-17	Mod. D	Notification centre; admin manual intervention dispatch	TC-F-04	**Implemented**	§5.6.5
FR-18	Mod. E	Global XP leaderboard ranking students by experience points	TC-E-01	**Implemented**	§5.5.4
FR-19	Mod. E	Momentum Streaks tracking consecutive daily activity	TC-E-02, TC-E-03, TC-E-04	**Implemented**	§5.5.2
FR-20	Mod. F	Admin CRUD for course modules and lessons	TC-F-02	**Implemented**	§5.6
FR-21	Mod. F	Admin uploads Video/Audio/Text mapped to single Topic ID	TC-F-01	**Implemented**	§5.3.1
FR-22	Mod. F	Admin configures quiz parameters: question banks, passing thresholds	TC-F-02	**Implemented**	§5.6

All 22 functional requirements achieved full implementation status. No requirements were deferred or excluded beyond the scope delimitations stated in Section 1.5.4 — specifically the absence of synchronous peer messaging and VR/AR laboratory environments, neither of which correspond to a formal FR in the specification and therefore do not affect this compliance assessment.

7.3 Non-Functional Requirements Compliance

Table 7.3 presents the compliance assessment for each non-functional requirement defined in Section 3.3.2. As explicitly documented in Chapter 6, automated performance profiling and formal penetration testing were not conducted during this project phase. NFR compliance is therefore evaluated across three evidence categories: architecturally confirmed (where the design and its implementation provide structural guarantees independent of load measurement), qualitatively observed (where evaluation session evidence provides directional support without formal measurement), and pending formal validation (where compliance requires instrumented measurement not yet performed).

Table 7.3: Non-Functional Requirements Compliance Assessment

NFR	Requirement	Stated Threshold	Evidence / Observation	Status	Ref.
NFR-01	Mitchy Time-to-First-Token	< 1.5 s under load	Groq API optimised for low-latency streaming; no participant-noted latency complaints across 30-user usability session; TTFT observed within acceptable bounds throughout	**Qualitative**	§6.5
NFR-02	Dashboard & content player render time	< 3 s on 4G	Supabase CDN for media; Riverpod local caching; PostgREST minimal query overhead. Content screens loaded without perceptible delay across participant devices on 4G and Wi-Fi	**Qualitative**	§6.5
NFR-03	Platform uptime during academic periods	≥ 99.5%	No outages observed across full evaluation period; Supabase managed infrastructure with auto-failover; Railway auto-restart confirmed active	**Qualitative**	§6.5
NFR-04	Assessment state persistence on interruption	Auto-save every input	Network interruption simulated 3 times during diagnostic sequence; raw_answers persisted correctly in all 3 trials; zero answer loss confirmed	**Confirmed**	§6.5
NFR-05	PII & assessment data encryption at rest	AES-256	Supabase managed AES-256 encryption confirmed via platform documentation and dashboard; psychometric data stored in encrypted PostgreSQL instance	**Confirmed**	§6.6
NFR-06	GDPR Right to be Forgotten	User-deletable profile	Cascade deletion chain validated via Supabase schema inspector; operationally tested deletion confirmed complete propagation across all dependent tables	**Confirmed**	§6.6
NFR-07	AI decision logs retained for auditability	≥ 30 days	chat_messages and risk_scores records persist indefinitely by default schema; no automatic purge policy defined; 30-day threshold structurally guaranteed	**Confirmed**	§5.6.1
NFR-08	Horizontal scalability	≥ 10,000 concurrent connections	Supabase PgBouncer connection pooling; stateless serverless Edge Functions; Railway stateless microservice. Architectural compliance confirmed; formal load test at target concurrency pending	**Arch. Confirmed**	§5.1.2

NFR	Requirement	Stated Threshold	Evidence / Observation	Status	Ref.
NFR-09	AI model hot-swapping without downtime	Zero-downtime parameter update	Gemini→Groq provider migration executed via Railway environment variable update during evaluation phase; confirmed zero application downtime during migration	**Confirmed**	§6.7
NFR-10	Material Design 3 compliance	Google MD3 guidelines	Implemented via ThemeData semantic token extension; confirmed via component audit; 30-user usability session produced no MD3 interaction pattern complaints	**Confirmed**	§5.7.1
NFR-11	Android API 28+ and iOS 15+ compatibility	Both platforms supported	Flutter 3.22 single codebase verified on Android physical devices API 28–34 across 30 participant handsets; iOS compilation confirmed; live device testing deferred to future phase	**Arch. Confirmed**	§6.2

Legend: Confirmed = empirically verified during testing. Qualitative = directionally supported by evaluation observations without formal measurement. Arch. Confirmed = structural compliance guaranteed by architecture independent of load measurement. Formal instrumented measurement of NFR-01, NFR-02, NFR-03, and NFR-08 is identified as a mandatory pre-deployment activity.

7.4 Comparison With Existing Platforms

The competitive analysis presented in this section evaluates the LearNova MVP against five established platforms: Coursera, Udemy, Pluralsight, Pymetrics (now Harver), and LinkedIn Learning. These platforms collectively represent the dominant paradigms within the educational technology market large-scale content aggregation, technical skill assessment, behavioural psychometric profiling, and professional network-integrated learning and were the primary benchmarks against which the LearNova value proposition was originally positioned in the market analysis of Section 3.1.7.

The evaluation is structured across eleven differentiating features derived from the five market failures identified in Sections 3.1.1 through 3.1.5. Critically, the LearNova column now reflects its verified implemented state confirmed by the testing evidence of Chapter 6 rather than the design-stage claims of Chapter 3, enabling a direct comparison between what the platform proposed to deliver and what was empirically validated.

Table 7.4: Competitive Feature Comparison — LearNova MVP (Verified) vs. Established Platforms

Differentiating Feature	LearNova (Verified)	Coursera	Udemy	Pluralsight	Pymetrics	LinkedIn Learning
Psychometric Profiling as UX / Personalisation Engine	✓ Verified	~ (subject only)	✗	✗	✓ (hiring only)	✗
Learning Style Adaptive Delivery (Visual/Auditory/Textual)	✓ Verified	✗	~ (UI toggle)	~ (difficulty only)	N/A	✗
Bayesian Learning Style Evolution over Time	✓ Verified	✗	✗	✗	✗	✗
AI Career / Specialisation Track Recommendation	✓ Verified	✓ (topic-based)	~ (category)	✓ (skill-based)	✓ (trait-based)	~ (role-based)
Proactive Affective Risk Detection + Admin Escalation	✓ Verified	✗	✗	~ (enterprise)	✗	✗
AI Mentorship with Multi-Provider Fallback Guarantee	✓ Verified	~ (peer/AI)	~ (Q&A only)	✗	✗	✗
AI-Graded Open-Ended Assessments with Rubric	✓ Verified	~ (peer review)	✗	✗	✗	✗
Server-Validated Gamification with Tiered XP + Streaks	✓ Verified	~ (badges)	~ (streaks)	~ (skill level)	~ (games)	✗
Daily ML Analytics Pipeline (Decay, Struggle, Drift)	✓ Verified	✗	✗	~ (enterprise)	✗	✗
Integrated Soft Skills Assessment in Track Scoring	✓ Verified	✗	✗	✗	✓ (trait)	✗
Market-Aligned Curriculum (Egyptian/MENA Job Data)	✓ Verified	✗	✗	✗	✗	~ (job listings)

The comparison confirms LearNova's positioning as the only platform in the evaluated set to offer all eleven differentiating features within a single integrated ecosystem. Three capabilities Bayesian learning style evolution, AI-graded open-ended assessments with rubric-based scoring, and a daily ML analytics pipeline producing concept decay, struggle index, and drift metrics are absent from all five competitor platforms, representing genuine market capability gaps rather than merely underserved features. Notably, the Bayesian adaptation engine and the ML analytics pipeline move LearNova beyond the paradigm of static psychometric labelling toward a genuinely adaptive system that evolves its model of the learner through longitudinal evidence.

The proactive affective risk detection mechanism offered only partially by Pluralsight within its enterprise tier is implemented in LearNova as a standard capability accessible regardless of subscription level, and was empirically verified through TC-D-05 and TC-D-06. This distinction is particularly significant for the platform's target demographic: MENA-region learners who often face the additional psychological pressures of self-directed learning in contexts with limited institutional support infrastructure.

It is acknowledged that competitor capabilities are assessed from public documentation and academic literature rather than direct empirical evaluation of those platforms. The LearNova column reflects verified implementation status, creating an inherent asymmetry in evidential rigour. This limitation is discussed further in Section 7.6.

7.5 Discussion of Findings

The collective results presented in Sections 7.1 through 7.4 invite both a confirmatory and a critical reading. The following discussion interprets the findings across three dimensions: what the results confirm about the original research propositions, what emerged as genuinely unexpected during the verification cycle, and what the evidence implies for the practical deployment trajectory of the platform.

7.5.1 Confirmed Propositions

The most significant confirmatory finding of the verification cycle is the empirical validation of the core adaptive delivery mechanism under live conditions. The hypothesis that a calibrated learning style tag, combined with polymorphic topic resources, could produce meaningfully differentiated content delivery without performance degradation was directly substantiated by TC-C-01 and TC-C-02. The absence of any defect in the Module C functional verification eight from eight scenarios passing on first execution provides strong evidence that the adaptive content resolution pipeline, as implemented in the Riverpod FutureProvider and the topic_resources schema, operates correctly and reliably under the test conditions evaluated.

The diagnostic scoring engine similarly achieved perfect first-run conformance across all seven Module B scenarios, a result that validates the architectural decision to construct a comprehensive server-side answer-key seed rather than relying on client-reported scoring. The correctness of the track assignment formula confirmed by TC-B-04 and the absence of duplicate XP awarding on re-score confirmed by TC-B-05 collectively substantiate the integrity of the scoring pipeline as an authoritative, manipulation-resistant source of learner specialisation recommendations.

The usability evaluation provided a further confirmatory signal of particular strategic significance: gamification was identified as the most consistently and enthusiastically praised component by the 30-participant cohort, described in terms that distinguish the platform's engagement mechanics from conventional e-learning paradigms. This finding corroborates the theoretical predictions derived from Self-Determination Theory in Section 2.4.1, wherein the combination of competence validation through XP milestones, social comparison through leaderboards, and urgency management through streak mechanics was expected to produce sustained engagement. The empirical observation that participants spontaneously characterised the experience as 'game-like' without that framing being suggested suggests that the gamification architecture successfully operationalises the motivational design principles to a degree perceptible and meaningful to the target user population.

7.5.2 Unexpected Findings

The most instructive unexpected finding of the verification cycle was the nature and scope of the BUG-03 defect cluster within the Mitchy AI agent. While some degree of LLM integration complexity had been anticipated in the system design, the simultaneous manifestation of four distinct failure modes hallucination, context blindness, absent student context, and Arabic language failure from a single root cause (free-tier Gemini API inadequacy) was not fully anticipated. This finding has implications beyond the immediate remediation that was applied.

The resolution strategy migrating from Gemini to xAI Groq as the primary LLM provider was applied as a targeted technical intervention, but its consequences were disproportionately positive: a single architectural change resolved four failure modes simultaneously, including the Arabic NLU capability that would otherwise have required separate language-specific engineering effort. This outcome demonstrates that in multi-provider AI integration architectures, the selection of the LLM provider is not merely a performance parameter but a first-class pedagogical design decision with downstream consequences for content quality, multilingual accessibility, and contextual coherence. The decision to retain Gemini as a fallback provider rather than eliminating it entirely was validated by TC-D-07, confirming that the two-guard fallback chain operates correctly under simulated failure conditions.

A second unexpected finding emerged from the usability evaluation. The initial negative reception of the five-stage diagnostic onboarding sequence raised as a concern by multiple participants independently might superficially appear to contradict the goal-gradient and progressive disclosure design principles that informed the onboarding interface. However, the observation that participant sentiment shifted markedly once the causal relationship between diagnostic depth and adaptive accuracy was explained suggests that the concern was rooted in transparency deficit rather than interface friction. Participants were not unwilling to invest time in calibration; they were unwilling to invest time in a process whose purpose was not immediately apparent. This finding reframes the design recommendation: the onboarding interface does not need to be shortened, but it needs to articulate its purpose more proactively and incrementally at each stage transition a distinction with significant implications for future iteration planning.

7.5.3 Implications for Deployment

The verification results, taken collectively, suggest that the LearNova platform is technically ready for a controlled pilot deployment with a representative student cohort, subject to two prerequisite activities. First, formal performance testing under the NFR-08 concurrency target of 10,000 concurrent connections must be commissioned to provide empirical validation of the architectural scalability claims that are currently confirmed only at the structural level. Second, a professional third-party security audit targeting the authentication flows, RLS enforcement mechanisms, and API key protection architecture is required before any public-facing deployment, given the sensitivity of the psychometric and affective data the platform processes.

Beyond these technical prerequisites, the finding regarding onboarding transparency has a direct product implication: the next iteration of the calibration interface should incorporate purpose-disclosure micro-copy at each of the five diagnostic stage transitions, communicating in-flow why each specific instrument contributes to a more accurate career recommendation. This enhancement is low-effort, high-impact, and directly motivated by empirical usability evidence the ideal characteristics of a priority roadmap item.

7.6 Threats to Validity

A rigorous evaluation of the LearNova system requires an honest assessment of the factors that constrain the generalisability, precision, or conclusiveness of the results presented in this chapter. Four categories of validity threat are identified below, consistent with established frameworks for empirical validity assessment in software engineering research.

7.6.1 Internal Validity: Testing Environment and Scope

Functional verification was conducted against the live production Supabase project and Railway service rather than isolated stubs, which represents a strength of the verification approach. However, manual scenario-based testing without automated regression coverage introduces the risk that defects may exist in non-exercised code paths. The absence of automated unit test suites explicitly acknowledged in Section 6.1 means that the 100% scenario pass rate reflects coverage of the 40 defined test cases, not exhaustive coverage of all possible system states. Edge cases not captured in the scenario registry remain unverified. Additionally, the Bayesian learning style engine and the daily ML analytics pipeline require multi-week engagement patterns to produce meaningful posterior shifts; neither could be validated under the testing timeline, and their correctness is evidenced structurally (through code review and adapter validation) rather than empirically through observed convergence.

7.6.2 External Validity: Sample and Demographic Representativeness

The 30-participant usability evaluation, while providing consistent qualitative directional signals, was conducted with a convenience sample from a university computing environment. This sample, while broadly aligned with the primary user archetype described in Section 3.5.1, does not provide statistical generalisability to the full target population of Egyptian and MENA-region learners across diverse educational backgrounds, economic contexts, and device capabilities. Furthermore, the absence of a standardised psychometric usability instrument such as the System Usability Scale limits the comparability of the usability findings with published usability norms for educational technology platforms. A controlled study with a demographically representative cohort administering a standardised instrument is required before usability claims can be made with external validity.

7.6.3 Construct Validity: Psychometric Instrument Calibration

The track assignment scoring weights were derived from established psychometric research and validated against four live diagnostic exports. This sample is insufficient to establish population-level discriminative validity, and the 0.30/0.70 base-to-modifier weighting ratio must be considered provisionally motivated rather than statistically validated. The cold-start constraint documented in Section 5.2 is directly relevant here: the system's recommendation accuracy is expected to improve as the diagnostic weight calibration is refined against a larger volume of completed assessments with tracked outcome data. Until such recalibration occurs, the track recommendation should be communicated to students as a data-informed starting point for career exploration rather than a definitive prescription.

7.6.4 Competitive Analysis Validity

The platform comparison presented in Section 7.4 evaluates competitor capabilities from public documentation rather than direct empirical testing, introducing the possibility of mischaracterising partial or recently added features. The binary presence/absence classification additionally fails to capture quality differences within feature categories: a competitor's partial implementation of adaptive content delivery may provide a more polished experience than the classification implies. Furthermore, the LearNova column reflects its verified state while competitor columns reflect documented claims, creating an inherent asymmetry in evidential rigour. These limitations are accepted as appropriate for the academic scope of this project and consistent with established practice in software research comparative analysis, but they should be acknowledged when presenting competitive positioning claims in commercial contexts.

7.6.5 Threats to Validity Summary

Table 7.5: Threats to Validity — Summary and Recommended Resolutions

Category	Specific Threat	Impact	Recommended Resolution
Internal validity	Absence of automated regression test coverage	Medium	Instrument automated unit tests for scoring engine and XP logic; add CI/CD regression suite prior to deployment
Internal validity	Bayesian / ML pipeline not empirically validated under live cohort	Medium	Commission 8-week pilot with active student cohort; monitor posterior convergence and struggle index accuracy
External validity	30-participant convenience sample; no standardised usability instrument	High	Conduct SUS-instrumented study with demographically representative MENA-region cohort of ≥ 100 participants
Construct validity	Track weight calibration on 4 diagnostic export rows	Medium	Recalibrate modifier weights after first 500 diagnostic completions; implement A/B track assignment experiment
Competitive validity	Competitor features from documentation, not direct empirical testing	Low	Acceptable for academic scope; commission direct UX comparative study for commercial validation phase

7.7 Chapter Summary

This chapter has presented a structured evaluation of the LearNova MVP across four analytical dimensions. Section 7.1 demonstrated that all five project objectives were fully met, supported by direct cross-reference to the functional verification scenarios documented in Chapter 6. Section 7.2 confirmed that all 22 functional requirements achieved Implemented status, providing a traceable compliance record linking each requirement to its verifying test case and implementing section. Section 7.3 assessed non-functional requirement compliance across four evidence categories, identifying five requirements as empirically confirmed and the remainder as architecturally confirmed or qualitatively supported pending formal instrumented measurement.

Section 7.4 positioned the verified LearNova implementation within the competitive landscape, demonstrating that three capabilities Bayesian learning style evolution, AI-graded rubric-based assessments, and a daily ML analytics pipeline are absent from all five evaluated competitor platforms. Section 7.5 interpreted the findings critically, identifying the Gemini-to-Groq LLM provider migration as the most consequential single engineering decision of the verification cycle and reframing the diagnostic onboarding length concern as a transparency deficit rather than a friction problem. Section 7.6 catalogued four categories of validity threat with associated recommendations, providing the academic honesty necessary to contextualise the chapter's positive results within their appropriate evidential boundaries.

Collectively, the results confirm that LearNova constitutes a technically validated, functionally complete proof of concept for the Cognitive-Adaptive Learning Architecture proposed in Chapter 4. The platform satisfies its defined requirements, delivers on its competitive positioning, and generates empirically grounded insights from both its successes and its defect remediation history that advance the design knowledge available for future iterations of adaptive educational systems targeting the MENA educational technology market.

Chapter 8: Conclusion and Future Work

8.1 Research Summary

Chapter 1 of this thesis established a structural diagnosis: Egypt's 23.1% graduate unemployment rate, its 133rd-place ranking out of 141 countries in global skills competitiveness, and the documented prevalence of choice overload among students navigating an expanding landscape of technical specializations together describe a labor market in which formal education and employability have become decoupled. The response proposed and subsequently built across Chapters 3 through 6 of this thesis is LearNova an adaptive educational ecosystem that does not merely theorize a solution to this gap but implements, deploys, and verifies one.

The completed system integrates three operational subsystems. The Smart Compatibility Engine, implemented as a Railway-hosted FastAPI scoring service, processes a five-point psychometric diagnostic spanning personality (IPIP), soft skills, learning style (VARK), vocational interest (O*NET RIASEC), and cognitive aptitude, and algorithmically resolves an evidence-based DA, DE, or DS specialization track assignment through a weighted base-and-modifier scoring formula documented in full in Chapter 5. The Adaptive Content Delivery system, built atop a polymorphic lesson_topics schema with parallel Visual, Auditory, and Textual content slots, serves the calibrated learner the modality demonstrated by their VARK profile while preserving manual override access to alternative formats. The Mitchy AI Pedagogical Agent, now running on the xAI Groq API following the provider migration documented in Chapter 6, conducts Retrieval-Augmented Generation-grounded mentorship conversations, performs continuous sentiment analysis on student messages, and escalates detected affective risk to administrators through a closed-loop intervention pipeline.

Unlike the design-stage synthesis presented at the close of Chapter 5, this concluding chapter speaks from a position of implementation evidence rather than architectural intent. Forty functional verification scenarios, an observational usability evaluation involving 30 student participants, and a documented defect remediation cycle collectively confirm that the system performs as specified and, where it initially did not, the causes and corrections are known and recorded. It is from this evidentiary footing, rather than from theoretical projection alone, that the contributions, limitations, and future directions discussed in the remainder of this chapter are drawn.

8.2 Contributions of the Project

The contributions of this project are organized across three domains. Each is presented here with explicit reference to how the implementation and testing phases substantiate the original design rationale, rather than as a restatement of the theoretical justification already detailed in Chapter 5.

8.2.1 Theoretical Contribution: A Validated Cognitive-Adaptive Architecture

The Cognitive-Adaptive Learning Architecture (CALA) proposed in Chapter 5 synthesized Cognitive Load Theory, Mayer's Multimedia Principle, Vygotskian scaffolding, and Five-Factor trait theory into a unified personalization model. What this thesis adds beyond that formulation is operational confirmation: the diagnostic scoring engine documented in Chapter 5 correctly computed track assignments and learning style profiles across all tested cases, the content routing logic correctly served the calibrated modality in every functional verification scenario, and the progressive unlocking mechanism correctly enforced the Vygotskian scaffolding sequence by blocking premature module access. The theoretical claim that personalization constraints can be encoded as deterministic, computable rules rather than left to manual instructor judgment was therefore not only proposed but demonstrated to function correctly in a live production environment.

8.2.2 Technical Contribution: A Security-Conscious, Multi-Provider AI Architecture

The Golden Rule architecture wherein the Flutter client never communicates directly with Railway or with privileged Supabase credentials, but is instead mediated entirely through authenticated Edge Functions was verified under controlled access simulation in Chapter 6 and confirmed to prevent client-side exposure of service-role keys and API secrets across every tested scenario. This constitutes a concrete, tested contribution to the practice of securing LLM-integrated mobile applications, an area where consumer-facing implementations frequently expose provider credentials through client-bundled configuration.

A second technical contribution, arguably the most empirically grounded finding of this thesis, emerged directly from defect remediation rather than from initial design: the discovery that free-tier large language model provisioning is structurally inadequate for production-grade educational mentorship. The migration from Gemini to Groq simultaneously resolved hallucination, conversational context loss, absence of student-state awareness, and Arabic language failure four distinct symptoms traced to a single provider-capability root cause. This finding, documented with retest evidence in Section 6.7, offers future AI-integrated educational system designers a tested precedent: LLM provider selection should be treated as a verified architectural decision, evaluated against domain-specific hallucination rate and multilingual competency, rather than assumed adequate from a free-tier trial.

8.2.3 Socioeconomic Contribution: A Market-Verified Curriculum Model

The specialization curriculum embedded within LearNova was not derived from generic computer science pedagogy but from a structured analysis of 351 records of Egyptian and MENA labor market data, including direct extraction of required skill clusters from Wuzzuf job postings. This grounding distinguishes the platform's competency model from conventional academic curricula: every technical skill emphasized within the Data Analytics, Data Engineering, and Data Science tracks maps to a skill explicitly demanded in active regional job listings, as documented in Section

8.2 of the thesis research database. Combined with the artifact-based verification model wherein learner progress generates auditable portfolio evidence rather than opaque transcript credentials this establishes a contribution toward closing the information asymmetry between job-seeking graduates and the employers evaluating them.

Domain	Validated Contribution	Evidence Source
Theoretical	Personalization constraints encoded as deterministic rules function correctly under test; CALA framework operationally confirmed, not merely proposed	Functional verification — Section 6.3 (40/40 scenarios at final conformance)
Technical	Golden Rule key-isolation architecture confirmed secure under access simulation; LLM provider selection identified as a verified, tested architectural variable	Security verification — Section 6.6; defect remediation — Section 6.7 (BUG-03)
Socioeconomic	Curriculum grounded in 351-record regional labor market dataset; artifact-based competency signaling model implemented end-to-end	Wuzzuf job posting analysis — thesis research database; student_progress and topic_resources schema

Table 8.1: Project Contributions and Their Supporting Evidence

8.3 Reflection on Implementation and Testing Outcomes

The verification activities documented in Chapter 6 produced a quantifiable conformance record: 40 functional scenarios executed across the platform's six modules, of which 36 (90%) passed on first execution and the remaining four reached conformance following targeted defect remediation, yielding a final pass rate of 100%. This outcome warrants interpretation rather than simple restatement. The two defect clusters responsible for the residual non-conformance Google OAuth provider misconfiguration and the Mitchy AI agent pipeline were both integration-boundary failures rather than internal logic defects, suggesting that the platform's core domain logic, including the diagnostic scoring formula and the adaptive content routing engine, was sound from initial implementation and that risk concentrated, as is typical in distributed systems, at the seams between independently developed services.

The observational usability evaluation, conducted with 30 participants drawn from the platform's two primary target archetypes, surfaced a finding with direct implications for the project's theoretical claims about cognitive load and onboarding friction. Participants initially raised concern regarding the length of the five-stage diagnostic sequence an entirely expected reaction given the extraneous load risk Cognitive Load Theory predicts at any system entry point. What is notable is that this concern resolved into acceptance once participants understood the causal link between calibration depth and recommendation accuracy, indicating that the platform's onboarding friction is not an inherent design flaw but a communication gap: the diagnostic depth is justified, but its justification is not yet sufficiently visible within the interface itself. This is a solvable interface problem, not an architectural one, and is reflected accordingly in the future work priorities of Section 8.5.

The gamification system received the most consistently positive reception of any platform component, with participants spontaneously describing the experience as game-like rather than as conventional coursework a directional confirmation of the Self-Determination Theory-informed design rationale established in Chapter 2. Mitchy, evaluated only after the Chapter 6 remediation cycle, was described as helpful and contextually aware, corroborating the technical retest evidence that the context injection and provider migration fixes resolved the agent's prior deficiencies. Taken together, the testing and evaluation evidence indicates a system whose foundational architecture performs as designed, whose most significant defects were infrastructure-boundary issues rather than core logic failures, and whose primary remaining friction point is interface-level transparency rather than underlying mechanism.

8.4 Limitations of the Implemented System

A rigorous evaluation of this project requires explicit acknowledgment of the boundaries within which the implemented system operates. The following constraints are presented not as deficiencies of design intent but as the defined edges of the current implementation phase, each accompanied by the specific evidence that surfaced it.

8.4.1 Content Production Scalability

The polymorphic content strategy requires three instructionally coherent, modality-specific assets for every learning objective, creating a content production multiplier that scales linearly with curriculum breadth rather than benefiting from any automation. The current implementation depends on manual curation and alignment of Open Educational Resources, a process inherently bottlenecked by expert-led verification capacity. This constraint is the principal obstacle to the horizontal domain expansion envisioned in Section 8.5.5, and its resolution likely depends on generative AI-assisted synthesis pipelines capable of producing congruent multi-modal assets from a single authored source [29].

8.4.2 Diagnostic Cold-Start and Profile Reliability

Initial personalization accuracy depends entirely on the truthfulness and self-awareness of the learner's responses to the five-point diagnostic, a dependency that introduces the classical cold-start problem documented in recommender systems literature [16], [29]. A learner who over-reports traits such as conscientiousness during calibration would receive a track assignment and engagement strategy misaligned with their actual behavioral tendencies. Although the Bayesian learning style update mechanism progressively corrects this misalignment using implicit engagement data, the correction window spans several weeks, during which early personalization may underperform its design target.

8.4.3 Mobile-First Pedagogical Boundary

The strategic decision to prioritize mobile delivery, appropriate given regional technology adoption patterns, restricts the platform's capacity to support hands-on technical practice. Data analytics, data engineering, and data science competencies are conventionally developed within full-featured development environments Jupyter Notebooks, SQL editors, version control interfaces none of which translate cleanly to a touch-based mobile surface [61]. The web application client capable of hosting browser-based development tooling was architecturally specified and partially built during this project but was not completed within the available timeline; its absence represents the most concrete and immediately addressable boundary identified in this thesis, distinct from the more open-ended research questions discussed elsewhere in this section.

8.4.4 Absence of Longitudinal Outcome Evidence

The platform's central premise that psychometrically grounded specialization guidance improves learner persistence and graduate employability relative to unguided self-selection has not been tested through longitudinal comparative measurement. The usability evaluation assessed immediate experiential reception across a 30-participant cohort; it did not, and by its observational design could not, measure downstream outcomes such as completion rates, employment placement, or salary trajectory. This represents the most significant unresolved evidential gap separating LearNova's current status as a verified technical prototype from a validated educational intervention, and its resolution requires a study design exceeding the temporal and institutional scope available to an academic capstone project.

Boundary	Description	Path to Resolution
Content production scalability	Manual three-asset curation creates a linear, non-automated content production dependency	Generative AI-assisted multi-modal synthesis from a single authored source (§8.5.3)
Diagnostic cold-start reliability	Self-reported psychometric accuracy not externally validated; multi-week Bayesian correction window before personalization stabilizes	Richer implicit-signal collection during early content interaction to accelerate correction
Mobile-first pedagogical boundary	No support for hands-on coding, IDE interaction, or applied technical practice; designed web client not yet deployed	Completion and deployment of the web client with browser-based IDE integration (§8.5.1)
No longitudinal outcome evidence	Adaptive efficacy claims unverified by completion-rate or employment-outcome comparison	Institutional partnership for a full-cycle cohort study (§8.5.5)

Table 8.2: System Boundaries and Identified Paths to Resolution

8.5 Recommendations for Future Work

The directions presented below are each anchored to a specific source of evidence a limitation surfaced during implementation, a finding from the verification activities of Chapter 6, or a proposal volunteered directly by usability evaluation participants rather than presented as unsubstantiated aspiration. A priority classification accompanies each recommendation, defined as follows: Immediate denotes work that addresses an already-built but undeployed component; High denotes work directly requested by evaluation participants with clear technical feasibility; Medium denotes work that extends an existing but partially implemented subsystem; and Strategic denotes work requiring external institutional partnership beyond the team's current capacity.

8.5.1 Web Application Deployment and Browser-Based Development Environment

The highest-priority future work direction is the completion and deployment of the web application interface, which was architecturally designed and partially developed during this project but not finalized within the available timeline. This priority is doubly justified: it resolves the mobile pedagogical boundary documented in Section 8.4.3, and it directly answers a demand independently raised by multiple usability evaluation participants. The recommended implementation path extends the existing Flutter codebase through the Flutter Web compilation target, preserving a shared Dart codebase and synchronized Supabase state, while introducing a browser-native capability the mobile client cannot provide: an embedded, WebAssembly-compiled development environment for example, a Pyodide-based in-browser Python runtime enabling learners to write, execute, and debug code directly within the LearNova workspace rather than merely reading about programming concepts.

8.5.2 Voice-Based Mitchy Interaction and Hybrid Human Mentorship

Usability participants proposed extending Mitchy's interaction modality beyond text-based chat to spoken conversation, a direction consistent with the platform's broader multimodal philosophy and with research on the pedagogical value of spoken dialogue in tutoring contexts [25]. A voice interface combining text-to-speech response synthesis with automatic speech recognition for student input would lower interaction friction for technical vocabulary, particularly for the platform's Arabic-speaking user base, and would serve auditory and kinesthetic learning preferences that the current chat-only interface addresses only partially.

A related proposal raised during evaluation was the introduction of human instructor touchpoints alongside AI mentorship scheduled expert Q&A sessions, peer tutoring coordinated through Mitchy, or recorded instructor explanations embedded in the chat flow. This hybrid model acknowledges a limitation inherent to any fully autonomous AI tutoring system: however contextually sophisticated, an AI agent cannot fully replicate the adaptive responsiveness of expert human instruction for genuinely novel or ambiguous technical problems, and a structured human escalation path would strengthen the platform's pedagogical ceiling without abandoning the scalability advantages of AI-led mentorship.

8.5.3 Proprietary Content Production

Participants expressed a preference for original, LearNova-authored instructional content over curated third-party materials, a finding consistent with the documented risk that externally sourced Open Educational Resources can introduce terminological and pedagogical inconsistency across

the Visual, Auditory, and Textual variants of a single learning objective [77], [80]. Future development should prioritize building a proprietary content library beginning with the Literacy Phase foundation track authored to a unified instructional specification, ensuring that all three modality representations of a given concept are produced from the same pedagogical source rather than independently sourced and reconciled after the fact.

8.5.4 Federated Guild-Based Peer Learning

The Psychometric Guild architecture described in the platform's original design specification algorithmically formed peer cohorts grouped by complementary behavioral profile was not realized in the current implementation, which provides one-to-one AI mentorship and competitive leaderboard mechanics but no structured many-to-many peer learning framework. Several usability participants independently proposed guild-based competition and collaborative quest mechanics, directly validating this direction. Future work should investigate dynamic guild formation algorithms based on multidimensional profile compatibility, alongside peer-review protocols and reputation-based incentive structures integrated into the existing gamification engine, extending the platform's social constructivist learning capacity beyond its current AI-mediated, individually scoped design [4].

8.5.5 Institutional Partnership and Longitudinal Outcome Validation

The risk analysis infrastructure already implemented within LearNova the daily ML metrics pipeline, sentiment history tracking, and intervention logging constitutes a latent capability for institutional deployment as a student retention analytics service. Adapting this infrastructure into a white-label analytics layer, exposable through secure APIs for integration with university Learning Management Systems such as Moodle, would extend the platform's value proposition into a B2B institutional pathway while simultaneously creating the partnership context required to resolve the evidential gap documented in Section 8.4.4. A cohort study conducted in collaboration with a partner institution, tracking enrolled students through a complete diploma cycle and comparing completion and employment outcomes against a non-adaptive control group, represents the single highest-value research investment available to the project's continuation, as it alone can transform LearNova's current status from a verified technical prototype into an empirically validated educational intervention.

Future Work Direction	Grounding Evidence	Primary Benefit	Priority
Web client deployment + browser-based IDE	§8.4.3 boundary; direct UAT participant request	Closes mobile pedagogical ceiling	Immediate
Voice interface + hybrid human mentorship for Mitchy	UAT participant proposals	Lowers interaction friction; Arabic accessibility	High
Proprietary original content library	UAT participant feedback on OER coherence	Instructional consistency across modalities	High
Federated Psychometric Guild peer learning	UAT guild/squad proposals; unrealized design spec	Extends social constructivist engagement	Medium
Institutional partnership + longitudinal cohort study	§8.4.4 evidential gap	Validates core platform claims empirically	Strategic

Table 8.3: Future Work Directions, Grounding Evidence, and Priority Classification

8.6 Closing Statement

Egypt's graduate unemployment crisis and the broader MENA skills mismatch documented in Chapter 1 are not problems of educational access. Access has expanded; outcomes have not kept pace. LearNova was built on the premise that the missing element is not more content but better-directed content guidance that knows the learner before it teaches them, and support that notices distress before a student disengages.

This thesis has shown that such a system can be built, not merely theorized. The diagnostic engine scores correctly. The content router serves the right format. Mitchy understands context, speaks Arabic, and has learned not to hallucinate. The gamification layer makes a five-stage psychometric assessment feel, to the people who tested it, less like an obstacle and more like a beginning. None of this resolves the deeper labor market crisis that motivated the project, and none of it should be mistaken for a finished product the limitations recorded in Section 8.4 are real, and the longitudinal evidence required to prove LearNova's impact at scale does not yet exist. What does exist is a working, tested architecture and a clear, evidence-grounded path to extend it. That is the contribution of this thesis, and it is offered as a foundation rather than a conclusion in the final sense of the word.

References

References

A. Academic Literature & Learning Theory

1. J. Sweller, "Cognitive load during problem solving: Effects on learning," *Cognitive Science*, vol. 12, no. 2, pp. 257–285, 1988.
2. R. E. Mayer, *Multimedia Learning*, 2nd ed. Cambridge: Cambridge University Press, 2009.
3. A. Paivio, "Dual coding theory: Retrospect and current status," *Canadian Journal of Psychology*, vol. 45, no. 3, pp. 255–287, 1991.
4. L. S. Vygotsky, *Mind in Society: The Development of Higher Psychological Processes*. Cambridge, MA: Harvard University Press, 1978.
5. A. Bandura, "Self-efficacy: Toward a unifying theory of behavioral change," *Psychological Review*, vol. 84, no. 2, pp. 191–215, 1977.
6. X. Wang et al., "Optimizing cognitive load and learning adaptability with adaptive AI systems," *PMC*, 2025. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC11522668/.
7. M. Al-Shehri, "The Effect of Multimedia on Vocabulary Learning and Retention," *ResearchGate*, 2024. [Online]. Available: https://www.researchgate.net/publication/382711394.
8. S. Chen et al., "Extracting typing game keystroke patterns as potential indicators of cognitive load," *Frontiers in Computer Science*, 2024. [Online]. Available: https://www.frontiersin.org/journals/computer-science/articles/10.3389/fcomp.2024.1412458/.
9. Y. Zhang, "Predicting Online Education Dropout: A New Machine Learning Approach," *Springer Open*, 2025. [Online]. Available: https://www.scribd.com/document/884421230/.
10. K. Hamari et al., "How competitive, cooperative, and collaborative gamification impacts student learning," *ResearchGate*, 2024. [Online]. Available: https://www.researchgate.net/publication/394666505.
11. F. Olszewski, "Social gamification elements on user engagement," *TU Wien Repository*, 2025. [Online]. Available: https://repositum.tuwien.at/bitstream/20.500.12708/216475/1/Olszewski.pdf.
12. R. M. Ryan and E. L. Deci, "Self-determination theory and the facilitation of intrinsic motivation," *American Psychologist*, vol. 55, no. 1, pp. 68–78, 2000.
13. Z. Liu, "The rise of adaptive learning in medical education," *iatroX*, 2025. [Online]. Available: https://www.iatrox.com/blog/adaptive-learning-medical-education.

14. P. A. Kirschner, "Challenging Cognitive Load Theory: The Role of Educational Technology," *MDPI*, 2025. [Online]. Available: https://www.mdpi.com/2076-3425/15/2/203.

15. T. Anderson, "Cognitive Load Management Through Adaptive AI learning Systems," *ResearchGate*, 2025. [Online]. Available: https://www.researchgate.net/publication/394680189.

16. S. Johnson, "The effectiveness of multimedia input on vocabulary learning," *Taylor & Francis*, 2022. [Online]. Available: https://www.tandfonline.com/doi/full/10.1080/17501229.2022.2131791.

17. U. Qualio, "How Effective Is Multimodal Learning?," *uQualio Blog*, 2025. [Online]. Available: https://uqualio.com/post/the-future-of-education.

18. H. Pashler et al., "A Learning Style Model Designed for Online Learning Environments," *ResearchGate*, 2018. [Online]. Available: https://www.researchgate.net/publication/325971166.

19. M. Kharb, "A comparative study of the learning styles among medical students," *ResearchGate*, 2016. [Online]. Available: https://www.researchgate.net/publication/309877874.

20. A. Pradhan, "Learning Styles Preferences Among Students," *IJFMR*, 2023. [Online]. Available: https://www.ijfmr.com/papers/2023/6/9015.pdf.

B. Technical Implementation & Algorithms (AI/ML)

21. D. Kundu, A. Mehta, R. Kumar, N. Lal, A. Anand, A. Singh, and R. R. Shah, "Keystroke Dynamics Against Academic Dishonesty in the Age of LLMs," *arXiv*, 2024. [Online]. Available: https://arxiv.org/html/2406.15335v1.

22. A. Mehta, R. Kumar, A. Singla, K. Bisht, Y. K. Singla, and R. R. Shah, "Detecting LLM-Assisted Academic Dishonesty using Keystroke Dynamics," *arXiv*, 2025. [Online]. Available: https://arxiv.org/html/2511.12468v1.

23. M. Brown, "A Review on Smartphone Keystroke Dynamics as a Digital Biomarker," *PMC*, 2023. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC10296416/.

24. Y. Chen, "MOOC Dropout Prediction via a Dilated Convolutional Attention Network," *MDPI*, 2022. [Online]. Available: https://www.mdpi.com/2227-9709/12/4/127.

25. S. Gupta, "Integrating Sentiment Analysis with Learning Analytics," *IJCESEN*, 2023. [Online]. Available: https://www.ijcesen.com/index.php/ijcesen/article/download/781/435/2671.

26. R. Kumar, "Dropout prediction of MOOCs students based on deep learning," *IEEE Xplore*, 2024. [Online]. Available: https://ieeexplore.ieee.org/document/10337466/.

27. X. Li, "MOOCs dropout prediction based on Hidden Markov Model," *Yokohama Publishers*, 2025. [Online]. Available: http://yokohamapublishers.jp/online-p/JNCA/Open/vol25/jncav25n4p879.pdf.

28. A. Patel, "Predicting Student Dropouts in MOOC Platforms Using Machine Learning," *IJPREMS*, 2024. [Online]. Available: https://www.ijprems.com/ijprems-paper/predicting-student-dropouts.

29. D. Wilson, "Machine Learning and Generative AI in Learning Analytics," *MDPI*, 2025. [Online]. Available: https://www.mdpi.com/2076-3417/15/15/8679.

30. K. Lee, "Predicting Student Dropout Risk in Online Learning using Stacked Ensemble," *IJCA*, 2024. [Online]. Available: https://www.ijcaonline.org/archives/volume187/number40/.

C. Global Employment & Skills Reports

31. World Economic Forum, "Future of Jobs Report 2025," *WEF*, Geneva, 2025. [Online]. Available: https://www.weforum.org/reports/future-of-jobs-report-2025/.

32. International Labour Organization (ILO), "Global Employment Trends for Youth 2024: The Arab States," *ILO*, 2024. [Online]. Available: https://www.ilo.org.

33. World Bank Group, "Youth Aspirations and the Reality of Jobs in the MENA Region," *World Bank*, 2023. [Online]. Available: https://documents.worldbank.org.

34. Brookings Institution, "Youth Unemployment in the MENA Region: A Structural Analysis," *Brookings Doha Center*, 2025. [Online]. Available: https://www.brookings.edu.

35. Gallup, "State of the Global Workplace Report 2025," *Gallup*, 2025. [Online]. Available: https://www.gallup.com/workplace/349484/state-of-the-global-workplace.aspx.

36. McKinsey & Company, "The State of AI in 2025: Generative AI's Breakout Year," *McKinsey Global Institute*, 2025.

37. Korn Ferry, "The Global Talent Crunch: Future of Work," *Korn Ferry Insights*, 2025. [Online]. Available: https://www.kornferry.com/insights.

38. BCG, "Alleviating the Heavy Toll of the Global Skills Mismatch," *Boston Consulting Group*, 2020. [Online]. Available: https://web-assets.bcg.com/c6/ab/801d122d4ba3961c9452d66f830e/bcg-alleviating-the-heavy-toll.pdf.

39. Federal Reserve Bank of St. Louis, "The Jobs and Degrees Underemployed College Graduates Have," *St. Louis Fed*, 2025. [Online]. Available: https://www.stlouisfed.org/open-vault/2025/aug/jobs-degrees-underemployed.

40. Strada Institute, "Talent Disrupted: College Graduates, Underemployment, and the Way Forward," *Strada Education*, 2024. [Online]. Available: https://static1.squarespace.com.

41. Lumina Foundation, "2024 Graduate Employability Report," *Lumina Foundation*, 2024. [Online]. Available: https://www.luminafoundation.org.

42. Cengage Group, "Higher Education's Value Shift: 2024 Employability Report," *Cengage*, 2024. [Online]. Available: https://www.prnewswire.com/news-releases/higher-educations-value-shift.

43. HackerRank, "2024 Developer Skills Report," *HackerRank*, 2024. [Online]. Available: https://www.hackerrank.com/research/developer-skills/2024.

44. CodeSignal, "University Ranking Report 2024," *CodeSignal*, 2024. [Online]. Available: https://codesignal.com/university-ranking-report-2024/.

45. Bayt.com, "The Middle East Job Index Survey," *Bayt.com*, Nov. 2025. [Online]. Available: https://www.bayt.com.

D. Gamification & Behavioral Mechanics

46. C. D. Orji, "A Qualitative Investigation of Team-Based Gamified Learning in an Online Environment," *ERIC Institute of Education Sciences*, 2023. [Online]. Available: https://files.eric.ed.gov/fulltext/EJ1338444.pdf.

47. R. M. Ryan, "As it unfolds: Exploring the impact of team-based gamification on performance, confidence, and engagement," *ResearchGate*, 2024. [Online]. Available: https://www.researchgate.net/publication/386301953.

48. A. T. Klock, "A Systematic Review of the Effect of Gamification on Adherence," *Lirias KU Leuven*, 2024. [Online]. Available: https://lirias.kuleuven.be/retrieve/610140/.

49. L. Facey-Shaw, "The impact of gamification in educational settings on student learning outcomes: a meta-analysis," *ResearchGate*, 2020. [Online]. Available: https://www.researchgate.net/publication/343085892.

50. A. Antonaci, "Systematic literature review about gamification in MOOCs," *Taylor & Francis Online*, 2020. [Online]. Available: https://www.tandfonline.com/doi/full/10.1080/02680513.2020.1798221.

51. Trophy.so, "How Streaks Leverages Gamification to Boost Retention: A Case Study," *Trophy Blog*, Jan. 2026. [Online]. Available: https://trophy.so/blog/streaks-gamification-case-study.

52. K. Werbach, "The Contribution of the Gamification Approach to Development," *ResearchGate*, 2025. [Online]. Available: https://www.researchgate.net/publication/394666505.

53. F. Bellotti, "Assessment in and of Serious Games: An Overview," *Advances in Human-Computer Interaction*, vol. 2013, Art. no. 136864, 2013.

E. Competitor Analysis & Benchmarking (Coursera/Pymetrics)

54. Pluralsight, "Pluralsight Acquires Live Mentoring Service HackHands," *Pluralsight Newsroom*, Jan. 2026. [Online]. Available: https://www.pluralsight.com/newsroom/press-releases/.

55. Pluralsight, "Iris: The AI Assistant for Tech Skills," *Pluralsight Product Platform*, Jan. 2026. [Online]. Available: https://www.pluralsight.com/product/iris.

56. Oxford University Careers Service, "The Pymetrics Games: Overview and Practice Guidelines," *Oxford.ac.uk*, Jan. 2026. [Online]. Available: https://www.careers.ox.ac.uk/article/the-pymetrics-games-overview-and-practice-guidelines.

57. Leland, "Blackstone Pymetrics Assessments: Interview Questions & Tips (2025)," *JoinLeland*, Jan. 2026. [Online]. Available: https://www.joinleland.com/library/a/blackstone-pymetrics.

58. Prismocean, "The Free Big Five OCEAN Personality Test," *Prismocean.com*. [Online]. Available: https://prismocean.com/big-five-test. (.

59. BitDegree, "Global EdTech Report 2025," *BitDegree.org*, 2025. [Online]. Available: https://www.bitdegree.org.

60. Reddit, "What pymetrics characteristics do financial institutions look out for?," *r/FinancialCareers*, Jan. 2026. [Online]. Available: https://www.reddit.com/r/FinancialCareers/.

F. Specific Technical Skill Reports (Cloud/AI/DevOps)

61. JetBrains, "The State of Developer Ecosystem 2024: Learning Curve Report," *JetBrains Research*, 2024. [Online]. Available: https://lp.jetbrains.com/cs-learning-curve-report-2024/.

62. Strategic Education Inc., "National Survey Finds Half of U.S. Workers Worried about Gaining Needed Skills," *SEI News*, 2024. [Online]. Available: https://sei.strategiceducation.com/resources/.

63. Federal Reserve Bank of Atlanta, "Measuring Employer Demand for AI Skills by Educational Requirements," *Workforce Currents*, May 2025. [Online]. Available: https://www.atlantafed.org/cweo/workforce-currents/2025/05/21/.

64. University of Rhode Island, "Cloud Computing in Higher Education: How AWS Powers Teaching," *URI ITS*, Dec. 2025. [Online]. Available: https://its.uri.edu/2025/12/05/cloud-computing-in-higher-education/.

65. Refont e-Learning, "Best Kubernetes Training Programs for DevOps Engineers in 2025," *Refont Blog*, 2025. [Online]. Available: https://www.refontelearning.com/blog/best-kubernetes-training-programs/.

66. HR Dive, "AI Jobs and Master's Degrees: The Shifting Requirement," *HR Dive*, 2025. [Online]. Available: https://www.hrdive.com/news/AI-jobs-masters-degrees/720285/.

67. IACIS, "Issues in Information Systems: Cloud Computing Curriculum," *IACIS*, vol. 24, no. 2, pp. 265–274, 2023. [Online]. Available: https://iacis.org/iis/2023/2_iis_2023_265-274.pdf.

68. Software Oasis, "The Decline in Degree Requirements for Software Engineers," *Software Oasis*, 2025. [Online]. Available: https://softwareoasis.com/decline-in-degree-requirements/.

69. UC San Diego, "Cloud Computing Architecture Using Amazon Web Services," *UCSD Extended Studies*, 2025. [Online]. Available: https://extendedstudies.ucsd.edu/courses/cloud-computing-architecture-using-amazon.

G. Economic Validation & Job Market Specifics

70. BestColleges, "Almost Half of Job-Seeking Graduates Regret Their Major," *BestColleges News*, 2025. [Online]. Available: https://www.bestcolleges.com/news/almost-half-job-seeking-graduates-regret-their-major/.

71. Higher Ed Dive, "Americans Believe Their College Major Didn't Prepare Them for the Market," *Higher Ed Dive*, Jan. 2025. [Online]. Available: https://www.highereddive.com/news/americans-believe-their-college-major-didnt-prepare-them-for-market/754144/.

72. CNN Business, "The Disconnect: Job Market Realities for Recent Grads," *CNN Economy*, Jan. 26, 2025. [Online]. Available: https://www.cnn.com/2025/01/26/economy/job-market-recent-grads-colleges.

73. Applauz, "The Real Cost of Employee Turnover in 2025," *Applauz Resources*, 2025. [Online]. Available: https://www.applauz.me/resources/costs-of-employee-turnover.

74. HR Morning, "The Real Cost of Employee Turnover: A 2025 Update," *HR Morning*, 2025. [Online]. Available: https://www.hrmorning.com/articles/real-cost-employee-turnover/.

75. Washington Center for Equitable Growth, "Turnover Costs and the Labor Market," *Equitable Growth Issue Brief*, Dec. 2020. [Online]. Available: https://equitablegrowth.org/wp-content/uploads/2020/12/122120-turnover-costs-ib.pdf.

76. PeopleKeep, "Employee Retention: The Real Cost of Losing an Employee," *PeopleKeep Blog*, 2025. [Online]. Available: https://www.peoplekeep.com/blog/employee-retention-the-real-cost-of-losing-an-employee.

77. Quality Matters, "Research on Video Length and Student Engagement," *QM Research*, 2025. [Online]. Available: https://www.qualitymatters.org/qa-resources/resource-center/articles-resources/research-video-length.

78. Arist, "Dynamic Training vs. Static eLearning: 2025 Retention Benchmarks," *Arist Blog*, 2025. [Online]. Available: https://www.arist.co/post/dynamic-training-vs-static-elearning-2025-retention-engagement-benchmarks.

79. NCBI, "The Impact of Microlearning on Retention," *PubMed Central*, 2020. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC7334715/.

80. ScienceDirect, "Adaptive Learning Systems: A Review of 2023 Architectures," *Computers & Education*, 2023. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0360131523000313.

81. American Society for Engineering Education (ASEE), "The Myth of the Six-Minute Rule in Video Learning," *ASEE Conference Proceedings*, 2024. [Online]. Available: https://www.scribd.com/document/869669254/.

H. Technical Frameworks & Development Standards

82. Google, "Flutter 3.19 Release Notes: Impeller and Graphics Performance," *Flutter.dev*, 2025. [Online]. Available: https://docs.flutter.dev/release/release-notes.

83. Remi Rousselet, "Riverpod: A Reactive Caching and Data-binding Framework," *Riverpod.dev*, v2.4.9. [Online]. Available: https://riverpod.dev.

84. Supabase, "Row Level Security (RLS) Policies," *Supabase Documentation*, 2025. [Online]. Available: https://supabase.com/docs/guides/auth/row-level-security.

85. PostgreSQL Global Development Group, "PostgreSQL 16: JSONB Data Types and Indexing," *PostgreSQL.org*, 2025. [Online]. Available: https://www.postgresql.org/docs/16/datatype-json.html.

86. Material Design, "Material 3: Navigation and Adaptive Layouts," *m3.material.io*, 2025. [Online]. Available: https://m3.material.io/foundations/adaptive-design.

87. Google DeepMind, *Gemini API Documentation*, Google LLC, 2025, available at https://ai.google.dev/docs

88. Unity Technologies, "Unity for Education: XR Development Curriculum," *Unity.com*, 2024. [Online]. Available: https://unity.com/education.

I. Curriculum Design Sources & Skill Validation

89. UC San Diego, "Cloud Computing Architecture Using Amazon Web Services," *UCSD Extended Studies*, 2025. [Online]. Available: https://extendedstudies.ucsd.edu/courses/cloud-computing-architecture-using-amazon.

90. Refont e-Learning, "Best Kubernetes Training Programs for DevOps Engineers in 2025," *Refont Blog*, 2025. [Online]. Available: https://www.refontelearning.com/blog/best-kubernetes-training-programs/.

91. Software Oasis, "The Decline in Degree Requirements for Software Engineers," *Software Oasis*, 2025. [Online]. Available: https://softwareoasis.com/decline-in-degree-requirements/.

92. University of Rhode Island, "Cloud Computing in Higher Education: How AWS Powers Teaching," *URI ITS*, Dec. 5, 2025. [Online]. Available: https://its.uri.edu/2025/12/05/cloud-computing-in-higher-education-how-aws-powers-teaching-learning-and-research/.

93. Federal Reserve Bank of Atlanta, "Measuring Employer Demand for AI Skills by Educational Requirements," *Workforce Currents*, May 21, 2025. [Online]. Available: https://www.atlantafed.org/cweo/workforce-currents/2025/05/21/by-degrees-measuring-employer-demand-for-ai-skills-by-educational-requirements.

94. IACIS, "Issues in Information Systems: Cloud Computing Curriculum," *IACIS*, vol. 24, no. 2, pp. 265–274, 2023. [Online]. Available: https://iacis.org/iis/2023/2_iis_2023_265-274.pdf.

J. Raw Qualitative Data (User Voices)

95. Reddit, "In second year of CS undergrad, can I start Kubernetes?," *r/kubernetes*, 2024. [Online]. Available: https://www.reddit.com/r/kubernetes/comments/12ar8ya/in_second_year_of_cs_undergrad_can_i_start/.

96. Reddit, "What pymetrics characteristics do financial institutions look out for?," *r/FinancialCareers*, 2025. [Online]. Available: https://www.reddit.com/r/FinancialCareers/comments/js9c1y/what_pymetrics_characteristics_do/.

97. Wuzzuf, "Search Query: AI Engineer Jobs in Egypt," *Wuzzuf.net*, Accessed Nov. 1, 2025. [Online]. Available: https://wuzzuf.net/search/jobs/?q=AI+Engineer&a=hpb.

98. Wuzzuf, "Search Query: Tech Jobs in Cairo," *Wuzzuf.net*, Accessed Nov. 1, 2025. [Online]. Available: https://wuzzuf.net/search/jobs/?q=Tech&a=hpb. (.

99. Prismocean, “Big Five Personality Test: Question Bank,” *Prismocean.com*, Accessed Nov. 1, 2025. [Online]. Available: https://prismocean.com/big-five-test.

K. Software Quality, Testing & Usability Standards

100. R. W. Picard, *Affective Computing*. Cambridge, MA: MIT Press, 1997.

101. J. Brooke, “SUS: A ‘Quick and Dirty’ Usability Scale,” in *Usability Evaluation in Industry*, P. W. Jordan, B. Thomas, B. A. Weerdmeester, and A. L. McClelland, Eds. London: Taylor & Francis, 1996, pp. 189–194.

102. International Organization for Standardization, *Systems and Software Engineering — Systems and Software Quality Requirements and Evaluation (SQuaRE) — Product Quality Model*, ISO/IEC Standard 25010:2023. Geneva: ISO, 2023.

103. C. Kaner, J. Bach, and B. Pettichord, *Lessons Learned in Software Testing: A Context-Driven Approach*. New York: John Wiley & Sons, 2002.

104. G. J. Myers, C. Sandler, and T. Badgett, *The Art of Software Testing*, 3rd ed. Hoboken, NJ: John Wiley & Sons, 2011.

105. J. Nielsen, *Usability Engineering*. San Francisco, CA: Morgan Kaufmann Publishers, 1994.